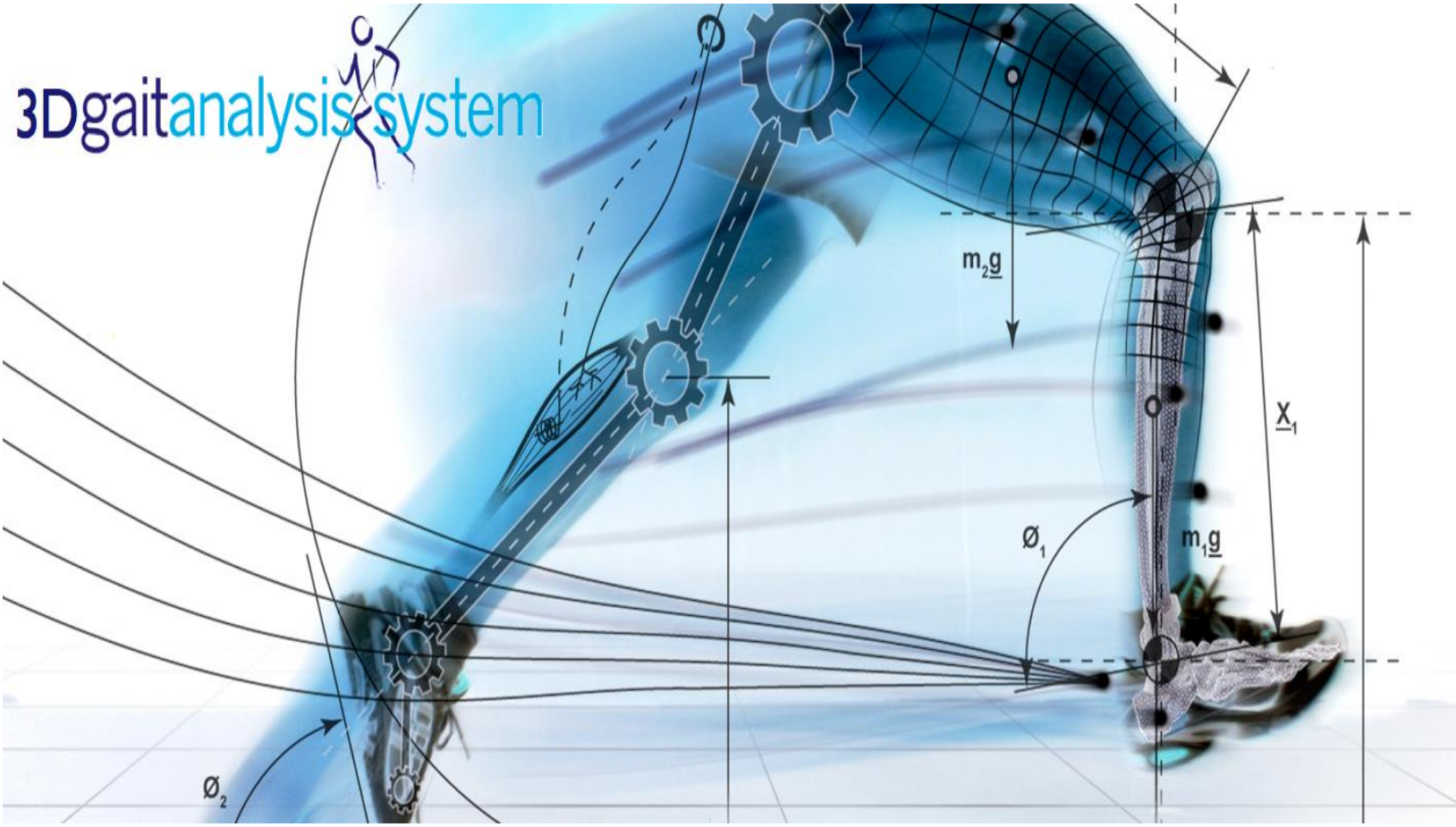
PHYSICS TO MEDICAL SCIENCES

SECOND EDITION



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Preface

For the second edition of Physics to Medical Sciences, most of the material is kept, but omit some sections, we thought that it is hard for the targeted students. We cleaned examples from errors and some examples has been added, and the end of the chapter problems have been increased substantially.

Because of the growing demand on Biophysics book motivated us to write this textbook. This book comes after several years of teaching this course to the students of college of medicine, Pharmacy, medical Sciences at Al-Azhar University-Gaza. We gather our thoughts and notes to come up with this book. The book is intended to cover the basic concepts that the medical students need. We tried to incorporate physics and its application to the human body or to the apparatus that use these physical phenomenon. The book covers a wide variety of topics that are related to the medical students such as biostatic, fluid dynamics, and bioelectricity. Furthermore, the design of this textbook came to accommodate the actual abilities of freshmen students, because of that we didn't extend the scope of the textbook further. However, we would like to introduce our humble work to our students and we would like to hear from our colleagues and student any suggestions and remarks.

Mistakes and transcription errors may be reduced but are unlikely to be eliminated. All problems have been rechecked from the first edition, and many (too many, of course) errors have been corrected.

Any corrections or suggestions for improvements will be appreciated. You can contact me on:

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Chapter 1

Units, Dimensions, and Vectors



Like other sciences, physics is based on experimental observations and quantitative measurements. These observations have described by numbers and units. Numbers give us how large our measurement was, and the units tell us the nature (the flavor) of this measurement. Because of that it is wise to know how to handle these number and units with care. Many of physical observations cannot be described by a value, it also needs a direction. Therefore, we have to furnish our background knowledge about vectors, their properties, and mathematic. Based on these requirements we designed this chapter to include the following sections: section 1.1; we will introduce the units, their prefixes, standard units, and units' conversions, in section 1.2; we'll discuss dimensional analysis and its importance in physics, we need to learn a tool to convert from one system of unit to another, that's what we are going to encounter in section 1.3; we often need to work with physical quantities which have both numerical and directional properties, sections 1.4, 1.5, and 1.6 are devoted to handle vector representation, properties, addition, subtraction, and vector multiplication. Vector quantities are used throughout this text, and it is therefore imperative that you master both their graphical and their algebraic properties.

1.1 Units

In everyday practice we use units to express the nature of things i.e. we go to the market and we ask about certain items how much it costs! How much is this item? We expect the answer in our currency (Palestinian Pounds, Insha'Allah) even though the salesman's didn't mention! However, if there is a listener from other country uses a different currency he would understand something different! Because of that, we should be clear when say numbers. The units come along with numbers to give them a flavor. This flavor gives the listener the nature of thing you are talking about. For example, if you went to the grocery store and asked the salesperson for some cheeses by saying please give me one cheese! The salesperson would understand this phrase in many different ways according to what he used to sell out: it might be one kilogram, or one piece, or whatever. As you can notice mentioning the nature of the numerical quantity is very important.

Standard Units

The laws of physics are expressed in terms of basic quantities that require a clear definition. In mechanics, the three basic quantities are length (L), mass(M), and time(T). All other quantities in mechanics can be expressed in terms of these three. If we are to report the results of a measurement to someone who wishes to reproduce this measurement, a standard must be defined. It would be meaningless if a visitor from another country were to talk to us about a length of 8 "$stadion$" if we do not know the meaning of the unit "$stadion$". On the other hand, if someone familiar with our system of measurement reports that a wall is 2 meters high and our unit of length is defined to be 1 meter, we know that the height of the wall is twice our basic length unit. Likewise, if we are told that a person has a mass of 75 kilograms and our unit of mass is defined to be 1 kilogram, then that person is 75 times as massive as our basic unit. Whatever is chosen as a standard must be readily accessible and possess some property that can be measured reliably: measurements taken by different people in different places must yield almost the same result.

In 1960, an international committee established a set of standards for time, length, mass, and other basic quantities. The system established is an adaptation of the metric system, and it is called the SI system of units. (The abbreviation SI comes from the system's French name "System International.") In this system, the units of length, mass, and time are the meter, kilogram, and second, respectively.

Prefixes

Sometimes the numerical value of our physical quantities is too large or in the contrary is too small, which makes the numbers bothersome to deal with and carry. For example, the distance between Gaza and Cairo is 350000 meter, or the mean radius of earth is 637000000 meter. Look, these numbers are not easy to carry and deal with. So, it is better to use prefixes: which are abbreviations come in front of the units to make them handy. See table 1.1, for the commonly used abbreviations in the field of medicine.

Table 1.1: Prefixes and their abbreviations

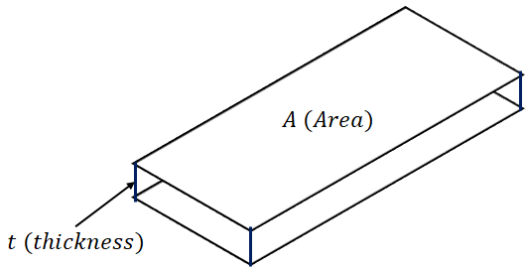
Power	Prefix	Abbreviation	Power	Prefix	Abbreviation
10^{12}	Tera	T	10^{-2}	centi	c
10^{9}	Giga	G	10^{-3}	milli	m
10^{6}	Mega	M	10^{-6}	mirco	μ
10^{3}	kilo	k	10^{-9}	nano	n
10^{-1}	deci	d	10^{-12}	pico	p

Example 1.1

A Painter used 5 liters of yellow pigment to paint certain wall. If his painting thickness on the wall is about 0.1 mm. How many square meters is the painter paint?

Solution

A Paint of a volume V covers an area A with thickness t, as in figure



We know that the volume equals

$$Volume = Area \times thickness$$

Rearrange for the unknown, which is the area A, that is

$$A = \frac{V}{t} = \frac{5000\ cm^3}{0.1\ mm} = \frac{5000 \times (10^{-2}m)^3}{0.1 \times 10^{-3}m} = 50\ m^2$$

Example 1.2

Express a speed of 50 kilometers per hour as meters per second

Solution

$$50\frac{km}{hr} = 50\frac{1000\ m}{60\times60\ sec} = 13.889\frac{m}{sec} \cong 14m/sec$$

Example 1.3

Convert a concentration of $220\ mg/dl$ to $grams/liter$

Solution

The letter m is abbreviation for (10^{-3}), the letter d is the abbreviation for $deci\ (10^{-1})$, and the litter l for liters, so

$$220\frac{mg}{dl} = 220\frac{10^{-3}g}{10^{-1}l} = 2.2\ g/l$$

1.2 Dimensional Analysis

In solving problems in physics, there is a useful and powerful procedure called dimensional analysis. This procedure, which should always be used, will help minimize the need for rote memorization of equations. Dimensional analysis makes use of the fact that dimensions can be treated as algebraic quantities. Mass, Length, and Time are the principal quantities (there are other quantities, i.e. charge, temperature, and amount of substance) have the following notations: $[length] = [L];\ [mass] = [M];\ [time] = [T]$ and all other quantities are derivable from these quantities for instance $[velocity] = \frac{[\text{length}]}{[Time]} = [L][T]^{-1}$ and $[force] = [M][L][T]^{-2}$, other physical quantities are listed in table 1.2. That is, quantities can be added or subtracted only if they have the same dimensions. Furthermore, each term on both sides of an equation must have the same dimensions.

By following these simple rules, you can use dimensional analysis to help determine whether an expression has the correct form or not. The relationship can be correct only if the dimensions are the same on both sides of the equation.

Thus, dimensions and units must be handled consistently in any algebraic calculation. To be added, two quantities must have the same dimensions and units. (Adding a volume and a mass is guaranteed to be wrong.) The factors in a multiplication or division may have different units, and the combined quantity will have units of the product or ratio of the factors. Equations involving physical quantities must have the same dimensions on both sides, and the dimensions must be the correct ones for the quantity calculated.

Verifying dimensional consistency is often called "checking the units, or Sanity Check," and is a powerful technique for uncovering errors in calculations. For purposes of checking consistency, dimensions or units may be considered algebraic quantities. Some examples of this procedure are:

Table 1.2: A list of the physical quantities along with their notation and units

Quantity	Description	Dimension	Units
Velocity	$\boldsymbol{Displacement/Time}$	$[\boldsymbol{v}] = [\boldsymbol{LT^{-1}}]$	$\boldsymbol{m/sec}$
Acceleration	$\boldsymbol{Displacement/Time^2}$	$[\boldsymbol{a}] = [\boldsymbol{LT^{-2}}]$	$\boldsymbol{m/sec^2}$
Momentum	$\boldsymbol{Mass \times velocity}$	$[\boldsymbol{p}] = [\boldsymbol{MLT^{-1}}]$	$\boldsymbol{Kg\ m/sec}$
Force	$\boldsymbol{Momentum/Time}$	$[\boldsymbol{F}] = [\boldsymbol{MLT^{-2}}]$	$\boldsymbol{Newton}$
Pressure	$\boldsymbol{Force/Area}$	$[\boldsymbol{P}] = [\boldsymbol{ML^{-1}T^{-2}}]$	$\boldsymbol{Pascal}$

Example 1.4

Checking dimensions for the famous $(Energy\ (E) - mass\ (m))$ formula $E = mc^2$, where c is the speed of light.

Solution

$$Energy = Mass \times speed^2$$

Energy is given in Joules, thus

$$Energy = Force \times Length = MLT^{-2} \times L = ML^2T^{-2}$$

$$Mass \times speed^2 = M \times (\boldsymbol{LT^{-1}})^{\boldsymbol{2}} = M \times L^2T^{-2} = ML^2T^{-2}$$

Thus, the right hand side is equivalent to left hand side, which implies the equation is dimensionally correct.

Example 1.5

a. Check the following equation check if an equation is correct,

$$x = v\,t^2$$

Where x is the displacement, v is the velocity, and t is the time

Solution

$$[L] = [L][T]^{-1} \times [T]^2 \Rightarrow [L] = [L][T]$$

The equation is dimensionally incorrect

b. Determine the unknown power in the following equation

$$x = v\,t^n$$

Solution

$$[L] = [L][T]^{-1} \times [T]^n \Rightarrow [L][T]^0 = [L][T]^{n-1}$$

Notice that on the left hand side there is no time unites, $T^0 = 1$, so the dimension of time is zero, so,$n - 1 = 0 \rightarrow n = 1$

So the equation $x = v\,t$ is dimensionally correct.

Example 1.6

Suppose we are told that the acceleration a of a particle moving with uniform speed v in a circle of radius r is proportional to some power of r^n, and some power of v^m. How can we determine the values of n and m?

Solution

Since $\boldsymbol{a}$ is proportional to the $\boldsymbol{r^n}$ and to $\boldsymbol{v^m}$, thus we can write

$a \propto r^n v^m$ or $a = kr^n v^m$

Where k is a dimensionless constant of proportionality knowing the dimensions of $a, r,$ and v. The unit of acceleration is $[a] = [L][T]^{-2}$, the unit of velocity is $[v] = [L][T]^{-1}$, and the unit of length is $[r] = [L]$. Apply these in the equation we have

$$[L][T]^{-2} = ([L])^n \left([L][T]^{-1}\right)^m = [L]^n [L]^m [T]^{-m} = [L]^{n+m} [T]^{-m}$$

Equating the powers of the left hand side units to the right hand side units, gives

$n + m = 1,\ and \quad -2 = -m \Rightarrow m = 2$

$Thus,\ \ n + 2 = 1 \Rightarrow n = -1$

So, our equations becomes

$$a = kr^{-1}v^2 = k\frac{v^2}{r}$$

Example 1.7

If the frequency (f) of a simple harmonic motion of a pendulum can be written as,

$$f = m^\alpha g^\beta l^\gamma$$

Here m is the mass of the oscillating pendulum, g is the acceleration due gravity, and l is the length of the pendulum. Find $\alpha, \beta,$ and γ.

Solution

In dimensions the above equations can be rewritten as,

$$[T]^{-1} = [M]^\alpha [LT^{-2}]^\beta [L]^\gamma$$

Expand and collect powers, we have

$$[T^{-1}] = [M]^\alpha [L^{\beta+\gamma}][T^{-2\beta}]$$

Now equate powers of the similar dimensions for the left hand side (LHS) to the right hand side (RHS),

$$[M]^0[L]^0[T^{-1}] = [M]^\alpha[L^{\beta+\gamma}][T^{-2\beta}]$$

We have

$$\alpha = 0, \beta + \gamma = 0, -2\beta = -1$$

Thus,

$$\beta = \frac{1}{2} \; and \; \gamma = -\frac{1}{2}$$

Our equation can be written as

$$f = \sqrt{\frac{l}{g}}$$

Does the frequency depend on the mass of the bob m,

I think it is clear now, the answer is NO

If the mass of the bob is $30\ g$, and l is equal to $39.4\ cm$, take $g = 9.807\ m/s^2$, find the time period of this motion in milliseconds, microseconds, and seconds.

$$f = \sqrt{\frac{g}{l}} = \sqrt{\frac{9.807 m \cdot s^{-2}}{0.394 m}} = \sqrt{24.891} = 4.989\ s^{-1}$$

$$T = \frac{1}{f} = \frac{1}{4.989} = 0.2004 \quad \text{sec},$$

1.3 Units Conversions

Sometimes it is necessary to convert units from one system to another. Conversion factors between the SI units and conventional units of length are as follows: One mile, abbreviated as $1\ mi$, is equal to $1.609\ km$, one feet, abbreviated $1\ ft$, is equal to $0.3048\ m$, and one inch, abbreviated as $1\ in.$, is equal to $0.0254\ m$ or $2.54\ cm$. Units can be treated as algebraic quantities that can cancel each other. For example, suppose we wish to convert $15.0\ in.$ to centimeters. We know that one inch $1\ in.$ is equal to $2.54\ cm$, we can find that

$$15.0\,in. = 15.0\,in.\left(\frac{2.54 cm}{in}\right) = 15.0 \times 2.54\ cm = 38.1\ \text{cm}$$

Example 1.8

A rectangular building lot is $100\ ft$ by $150\ ft$. Determine the area of this lot in m^2.

Solution

We know that: $1\ feet = 1ft = 0.3048\ m$

Thus the area in square meters is

$$A = \left(100\ ft \times 0.3048 \frac{m}{ft}\right) \times \left(150\ ft \times 0.3048 \frac{m}{ft}\right) = 100 \times 150 \times (0.3048)^2\ \ m^2$$

$$A = 1393.5456\ \text{m}^2$$

Example 1.9

Suppose your hair grows at the rate $1/32\ in.$ per day. Find the rate at which it grows in nanometers per second. Since the distance between atoms in a molecule is on the order of $0.1\ nm$, your answer suggests how rapidly layers of atoms are assembled in this protein synthesis.

Solution

Apply the following conversions

$$1\,in. = 2.54\,cm,$$

$$one\ \ day = 24 hour \times 60 \frac{\text{min}}{hour} \times 60 \frac{\text{sec}}{\text{min}} = 86400\,\text{sec},$$

$$1\,\text{m} = 10^9\ \ nm,\ and\ \ \ 1\ \ cm = 10^{-2}\ \ m$$

$$\left(\frac{1}{32}\frac{in}{day}\right) = \frac{1}{32}\frac{in}{day} = \frac{1}{32}\frac{2.54\ cm}{86400\,\text{sec}} = \frac{1}{32}\frac{2.54 \times 10^{-2} m}{86400\ \text{sec}}$$

$$\left(\frac{1}{32}\frac{in}{day}\right) = \frac{1}{32}\frac{2.54 \times 10^{-2} \times 10^9\ \ nm}{86400\ \text{sec}} = 9.19\ \ nm/\text{sec}$$

This means the proteins are assembled at a rate of many layers of atoms each second!

1.4 Vector and Scalar Quantities

Imagine yourself in physic land and you are wandering where the "Physics Supermarket" is, you have no clue on that. So, you start asking about the location of this Supermarket and finally you found Mr. Vector to give you directions. Mr. Vector gave you the following directions: you go in this street then turn to East and walk four buildings blocks and then turn to the west for two building blocks you will find "Physics Supermarket". You followed his instructions and you are there at the front door of "Physics Supermarket". You are about to come in and thinking what are about to buy from there. Anyhow, you are in "Physics Supermarket" and started to look around, then the Salesperson "Mr. Scalar" approaches you and ask you if you want any help! You told him you want to know what the volume of a sphere is, and what the length of this meter stick. From the above example, we know that some physical quantities are scalar quantities whereas others are vector quantities.

A scalar quantity is specified by a single value with an appropriate unit and has no direction

From Mr. Vector we know that to reach somewhere you need to have both direction and magnitude to the assigned destination.

A vector quantity has both magnitude, direction, and units

Another example of a vector quantity is displacement. Suppose a particle moves from some point A to some point B along a straight path, as shown in Figure 1.1. We represent this displacement by drawing an arrow from A to B, with the tip of the arrow pointing away from the starting point. The direction of the arrowhead represents the direction of the displacement, and the length of the arrow represents the magnitude of the displacement. If you walk from the tent (location A) to the hole (location B), as shown in Figure 1.1a, the vector $\vec{D}$ (or in bold representation $\boldsymbol{D}$), representing your displacement, is drawn as the arrow that originates at point A and ends at point B. The arrowhead marks the end of the vector. The direction of the displacement vector $\vec{D}$ is the direction of the arrow. The length of the arrow represents the magnitude D of vector $\vec{D}$. If you travel along some other path from A to B, such as the broken line in Figure 1.1, its displacement is still the arrow drawn from A to B. Otherwise, you're started at the location A and while going there you stopped for a while at the location C, Figure

1.1b, and then you continue your journey to B, still your displacement is given by the vector $\vec{D}$.

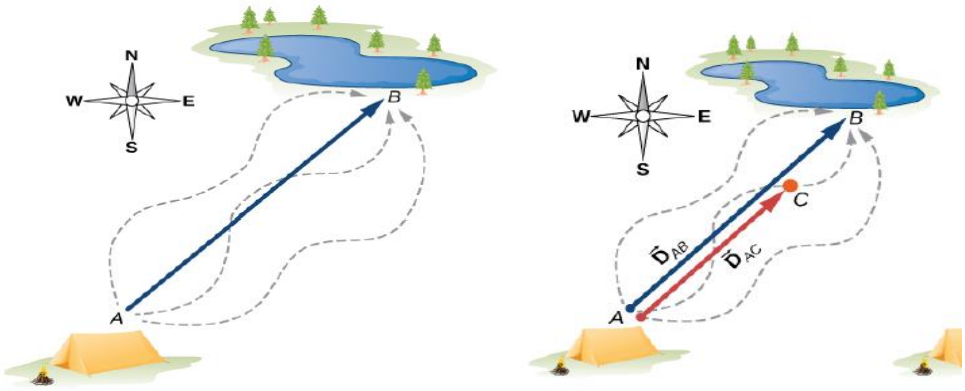


Figure 1.1: a. The displacement vector from point A (the initial position at the campsite) to point B (the final position at the fishing hole) is indicated by an arrow with origin at point A and end at point B. The displacement is the same for any of the actual paths (dashed curves) that may be taken between points A and B. **b**. Stopping to rest at point C while walking from camp (point A) to the pond (point B).

The term vector is used by scientists to indicate a quantity (such as displacement or velocity or force) that has both magnitude and direction. A vector is often represented by an arrow or a directed line segment. The length of the arrow represents the magnitude of the vector and the arrow points in the direction of the vector. We denote a vector by printing a letter in boldface ($\boldsymbol{v}$) or by putting an arrow above the letter $\vec{v}$.

1.5 Properties of Vectors

A. Equality of Two Vectors

For instance, suppose a particle moves along a line segment from point $\boldsymbol{M}$ to point $\boldsymbol{N}$. The corresponding displacement vector $\boldsymbol{A}$, shown in Figure 1.2, has initial point $\boldsymbol{M}$ (the tail) and terminal point $\boldsymbol{N}$ (the tip) and we indicate this by writing $\boldsymbol{A} = \overrightarrow{MN}$. Notice that the vector $\boldsymbol{B} = \overrightarrow{LK}$ has the same length and the same direction as even though it is in a different position. We say that $\boldsymbol{A}$ and $\boldsymbol{B}$ are equivalent (or equal) and we write $\boldsymbol{A} = \boldsymbol{B}$.

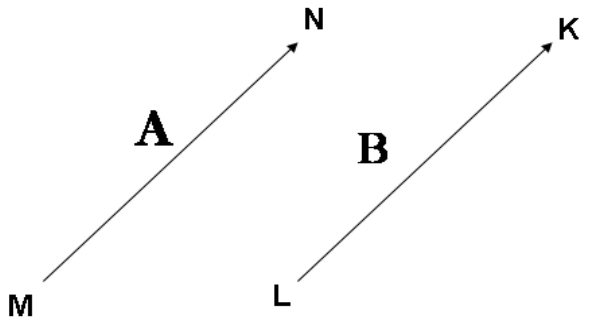


Figure 1.2: Equivalent (equal) vectors

B. Negative of a Vector

The negative of the vector A is defined as the vector that when added to A gives zero for the vector sum. That is, $\boldsymbol{A} + (-\boldsymbol{A}) = \boldsymbol{0}$. The vectors $\boldsymbol{A}$ and $-\boldsymbol{A}$ have the same magnitude but point in opposite directions, figure 1.3. The zero vectors, denoted by $\boldsymbol{0}$, have length 0. It is the only vector with no specific direction.

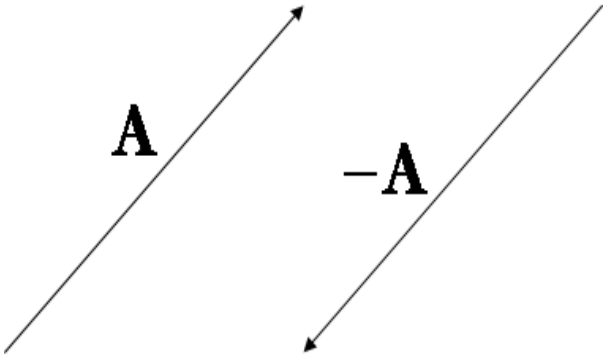


Figure 1.3: Vector $\boldsymbol{A}$ and the negative vector $-\boldsymbol{A}$

C. Multiplying a Vector by a Scalar

If vector $\boldsymbol{A}$ is multiplied by a positive scalar quantity m, then the product $m\boldsymbol{A}$ is a vector that has the same direction as $\boldsymbol{A}$ and magnitude $m\boldsymbol{A}$. If vector $\boldsymbol{A}$ is multiplied by a negative scalar quantity $-m$, then the product $-m\boldsymbol{A}$ is directed opposite $\boldsymbol{A}$. For example, the vector $\boldsymbol{5A}$ is five times as long as $\boldsymbol{A}$ and points in the same direction as $\boldsymbol{A}$; the vector $-\frac{1}{3}\boldsymbol{A}$ is one-third the length of $\boldsymbol{A}$ and points in the direction opposite $\boldsymbol{A}$.

D. Adding Vectors

Graphical Method (Geometrical method)

To add vector $\boldsymbol{B}$ to vector $\boldsymbol{A}$, first draw vector A, with its magnitude represented by a convenient scale, on graph paper and then draw vector $\boldsymbol{B}$ to the same scale with its tail starting from the tip of $\boldsymbol{A}$, as shown in Figure 1.4a. First translate the vector $\boldsymbol{B}$ until the tail of the vector $\boldsymbol{B}$ touches the head of the vector $\boldsymbol{A}$, figure 1.4b. The resultant vector $\boldsymbol{R} = \boldsymbol{A} + \boldsymbol{B}$ is the vector drawn from the tail of $\boldsymbol{A}$ to the tip of $\boldsymbol{B}$. This procedure is known as the triangle method of addition.

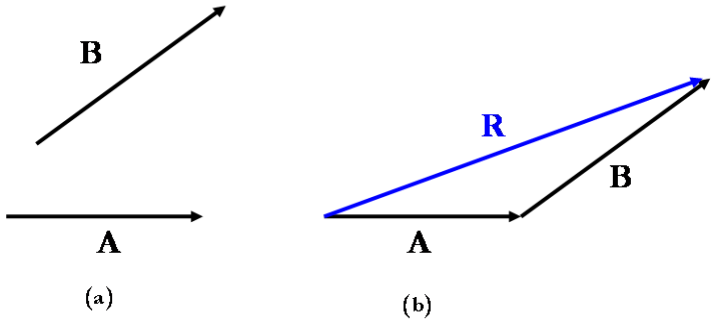


Figure 1.4: The vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ need to be added, b. the translations of the vector $\boldsymbol{B}$ to touch the vector $\boldsymbol{A}$, then the resultant vector $\boldsymbol{R}$ is the vector that runs from the tail of $\boldsymbol{A}$ to the tip of $\boldsymbol{B}$.

When two vectors are added, the sum is independent of the order of the addition. This can be seen from the geometric construction in Figure 1.5 and is known as the commutative law of addition:

$$\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A} \tag{1.1}$$

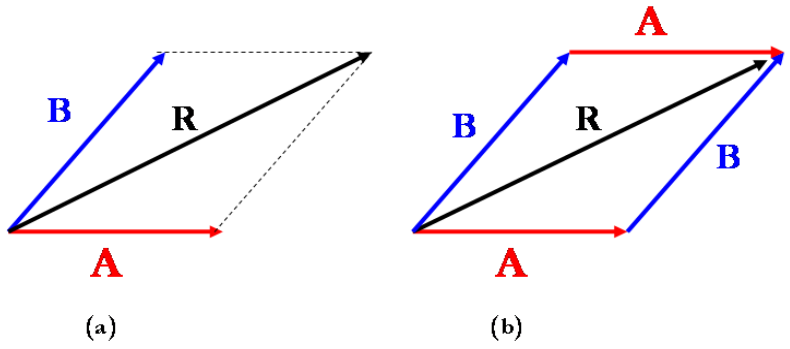


Figure 1.5: (a) In this construction, the resultant $\boldsymbol{R}$ is the diagonal of a parallelogram having sides $\boldsymbol{A}$ and $\boldsymbol{B}$. (b) this construction shows that $\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$, in other words, that vector addition is commutative.

D. Subtracting Vectors

The operation of vector subtraction makes use of the definition of the negative of a vector. We define the operation A-B as vector -B added to vector A:

$$\boldsymbol{A - B = A + (-B)} \tag{1.2}$$

The geometric construction for subtracting two vectors in this way is illustrated in Figure 1.6a. Another way of looking at vector subtraction is to note that the difference $\boldsymbol{A - B}$ between two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ is what you have to add to the second vector to obtain the first. In this case, the vector $\boldsymbol{A - B}$ points from the tip of the second vector to the tip of the first, as Figure 1.6b shows.

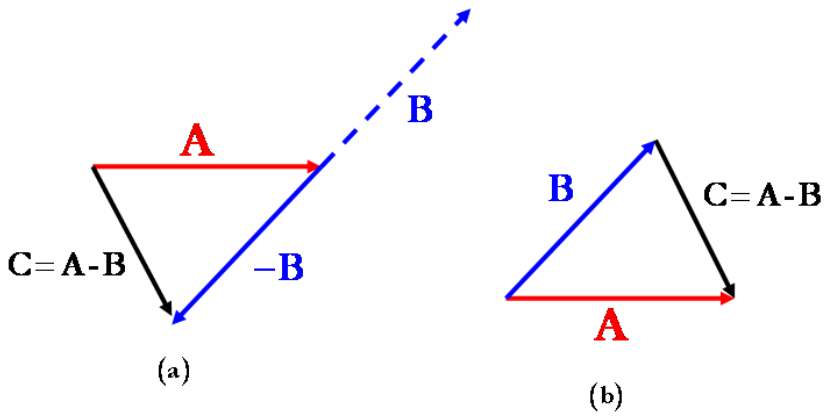
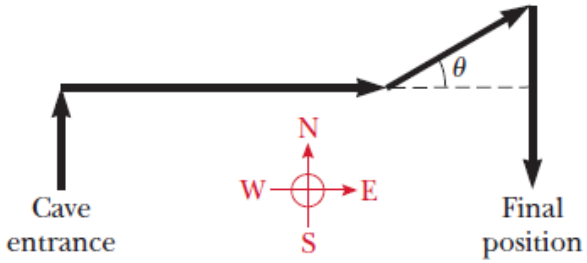


Figure 1.6: (a) This construction shows how to subtract vector $\boldsymbol{B}$ from vector $\boldsymbol{A}$. (b) A second way of looking at vector subtraction. The difference vector $\boldsymbol{C = A - B}$ is the vector that we must add to $\boldsymbol{B}$ to obtain $\boldsymbol{A}$.

Example 1.10

While exploring a cave, a spelunker starts at the entrance and moves the following distances in a horizontal plane. She goes $75.0\,m$ north, $250\,m$ east, $125\,m$ at an angle $\theta = 30.0°$ north of east, and $150\,m$ south. Find her resultant displacement from the cave entrance. Example figure suggests the situation but is not drawn to scale



Solution

Let $+x$ be East and $+y$ be North. We can sum the total x and y displacements of the spelunker as

$$\Sigma x = 250\,m + (125\,m)cos\,30° = 358\,m$$
$$\Sigma y = 75\,m + (125\,m)sin\,30° - 150\,m = -12.5\,m$$

The total displacement is then

$$d = \sqrt{(\Sigma x)^2 + (\Sigma y)^2} = \sqrt{(358\,m)^2 + (-12.5\,m)^2} = 358.218\,m$$

at an angle of

$$\theta = tan^{-1}\left(\frac{\Sigma y}{\Sigma x}\right) = tan^{-1}\left(\frac{-12.5}{358}\right) = -2.00^{\circ}$$

Or

$$\vec{d} = 358.218\,m\ at\ -2.00^{\circ}\ south\ of\ East$$

Algebraic Method

Handling vectors with graphical method could bothersome especially if we are going to deal with a significant number of vectors. So, the geometric method of adding vectors are not recommended whenever great accuracy is required or in three-dimensional problems. So, it would be more convenient if we could represent vectors algebraically and then deal with them on algebraic basis. In this section, we describe a method of adding vectors that makes use of the projections of vectors along coordinate axes. These projections are called the components of the vector. Any vector can be completely described by its components. Consider a vector

$\boldsymbol{A}$ lying in the xy plane and making an arbitrary angle θ with the positive x axis, as shown in Figure 1.7.

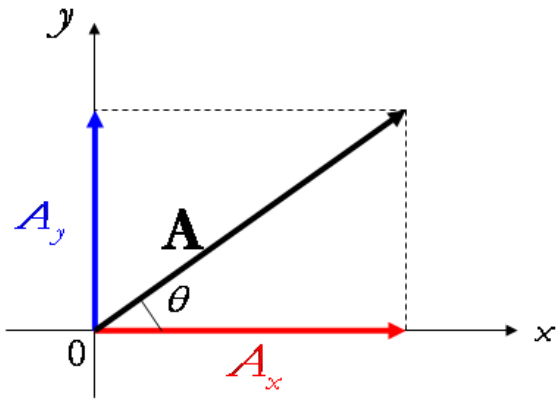


Figure 1.7: Position vector in x-y plane

This vector can be expressed as the sum of two other vectors $\boldsymbol{A_x}$ and $\boldsymbol{A_y}$. From Figure 1.7, we see that the three vectors form a right triangle and that $= \boldsymbol{A_x} + \boldsymbol{A_y}$. We shall often refer to the "components of a vector $\boldsymbol{A}$," written $\boldsymbol{A_x}$ and A_y (without the boldface notation). The component A_x represents the projection of $\boldsymbol{A}$ along the x axis, and the component Ay represents the projection of A along the y axis. These components can be positive or negative. The component A_x is positive if $\boldsymbol{A_x}$ "points" in the positive x direction and is negative if $\boldsymbol{A_x}$ "points" in the negative x direction. The same is true for the component A_y. From Figure 1.7 and the definition of sine and cosine, we see that

$\cos(\theta) = A_x / A$ and that $\sin(\theta) = A_y / A$, note that $A = |\boldsymbol{A}|$.

Hence, the components of A are

$$A_x = A\cos(\theta) \tag{1.3}$$

$$A_y = A\sin(\theta) \tag{1.4}$$

These components form two sides of a right triangle with a hypotenuse of length A. Thus, it follows that the magnitude and direction of $\boldsymbol{A}$ are related to its components through the expressions

$$A = \sqrt{A_x^2 + A_y^2} \tag{1.5}$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right) \tag{1.6}$$

Note that the signs of the components Ax and Ay depend on the angle θ . For example, if $\theta = 120°$, then A_x is negative and A_y is positive. If $\theta = 225°$, then both A_x and A_y are negative. Figure 1.8 summarizes the signs of the components when $\boldsymbol{A}$ lies in the various quadrants. When solving problems, you can specify a vector $\boldsymbol{A}$ either with its components A_x and A_y or with its magnitude and direction $\boldsymbol{A}$ and θ .

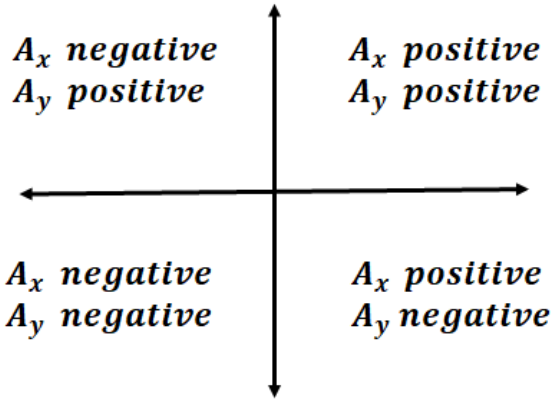


Figure 1.8: The signs of the components of a vector A depend on the quadrant in which the vector is located.

1.6 Unit Vectors

Assume that you have visited certain city and you are walking in its street: the only thing keep you tracking where you are is the street's names. These names it tells you are going south or west or east or anything in between. Also, in vectors we need tags that help us identifying our existence on the axis: is it on the x-axis or the y-axis or the z-axis if we are talking about three dimensional space. So it is useful to express vectors in terms of unit vectors.

A unit vector is a dimensionless unitless vector having a magnitude of exactly 1.

Unit vectors are used to specify a given direction and have no other physical significance. They are used solely as a convenience in describing a direction in space. We shall use the symbols

$i, j,$ and k to represent unit vectors pointing in the positive $x, y,$ and z directions, respectively. The unit vectors $i, j,$ and k form a set of mutually perpendicular vectors in a right-handed coordinate system, figure 1.9a. The magnitude of each unit vector equals 1; that is

$$|i| = |j| = |k| = 1 \tag{1.7}$$

Consider a vector $\boldsymbol{A}$ lying in the xy plane, as shown in Figure 1.9b. The product of the component Ax and the unit vector i is the vector $Ax\ i$, which lies on the x axis and has magnitude $|A_x|$ (The vector $Ax\ i$ is an alternative representation of vector $\boldsymbol{A_x}$.) Likewise, $Ay\ j$ is a vector of magnitude $|A_y|$ lying on the y axis (Again, vector $Ay\ j$ is an alternative representation of vector $\boldsymbol{A_y}$.). Thus, the unit–vector notation for the vector **A** is

$$\vec{A} = A_x i + A_y j \tag{1.8}$$

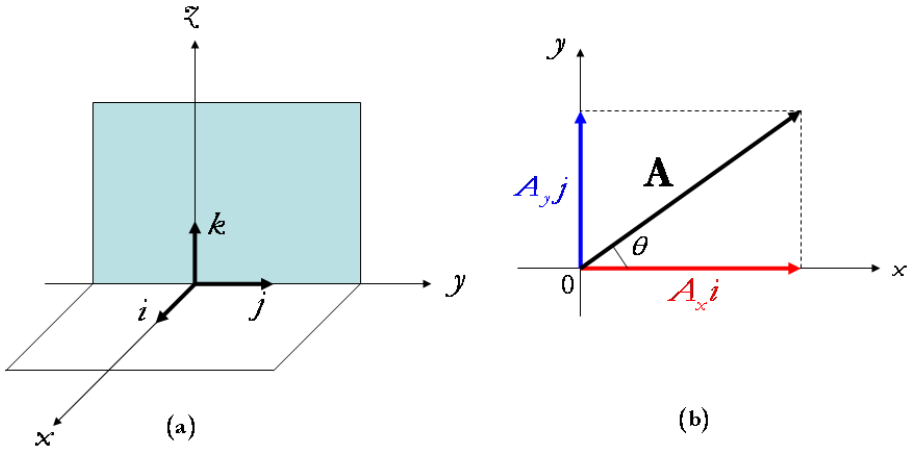


Figure 1.9: (a) The unit vectors $i, j,$ and k are directed along the $x, y,$ and z axes, respectively. (b) Vector $\vec{A} = A_x i + A_y j$ lying in the xy plane has components A_x and A_y.

Now let us see how to use components to add vectors when the geometric method is not sufficiently accurate. Suppose we wish to add vector **B** to vector **A**, where vector **B** has components B_x and B_y. All we do is add the x and y components separately. The resultant vector $\boldsymbol{R} = \boldsymbol{A} + \boldsymbol{B}$ is therefore or

$$\boldsymbol{R} = (A_x i + A_y j) + (B_x i + B_y j)$$

Or,

$$\boldsymbol{R} = (A_x + B_x)i + \big(A_y + B_y\big)j$$

Because $\boldsymbol{R} = R_x i + R_y j$, we see that the components of the resultant vector are

$$R_x = (A_x + B_x)$$

$$R_y = \big(A_y + B_y\big)$$

We can check this addition by components with a geometric construction, as shown in Figure 1.10. Remember that you must note the signs of the components when using either the algebraic or the geometric method.

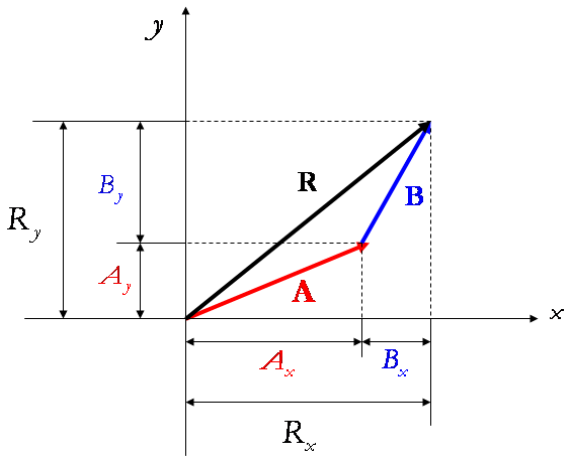


Figure 1.10: This geometric construction for the sum of two vectors shows the relationship between the components of the resultant R and the components of the individual vectors.

In general, if $A \neq 0$, then the unit vector that has the same direction as $\boldsymbol{A}$ is

$$\widehat{\boldsymbol{A}} = \frac{\boldsymbol{A}}{|\boldsymbol{A}|} \qquad (1.9.a)$$

Or in more compact form

$$\widehat{\boldsymbol{A}} = \frac{\vec{A}}{A} \qquad (1.9b)$$

Let us conform on the following: the vector can be assigned in either notations $\boldsymbol{A} \equiv \vec{A}$, and the magnitude of the vector also can be assigned in either notations $|\boldsymbol{A}| \equiv A$.

There is one more piece of notation we shall use when writing vectors. If **A** is any vector, we shall write $\hat{\mathbf{A}}$ to represent a unit vector in the direction of A. This notation gives us another way of writing the vector $\boldsymbol{A}$: we can write it as $A\widehat{A}$, so that it is the length a multiplied by the unit vector $\hat{A}$. A unit vector in the direction of the vector $\boldsymbol{A}$ is written as $\widehat{A}$, so that

$$\vec{A} = |A|\hat{A} \qquad (1.10)$$

Example 1.11

Find the unit vector in the direction of the vector $A = 2i - j - 2k$

Solution

$$\hat{A} = \frac{\boldsymbol{A}}{A} = \frac{2i - j - 2k}{\sqrt{4+1+4}} = \frac{1}{\sqrt{3}}(2i - j - 2k)$$

Example 1.12

Find the sum of two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ lying in the xy plane and given by

$$\boldsymbol{A} = 2.0\,i + 2.0\,j\ m, and\ \ \boldsymbol{B} = 2.0\,i - 4.0\,j\ \ m$$

Then find the magnitude and direction of the vector sum

Solution

First of all, notice the units of each vector "they are in meters" if they are not in the same units you have to convert one to another units to make units consistent before adding!

Adding to two or many vectors is so simple: add all x-components, then all y-components, then all z −components if there is, then you have the resultant vector.

Let name the resultant vector $\boldsymbol{R}$, so

$R_x = \sum_x (2.0 + 2.0) = 4.0\ \ m$ is the sum of all x-components

$R_y = \sum_y (2.0 - 4.0) = -2.0\ m$ is the sum of all y-components

We know that the resultant vector is given as,

$$\boldsymbol{R} = R_x i + R_y j$$

Thus,

$$\boldsymbol{R} = 4.0\ i - 2.0\ j \qquad m$$

The magnitude and direction of the vector $\boldsymbol{R}$ are given respectively as

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{16 + 4} = \sqrt{20}\ m$$

$$\theta = tan^{-1}\left(\frac{R_y}{R_x}\right) = tan^{-1}\left(\frac{-2}{4}\right) = -26.57^{\circ}$$

The resultant Vector is below the $x - axis$ by 27 degrees (clockwise rotation).

Example 1.13

A particle undergoes three consecutive displacements:

$d_1 = 15i + 30j + 12k\ cm$, $d_2 = 23i - 14j - 5k\ \ cm$, and $d_3 = 0.13i - 0.15j \quad m$

Find the components of the resultant displacement and its magnitude.

Solution

Inspecting the three vectors we found that the first two displacement vectors are in centimeters and the third vector is in meters. First of all we need to convert the third vector to be in centimeters (it is easier to convert one vector than two). That is $d_3 = 0.13i - 0.15j \quad m$

Now the resultant displacement vector is (like the previous example)

$$D_x = \sum_x (15 + 23 + 13) = 51 \ \ cm$$

$$D_y = \sum_y \big(30 + (-14) + (-15)\big) = 1 \ \ cm$$

$$D_z = \sum_z (12 + (-5) + 0) = 7 \ \ cm$$

$$\boldsymbol{D} = 51i + j + 7k$$

$$D = \sqrt{51^2 + 1 + 49} = 51.49\ cm$$

Example 1.14

Consider two vectors $A = 3i - 2j$ and $B = -i - 4j$.

Calculate: $\boldsymbol{A + B, A - B, |A + B|, |A - B|}$, and the direction of $\boldsymbol{A + B}$ and $\boldsymbol{A - B}$,

Solution

$$\boldsymbol{A + B = \big(3 + (-1)\big)i + \big((-2) + (-4)\big)j = 2i - 6j}$$

At the beginning find $\boldsymbol{-B}$ then add it to $\boldsymbol{A}$, that is

$$-B = i + 4j$$

Then,

$$\boldsymbol{A + (-B) = (3 + 1)i + \big((-2) + 4\big)j = 4i + 2j}$$

Another method:

$$\boldsymbol{A} = 3i - 2j$$

$$\boldsymbol{-B} = +i + 4j$$

$$\overline{\boldsymbol{A - B} = \ 4i + 2j}$$

It is like any algebraic addition operation.

$$|\boldsymbol{A + B}| = \sqrt{4 + 36} = \sqrt{40}$$

$$|\boldsymbol{A - B}| = \sqrt{16 + 4} = \sqrt{20}$$

The directions are

$$\theta_1 = tan^{-1}\left(\frac{-6}{2}\right) = -71.57^{\circ}$$

$$\theta_2 = tan^{-1}\left(\frac{2}{4}\right) = 26.57^{\circ}$$

Example 1.15

A displacement vector lying in the xy plane has a magnitude of $50.0\ m$ and is directed at an angle of $120°$ to the positive x axis. Find the x and y components of this vector and express the vector in unit vector notation.

Solution

All we have is the following: $|A| = 50.0\ m$, and $\theta = 120°$ to the positive x-axis. But we know that the vector have two components namely: A_x and A_y which are related to vector magnitude $|A|$ and the inclination angle θ by

$$A_x = |A|cos(\theta),$$

And

$$A_y = |A|sin(\theta),$$

Thus,

$$A_x = 50.0\cos(120°) = -25.0\ m$$

$$A_y = 50.0\ sin(120°) = 43.3\ m$$

The vector $\boldsymbol{A}$ can be written as

$$A = A_x i + A_y j$$

$$A = -25.0\ i + 43.3\ j \quad m$$

1.6 Multiplication of Vectors

Like scalars, vectors of different kinds can be multiplied by one another to generate quantities of new physical dimensions. Because vectors have direction as well as magnitude, the vector multiplication cannot follow exactly the same rules as the algebraic rules of scalar multiplication. We must establish new rules of multiplication for vectors.

Scalar Product (Dot Product)

One of the ways in which two vectors can be combined is known as the scalar product. When we calculate the scalar product of two vectors the result, as the name suggests is a scalar, rather than a vector.

In this unit you will learn how to calculate the scalar product and meet some geometrical applications.

Study the two vectors **a** and **b** drawn in Figure 1.11. Note that we have drawn the two vectors so that their tails are at the same point. The angle between the two vectors has been labeled θ .

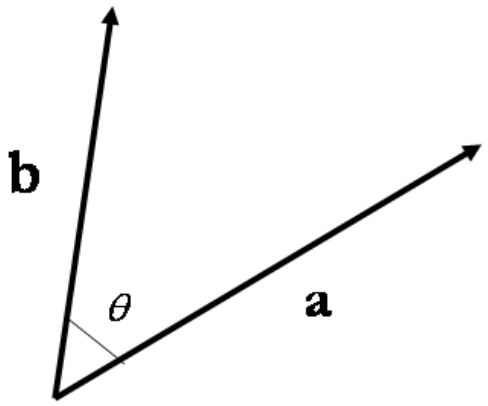


Figure 1.11: Two vectors, a and b, drawn so that the angle between them is θ.

The scalar product of **a** and **b** is defined to be

$$\boldsymbol{a} \cdot \boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}| \cos(\theta) \tag{1.11}$$

where,

$|\boldsymbol{a}|$is the modulus, or magnitude of $\boldsymbol{a}$,

$|\boldsymbol{b}|$ is the modulus of $\boldsymbol{b}$, and

θ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$.

Properties of the Scalar Product

1. The scalar product is commutative

$$\begin{cases} \boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|cos(\theta) \\ \boldsymbol{b}\cdot\boldsymbol{a} = |\boldsymbol{b}||\boldsymbol{a}|cos(\theta) \\ \boldsymbol{a}\cdot\boldsymbol{b} = \boldsymbol{b}\cdot\boldsymbol{a} \end{cases} \tag{1.12}$$

2. The scalar product is distributive over addition

$$\begin{cases} \boldsymbol{a}\cdot(\boldsymbol{b}+\boldsymbol{c}) = \boldsymbol{a}\cdot\boldsymbol{b} + \boldsymbol{a}\cdot\boldsymbol{c} \\ or \\ (\boldsymbol{b}+\boldsymbol{c})\cdot\boldsymbol{a} = \boldsymbol{b}\cdot\boldsymbol{a} + \boldsymbol{c}\cdot\boldsymbol{a} \end{cases} \tag{1.13}$$

3. For two perpendicular vectors

$$\begin{cases} \boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|cos(\theta) = |\boldsymbol{a}||\boldsymbol{b}|cos\left(\frac{\pi}{2}\right) = 0 \\ \boldsymbol{a}\cdot\boldsymbol{b} = 0 \end{cases} \tag{1.14}$$

The scalar product of two vectors given in Cartesian form

We now consider how to find the scalar product of two vectors when these vectors are given in Cartesian form,

$$\boldsymbol{A} = A_x i + A_y j + A_z k \text{ and } \boldsymbol{B} = B_x i + B_y j + B_z k$$

where $i,\ j$ and k are unit vectors in the directions of the x, y and z axes respectively.

Suppose you have two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ as follow:

$$\boldsymbol{A}\cdot\boldsymbol{B} = \left(A_x i + A_y j + A_z k\right)\cdot\left(B_x i + B_y j + B_z k\right)$$

To find the outcome of this scalar product we need to find the following set of unit vectors scalar product: $i\cdot i,\ i\cdot j,\ i\cdot k$ and similar sequences for j and k.

To find these scalar products of these unit vectors all we need just one formula that is:

$$\boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|cos(\theta)$$

$$\boldsymbol{i}\cdot\boldsymbol{i} = |\mathbf{1}||\mathbf{1}|cos(0) = 1$$

$$\boldsymbol{i}\cdot\boldsymbol{j} = |\mathbf{1}||\mathbf{1}|cos\left(\frac{\pi}{2}\right) = 0$$

$$\boldsymbol{i}\cdot\boldsymbol{k} = |\mathbf{1}||\mathbf{1}|cos\left(\frac{\pi}{2}\right) = 0$$

Thus, we can conclude the following:

$$\left.\begin{matrix} i\cdot i = j\cdot j = k\cdot k = 1 \\ i\cdot j = i\cdot k = j\cdot k = 0 \end{matrix}\right\} \tag{1.15}$$

The scalar product of the vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ can be written as

$$\boldsymbol{A}\cdot\boldsymbol{B} = A_xB_x + A_yB_y + A_zB_z \tag{1.16}$$

The angle, θ, between the two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ can be found from

$$\boldsymbol{A}\cdot\boldsymbol{B} = A \times B \times cos(\theta)$$

$$\theta = cos^{-1}\left(\frac{\boldsymbol{A}\cdot\boldsymbol{B}}{|\boldsymbol{A}||\boldsymbol{B}|}\right) \tag{1.17}$$

Example 1.16

Consider the two vectors **a** and **b**. Suppose **a** has modulus 4 units, **b** has modulus 5 units, and the angle between them is 60°. Find the scalar product of the two vectors.

Solution

$$\boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|cos(\theta)$$

$$\boldsymbol{a}\cdot\boldsymbol{b} = 4 \times 5 \times cos(60) = 10\ units$$

Example 1.17

Show that the $\mathbf{a}\cdot\mathbf{b} = \mathbf{b}\cdot\mathbf{a}$ using the following vectors:

$$\boldsymbol{a} = 3i - 2j + 7k \text{ and } \boldsymbol{b} = -5i + 4j - 3k$$

Solution

$$\mathbf{a}\cdot\mathbf{b}=(3\times-5)+(-2\times4)+(7\times-3)=-15-8-21=-44$$
$$\mathbf{b}\cdot\mathbf{a}=(-5\times3)+(4\times-2)+(-3\times7)=-15-8-21=-44$$

Example 1.18

Suppose we wish to test whether or not the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ are perpendicular, where, $\boldsymbol{a} = 3i + 2j - k$ and $\boldsymbol{b} = i - 2j - k$

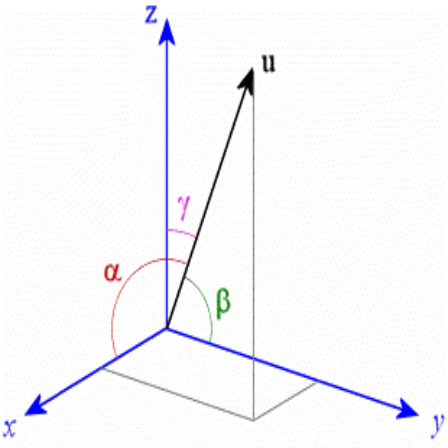
Solution

$$\mathbf{a}\cdot\mathbf{b}=(3\times1)+(2\times-2)+(-1\times-1)=3-4+1=0$$

Which means that the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ are perpendicular.

Example 1.19

For the vector $\boldsymbol{u} = 3i + 2j + 5k$ find the inclination angles (α, β, γ), where α is the angle between $\boldsymbol{u}$ and the *x*-axis, β is the angle between $\boldsymbol{u}$ and the *y*-axis and γ is the angle between **u** and the *z*-axis, as shown in figure.



Solution

We use dot product properties to find the angles,

With x-axis

$$\boldsymbol{u}\cdot\boldsymbol{i} = (3i + 2j + 5k)\cdot i = 3$$

$$\boldsymbol{u}\cdot\boldsymbol{i} = |\boldsymbol{u}||\mathbf{1}|cos(\alpha)$$

Thus,

$$\alpha = cos^{-1}\left(\frac{\boldsymbol{u}\cdot\boldsymbol{i}}{|\boldsymbol{u}||\mathbf{1}|}\right) = cos^{-1}\left(\frac{\mathbf{3}}{\sqrt{9+4+25}}\right) = 60.88^{\circ}$$

With y-axis

$$\boldsymbol{u}\cdot\boldsymbol{j} = (3i + 2j + 5k)\cdot j = 2$$

$$\boldsymbol{u} \cdot \boldsymbol{j} = |\boldsymbol{u}||\boldsymbol{1}|cos(\beta)$$

Thus,

$$\beta = cos^{-1}\left(\frac{\boldsymbol{u} \cdot \boldsymbol{j}}{|\boldsymbol{u}||\boldsymbol{1}|}\right) = cos^{-1}\left(\frac{\boldsymbol{2}}{\sqrt{9+4+25}}\right) = 71.07^{\circ}$$

With the z-axis

$$\boldsymbol{u} \cdot \boldsymbol{k} = (3i + 2j + 5k\,) \cdot k = 5$$

$$\boldsymbol{u} \cdot \boldsymbol{k} = |\boldsymbol{u}||\boldsymbol{1}|cos(\gamma)$$

Thus,

$$\gamma = cos^{-1}\left(\frac{\boldsymbol{u} \cdot \boldsymbol{j}}{|\boldsymbol{u}||\boldsymbol{1}|}\right) = cos^{-1}\left(\frac{\boldsymbol{5}}{\sqrt{9+4+25}}\right) = 35.8^{\circ}$$

From this example we have reached to general formula that you can use for future problems, which is

$$\alpha = cos^{-1}\left(\frac{A_x}{A}\right), \beta = cos^{-1}\left(\frac{A_y}{A}\right), \text{ and } \gamma = cos^{-1}\left(\frac{A_z}{A}\right)$$

A is the magnitude of the vector $\boldsymbol{A}$, and A_x, A_y, and A_z are its components.

Cross product (Vector Product)

Given any two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$, the vector product $\boldsymbol{A} \times \boldsymbol{B}$ is defined as a third vector $\boldsymbol{C}$ which is perpendicular to plane that contains both vectors, the magnitude of which $is\ AB\ sin\theta$, where θ is the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$. That is, if $\boldsymbol{C}$ is given by

$$\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B} \tag{1.18}$$

Then its magnitude is

$$|\boldsymbol{C}| = |\boldsymbol{A}||\boldsymbol{B}|sin(\theta) \tag{1.19}$$

The quantity $AB\ sin\theta$ is equal to the area of the parallelogram formed by $\boldsymbol{A}$ and $\boldsymbol{B}$, as shown in Figure 1.12. The direction of $\boldsymbol{C}$ is perpendicular to the plane formed by $\boldsymbol{A}$ and $\boldsymbol{B}$, and the best way to determine this direction is to use the right-hand rule illustrated in Figure 1.11. The four fingers of the right hand are pointed along $\boldsymbol{A}$ and then "wrapped" into $\boldsymbol{B}$ through the angle θ. The direction of the erect right thumb is the direction of $\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$. Because of the notation, $\boldsymbol{A} \times \boldsymbol{B}$ is often read "$\boldsymbol{A}\ cross\ \boldsymbol{B}$"; hence, the term cross product. Figure 1.13 for give some examples between the vectors a and b.

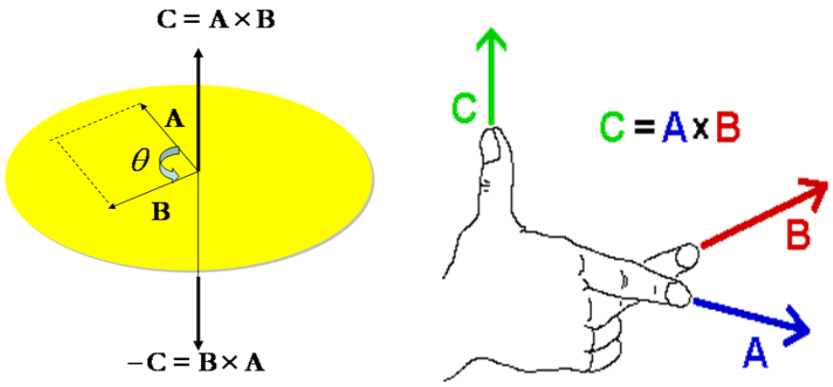


Figure 1.12: The vector product $\mathbf{A} \times \mathbf{B}$ is a third vector C having a magnitude AB sin θ equal to the area of the parallelogram shown. The direction of C is perpendicular to the plane formed by A and B, and this direction is determined by the righthand rule.

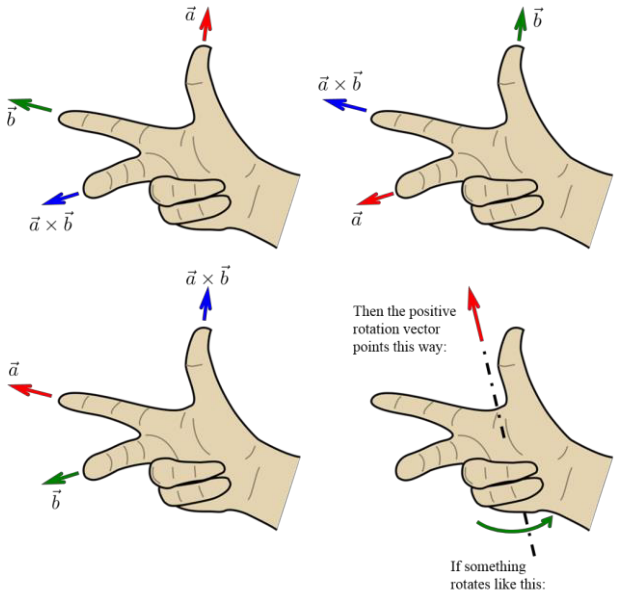


Figure 1.13: show examples of cross product between vectors a and b

Properties of the vector product

1. Unlike the scalar product, the vector product is not commutative. Instead, the order in which the two vectors are multiplied in a cross product is important:
   $\boldsymbol{A \times B = -B \times A}$ see figure 1.12,
2. If $\boldsymbol{A}$ is parallel to $\boldsymbol{B}$ ($\theta = 0^{\circ}, or\ \theta = 180^{\circ}$), then $\boldsymbol{A \times B}$ =0; therefore, it follows that $\boldsymbol{A \times A = 0}$,
3. If $\boldsymbol{A}$ is perpendicular to $\boldsymbol{B}$, then $\boldsymbol{|A \times B| = |A||B|}$
4. The vector product obeys the distributive law:
   $\boldsymbol{A \times (B + C) = A \times B + A \times C}$
5. The unit vectors cross product follow the following rules
   Using property 2, we have

   $$i \times i = 0;\ \ j \times j = 0;\ \ k \times k = 0;$$

   And using the definition and property 1, we have

   $$i \times j = k;\ but\ \ j \times i = -k;$$
   $$j \times k = i;\ but\ \ k \times j = -i;$$
   $$k \times i = j;\ but\ i \times k = -j;$$

The cross product of any two vectors A and B can be expressed in the following determinant form:

$$A \times B = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = i \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - j \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + k \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix} \qquad (1.20)$$

Example 1.20

Two vectors lying in the xy plane are given by the equations $\mathbf{A} = 2i + 3j$ and $\mathbf{B} = -i + 2j$. Find $\mathbf{A \times B}$ and verify that $\mathbf{A \times B = -B \times A}$

Solution

$$\mathbf{A}\times\mathbf{B}=\begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ -1 & 2 & 0 \end{vmatrix}=i\begin{vmatrix} 3 & 0 \\ 2 & 0 \end{vmatrix}-j\begin{vmatrix} 2 & 0 \\ -1 & 0 \end{vmatrix}+k\begin{vmatrix} 2 & 3 \\ -1 & 2 \end{vmatrix}$$

$$\mathbf{A}\times\mathbf{B}=k\left(4-\left(-3\right)\right)=7k$$

$$\mathbf{B}\times\mathbf{A}=\begin{vmatrix} i & j & k \\ -1 & 2 & 0 \\ 2 & 3 & 0 \end{vmatrix}=i\begin{vmatrix} 2 & 0 \\ 3 & 0 \end{vmatrix}-j\begin{vmatrix} -1 & 0 \\ 2 & 0 \end{vmatrix}+k\begin{vmatrix} -1 & 2 \\ 2 & 3 \end{vmatrix}$$

$$\mathbf{A}\times\mathbf{B}=k\left(-3-\left(4\right)\right)=-7k$$

So,

$$\boldsymbol{A}\times\boldsymbol{B}=-\boldsymbol{B}\times\boldsymbol{A}$$

Example 1.21

Use the scalar triple product to show that the vectors $\mathbf{A}=i+4j-7k$, $\mathbf{B}=2i-j+4k$ and $\mathbf{C}=-9j+18k$ are coplanar (lies in the same plane).

Solution

$$\mathbf{A}\cdot\left(\mathbf{B}\times\mathbf{C}\right)=\begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix}=1\begin{vmatrix} -1 & 4 \\ -9 & 18 \end{vmatrix}-4\begin{vmatrix} 2 & 4 \\ 0 & 18 \end{vmatrix}+\left(-7\right)\begin{vmatrix} 2 & -1 \\ 0 & -9 \end{vmatrix}$$

$$=\left(-18-\left(-36\right)\right)-4\left(36\right)-7\left(-18\right)=0$$

That means the vector $\boldsymbol{A}$ is perpendicular to the vector $\mathbf{D}=\mathbf{B}\times\mathbf{C}$ which means that the vector $\boldsymbol{A}$ lies in the same plane that the vectors $\boldsymbol{B}$ and $\boldsymbol{C}$ lies, that is, they are coplanar vectors.

Example 1.22

A student claims that she has found a vector $\boldsymbol{A}$ such that

$$(2i-3j=4k)\times\boldsymbol{A}=4i+3j-k$$

Do you believe this claim? Explain.

Solution

If her claim is true the vector $4i + 3j - k$ must be perpendicular to the plane that contains the vectors: $(2i - 3j = 4k) and \boldsymbol{A}$. We can say in other words, the vector $4i + 3j - k$ is perpendicular on both$(2i - 3j = 4k) and \boldsymbol{A}$. To test this fact whether true or not, take the scalar product,

$$(2i - 3j = 4k) \cdot (4i + 3j - k) = -5 \neq 0$$

Since that scalar product does not equal to zero the two vectors are not perpendicular to each other! Thus, the student claim is false, there is no such a vector $\boldsymbol{A}$ that gives the vector $4i + 3j - k$.

Example 1.23

For the vectors $\boldsymbol{A} = -3i + 7j - 4k$ and $\boldsymbol{B} = 6i - 10j + 9k$, find the angle between the vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ using vector product.

Solution

$$|\mathbf{A}| = \sqrt{9+49+16} = \sqrt{74} \quad and \quad |\mathbf{B}| = \sqrt{36+100+81} = \sqrt{217}$$
$$|\mathbf{A}||\mathbf{B}| = \sqrt{74}\sqrt{217} = 126.720$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} i & j & k \\ -3 & 7 & -4 \\ 6 & -10 & 9 \end{vmatrix} = i(63-40) - j(-27+24) + k(30-42)$$
$$\mathbf{A} \times \mathbf{B} = 23i + 3j - 12k$$

$$|\mathbf{A} \times \mathbf{B}| = \sqrt{23^2 + 3^2 + 12^2} = 26.115$$

$$\theta = sin^{-1}\left(\frac{26.115}{126.720}\right) = 11.89^{\circ}$$

Summary

- The three fundamental physical quantities of mechanics are length, mass, and time, which in the SI system have the units meters (m), kilograms (kg), and seconds (s), respectively.
- Dimensions can be treated as algebraic quantities. By making estimates and making order-of-magnitude calculations, you should be able to approximate the answer to a problem when there is not enough information available to completely specify an exact solution.
- Vector quantities have both magnitude and direction and obey the laws of vector addition.
    - We can add two vectors **A** and **B** graphically, using either the triangle method or the parallelogram rule. As alternative, the algebraic method is nice and easy method used by considering the x *and* y components of the vectors.
    - If a vector **A** has an x component **Ax** and a y component **Ay**, the vector can be expressed in unit–vector form as $\mathbf{A} = A_x i + A_y j$.
    - In this notation, i is a unit vector pointing in the positive x direction, and j is a unit vector pointing in the positive y direction.
    - Because i and j are unit vectors, $|i| = |j| = 1$. We can find the resultant of two or more vectors by resolving all vectors into their x and y components, adding their resultant x and y components, and then using the Pythagorean Theorem to find the magnitude of the resultant vector.
    - Given two vectors A and B, the dot product $\mathbf{A} \cdot \mathbf{B}$ a scalar and is given by,

      $$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

      and the angle, θ, between the two vectors A and B can be found from

      $$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos(\theta)$$
      $$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}$$

- Given two vectors A and B, the cross product is $\mathbf{A}\times\mathbf{B}$ a vector C having a Magnitude

  $|\mathbf{C}| = |\mathbf{A}||\mathbf{B}|\sin(\theta)$

  Where θ is the angle between $\boldsymbol{A}\ and\ \boldsymbol{B}$. The direction of the vector $\mathbf{C} = \mathbf{A}\times\mathbf{B}$ is perpendicular to the plane formed by $\boldsymbol{A}$ and $\boldsymbol{B}$, and this direction is determined by the right-hand rule, figure 1.12

Problems

1. The speed limit on some interstate highways is roughly $100\ km/h$. (a) What is this in meters per second? (b) How many miles per hour is this?
2. One gallon of paint $(volume = 3.78 \times 10^{-3} m^3)$ covers an area of $25.0\ m^2$. What is the thickness of the paint on the wall?
3. Assume that it takes $7.00\ min$ to fill a $30.0 - gal$ gasoline tank. (a) Calculate the rate at which the tank is filled in gallons per second. (b) Calculate the rate at which the tank is filled in cubic meters per second. (c) Determine the time, in hours, required to fill a 1-cubic-meter volume at the same rate. $(1\ U.S.gal = 231\ in^3)$
4. A micro-laser volume is about $36\mu m^3$ express the laser volume in:

   a. mm^3 b. m^3
5. Iron density is about $8.6\ g/cm^3$ express the density in kg/m^3.
6. Newton's law of universal gravitation is represented by

$$F = G\frac{mM}{r^2}$$

   Here $\boldsymbol{F}$ is the gravitational force, M and m are masses, and r is a length. Force has the SI units $kg\ m/s^2$. What are the SI units of the proportionality constant G?

7. the smallest meaningful measure of length is called the "Plank length", and is defined in terms of the three fundamental constants of nature, the speed of light $c = 3 \times 10^8\ m/s$, the gravitational constant $G = 6.67 \times 10^{-11}\frac{m^3}{kg \cdot s^2}$, and Plank's

constant$h = 6.63 \times 10^{-34} kg.m^2/s$. The Plank length λ_p is given by the following combination of these three constants: $\lambda_p = \sqrt{\frac{Gh}{c^3}}$

Show that the dimensions of λ_p are length $[L]$, and find the order of the magnitude of λ_p.

8. We express the distance through which the person falls, before springing back, by the equation

$$S = Cg^{\alpha}t^{\beta}m^{\gamma}$$

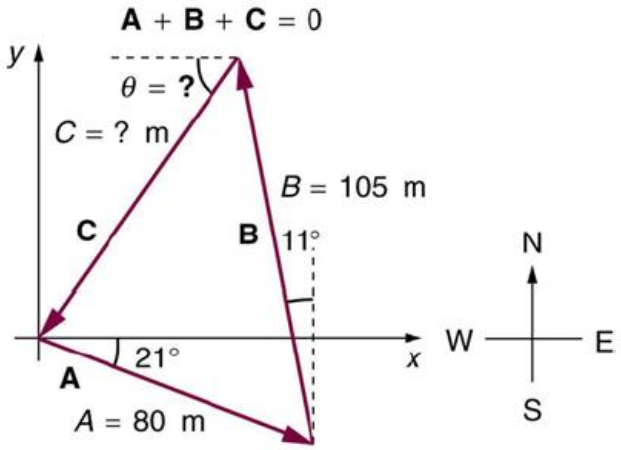
Where $\alpha, \beta, and\ \gamma$ are unknown parameters to be determined, C is an overall constant that cannot be determined using this method, m is person's mass, and g is the acceleration due gravity. Find the $\alpha, \beta, and\ \gamma$ using dimension analysis.

9. The velocity of sound waves through any material depends on its density, ρ, and its modulus of elasticity, E. Thus the velocity v has the following form

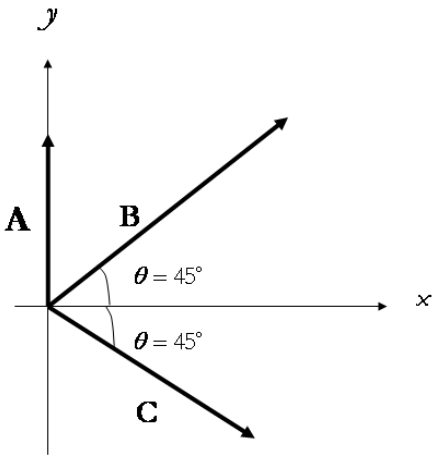
$$v = kE^{x}\rho^{y}$$

Where k is a dimensionless constant and x, y are indices need to find using Dimension analysis. Hint: E dimensions are $ML^{-1}T^{-2}$

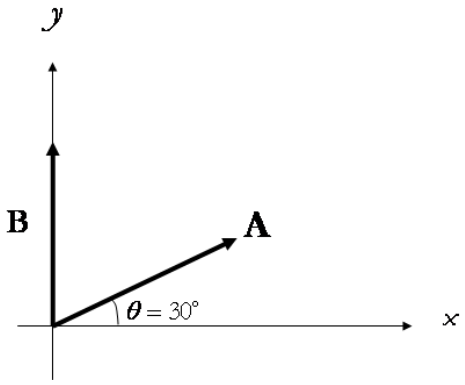
10. What are the components of a vector **A** in the xy plane if its direction is 250° counterclockwise from the positive x-axis and its magnitude is 7.3 units?
11. A new landowner has a triangular piece of flat land she wishes to fence. Starting at the west corner, she measures the first side to be $80.0\ m$ long and the next to be $105\ m$. These sides are represented as displacement vectors **A** from **B** in Figure below. She then correctly calculates the length and orientation of the third side **C**. What is her result?



12. Find $|\mathbf{A}|$, $\mathbf{A+B}$, *and* $\mathbf{A-2B}$ for the flowing sets of vectors:
    a. $\mathbf{A} = i - 2j + k$, $\mathbf{B} = j + 2k$
    b. $\mathbf{A} = 3i - 2k$, $\mathbf{B} = i - j + k$
13. If $\mathbf{v}$ lies in the first quadrant and makes an angle $\pi/3$ with the positive-axis and $|\mathbf{v}| = 3.0\ units$, find in component form.
14. If $\mathbf{A} = 6i - 8j\ units$, $\mathbf{B} = -8i + 3j\ units$, *and* $\mathbf{C} = 26i + 19j\ units$, determine **a** and **b** such that $a\mathbf{A} + b\mathbf{B} + \mathbf{C} = 0$
15. Find a unit vector that has the same direction as the given vectors
    $\mathbf{A} = 8i - 2j + 4k$
    $\mathbf{B} = 3i - 4j + 5k$
16. Given $M = 6i + 2j - k$ and $N = 2i - j - 3k$, calculate the vector product $\mathbf{M \times N}$.
17. If $|\boldsymbol{A} \times \boldsymbol{B}| = \boldsymbol{A} \cdot \boldsymbol{B}$ what is the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$?
18. Three coplanar vectors $\mathbf{A} = 4i - j$; $\mathbf{B} = -3i + 2j$; *and* $\mathbf{C} = -3j$ Find the resultant vector
19. Three vectors are oriented as shown in figure, where $|\boldsymbol{A}| = 20.0\ units$, $|\boldsymbol{B}| = 40.0\ units$, and $|\boldsymbol{C}| = 30.0\ units$. Find (a) the x and y components of the resultant vector (expressed in unit vector notation) and (b) the magnitude and direction of the resultant vector.



20. Each of the displacement vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ as in figure has a magnitude of $3.0\ m$. Find a. $\boldsymbol{A}+\boldsymbol{B}$, b. $\boldsymbol{A}-\boldsymbol{B}$, c. $\boldsymbol{B}-\boldsymbol{A}$, d. $\boldsymbol{A}-\boldsymbol{2B}$. Report all angles counterclockwise from the positive x axis.



21. For what values of b are the vectors $\mathbf{A} = -6i + bj + 2k$, *and* $\mathbf{B} = bi + b^2 j + bk$ perpendicular?
22. Is the vector $(i+j+k)$ a unit vector? Justify your answer. (b) Can a unit vector have any components with magnitude greater than unity? Can it have any negative components? In each case justify your answer. (c) If $\vec{A} = a(3i+4j)$, where a is a constant, determine the value of a that makes a unit vector.
23. If $\vec{a}-\vec{b}=2\vec{c}, \vec{a}+\vec{b}=4\vec{c}, and\ \vec{c}=3i+4j$, then what are the vectors $\vec{a}$ and $\vec{b}$?

Then Find $\vec{a}\cdot\vec{b}\ \ and\ \vec{a}\times\vec{b}$

24. Find the angle between the vectors using the dot product.

a. $\mathbf{A} = i + 2j, \quad \mathbf{B} = 2i - j$

b. $\mathbf{A} = 3i + 2j + 3k, \quad \mathbf{B} = i - j + 2k$

c. $\mathbf{A} = 3i - j + 2k, \quad \mathbf{B} = i - j + k$

25. Find the angle between the vectors using the cross product.

a. $\mathbf{A} = i - 2j - k, \quad \mathbf{B} = 2i - j + 3k$

b. $\mathbf{A} = 2i - 7j + 11k, \quad \mathbf{B} = 5i - j + 2k$

c. $\mathbf{A} = i - 3j + 2k, \quad \mathbf{B} = i - 7j + 3k$

26. Given tow vectors $\mathbf{A} = -i + 2j - 5k$, *and* $\mathbf{B} = 2i + 3j - 2k$, obtain the following Find the magnitude of each vector, write an expression for the vector sum, using the unit vector, find the magnitude of the vector sum, and find the angle between the vector ***A*** and ***B*** using: the scalar product and the vector product

27. Find the scalar product and the vector product for the two vectors $\mathbf{A} = 2i - 5j$ *and* $\mathbf{B} = 5i + 2j$ Find the scalar product and the vector product for the two vectors $\mathbf{A} = 2i - 5j$ *and* $\mathbf{B} = 4i - 10j$

28. Given two vectors $\mathbf{A} = -i + 2j - 5k$, *and* $\mathbf{B} = 2i + 3j - 2k$, obtain the following

    1. $\mathbf{A} \cdot \mathbf{B}$, 2. $\mathbf{A} \times \mathbf{B}$, 3. $\hat{A}$, 4. The angle between the two vectors ϑ, 5. The magnitude of the vector sum $\mathbf{A} + \mathbf{3B}$

29. For the following three vectors $\mathbf{A} = -i + 2j$, $\mathbf{B} = i - 2k$, *and* $\mathbf{C} = -2j + 2k$ find,

    1. The angle between the vectors $\mathbf{D} = \mathbf{A} + \mathbf{B}$, *and* $\mathbf{T} = \mathbf{A} - \mathbf{C}$

    2. find the vectors $\mathbf{E} = \mathbf{A} \times \mathbf{C}$, *and* $\mathbf{F} = \mathbf{B} \times \mathbf{C}$, and the unit vectors $\hat{E}$

30. When two vectors ***A*** and ***B*** are drawn from a common point, the angle between them is θ. Show that the magnitude of their vector sum is given by $\sqrt{A^2 + B^2 + 2AB\cos\theta}$. If ***A*** and ***B*** have the same magnitude, under what condition will their vector sum have the same have the magnitude of the vector ***A*** or ***B***?

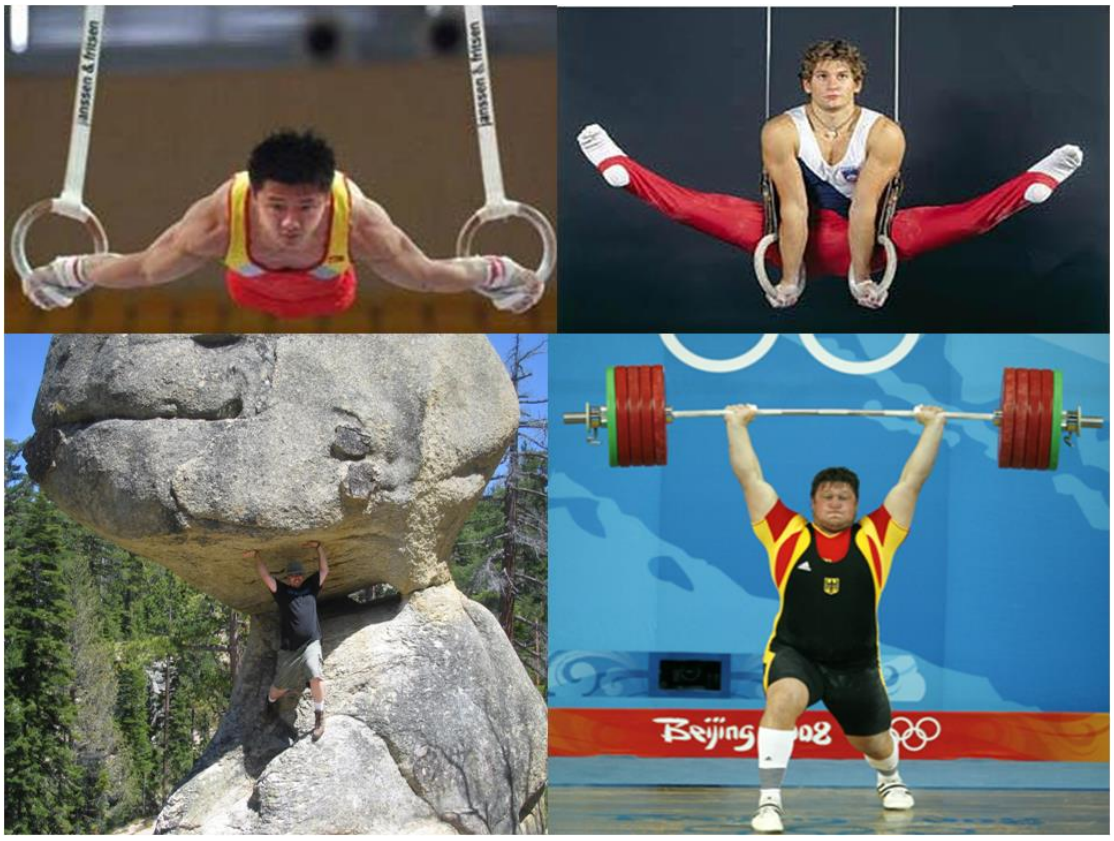
31. Find the two unit vectors that are perpendicular to the plane formed by the two vectors

$$\boldsymbol{A} = 3i - 2j \text{ and } \boldsymbol{B} = 4i - 3j + 2k$$

32. Find a vector in the x-y plane that is perpendicular to the vector $A = 10i - 7j$

Chapter 2

Biostatics



Usually the people concerned about studying objects in motion. Sometimes, however, we are more interested in avoiding motion or at least certain kinds of motion. When this is the case, we are studying statics. Static principles are of most interest to those who want to determine that their structures are going to hold when assembled. This is an important point for architects, civil engineers, mechanical engineers, and carpenters. As you will see in this chapter, static principles are also important for analyzing the mechanical functionality of living organisms and physical behavior of bones and organs such as the lungs and the heart. This is extremely important for development of prosthetic devices such as artificial limbs and mechanical hearts.

2.1 Basic concepts

There are few conceptual points and basic definitions for us to start with, here and then we will apply these roles by which they are interrelated for determining the conditions of equilibrium.

Rigid Body: is an idealized object extended in space that does not change its shape or size under the action of force. Real objects are made up of a large number of particles held together by internal forces and they may vibrate or bend when they are subjected to forces. However objects like baseball, bones, and steel beams are rigid enough so that their deformations can be neglected.

Force: is a push or pull exerted on an object which tends to change the state of motion of the body. The magnitude of the force is a measure of how hard the push or pull is. In addition, force is a vector quantity has both magnitude and direction and it is often denoted **F**. In general, force is measured in the units of Newton (N) or pound (lb). There is another useful concept for force to know, *Force Components*. Each force can be thought of as being composed of a horizontal component and a vertical component. (Actually, any two mutually perpendicular directions will work.) The components of any force are the perpendicular projections onto the horizontal and vertical lines, see Fig. 2.1, and remember what you have learned in chapter 1 about vector analysis.

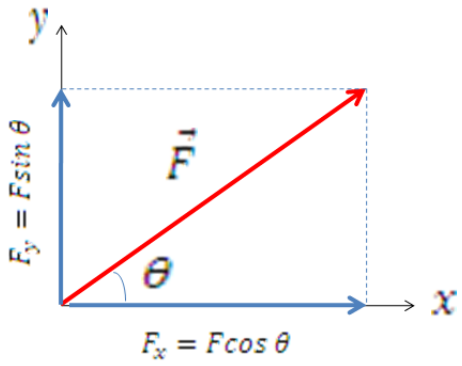


Figure 2.1: Force vector and its components

Mass: As the force tends to change the state of motion of an object, it resists this change. Mass is a quantitative measure of the resistance to a change of motion. It is measured in kilogram, kg.

Weight: Every mass exerts an attractive force on every other masses, this is called the gravitational force. The weight of an object is the force exerted on the object by the mass of the earth. It is a vector with magnitude $W = mg$ where g is the acceleration due to gravity and it points vertically down.

Newton's third law: Whenever a particle A exerts a force on another particle B, B simultaneously exerts a force on A with the same magnitude in the opposite direction. The strong form of the law further postulates that these two forces act along the same line. This law is often simplified into the sentence "*Every action has an equal and opposite reaction.*" This law is extremely useful in analyzing static objects, since it tells us that the sum of all the forces acting on a stationary object must be zero. One of the simple illustrations of this law is given in the answer of the following question that comes always to our mind: How does a building stay up? There is a huge downwards force on it, due to gravity, the weight, and this is exactly balanced by the upwards force supplied by the ground. *Action* (the weight) and *reaction* (the up thrust) are equal and opposite

Torque: is defined as the tendency of a force to produce rotation about an axis, means no axis of rotation no torque. Torque, which is usually denoted by $\boldsymbol{\tau}$, the Greek letter tau is given by the vector product of the force $\boldsymbol{r}$ and $\boldsymbol{F}$, that is

$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F} \quad \textbf{(2.1)}$$

$$= \boldsymbol{rF}\sin\boldsymbol{\theta} \quad \textbf{(2.2)}$$

Where $\boldsymbol{r}$ is the distance between the axis of rotation and the point of contact of the force, $\boldsymbol{F}.\boldsymbol{\theta}$ is the absolute angle between $\boldsymbol{r}$ and $\boldsymbol{F}$.

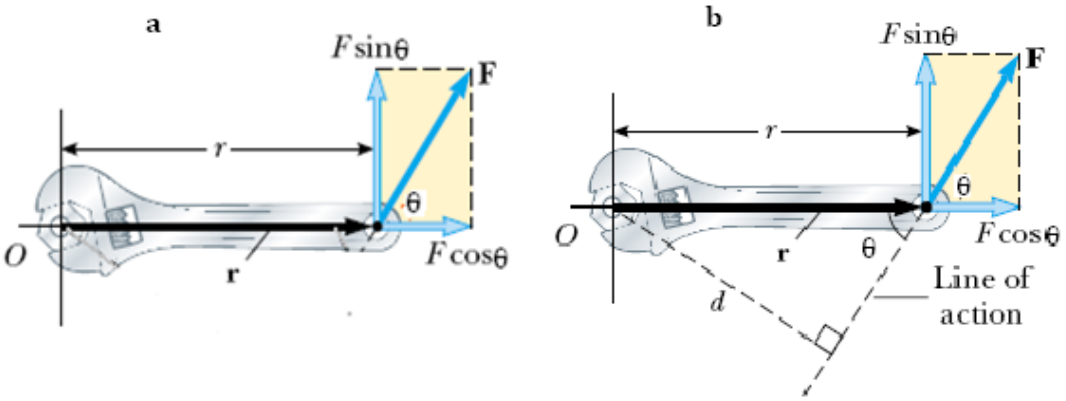


Figure 2.2: The torque on a rigid body about point P has the value $\tau = rFsin\theta$. The component $F\ sin\ \theta$ tends to rotate the wrench about O (a). The torque can be expressed in terms of the moment arm d (b).

The body will rotate around the point O which is called is the axis of rotation of the body. Notice that the component of $\boldsymbol{F}$ that is parallel to $\boldsymbol{r}$, $F_{||}$ $(Fcos\theta)$ tends to stretch the body. Since the body is rigid and is only free to rotate around O, this force has no effect. Only the component of $\boldsymbol{F}$ that is perpendicular to $\boldsymbol{r}$, $\boldsymbol{F}_{\perp}$ $(Fsin\theta)$ causes the body to rotate (see Fig. 2.2a). The SI unit for torque is Newton meters $(N\ m)$. In British Units System, it is measured in foot pounds $(ft \cdot lb)$ (also known as 'pound feet').

Another way of expressing the torque equation is in terms of the moment arm $\boldsymbol{d}$ which is defined as the *perpendicular distance from the line of action of the force to the axis of rotation* O. We defined the torque as

$$\tau = rFsin\theta = (rsin\theta)F = \ dF \quad (2.3)$$

The choice of looking at Moments in terms of the moment arm $\boldsymbol{d}$ or $\boldsymbol{F}_{\perp}$ depends on the problem and how it is presented.

Suppose you need to unscrew a large nut, see Fig.2.2. To maximize the torque you use the longest available wrench and exert as large force as you can. When $\theta = 90$ then $sin\theta = 1$, the largest value for the sin. Consequently, you should pull at right angle to the wrench to get the maximum torque. For the same wrench and same value of F you get less torque when the angle between $\boldsymbol{r}$ and $\boldsymbol{F}$ has value other than 90. The torque is zero when $\boldsymbol{r}$ and $\boldsymbol{F}$ are parallel $(\theta = 0)$ or antiparallel $(\theta = 180)$. Or, if you are not strong enough to pull with the required force, F. Increase the moment arm length, and by this increase you substitute the shortage in your force.

To understand the concept of the torque, imagine yourself pushing a door to open it. The force of your push $\boldsymbol{F}$ causes the door to rotate about its hinges (the pivot point). How hard you need to push depends on your distance from the hinges $\boldsymbol{r}$ (and several other things, but let's ignore them now). The closer you are to the hinges (i.e. the smaller $\boldsymbol{r}$ is), the harder it is to push. This is what happens when you try to push open a door on the wrong site. The torque you created on the door is smaller than it would have been if you pushed the correct site (away from its hinges). Note that the applied force, $\boldsymbol{F}$, and the moment arm, $\boldsymbol{r}$, are independent of the object. Furthermore, a force applied at the pivot point will cause no torque since the moment arm would be zero ($\boldsymbol{r} = 0$).

Direction of Torque

It is simple to specify the torque direction for situation where the axis of rotation is given. Draw two-dimensional picture with the force $\boldsymbol{F}$ and the moment arm $\boldsymbol{r}$ vectors lying in a plane perpendicular to the axis of rotation. The torques that tend to produce counterclockwise rotation are said to be vectors directed along the axis outward of the page and said to be positive. Similarly torques causing clockwise rotation are directed along the axis into the page and conventionally taken to be negative.

A more generalized definition of the direction is needed for objects free to move about any axis like baseball. The direction of torque is perpendicular to the plane formed by $\boldsymbol{r}$ and $\boldsymbol{F}$, and the best way to determine this direction is to use the *right-hand rule* illustrated in Fig.2.3. (This concept is explained in chapter 1) The four fingers of the right hand are pointed along $\boldsymbol{r}$ and then "wrapped" into $\boldsymbol{F}$ through the angle θ. The direction of the erect right thumb is

the direction of torque. Fig.2.3a illustrate positive direction of rotation while (b) illustrate negative direction.

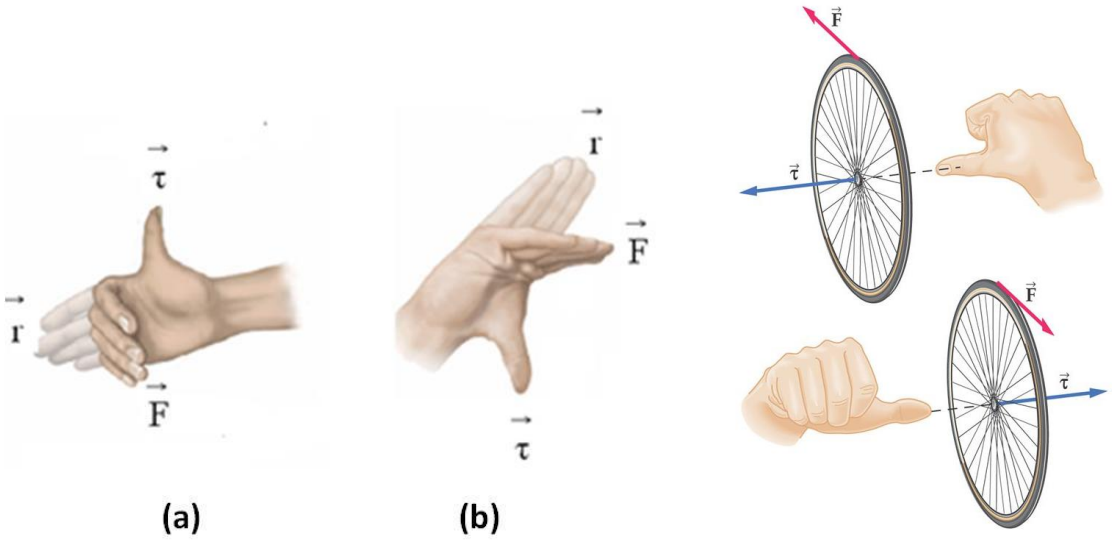


Figure 2.3: The direction of τ is perpendicular to the plane formed by $\vec{r}$ and $\vec{F}$, and this direction is determined by the right hand rule. (a) Positive direction (b), negative direction

Example 2.1

A force $F = (2.00i + 3.00j)\ N$ is applied to an object that is pivoted about a fixed axis aligned along the z −coordinate axis. If the force is applied at a point located at $r = (4.00i + 5.00j)\ m$, find the torque vector τ.

Solution

The torque vector is defined by means of a cross product in Equation 2.1:

$$\tau = r \times F$$

This can be solved, as shown in chapter 1, by the determinant method

$$\tau = \begin{vmatrix} i & j & k \\ 4 & 5 & o \\ 2 & 3 & 0 \end{vmatrix} = (12 - 10)k = 2\,k \qquad N.m$$

Notice that both r and F are in the xy plane. As expected, the torque vector is perpendicular to this plane, having only a z component. As it has positive value it causes a counterclockwise rotation.

Example 2.2

Find the net torque on the wheel in the figure below about the axle through O if a is $10.0\ cm$ and b is $25.0\ cm.$

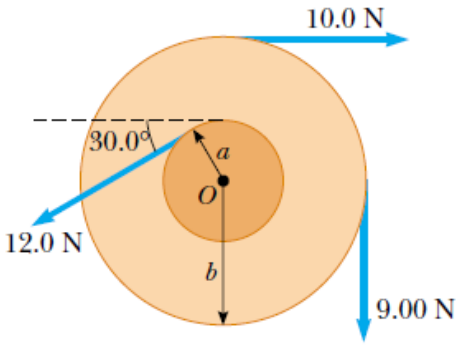


Figure 2.4: Two cylinders pivoted on the axel O, and three tangential forces acting on the cylinders

Solution

We chose to the following signs for torque direction

$$c.w.\ \ gets\ \ negative\ sign \text{ and } \ c.c.w. gets\ positive\ sign$$

Since all forces are tangential, the force arm is the radius of each cylinder, that is

$$\tau = -10 \times 0.25\ c.w. - 9 \times 0.25\ c.w. + 12 \times 0.1 c.c.w$$

$$\tau = -3.55\ Nm \quad c.w.$$

2.2 Static equilibrium

Static equilibrium refers to objects that are not moving. There are many objects that we would like to have in static equilibrium; your kitchen cabinets or bedroom wall, for example. When an object is not moving from one place to another, we also say it is not *translating*. Logically in order for a body not to move there can be no net force acting on it. If the forces add to zero, the object will not be translating.

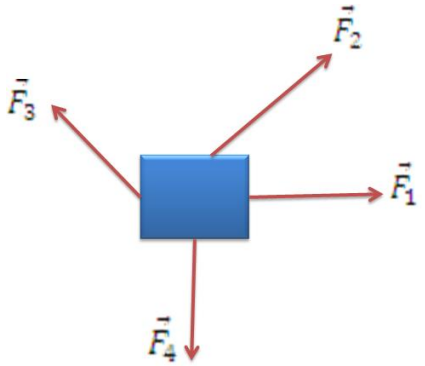


Figure 2.5: Forces pulling on a box in different directions

$$\sum \boldsymbol{F} = \mathbf{0} \qquad \textbf{(2.4)}$$

Forces can still add to zero even if none of them are in exactly opposite directions. Consider the forces from the ropes that are pulling on the box shown, Fig. 2.5. These forces can add to zero if the force vectors add to zero. Force components are particularly useful for those forces where the sum of all the horizontal components must be zero, and the sum of all the vertical components must be zero, independently. In other words,

$$\sum \boldsymbol{F_x} = \mathbf{0}, \qquad \sum \boldsymbol{F_y} = \mathbf{0} \qquad \textbf{(2.5)}$$

Example 2.3

A traffic light weighing 125 N hangs from a cable tied to two other cables fastened to a support. The upper cables make angles of 37.0° and 53.0° with the horizontal. Find the tension in the three cables.

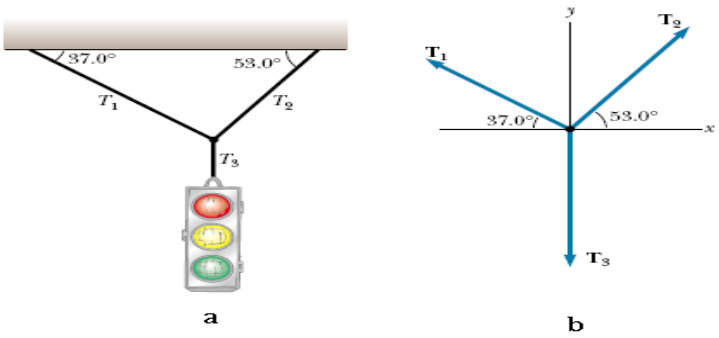


Figure 2.6: (a) A traffic light suspended by cables. (b) Free-body diagram for the knot where the three cables are joined.

Solution

Figure 2.6 shows the type of drawing we might make of this situation. We then construct a free-body diagram for the knot that holds the three cables together, as seen in figure 2.6b. This knot is a convenient object to choose because all the forces we are interested in act through. Because the system is not translating (equilibrium), we know that the net force on the light and the net force on the knot are both zero. The force T_3 exerted by the vertical cable which supports the light equals to its weight, that is $T3 = Fg = 125\ N.$

Next, we choose the coordinate axes shown in figure 2.6b and resolve the forces acting on the knot into their components:

$$F_x = -T_1 \cos 37 + T_2 \cos 53 = 0 \quad (1)$$

$$F_y = T_1 \sin 37 + T_2 \sin 53 - 125 = 0 \quad (2)$$

From (1) we see that the horizontal components of T_1 and T_2 must be equal in magnitude, and from (2) we see that the sum of the vertical components of T_1 and T_2 must balance the weight of the light. We solve (1) for T_2 in terms of T_1 to obtain $T_2 = \frac{cos37}{cos53}\ T_1 = 1.33T_1$

This value for T_2 is substituted into (2) to yield

$$T_1 \sin 37 + (1.33T_1) \sin 53 - 125 = 0 \quad (3)$$

$$T_1 = 75.1N$$

$$T_2 = 99.9N$$

This problem is important because it combines what we have learned about vectors with the new topic of forces.

Indeed the condition of zero net force is a sufficient condition for equilibrium for a point particle since the only possible motion is the *translational*. However, when we are dealing with macroscopic rigid bodies we have to augment this description because it is possible for the forces to add to zero, yet the object is not in equilibrium, but it might *rotate!* Also, when the net force on body is not zero, which might cause other types of motion, for example: circular or vibrational. In figure 2.7, we have two equal and opposite forces acting on a box,

the two forces are not aligned (The line of action is different) and the box will twist or rotate. Thus sum of the *torques* is not zero.

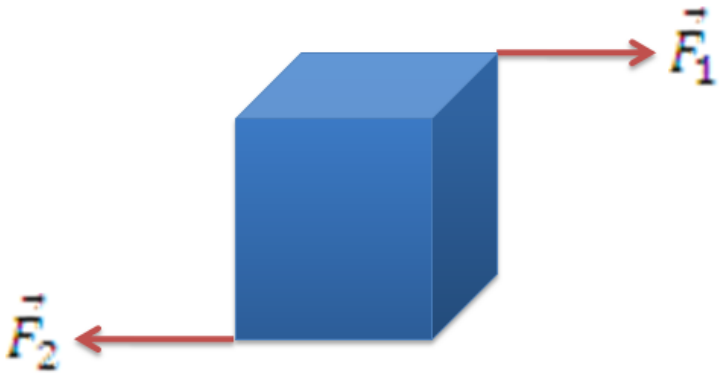


Figure 2.7: equal but opposite forces acting on box with aligned lines of action (top), not aligned lines (bottom)

$$\sum \boldsymbol{\tau} = \mathbf{0} \qquad \textbf{(2.6)}$$

In summary, in order for a rigid body to be static we must therefore require that

i. **THE SUM OF ALL THE EXTERNAL FORCES ACTING ON AN OBJECT IN STATIC EQUILIBRIUM MUST ADD TO ZERO. (**No Translational Motion**)**

$$\sum \boldsymbol{F} = \mathbf{0} \quad \text{i.e.} \quad \sum \boldsymbol{F_x} = \mathbf{0}, \ \sum \boldsymbol{F_y} = \mathbf{0} \qquad \textbf{(2.7)}$$

ii. **THE SUM OF ALL THE TORQUES ACTING ON AN OBJECT IN STATIC EQUILIBRIUM MUST ADD TO ZERO. (N**o Rotational Motion**)**

$$\sum \boldsymbol{\tau} = \mathbf{0} \qquad \textbf{(2.8)}$$

The question arises what reference point should be associated with the torque equation. In principle we should require that $\sum \boldsymbol{\tau}$ vanishes for any point of reference. This is important to notice in problem solutions because

1. There is only a single torque equation associated with each rigid body in a statics problem. Nothing is gained by writing torque equations for several points of reference.
2. We have complete freedom in choosing our point of reference. With a good choice there can be significant computational simplifications. A good choice will

- be a point about which unknown forces in the problem have no torque. Such reference points produce a torque equation with few unknowns.
- be a point which makes it geometrically simple to calculate the torque associated with the remaining forces.

With this said, it is important not to panic about choosing the ``correct'' reference point. There is no ``wrong'' choice, but there is a bad choice. You will be able to work your way through a problem with any reference point chosen. A good choice can only make the algebraic manipulation simpler with less chance of error. Here as in other areas of physics it is interesting to note that there is a sense in which our brains already knows all this material. Use the knowledge of physics that you have built up since you were born to guess the solutions and to check whether your calculated solutions make any sense.

Various states of static equilibrium are experienced throughout one's life. Think of the seesaw at a playground (Fig. 2.8). Two or more individuals sit upon a board which has been fixed to a fulcrum (The pivot point of a lever) which allows rotation. If each of the individuals on the seesaw weights exactly the same amount and sit at exactly the same distance from the fulcrum the teeter-totter will not move. A state of equilibrium has been achieved. The two will remain at rest until an action takes them out of equilibrium.

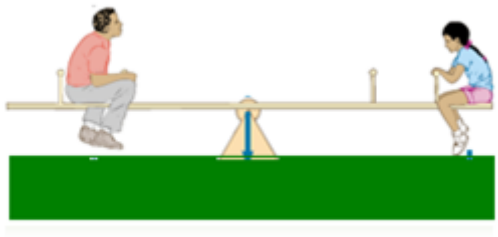


Figure 2.8: Seesaw is an example of day life equilibrium

A large balanced rock at the Garden in Colorado Springs, Colorado is an example of stable equilibrium, see Fig. 2.9. The second condition can be satisfied only when the center of gravity of the rock is directly over the support point.

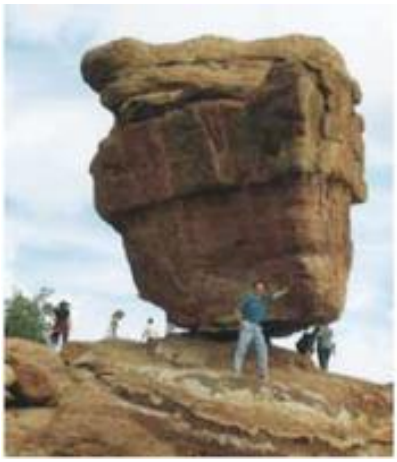
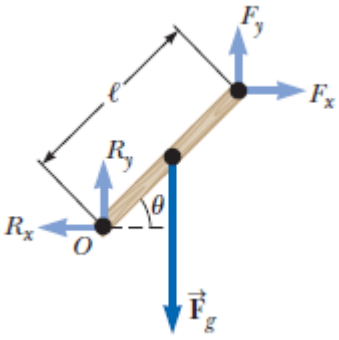


Fig. 2.9: A large balanced rock at the Garden in Colorado Springs, Colorado.

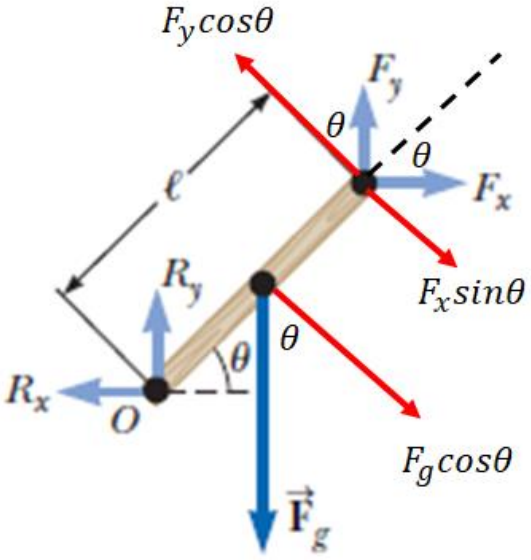
Example 2.4

What are the necessary conditions for equilibrium of the object shown in Figure? Calculate torques about an axis through point O.



Solution

Use distances, angles, and forces as shown in figure. Then use the conditions of equilibrium, which are:



From the first law of equilibrium we have,

$$\sum F_x = 0 \rightarrow F_x - R_x = 0,$$

$$\sum F_y = 0 \rightarrow F_y + R_y - F_g = 0,$$

From the second law of equilibrium we have,

$$\sum \tau = 0 \rightarrow F_y l\, cos\theta(c.c.w) - F_x l\, sin\theta(c.w.) - F_g \left(\frac{l}{2}\right) cos\theta(c.w.) = 0$$

Example 2.5

A $400.0 - N$ sign hangs from the end of a uniform strut. The strut is $4.0\ m$ long and weighs $600.0\ N$. The strut is supported by a hinge at the wall and by a cable whose other end is tied to the wall at a point $3.0\ m$ above the left end of the strut. Find the tension in the supporting cable and the force of the hinge on the strut.

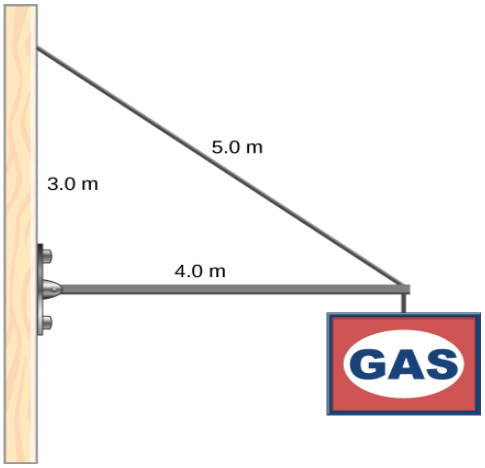


Fig. 2.10: A uniform beam pinned to wall and supported by a cable.

Solution

We first apply static equilibrium to the beam and determine the external support reactions acting on the beam at point A. The forces that acting on the beam are the downward force due to the load at point B, a tension force T due to the supporting rod and a force $\vec{A}$ due to the hinge.

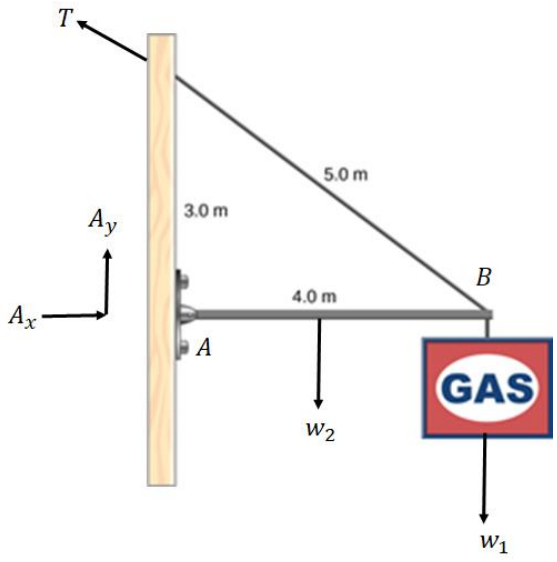


Figure 2.11: example 2.5

Applying the first equilibrium condition, see figure 2.11

$$\sum F_x = A_x - T\cos\theta = 0 \quad (1)$$

$$\sum F_y = A_y + T\sin\theta - w_1 - w_2 = 0 \quad (2)$$

As we have three unknowns and two equations, thus we need a third equation

Applying the second condition of equilibrium

$$\sum \tau = -400N \times 4m - 600N \times 2m + T\sin\theta \times 4m = 0 \quad (3)$$

Solving for T

$$T = \frac{600 \times 2 + 400N \times 4m}{4m \times \left(\frac{3}{5}\right)} = 1166.67\ \text{N}$$

$$A_x = T\cos\theta = 1166.67 \times \frac{4}{5} = 933.336\ N$$

$$A_y = w_1 + w_2 - T\sin\theta = 400 + 600 - 1166.67 \times \frac{3}{5} = 300N$$

$$A = \sqrt{933.336^{\,2} + 300^2} = 1204.62\ N$$

$$\varphi = tan^{-1}\left(\frac{300}{1166.67}\right) = 14.42^{\circ}$$

Example 2.6

A uniform $40.0\ \mathrm{N}$ board supports a father and daughter weighing $800\ \mathrm{N}$ and $350\ \mathrm{N}$, respectively, as shown in Fig.2.12. If the support (called the fulcrum) is under the center of gravity of the board and if the father is 1.00 m from the center, (a) Determine the magnitude of the upward force N exerted on the board by the support. (b) Determine where the child should sit to balance the system.

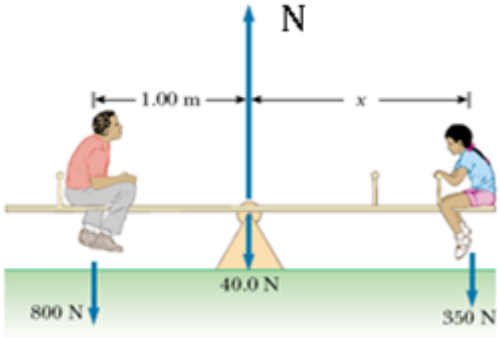


Figure 2.12: system in equilibrium.

Solution

(a) First note that, in addition to the support reaction, N, the external forces acting on the board are the downward forces exerted by each person and the force of gravity acting on the board. We know that the board's center of gravity is at its geometric center because we were told the board is uniform. Because the system is in static equilibrium, the upward force n must balance all the downward forces. From equilibrium condition $\sum F_y = 0$, we have,

$$N - 800 - 350 - 40.0 = 0$$

$$N = 1190\ Newton$$

The equation $\sum F_x = 0$ also applies, but we do not need to consider it because no forces act horizontally on the board.)

(b) To find this position, we must invoke the second condition for equilibrium. Taking an axis perpendicular to the page through the center of gravity of the board as the axis for our torque equation (this means that the torques produced by N and the force of gravity acting on the board about this axis are zero), we see from $\sum \tau = 0$ that

$$(800\ N)\ (1.00\ m) - (350\ N)\ x = 0$$

$$X = 2.29\ m$$

2.3 Equilibrium consideration for human body

2.3.1 The elbow joint

The two most important muscles producing elbow movement are the biceps and the triceps. The contraction of the triceps causes an extension, or opening, of the elbow, while contraction of the biceps closes the elbow. In our analysis of the elbow, we will consider the action of only these two muscles. This is a simplification, as many other muscles also play a role in elbow movement. Some of them stabilize the joints at the shoulder as the elbow moves, and others stabilize the elbow itself.

Fig. 2.13 shows a weight *W* held in the hand with the elbow bent at a 90 degrees angle. The dimensions shown in the figure are reasonable for a human arm, but they will, of course, vary from person to person. The weight pulls the arm downward. Therefore, the muscle force acting on the lower arm must be in the up direction. Accordingly, the prime active muscle is the biceps. The position of the upper arm is fixed at the shoulder by the action of the shoulder muscles.

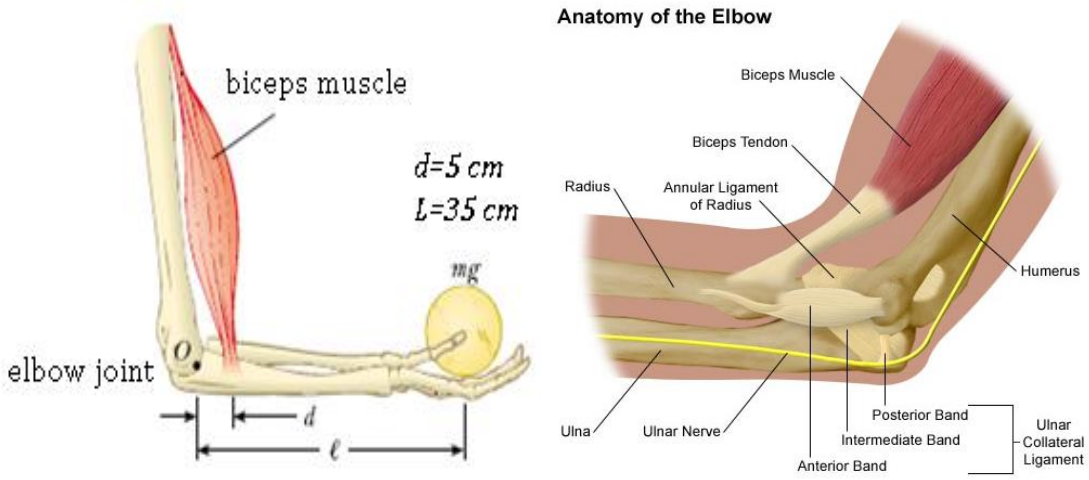


Figure 2.13: A simplified drawing shows weight held in the hand, the elbow joint with biceps muscle.

Example 2.7

A model for the forearm holding 12 N weight in hand in the position shown in figure 2.14 is a pivoted bar supported by a cable. The weight of the forearm is 12 N and can be treated as concentrated at the point shown. Find the tension T exerted by the biceps muscle and the force E exerted by the elbow joint.

Solution

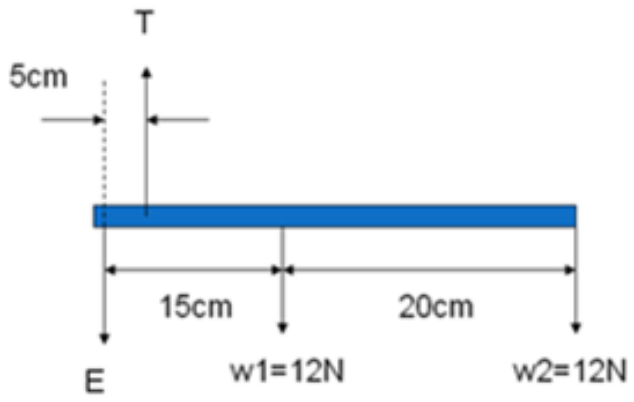


Figure 2.14: Free body diagram FBD for a forearm carrying 12 N weight in hand

We will perform our calculations under the conditions of equilibrium. The tension *T* and the weight *w* has no horizontal components. Since the net horizontal force must be zero, the force E exerted by the joint cannot have a horizontal component. We assume that E is directed downward, if it turns out to be negative this will mean it is in the opposite direction.

Applying the first condition of equilibrium

$$\sum F_y = T - E - W_1 - W_2 = 0$$

$$T - E - 12 - 12 = T - E - 24 = 0 \qquad (1)$$

This contains both unknowns, *T* and *E*.

Applying the second condition of equilibrium, $\sum \tau = 0$

Calculating the torques about the pivot point (the elbow) E produces no torque because it passes through the pivot point $(r = 0)$, thus we have

$$T \times (0.05m) - 12N \times (0.15m) - 12N \times (0.15 + 0.2)m = 0 \qquad (2)$$

Solving equations (1) and (2) for T gives

$$T = \frac{12N \times (0.15m) + 12N \times (0.35m)}{0.05m} = 120N$$

Using this value in equation 1 gives

$$120 - E - 24 = 0 \Rightarrow E = 96\ N$$

Note: for athlete person the distance between the biceps muscle and the elbow joint is bigger than that of an average person. So, consider this distance for an athlete is just 5.5 cm instead of 5 cm as in the example, and repeat you calculation. (I'm waiting!)

Ok, yes you are right the tension force is 109.1 N and the force exerted on the elbow joint is 85.1 N. This means the athlete persons feels objects lighter than an average person, he has less tension in his muscles and less force exerted on his elbow joint.

2.3.2 The Arm

In human anatomy, the arm is the part of the upper limb between the shoulder joint and the elbow joint. In common usage, the arm extends to the hand. It can be divided into the upper arm, which extends from the shoulder to the elbow, the forearm which extends from the elbow to the hand, and the hand. Anatomically the shoulder girdle with bones and corresponding muscles is by definition a part of the arm.

Example 2.8

The illustration in figure 2.15 shows an athlete's outstretch arm holding $w = 5\text{kg}$. The tension F_M in the deltoid muscle is applied at an angle of 17 degree. The mass of the arm is 4.5 kg. Determine the tension F_M in the muscle, which is $15\ \text{cm}$ away from the center of mass, and the force exerted by the shoulders joint F_J and its direction. Consider $\text{g} = 10\ \text{m.s}^{-2}$

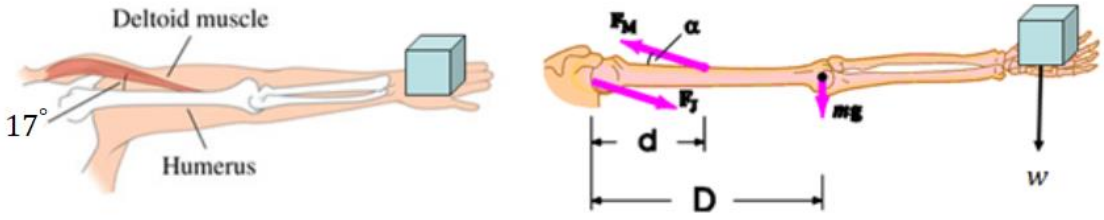
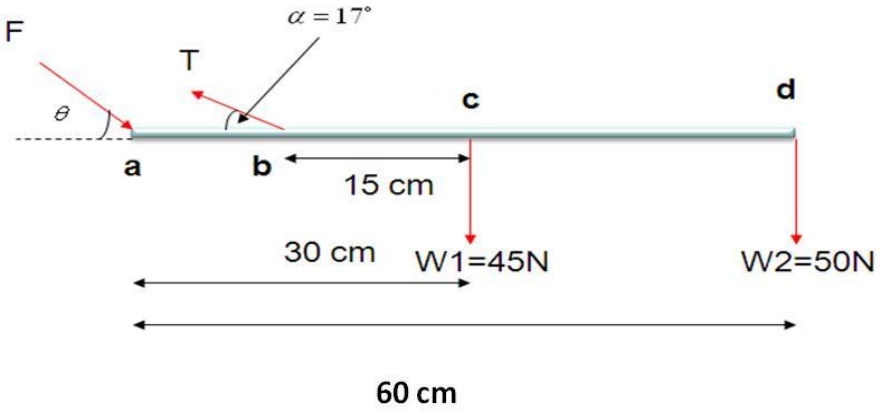


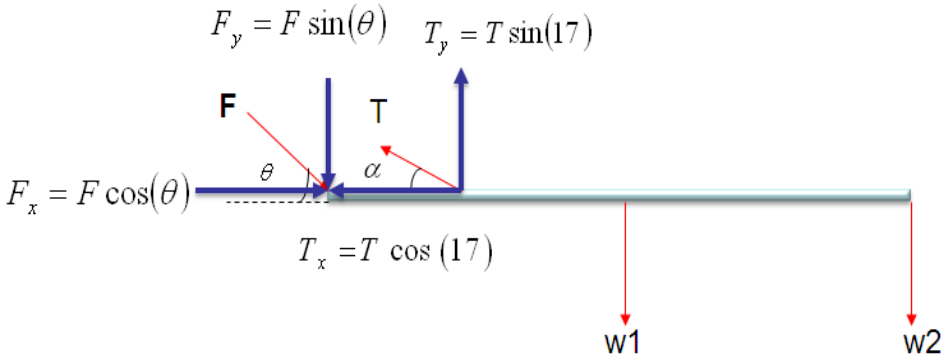
Figure 2.15: A person strengthens his shoulder by doing dumbbell exercises

Solution

It is advised to redraw the free body diagram before starting your analysis, that is



Now, it is much easier to handle the force applied on the outstretched arm, projecting force on the x and y axis, we have



Since the outstretched arm is at equilibrium start applying the laws of equilibrium, that is, apply the first law of equilibrium

$\sum F_x = 0, and \quad \sum F_y = 0$, notice the arrow head of each vector or vector's component.

$$F_x - T\cos(17) = 0 \Rightarrow F_x = T\cos(17) \qquad 1$$

and

$$-F_y + T\sin(17) - w_1 - w_2 = 0 \Rightarrow F_y = T\sin(17) - w_1 - w_2 \qquad 2$$

Here, we have two equations but with three unknown, namely: $F_x, F_y, and \ T$. Since the number of unknown is greater than the number of equations we cannot solve this system of equation. Thus, we have to invoke the second condition of equilibrium seeking help. That is,

$\sum \tau = 0$, notice that if a force or a force's component points upward the rotation is c.c.w (counter clockwise) with positive sign, and if it points down the rotation is c.w. (clockwise) with negative sign.

$$-w_1(0.3m)(cw.) - w_2(0.60m)(cw) + T_y(0.15m)(cc w) = 0, \quad 3$$

Rearranging 3,

$$-w_1(0.3m) - w_2(0.60m) + T\sin(17)(0.15m) = 0, \quad 4$$

Solve equations (1), (2), and (4) for T, we have

$$T = \frac{+w_1(0.3m) + w_2(0.60m)}{\sin(17)(0.15m)} = \frac{45\times 0.3 + 50\times 0.60}{0.15\times\sin(17)} = 991.888N$$

Now, insert the value of T in equation (1) and (2) and then solve for the force components, that is

From equation (1), we have

$F_x = T\cos(17) = 948.547N$,

and from equation (2), we have

$F_y = T\sin(17) - w_1 - w_2 = 194.999N$

and

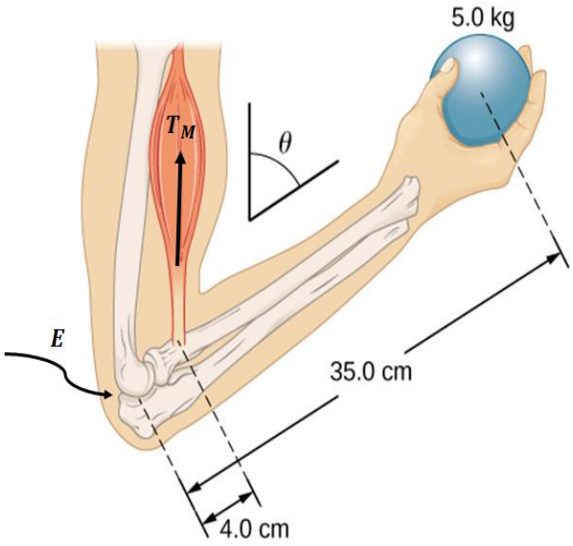
$|F| = \sqrt{F_x^2 + F_y^2} = 968.38N$

and its direction is

$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = 11.6163^\circ$.

Example 2.9

The forearm shown below is positioned at an angle θ with respect to the upper arm, and a $5.0 - kg$ mass is held in the hand. The total mass of the forearm and hand is $3.0\ kg$, and their center of mass is $15.0\ cm$ from the elbow. (a) What is the magnitude of the force that the biceps muscle exerts on the forearm for $\theta = 60°$ (b) What is the magnitude of the force on the elbow joint for the same angle? (c) How do these forces depend on the angle θ?



Solution

We have two equations for the first equilibrium condition, one equation for each component of the net force acting on the forearm. Figure 2.16, shows forces components.

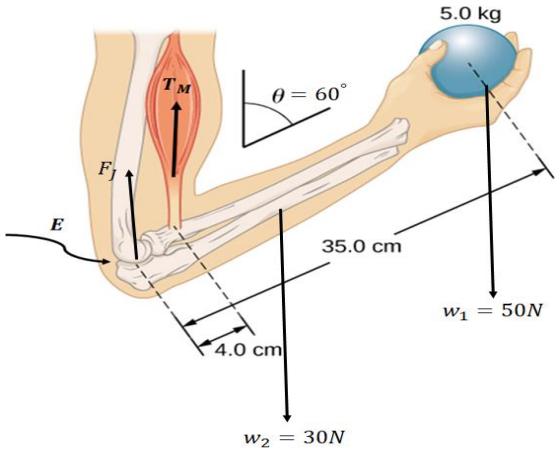


Figure 2.16: Free-body diagram for the forearm: The pivot is located at point E (elbow)

Solution

We see from the free-body diagram that the net force satisfies the equation

$$\sum F_y = 0 = +F_J + T_M - w_1 - w_2 = 0$$

Invoke the second law of equilibrium, we have, see figure 2.17

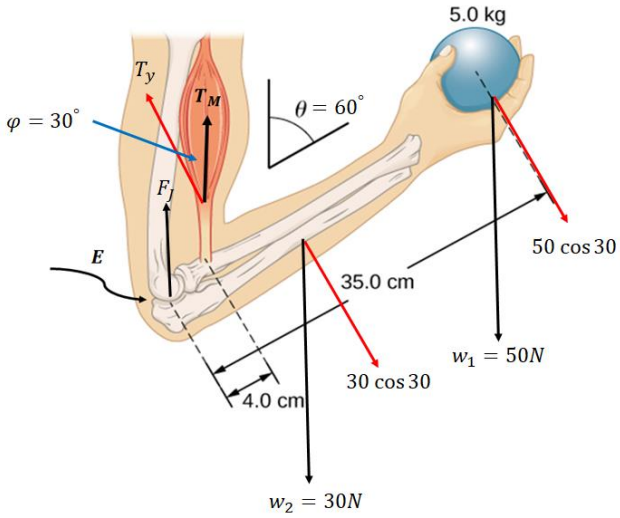


Figure 2.17: torque diagram

$$\sum \tau = 0 = +T_y \times 0.04 - 30\ cos30 \times 0.15 - 50cos30 \times 0.35 = 0$$

$$\sum \tau = 0 = +T_M cos30 \times 0.04 - 30\ cos30 \times 0.15 - 50cos30 \times 0.35 = 0$$

$$T = \frac{30 \times 0.15 + 50 \times 0.35}{0.04} = 550N$$

Our torque equation will look like the one just that we obtain in example 2.7, except that each term will have its lever arm reduced by the same factor, which will cancel out, thus the same result is obtained.

$$F_J = w_1 + w_2 - T_M = 50 + 35 - 550 = -465N$$

You need to flip the direction of the force F_J.

2.3.3 The Ankle

In human anatomy, the ankle joint is formed where the foot and the leg meet. The ankle, or talocrural joint, is a synovial hinge joint that connects the distal ends of the tibia and fibula in the lower limb with the proximal end of the talus bone in the foot, figure 2.18. The articulation

between the tibia and the talus bears more weight than between the smaller fibula and the talus. The term "ankle" is used to describe structures in the region of the ankle joint proper

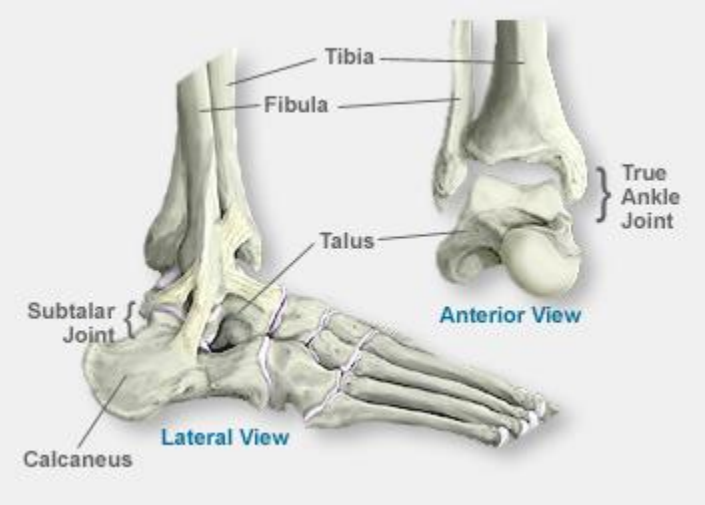


Figure 2.18: the anatomy of the ankle

Example 2.10

A ballerina stands on her tiptoe; the position of the foot is as shown below. The total weight of her body $500\ \mathrm{N}$ is supported by the force n exerted by the floor on the toe. A free-body-diagram is shown in Figure, where T is the force exerted by the Achilles tendon on the foot and R is the force exerted by the tibia on the foot. Find the values of $T, and\ \ R$, take $\theta = 20^{\circ}$.

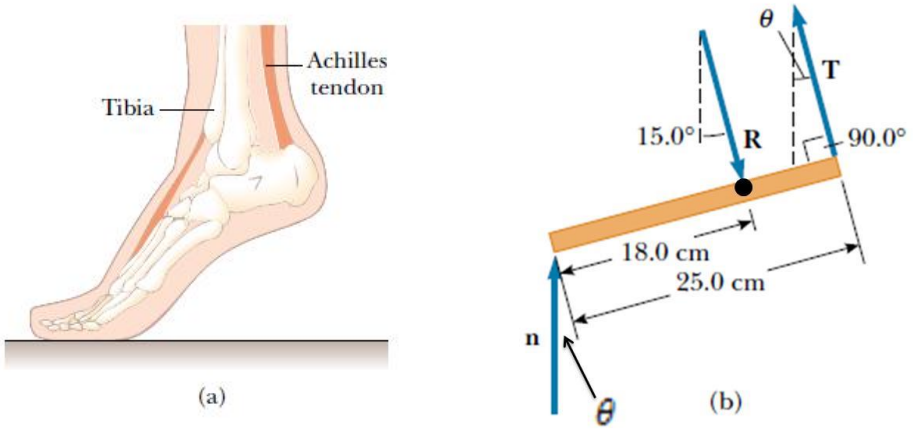


Figure 2.19: a man stands of his tiptoe and the free body diagram

Solution

The sum of forces has to equal to zero,

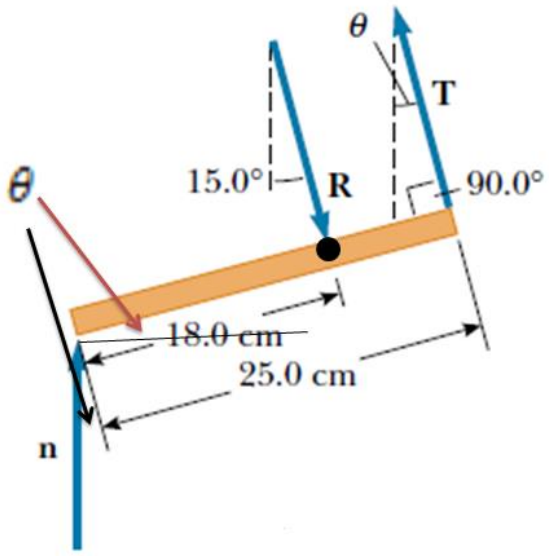
$$\sum f_x = R\ sin\ 15 - T \sin 20 = 0 \Rightarrow R_x = T \sin 20 \qquad (1)$$

$$\sum f_y = n - Rcos\ 15 + T \cos 20 = 0\ \Rightarrow R_y = n - Tcos20 \qquad (2)$$

We know that $n = 500N$, equation (2) becomes,

$$R_y = 500 - Tcos20 \qquad (3)$$

Now, we have two equations with three unknown namely: $R\ and\ T$. Thus we have to invoke the second condition of equilibrium to be able to solve, from figure below, we have



Invoke the second law of equilibrium, that is

$$\sum \tau = -500 \cos 20 \times (0.18m)\ c.w. + T\ \times (0.25 - 0.18)m = 0 \qquad (4)$$

$$-500 \cos 20 \times (0.18m)\ c.w. + T\ \times (0.25 - 0.18)m = 0$$

$$T = \frac{500N \times 0.18m \times \cos 20}{0.07m} = 1208\ N$$

From equation (1),

$$R_x = T \sin 20 = 1208 \sin 20 = 413\ N$$

From equation 3,

$$R_y = 500 - Tcos20 = 500 - 1135 = -635N$$

$$R = 757\ N$$

2.3.4 The Back

When the trunk is bent forward, the spine pivots mainly on the fifth lumbar vertebra, figure 2.20. We will analyze the forces involved when the trunk is bent at 60° from the vertical with the arms hanging freely. The pivot point A is the fifth lumbar vertebra. The lever arm AB represents the back.

Example 2.11

The weight of the head is W_1, the weight of the trunk W_2 is the weight of the arms, and W_3 is the weight of the trunk. The tension in the erector spinal's muscle at C maintains the position of the back. The angle between the spine and this muscle is about 12◦. For a $w = 70kg\ man's\ mass$, $W_1 = 0.07w$ (head) with $72\ cm$ from the fifth lumber vertebra, $W_2 = 0.12w$ with $48\ cm$ from the fifth vertebra (arms), and $W_3 = 0.46w$ (trunk) with 36 cm from the fifth vertebra.

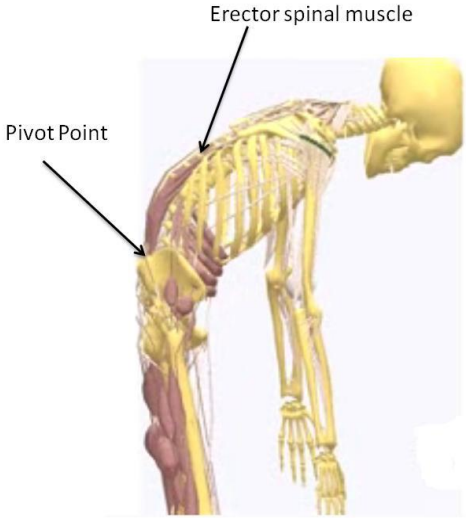
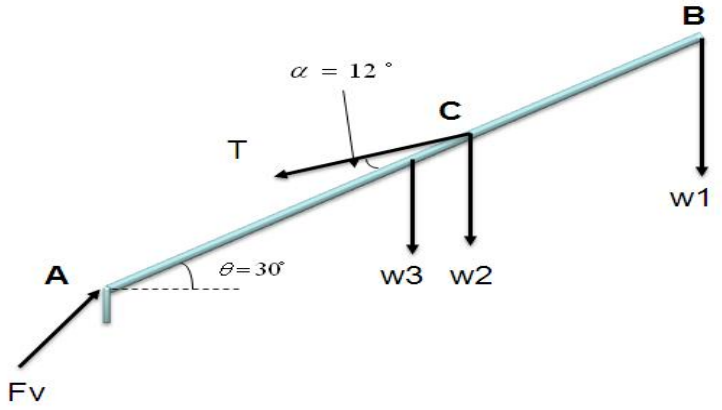


Figure 2.20: The bent back

Solution

Before starting the analysis, draw the free body diagram, which is



Resolve the force, we have

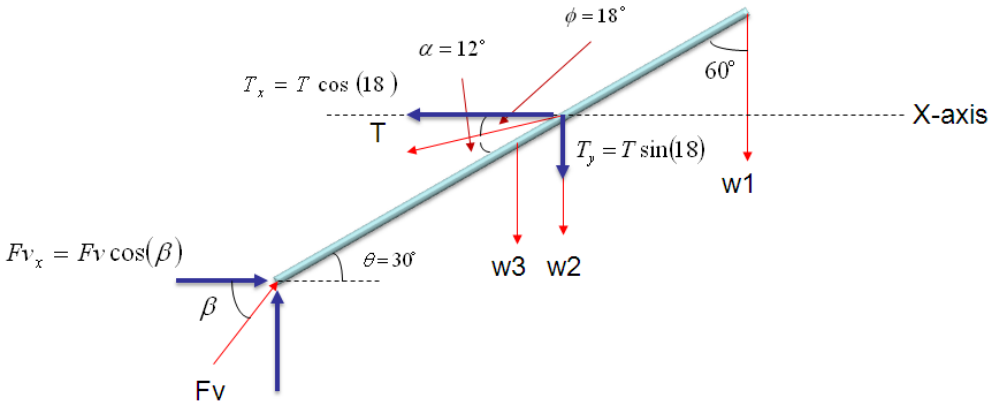


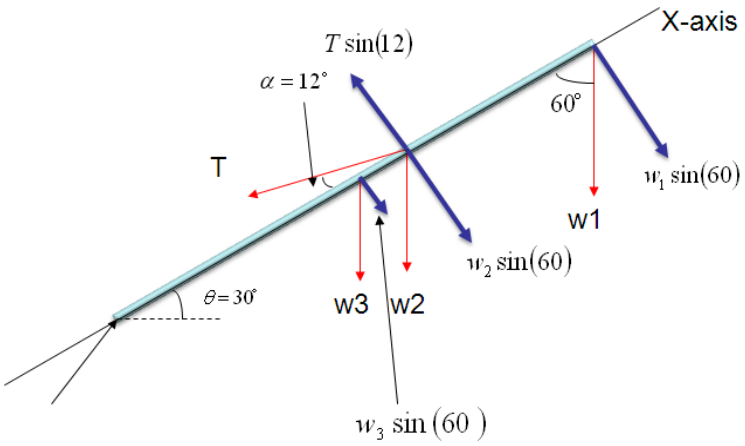
Figure 2.21: FBD for the bent back of 70 kg man shown in Fig. 2.20

Since the body is at equilibrium, apply the conditions of equilibrium

$$\sum F_x = 0 \Rightarrow Fv_x - T\cos(18) = 0 \Rightarrow Fv_x = T\cos(18) \quad 1$$

$$\sum F_y = 0 \Rightarrow Fv_y = T\sin(18) + w_1 + w_2 + w_3 \quad 2$$

Also, we have three unknowns and two equations, which means! We have to invoke the second condition of equilibrium.



$\sum \tau = 0$,

Calculating the torques around the pivot point A simplifies the calculations

$$(T\sin(12))(0.48m)(ccw) - (w_1\sin(60))(0.72m)(cw) - (w_2\sin(60))(0.48m)(cw) - (w_3\sin(60))(0.36)(cw) = 0 \quad 3$$

Solving equation 3 for **T**, we have

$T = 2.374\, w$

Substitute this value into equations 1 and 2, to find the force components and its direction

If you did so, you will have,

$$F_{v_x} = 2.258\, w, F_{v_y} = 1.384\, w$$

$$F_v = 2.648\, w, \beta = 31.50^{\circ}$$

As you can the tension, in the back muscles, increases with body weight increase and also the compression on the fifth lumber vertebra. This example indicates that large forces are exerted on the fifth lumbar vertebra. It is not surprising that backaches originate most frequently at this point. It is evident too that the position shown in the figure is not the recommended way of lifting a weight. Now, we want to explore what is the best position for the back.

2.3.5 The Hip

The hip (pelvis) is not a single bone, but several bones that are fused together. The hip joint and its simplified lever representation, with dimensions that are typical for a male body are shown in Figure 2.22(a). The hip is stabilized in its socket by a group of muscles, which is represented in Figure 2.22(a), as a single resultant force M. There are only two forces act on the body. The body weight W_b acts downward, which is in the midline in the hip (Figure 2.22). The foot feels an upward normal force from the floor of magnitude N. There are no forces in the x direction, and these two forces in the y direction must balance in equilibrium, so $N = W_b$. In equilibrium the body cannot start to rotate, so the torques are zero. It is clear from the Fig. 2.22 that this occurs when the foot is directly below the hip, in the midline.

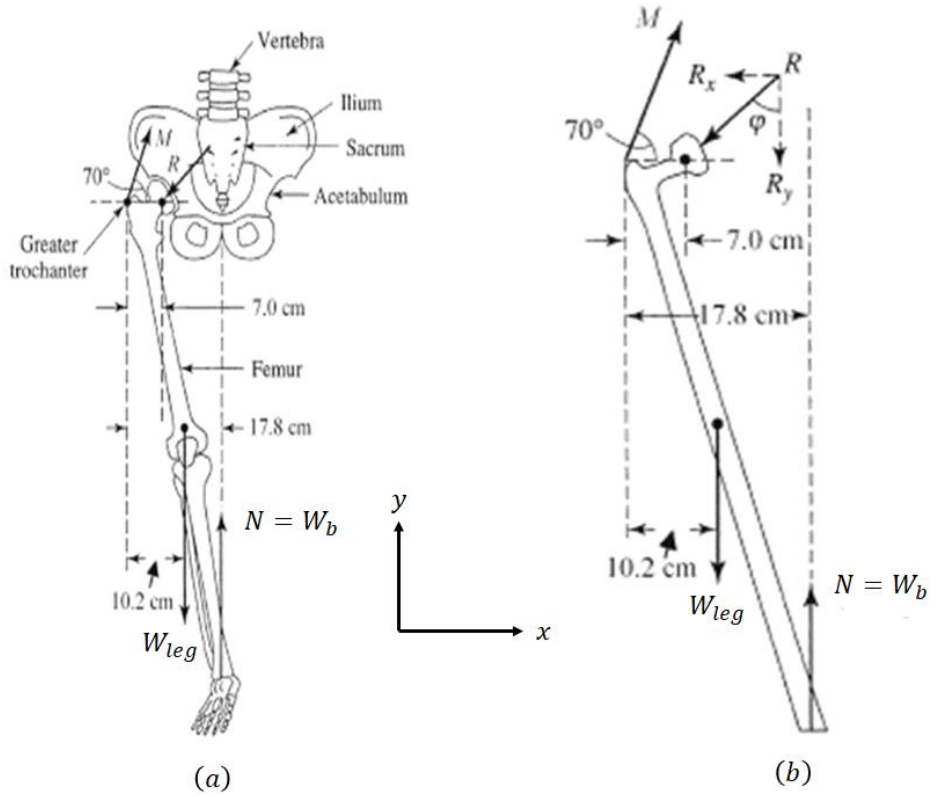


Figure 2.22: a. Anatomical diagram of the leg and hip for someone standing on one leg, or during slow walking, showing the forces on them and relevant dimensions, including the force exerted on the head of the femur by the acetabulum R and the net force exerted by the hip abductor muscles, b. Force diagram for a leg for someone standing on one foot

Equilibrium of the Individual Body Component

There are four external forces on the leg (Figure 2.22a, b):

(a) N is the normal force on the leg from the floor, and we know that $N = W_b$.

(b) W_{leg} is the weight of the leg, which is about $0.16W_b$,

(c) $\boldsymbol{R}$ is the reaction force on the leg from the hip, and it is normal to the hip socket. We will define the x and y components R_x and R_y,

(d) $\boldsymbol{M}$ is the force due to the hip abductor muscles. Its effective action is 70° to the horizontal, acting on the greater trochanter.

Example 2.11

Calculate the magnitude of the muscle force $\boldsymbol{M}$ and the force $\boldsymbol{R}$ at the hip joint when the person $800\ N$ weight is standing erect on one foot as in a slow walk, (shown in Fig. 2.22), with respect to the presented dimensions.

Solution

We have three equations with three unknowns: R_x, R_y, $and\ \boldsymbol{M}$. From the first law of equilibrium, we have

$$\sum F_x = Mcos70 - R_x = 0 \quad (1)$$

$$\sum F_y = Msin\,70 - R_y - W_{leg} + W_b = 0$$

Replace W_{leg} by $0.16W_b$, we have

$$\sum F_y = Msin\,70 - R_y - 0.16W_b + W_b = 0 \quad (2)$$

To apply the condition of equlibrium, we will choose the rotation axis to emanate from the center of the head of the femur because the reaction force from the acetabulum passes through this point. This makes the analysis easier. The relevant distances of interest are shown in figure 2.22b. Each force will induce torque as follow:

(a) The normal force causes a torque of $(17.8 - 7.0 = 10.8\ cm)W_b$. This is a positive torque because the normal force induces a counter clockwise rotation about the chosen z-axis (see Figure 22.2).

(b) The force of the weight of the leg is $3.2\ cm\ (= 10.2cm - 7.0cm)$ and this force tends to induce a clockwise rotation, so it contributes a torque of

$$-(3.2\ cm)W_{leg}\ =\ -(3.2\ cm)(0.16W_b)\ =\ -(0.5\ cm)W_b.$$

(c) With the choice of the axis, the torque from the reaction force from the hip is zero, because the distance vector and normal force are antiparallel.

(d) The component of the force from the hip abductor muscles normal to the horizontal distance vector (of magnitude 7.0 cm) is $M\ sin\ 70^{\circ}$, causes a clockwisw rotation, the torque is $-(7.0cm)Msin70$.

Thus the second condition of equilibrium gives,

$$\sum \tau = 10.8\ cm\ W_b - (3.2\ cm)(0.16W_b) - (7.0cm)Msin70 = 0$$

Solve for M,

$$M = \frac{10.8 - 0.5}{7.0\ sin70} W_b = 1.57W_b = 1.57 \times 800N = 1256N$$

Now, Use M in equation (1) and (2) to find R_x and R_y.

From equation (1),

$$R_x = Mcos70 = 1.57\ W_b\ cos70 = 0.54\ W_b = 432\ N$$

From equation (2),

$$R_y = 1.57W_b\ sin70 - 0.16W_b + W_b$$
$$R_y = (1.57sin70 - 0.16 + 1)W_b = 2.31W_b = 1848N$$

$$\theta = tan^{-1}\left(\frac{R_y}{R_x}\right) = tan^{-1}\left(\frac{2.31}{0.54}\right) \cong 13^{\circ}$$

Summary

Human body (or in general a rigid object) is in equilibrium if the following conditions hold:

i. **THE SUM OF ALL THE EXTERNAL FORCES ACTING ON AN OBJECT IN STATIC EQUILIBRIUM MUST ADD TO ZERO. (**No Translational Motion**)**

$$\sum \boldsymbol{F} = \mathbf{0} \quad \text{i.e.} \quad \sum \boldsymbol{F_x} = \mathbf{0}, \ \sum \boldsymbol{F_y} = \mathbf{0} \qquad \textbf{(2.7)}$$

ii. **THE SUM OF ALL THE TORQUES ACTING ON AN OBJECT IN STATIC EQUILIBRIUM MUST ADD TO ZERO. (N**o Rotational Motion**)**

$$\sum \boldsymbol{\tau} = \mathbf{0} \qquad \textbf{(2.8)}$$

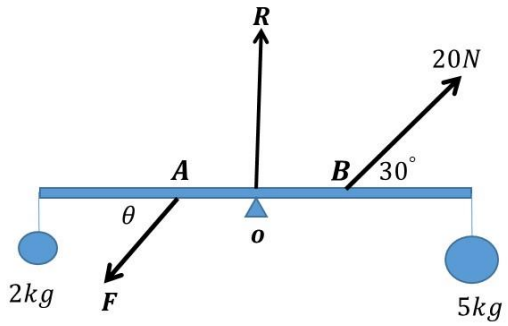
The first condition is the condition for translational equilibrium, and the second is the condition for rotational equilibrium. These equations allow us to manipulate and analyze human body state of equilibrium under many situations, as shown in the given examples. To able to go through these problems you need to identify forces, create a FBD, and then apply Equations 2.7 and 2.8 and solve for the unknowns.

The force of gravity exerted on an object can be considered as acting at a single point called the center of gravity. The center of gravity of an object coincides with its center of mass if the object is in a uniform gravitational field.

Problems

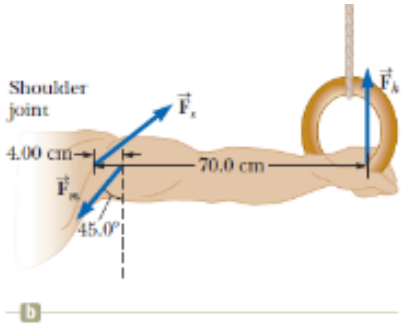
1. (a) When opening a door, you push on it perpendicularly with a force of $55.0\ N$ at a distance of $0.850m$ from the hinges. What torque are you exerting relative to the hinges? (b) Does it matter if you push at the same height as the hinges?
2. Two children push on opposite sides of a door during play. Both push horizontally and perpendicular to the door. One child pushes with a force of $17.5\ N$ at a distance of $0.600\ m$ from the hinges, and the second child pushes at a distance of $0.450\ m$. What force must the **second child** exert to keep the door from moving? Assume friction is negligible.
3. Two children of mass $20.0\ kg$ and $30.0\ kg$ sit balanced on a seesaw with the pivot point located at the center of the seesaw. If the children are separated by a distance of $3.00\ m$, at what distance from the pivot point is the small child sitting in order to maintain the balance

4. Find the force F that makes the shown system below in equilibrium, where the bar is weightless and is $60\ cm$ in length. The pivot O is in the middle of the bar, and the points A and B are locate $12\ cm$ to the left and $15\ cm$ to the right of the pivot, respectively. What is the magnitude of the reaction force on the pivot?

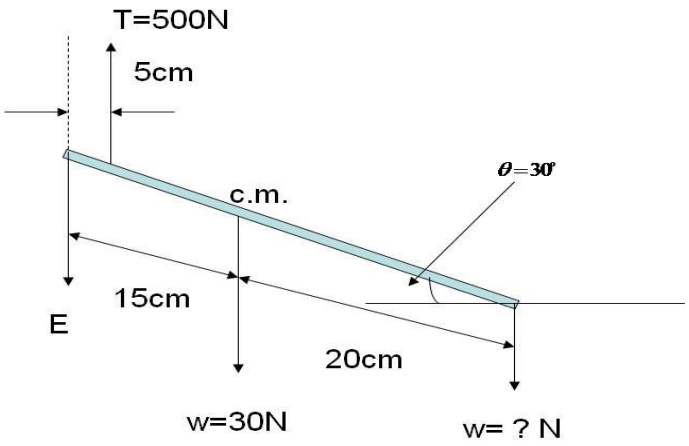


5. When a gymnast performing on the rings executes the iron cross, he maintains the position at rest shown in Figure P12.53a. In this maneuver, the gymnast's feet (not shown) are off the floor. The primary muscles involved in supporting this position are the latissimus dorsi ("lats") and the pectoralis major ("pecs"). One of the rings exerts an upward force $\boldsymbol{F_h}$ on a hand as shown in Figure P12.53b. The force $\boldsymbol{F_s}$ is exerted by the shoulder joint on the arm. The latissimus dorsi and pectoralis major muscles exert a total force $\boldsymbol{F_m}$on the arm. (a) Using the information in the figure, find the magnitude of the force $\boldsymbol{F_m}$ for an athlete of weight $750\ N$.

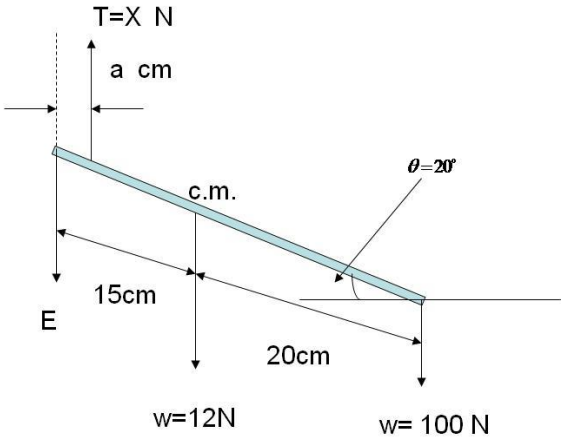


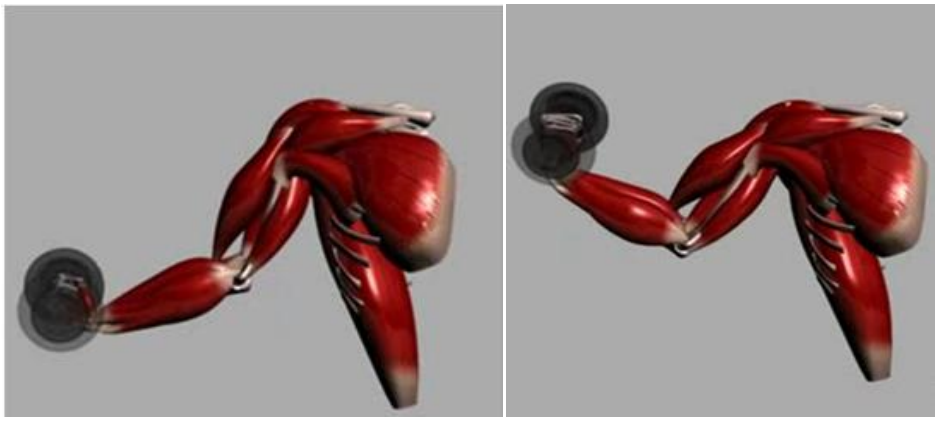


6. An athlete his arm is inclined by 30°, as shown in figure below. If the maximum tension can be tolerated by athlete is $500\ N$. Calculate the maximum Weight can be held in his hand



7. An athlete his arm is inclined by 20 deg, as shown in figure below. Find the maximum tension. Consider $a = 5$ cm





8. Repeat, Q2 with the athlete arm inclined 20 deg above the horizontal. Consider $a = 7$ cm

9. Compare the results of the solved example and with Q7 and Q8 result!

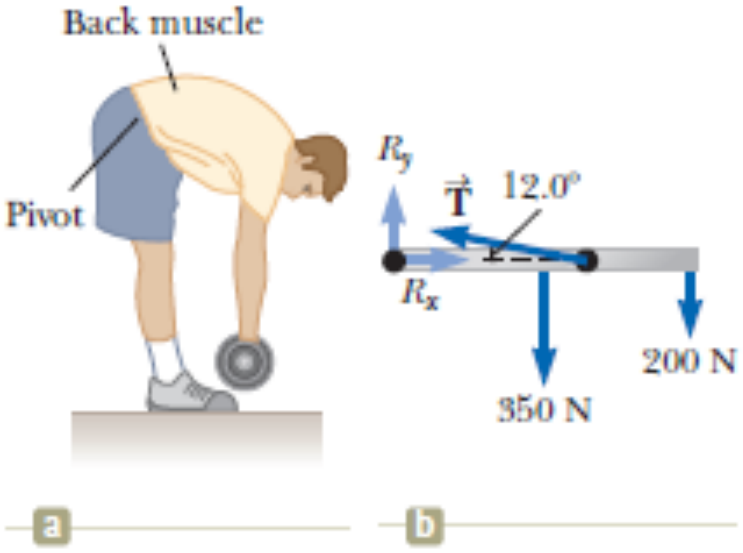
10. Assume a person bends forward to lift a load "with his back" as shown in Figure q6a. The spine pivots mainly at the fifth lumbar vertebra, with the principal supporting force provided by the erector spinal is muscle in the back. To see the magnitude of the forces involved, consider the model shown in Figure q6b for a person bending forward to lift a $200\,N$ object. The spine and upper body are represented as a uniform horizontal rod of weight $350\,N$, pivoted at the base of the spine. The erector spinal is muscle, attached at a point two-thirds of the way up the spine, maintains the position of the back. The angle between the spine and this muscle is $\theta = 12^{\circ}$. Find

    (a) the tension T in the back muscle and

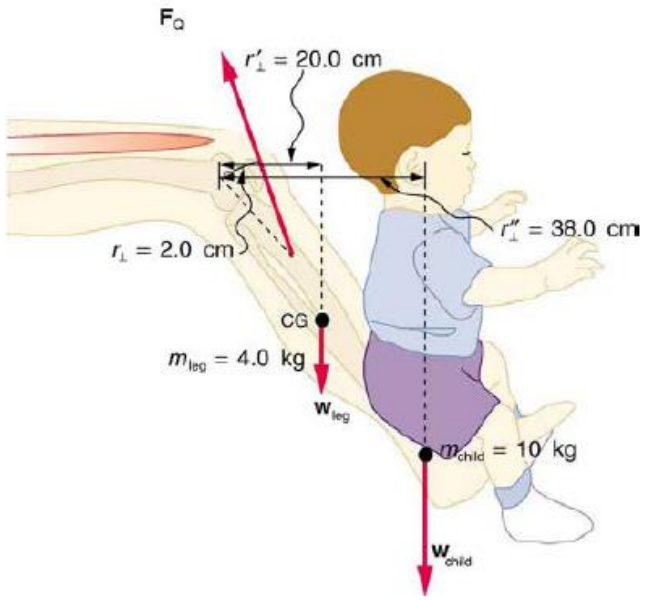
    (b) the compressional force in the spine.

    (c) Is this method a good way to lift a load? Explain your answer, using the results of parts (a) and (b).

    (d) Can you suggest a better method to lift a load



11. A father lifts his child as shown in Figure below. What force should the upper leg muscle exert to lift the child at a constant speed?



12. When a person stands on tiptoe (a strenuous position), the position of the foot is as shown in Figure q8a. The total weight of the body Fg is supported by the force n exerted by the floor on the toe. A mechanical model for the situation is shown in Figure 2q8b, where T is the force exerted by the Achilles tendon on the foot and R is the force exerted by the tibia on the foot. Find the values of T, R, and when $Fg = 700\ N$.

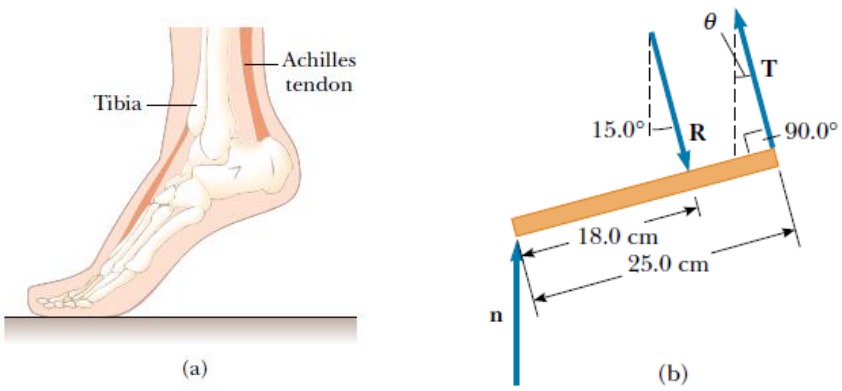
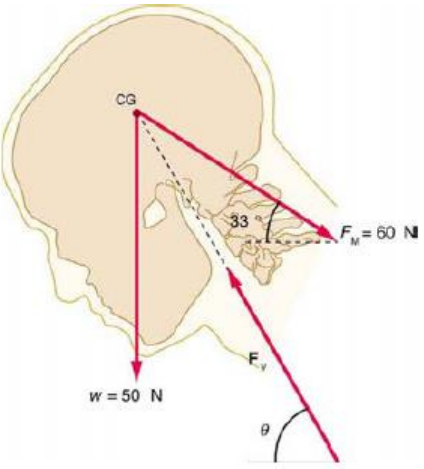


Figure q8: a man stands of his tiptoe and the free body diagram

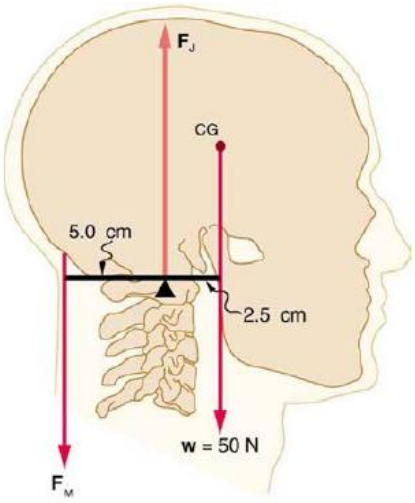
13. If $35\ kg$ is the maximum mass m that a person can hold in a hand when the arm is positioned with a $105°$ angle at the elbow as shown in Figure, what is the maximum force F_{max} that the biceps muscle exerts on the forearm? Assume the forearm and hand have a total mass of $2.0\ kg$ with a CG that is $15\ cm$ from the elbow, and that the biceps muscle attaches $5.0\ cm$ from the elbow.



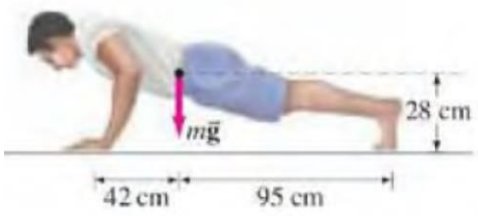
14. A person working at a drafting board may hold her head as shown in Figure, requiring muscle action to support the head. The three major acting forces are shown. Calculate the direction and magnitude of the force supplied by the upper vertebrae F_V to hold the head stationary, assuming that this force acts along a line through the center of mass as do the weight and muscle force.



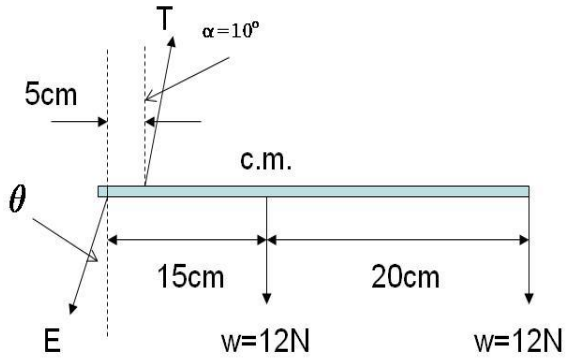
15. Even when the head is held erect, as in Figure 9.41, its center of mass is not directly over the principal point of support (the atlanto-occipital joint). The muscles at the back of the neck should therefore exert a force to keep the head erect. That is why your head falls forward when you fall asleep in the class. (a) Calculate the force exerted by these muscles using the information in the figure. (b) What is the force exerted by the pivot on the head?



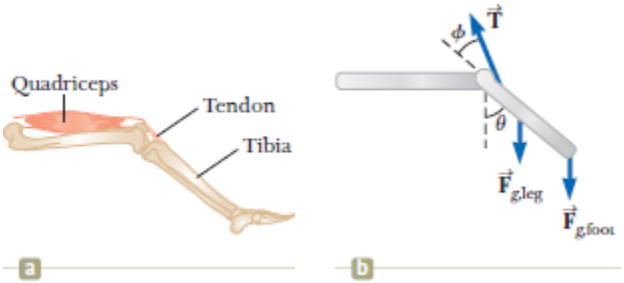
16. A man doing push-ups pauses in the position shown in Fig. 12-86. His mass $m = 68\ kg$. Determine the normal force exerted by the floor (a) on each hand; (b) on each foot.



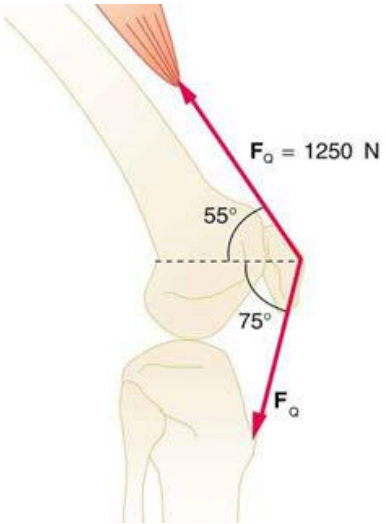
17. A realistic free body diagram model for the forearm. In the figure below, $w_1 = 12$ N find the tension T exerted by the muscle and the force exerted by the elbow joint E.



18. The large quadriceps muscle in the upper leg terminates at its lower end in a tendon attached to the upper end of the tibia (Figure below (a, b)). The forces on the lower leg when the leg is extended are modeled as in Figure b below, where $\boldsymbol{T}$ is the force in the tendon, $\boldsymbol{F_{g,leg}}$ is the gravitational force acting on the lower leg, and $\boldsymbol{F_{g,foot}}$ is the gravitational force acting on the foot. Find $\boldsymbol{T}$ when the tendon is at an angle of $\varphi = 25^{\circ}$ with the tibia, assuming $\boldsymbol{F_{g,leg} = 30.0\ N}$, $\boldsymbol{F_{g,foot} = 12.5\ N}$, and the leg is extended at an angle $\theta = 40^{\circ}$ with respect to the vertical. Also assume the center of gravity of the tibia is at its geometric center and the tendon attaches to the lower leg at a position one-fifth of the way down the leg.



19. The upper leg muscle (quadriceps) exerts a force of $1250\ N$, which is carried by a tendon over the kneecap (the patella) at the angles shown in Figure below. Find the direction and magnitude of the force exerted by the kneecap on the upper leg bone (the femur).



Chapter 3

Elastic Properties of Material



In mechanics we consider the effect of forces only on the motion of a body assuming that objects remain undeformed when external forces act on them. In reality, all objects are deformable. That is, it is possible to change the shape or the size of an object (or both) by applying external forces. As these changes take place, however, internal forces in the object resist the deformation. In this chapter, we will examine the effect of forces on the shape of the body. Depending on how the force is applied, the body may be stretched, compressed, bent, or twisted. The elastic properties of material are extremely important in determining the strength of materials and the optimal design of the object. We will review the theory of deformation in terms of stress, quantity that is proportional to the force causing a deformation; and strain, quantity that measures the degree of deformation, then examines the damaging effects of forces on bones and tissue.

3.1 Elasticity

Elasticity is the property of certain materials that enables them to return to their original size and shape after an applied force has been removed. Rubber is the first material that comes to mind when the word elasticity is mentioned. A solid rubber ball, for example, can be squeezed, stretched, bent, or twisted yet it will still revert to its original shape when the pressure or stress is removed.

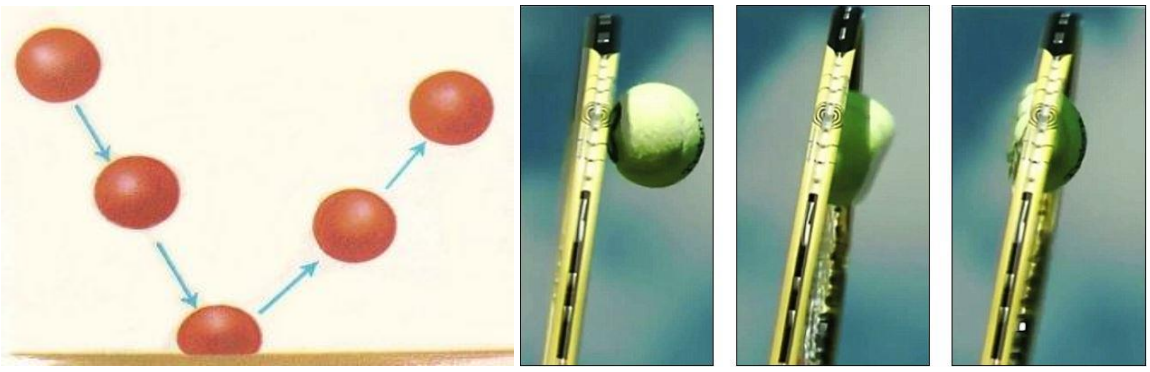


Figure 3.1: A rubber ball is badly deformed upon impact with the ground but always regains its original form. The other picture shows a ball turning to 'goo' after 140mph serve.

However, all solids possess this property to a certain extent, but in most cases it is barely noticeable. The only metal in which elasticity is very well developed is steel and even then it must be hardened steel. This is produced by rapidly cooling or *quenching* red-hot steel in cold

water, a process called tempering. Tempered steels are hard and brittle but very elastic. Springs are nearly always made of tempered steel which has been gently reheated, a process that destroys some of the elasticity but makes the steel softer and tougher.

The elastic properties of materials are due to forces acting between atoms or molecules. For example, the reason why rubber can be stretched so much is that it is built up of long molecular chains, most of which are folded like tangled ropes. When the material is stretched the chains simply straighten themselves out and when the force is removed they revert to their original tangled state. Many other materials, such as wood and silk, are built of chain molecules, but in most cases strong links between the chains prevent them from curling back upon themselves, so that elasticity is not pronounced.

Stress

Consider a long bar of cross-sectional area A and initial length L_i that is clamped at one end, as in Fig.3.2. When an external force is applied perpendicular to the cross section, internal forces in the bar resist distortion ("stretching"), but the bar attains an equilibrium in which its length L_f is greater than L_i and in which the external force is exactly balanced by internal forces. In such a situation, the bar is said to be stressed.

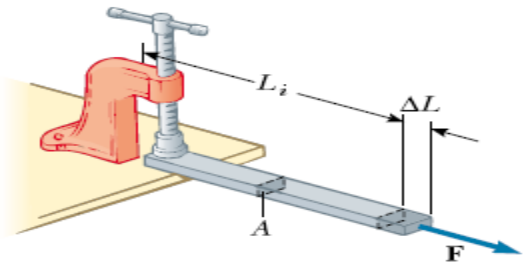


Figure 3.2: A long bar clamped at one end is stretched by an amount ΔL under the action of a force F.

Stress is a measure of the average amount of force exerted per unit area. It is a measure of the intensity of the total internal forces acting within a body across imaginary internal surfaces, as a reaction to external applied forces and body forces. It was introduced into the theory of elasticity by Cauchy around 1822. Stress is a concept that is based on the concept of continuum. In general, stress is expressed as

$$\sigma = \frac{F}{A} \tag{3.1}$$

Where σ is the average stress and is also called engineering or nominal stress, and F is the force acting over the area A.

The SI unit for stress is the Pascal (symbol Pa), which is a shorthand name for one Newton (Force) per square meter (Unit Area). The unit for stress is the same as that of pressure, which is also a measure of Force per unit area. Engineering quantities are usually measured in Megapascals (MPa) or Gigapascals (GPa). In English units, stress is expressed in pounds-force per square inch (psi) or kilopounds-force per square inch (ksi). If the force in Fig. 3.2 is reversed, the bar is compressed, and its length is reduced.

There are three kinds of stress commonly defined as:

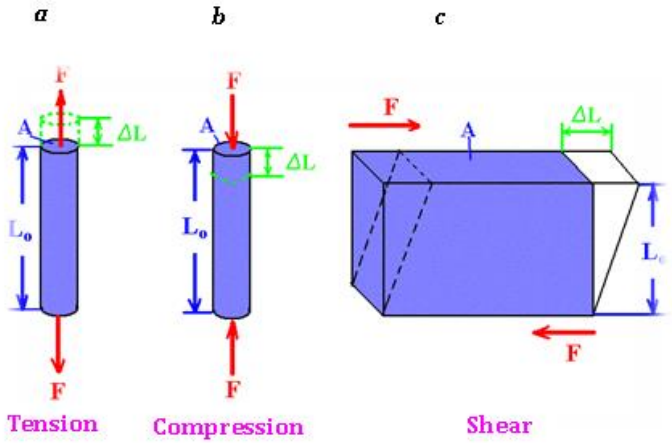


Figure 3.3: The three kinds of stress, (a) tensile stress or tension, (b) compression and (c) shear stress.

Tensile stress or tension which is the force per unit area producing elongation; however *Compression stresses* the force per unit area act to compress the object. *Shear stress* corresponds to scissor like force will be explained in detail later.

Strain

If we have two bars of the same material and the same diameter but with different lengths, the taller bar will stretch more than the short bar when they exposed to the same tension force. To

take this into account, we express the extension as a fractional change in length. We call this the **tensile strain** which we define as **the extension per unit length**.

$$\varepsilon = \frac{\Delta L}{l} \tag{3.2}$$

There are no units for strain; it's just a number. It can sometimes be expressed as a percentage.

You will find that the same is true for when we compress a material.

Stress- strain curve

When engineers design a bridge they need to know the behavior of the materials under forces which tend to stretch or compress i.e. tensile strength of the steel to be used in the construction. Similarly, the tensile strength of rubber is of great importance to manufacturers of tires and similar articles. The way in which this is determined involves adding weights to one end of a wire or strand of a substance to be tested, the other end being fixed as shown in Fig. 3.4. At first, the wire or strand increases its length by equal amounts if equal weights are added. For instance, if $10\ N$ weigh stretches a steel wire half a millimeter, a $20\ N$ weight will stretch it one millimeter, and so on. And when the weights are removed the wire or strand returns to its original length. The fact that *the extension of the wire is proportional to the force applied* was discovered as early as 1650 by Robert Hooke and is now known as ***Hooke's law***.

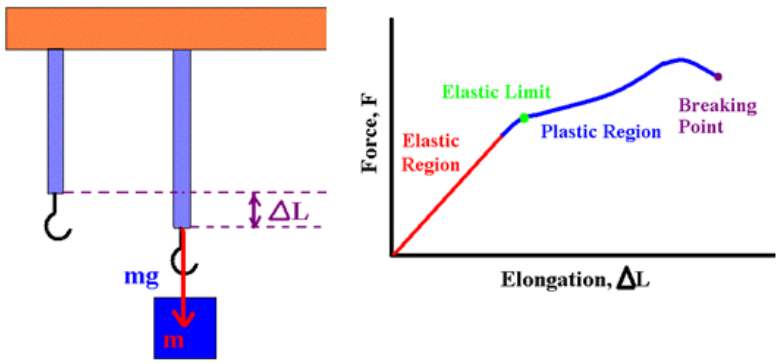


Figure 3.4: Stress-strain curve for elastic solid

$$F = -k\Delta L \tag{3.3}$$

Where $\boldsymbol{F}$ is the applied force, ΔL is the change in length, k is a constant called the spring constant, and the negative sign due the fact the that the spring's restoring force opposes the change in length. Figure 3.4 shows that in this region the relationship between $\boldsymbol{F}$ and ΔL is linear. So, it is called the Elastic Region. But this process does not go on indefinitely. There comes a region where the material does not respond to the applied force in linear way, this regionis called the Plastic Region, The wire or strand begins to stretch much further than Hooke's law would predict when extra weights added. Moreover, there is a point in this region when the weights are removed the material no longer reverts to its original length. The point at which this happens is called the Elastic Limit. The elastic limit of a substance is defined as the maximum stress that can be applied to the substance before it becomes permanently deformed. The amount of extension increases rapidly after the elastic limit has been reached until the wire suddenly parts at the Breaking Point. A material is said to be ductile (flexible) if it can be stressed well beyond its elastic limit without breaking. A brittle (fragile) material is one that breaks soon after the elastic limit is reached

3.2 Young modulus

The form of Hooke's law represented in Eq. 3.3 is not the most useful form because the value of *k* is different not just for each material but also for different cross-sectional areas. So we need to introduce a new quantity known as Young modulus or elastic modulus, which depends only on the material type. Young modulus relates the elastic deformation of a solid to the associated stresses. It is defined *as the ratio of stress over strain in the region in which Hooke's Law is obeyed for the material*. This can be experimentally determined from the slope of a stress-strain curve created during tensile tests conducted on a sample of the material.

$$Y = \frac{\sigma}{\varepsilon} \tag{3.4}$$

$$Y = \frac{F/A}{\Delta l/l} \tag{3.5}$$

From which one can find

$$F = \frac{YA}{l}\,\Delta l \tag{3.6}$$

Equation 3.6 is identical to the equation for a spring with a spring constant $\frac{YA}{l}$. Young's modulus is represented by the slope of the initial straight segment of the stress-strain diagram. More correctly, Y is a measure of stiffness (rigidity), having the same units as stress, Pascal or pounds per square inch.

Table3.1: typical values for elastic Modulus

substance	young modulus(N/m^2)	Shear modulus(N/m^2)	bulk modulus(N/m^2)
Tungsten	35×10^{10}	14×10^{10}	20×10^{10}
Steel	20×10^{10}	8.4×10^{10}	6×10^{10}
Copper	11×10^{10}	4.2×10^{10}	14×10^{10}
Brass	9.1×10^{10}	3.5×10^{10}	6.1×10^{10}
Aluminum	7×10^{10}	2.5×10^{10}	7×10^{10}
Glass	$6.5 - 7.8 \times 10^{10}$	$2.6 - 3.2 \times 10^{10}$	$5 - 5.5 \times 10^{10}$
Quartz	5.6×10^{10}	2.6×10^{10}	2.7×10^{10}
Water	–	–	0.21×10^{10}
mercury	–	–	2.8×10^{10}

Example 3.1

A vertical steel girder with a cross-sectional area of $0.15\ m^2$ has a $1550\ kg$ sign hanging from its end. (a) What is the stress within the girder? (b) What is the strain on the girder? (c) If the girder is $9.50\ m$ long, how much is it lengthened? (Ignore the mass of the girder itself.)

Solution

$A = 0.15\ m^2, m = 1550kg, l_0 = 9.5$ m,

$F = mg = \ 1550\ (9.8) = \ 15200\ N$

$$stress\ \sigma = \frac{F}{A} = \frac{mg}{A} = \frac{15200}{0.15} = 1.0 \times 10^5 N/m^2$$

$$Y = \frac{stress}{strain} = \sigma/\varepsilon$$

$$strain = \frac{stress}{Y} = \frac{1.0 \times 10^5}{200 \times 10^9} = 5.0 \times 10^{-7}$$

$$\Delta l = strain \times l_0 = 5 \times 10^{-7}(9.5) = 4.8 \times 10^{-6} m$$

Example 3.2

A 15 cm long animal tendon was found to stretch 3.7 mm by a force of 13.4 N. The tendon was approximately round with an average diameter of 8.5 mm. Calculate the elastic modulus of this tendon

Solution

$\Delta l = 3.7\ mm, l_0 = 15$ cm, 2r = 8.5 mm and F = 13.4 N

First we have to convert units to be in same unit system.

$\Delta l = 3.7 \times 10^{-3}\ m, l_0 = 0.15$ m, r = $4.25\ \times 10^{-3}$m

The cross sectional area A= $\pi r^2 = \pi \times (4.25\ \times 10^{-3})^2 = 5.7 \times 10^{-5} m^2$

$$\mathrm{Y} = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\varepsilon} = \frac{\mathrm{F/A}}{\Delta l/l_0} = \frac{13.4 \div 5.7\ \times 10^{-5}}{3.7 \times 10^{-3} \div 0.15\ \mathrm{m}} = 9.5 \times 10^6 N/m^2$$

Example 3.3

A nylon tennis string on a racquet is under a tension of $250\ N$. If its diameter is $1.00\ mm$, by how much is it lengthened from its un-tensioned length?

Solution

$$2r = 1\ mm, F = 250\ N,\ Y = 5 \times \frac{10^9 N}{m^2}, \Delta l = ?$$

$$A = \pi r^2 = \pi(0.5 \times 10^{-3})^2 = 7.85 \times 10^{-7} m^2$$

$$\boldsymbol{Y = \frac{stress}{strain} = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta l/l_0} = \frac{(250\text{N})/(7.9 \times 10^{-7} m^2)}{\Delta l/l_0} = 5 \times 10^9 N/m^2}$$

Solving for $\Delta l/l_0$

$$\frac{\Delta l}{l_0} = \frac{(250\text{N})/(7.9 \times 10^{-7} m^2)}{5 \times 10^9\ N/m^2} = 0.0636$$

3.3 Elastic Strain Energy

When we stretch a wire, a work done on the wire. We are stretching the bonds between the atoms. If we release the wire, we can recover the energy stored in the wire due to stretch, which is called the **elastic strain energy**.

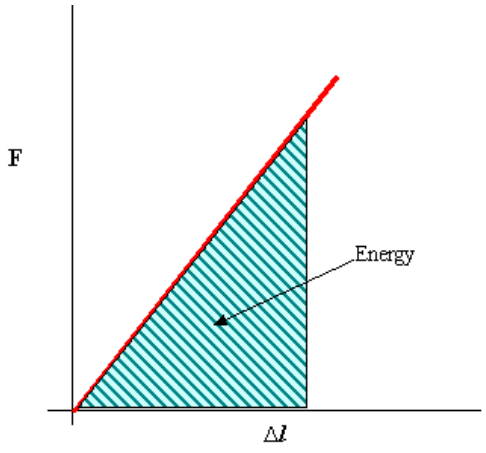


Figure 3.5: graphic representation of the elastic strain energy.

Ideally we recover all of it, but in reality a certain amount is lost as heat. This lost energy is called **hysteresis**. The energy is the **area** under the **force-extension** graph (the area of the triangle) (see Fig.3.5). So we can use this result to say:

$$\mathcal{E} = \frac{1}{2} F \Delta l \tag{3.7}$$

Substituting for F from eq. 3.5 we get

$$\mathcal{E} = \frac{YA}{2l} \Delta l^2 \tag{3.8}$$

Example 3.4

What is the elastic strain energy contained in a copper wire of diameter $0.8\,mm$ and $16\,cm$ length that has stretched by $4\,mm$ under a load of $400\,N$?

Solution

$$\mathcal{E} = \frac{1}{2} F \Delta l = 0.5 \times 400N \times 4 \times 10^{-3} m = 0.8\,J$$

Example 3.5

A scallop forces open its shell with an elastic material called Abductin (Biomaterial), whose elastic modulus is $2.0 \times 10^6\ N/m^2$. If this piece of Abductin is $3.0\,mm\ thick$, and has a cross-sectional area of $0.50\ cm^2$, how much potential energy does it store when compressed $1.0\,mm$?

Solution

$$Y = 2.0 \times 10^6 Nm^2, A = 0.50 cm^2 = 0.50 \times 10^{-4} m^2, \Delta l = 1.0 \times 10^{-3} m,$$

$l = 3 \times 10^{-3} m$ and $\mathcal{E} = \frac{1}{2} \frac{YA}{l} \Delta l^2$

$$= \frac{1}{2} \frac{(2.0 \times 10^6)(0.50 \times 10^{-4})}{3 \times 10^{-3}} (1.0 \times 10^{-3})^2 = 0.017J$$

3.4 Bone Fracture: Energy Considerations

Knowledge of the maximum energy that parts of the body can safely absorb allows us to estimate the possibility of injury under various circumstances. We shall first calculate the amount of energy required to break a bone of area A and length l. Assume that the bone remains elastic until fracture. Let us designate the breaking stress of the bone as σ_B. The corresponding force F_B that will fracture the bone is, from Eq. 3.6,

$$F_B = \sigma_B A = \frac{YA}{l}\Delta l \tag{3.9}$$

The compression Δl at the breaking point is, therefore,

$$\Delta l = \frac{\sigma_B l}{Y} \tag{3.10}$$

From Eq. 3.8, the energy stored in the compressed bone at the point of fracture is

$$\mathcal{E} = \frac{YA}{2l}\left(\frac{\sigma_B l}{Y}\right)^2$$

$$\mathcal{E} = \frac{Al{\sigma_B}^2}{2Y} \tag{3.11}$$

Example 3.6

Discuss the possibility of fracture of two leg bones that each have a length of about $90\ cm$ and an average area of about $6\ cm^2$ when a $70\ kg$ person jump from a height of $60\ cm$.

Solution

First we calculate the maximum energy the two legs can absorb before fracture:

The breaking stress of the bone σ_B is $10^9\ dyn/cm^2$, and Young's modulus for the bone is $14\ 10^{10}\ dyn/cm^2$. ($Hint$: $dyn = 10^{-5}N = 10\mu N, erg = 10^{-7}J,\ dyn/cm^2 \equiv 10^{-1}\ N/m^2$)

The total energy absorbed by the bones of one leg at the point of compressive fracture is, from Eq. 3.11,

$$\mathcal{E} = \frac{1}{2}\frac{Al{\sigma_B}^2}{Y}$$

$$= \frac{1}{2}\frac{(6 \times 10^{-4} m^2)(0.9m)(10^8 N/m^2)^2}{14 \times 10^9 N/m^2} = 192.86J$$

Now we have to calculate the energy gained by jumping

$$\mathcal{E} = mgh = 70\ (9.8)(0.60) = 411.6\,J$$

Now, this energy is evenly distributed on both legs, thus each leg absorbs energy equals to $411.6J/2 = 205.8J$. This number exceeds the allowed energy to be stored at the bone break, which will endanger the leg bones and it would cause bone fracture. If this person jumped on one leg this might lead to multiple fracture because of the huge difference between the allowed energy (less than $192.86J$) and the energy stored due to jumping ($411.6J$).

This usually doesn't happen! Why! It is certainly possible to jump safely from a height considerably greater than 60 cm if, on landing, the joints of the body bend and the energy of the fall is redistributed to reduce the chance of fracture, this can be done by experienced person like athletes. The calculation does however point out the possibility of injury in a fall from even a small height.

Example 3.7

What is the elastic strain energy contained in a copper wire of diameter $0.8\,mm$ and $16\,cm$ length that has stretched by $4\,mm$ under a load of $400\,N$?

Solution

Notice that, you have extra pieces of information you do not need!

$$\mathcal{E} = \frac{1}{2}F\Delta l = 0.5 \times 400N \times 4 \times 10^{-3} m = 0.8\,J$$

If this energy is less than the maximum energy can be stored in the wire, the wire will not break!

3.5 Shear Modulus: Elasticity of Shape

Another type of deformation occurs when an object is subjected to a force tangential to one of its faces while the opposite face is held fixed by another force (friction force between the object base and the holding surface) (Fig.3.6 right). The stress in this case is called *a shear stress*. If the object is originally a rectangular block, a shear stress results in a shape whose cross-section is a parallelogram.

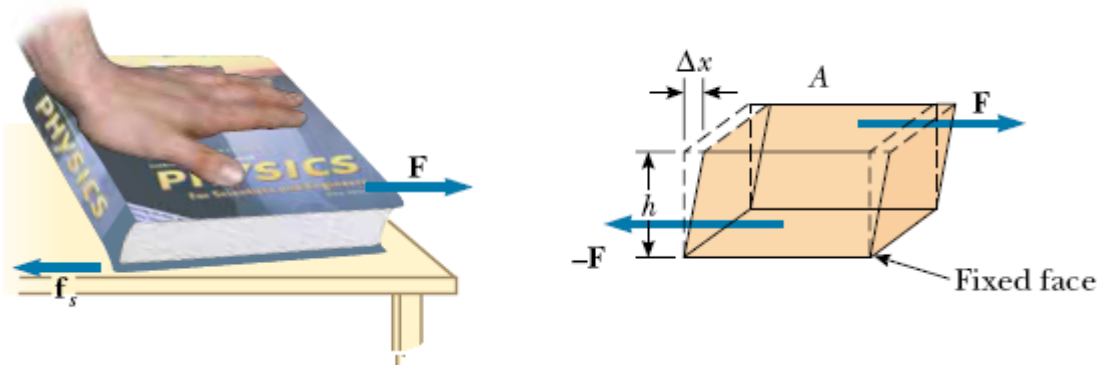


Figure 3.6: A shear deformation in which a rectangular block is distorted by two forces of equal magnitude but opposite directions applied to two parallel faces (right). A book under shear stress (left).

A book pushed sideways, as shown in Fig. 3.6b, is an example of an object subjected to a shear stress. To a first approximation (for small distortions), no change in volume occurs with this deformation.

We define the shear stress as F/A, the ratio of the tangential force to the area A of the face being sheared.

The shear strain is defined as the ratio $\Delta x/h = \tan\alpha$, where Δx is the horizontal distance that the sheared face moves and h is the height of the object. It is a measure of the angular deformation.

In terms of these quantities, the shear modulus is

$$S = \frac{shear\ stress}{shear\ strain} = \frac{\sigma_s}{\epsilon_S} = \frac{F/A}{\Delta x/h} \qquad (3.12)$$

Example 3.8

A skyscraper has an outer skin of brick-faced concrete panels attached to a structural frame by steel pins. Each pin is a cylinder of radius $0.01\,m$ and supports a mass of $1000\,kg$. (a) what is the shear stress on a pin? (b) what is the shear strain? (c) If the maximum shear stress for pins is $2 \times 10^8 Nm^{-2}$, how large a safety factor is built into design?

Solution

(a) The shear forces on a pin are equal to the weight it supports. Thus the stress is

$\sigma_s = F/A = mg/(\pi r^2) = 1000(9.8)/(\pi(0.01)^2) = 3.12 \times 10^7\ Nm^{-2}$

(b) With $S = \sigma_s/\varepsilon_s$

$\varepsilon_s = \sigma_s/S = (3.12 \times 10^7)/(8.4 \times 10^{10}) = 3.71 \times 10^{-4}$

Since $\varepsilon_s = \tan\alpha$, this corresponds to angular deformation $\alpha = 0.02^o$

c. The ratio of the maximum shear stress to its actual value is

$$\frac{3 \times 10^8}{3.12 \times 10^7} = 6.41$$

The pins can support over 6 times their load without breaking. Buildings are designed conservatively to allow for fatigue and abnormal stresses.

3.6 Bulk Modulus: Volume Elasticity

Bulk modulus characterizes the response of a substance to uniform squeezing or to a reduction in pressure when the object is placed in a partial vacuum. Suppose that the external forces acting on an object are at right angles to all its faces, as shown in Fig. 3.8, and that they are distributed uniformly over all the faces. Such a uniform distribution of forces occurs when an object is immersed in a fluid. An object subject to this type of deformation undergoes a change in volume but no change in shape.

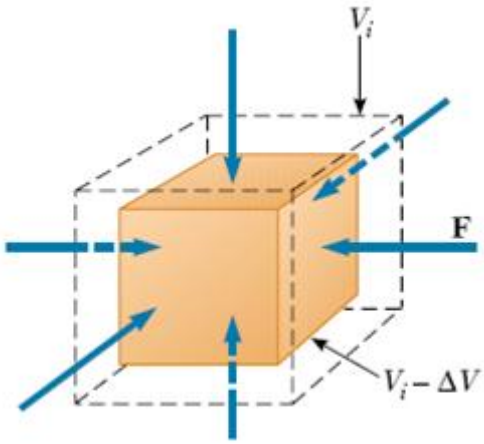


Figure 3.8: This cube is compressed on all sides by forces normal to its six faces**.**

The volume stress is defined as the ratio of the magnitude of the normal force F to the area A. The quantity $P = \frac{F}{A}$ is called the pressure. If the pressure on an object changes by an amount $\Delta P = \Delta F/A$, then the object will experience a volume change ΔV. The volume strain is equal to the change in volume ΔV divided by the initial volume V_i. Thus, from Eq. 3.4, we can characterize a volume ("bulk") compression in terms of the bulk modulus, which is defined as

$$B = (volume\ stress)/(volume\ strain) = -(F/A)/(\Delta V/V_i) \tag{3.13}$$

A negative sign is inserted in this defining equation so that *B* is a positive number. This maneuver is necessary because an increase in pressure (positive ΔP) causes a decrease in volume (negative ΔV) and vice versa. Table 3.1 lists bulk module for some materials. If you look up such values in a different source, you often find that the reciprocal of the bulk modulus is listed. The reciprocal of the bulk modulus is called the *compressibility* of the material. Note from Table 3.1 that both solids and liquids have a bulk modulus. However, no shear modulus and no Young's modulus are given for liquids because a liquid does not sustain a shearing stress or a tensile stress (it flows instead).

Example 3.9

How much pressure is needed to compress the volume of an iron block by 0.10%? Express answer in N/m^2, and compare it to atmospheric pressure ($1.0 \times 10^5\ N/m^2$).

Solution

$From\ table\ 3.1\ the\ bulk\ modulus\ for\ iron = \ 90 \times 10^9 N/m^2$

$$\frac{\Delta V}{V_0} = \frac{0.1}{100}$$

$$B = \frac{\Delta P}{\Delta V/V_0}$$

$$90 \times 10^9 = \frac{\Delta P}{10^{-3}}\ \ Solving\ for\ \Delta P\ we\ find$$

$$\Delta P = 90 \times 10^6\ N/m^2$$

Summary

Stress is a quantity proportional to the force producing a deformation;

$$\sigma = \frac{F}{A} \tag{3.1}$$

Where σ is the average stress and is also called engineering or nominal stress, and F is the force acting over the area A.

Strain is a measure of the degree of deformation.

$$\varepsilon = \frac{\Delta L}{l} \tag{3.2}$$

Strain is proportional to stress, and the constant of proportionality is the elastic modulus:

$$Y = \frac{F/A}{\Delta l/l} \tag{3.5}$$

If we released under tension wire, the wire can recover the energy stored in the wire due to stretch, which is called the elastic strain energy.

$$\mathcal{E} = \frac{1}{2} F \Delta l \tag{3.7}$$

Three common types of deformation are

1. The resistance of a solid to elongation under a load, characterized by Young's modulus Y;

2. The resistance of a solid to the motion of internal planes sliding past each other, characterized by the shear modulus S;

$$S = \frac{shear\ stress}{shear\ strain} = \frac{\sigma_s}{\epsilon_S} = \frac{F/A}{\Delta x/h} \tag{3.12}$$

3. The resistance of a solid or fluid to a volume change, characterized by the bulk modulus B.

$$B = (volume\ stress)/(volume\ strain) = -(F/A)/(\Delta V/V_i) \tag{3.13}$$

Problems

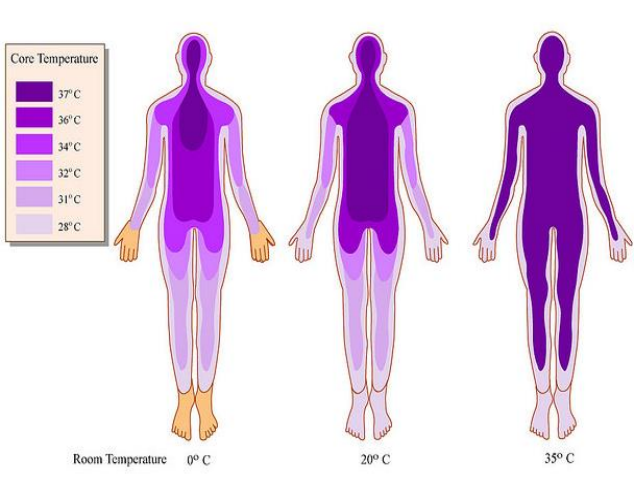
1. A $200 - \text{kg}$ load is hung on a wire with a length of $4\ \text{m}$, a cross-sectional area of $0.2 \times 10^{-4}\ \text{m}^2$, and a Young's modulus of $8 \times 10^{10}\ \text{N/m}^2$. What is its increase in length?
2. During a circus act, one performer swings upside down hanging from a trapeze holding another, also upside-down, performer by the legs. If the upward force on the lower performer is three times her weight, how much do the bones (the femurs) in her upper legs stretch? You may assume each is equivalent to a uniform rod 35.0 cm long and 1.80 cm in radius. Her mass is 60.0 kg
3. During a wrestling match, a 150 kg wrestler briefly stands on one hand during a maneuver designed to perplex his already moribund adversary. By how much does the upper arm bone shorten in length? The bone can be represented by a uniform rod 38.0 cm in length and 2.10 cm in radius
4. Assume that Young's modulus for bone is $1.5 \times 10^{10}\ \text{N/m}^2$ and that a bone will fracture if more than $1.50 \times 10^{8}\ \text{N/m}^2$ is exerted. (a) What is the maximum force that can be exerted on the femur bone in the leg if it has a minimum effective diameter of $2.50\ \text{cm}$? (b) If a force of this magnitude is applied compressively, by how much does the $25.0 - \text{cm} - \text{long}$ bone shorten?
5. TV broadcast antennas are the tallest artificial structures on Earth. In 1987, a 72.0-kg physicist placed himself and 400 kg of equipment at the top of one 610-m high antenna to perform gravity experiments. By how much was the antenna compressed, if we consider it to be equivalent to a steel cylinder 0.150 m in radius?

6. A steel wire $1\ mm$ in diameter can support a tension of $0.2\ kN$. Suppose you need a cable made of these wires to support a tension of $20\ kN$. The cable's diameter should be of what order of magnitude?
7. If the elastic limit of copper is $1.5 \times 10^8\ \mathrm{N/m^2}$, determine the minimum diameter a copper wire can have under a load of $10.0\ kg$ if its elastic limit is not to be exceeded.
8. (a) Find the minimum diameter of a steel wire 18 m long that elongates no more than 9 mm when a load of 380 kg is hung on its lower end. (b) If the elastic limit for this steel is $3 \times 10^8\ \mathrm{N/m^2}$, does permanent deformation occur with this load?
9. For safety in climbing, a mountaineer uses a $50\ m$ nylon rope that is $10\ mm$ in diameter. When supporting the $90\ kg$ climber on one end, the rope elongates by $1.60\ m$. Find Young's modulus for the rope material.
10. A pipe has an inner radius of $0.02\ m$ and an outer radius of $0.023\ m$. If subjected to a tension stress of $5 \times 10^7 \mathrm{Nm^{-2}}$, how large is the applied force?
11. A man leg can be thought of as a shaft of bone $1.2\ m$ long. If the strain is 1.3×10^{-4} when the leg supports his weight, by how much is his leg shortened?
12. What is the spring constant of a human femur under compression of average cross-sectional area $10^{-3} m^2$ and length $0.4\ m$?
13. Calculate the force a piano tuner applies to stretch a steel piano wire 8.00 mm, if the wire is originally 0.850 mm in diameter and 1.35 m long
14. The average cross-sectional area of a woman femur is $10^{-3} m^2$ and it is $0.4\ m$ long. The woman weights $750\ N$ (a) what is the length change of this bone when it supports half of the weight of the woman? (b) Assuming the stress-strain relationship is linear until fracture, what is the change in length just prior to fracture?
15. A 20.0-m tall hollow aluminum flagpole is equivalent in strength to a solid cylinder 4.00 cm in diameter. A strong wind bends the pole much as a horizontal force of 900 N exerted at the top would. How far to the side does the top of the pole flex??
16. A vertebra is subjected to a shearing force of 500 N. Find the shear deformation, taking the vertebra to be a cylinder 3.00 cm high and 4.00 cm in diameter
17. A disk between vertebrae in the spine is subjected to a shearing force of 600 N. Find its shear deformation, taking it to have the shear modulus of $1.0 \times 10^9 N/m^2$. The disk is equivalent to a solid cylinder 0.700 cm high and 4.00 cm in diameter.

18. A farmer making grape juice fills a glass bottle to the brim and caps it tightly. The juice expands more than the glass when it warms up, in such a way that the volume increases by 0.2% (that is, , $\frac{\Delta V}{V_0} = 2 \times 10^{-3}$)) relative to the space available. Calculate the force exerted by the juice per square centimeter if its bulk modulus is $1.8 \times 10^9 N/m^2$, assuming the bottle does not break. In view of your answer, do you think the bottle will survive?
19. Assume that a $50\ kg$ runner trips and falls on his extended hand. If the bones of one arm absorb all the kinetic energy (neglecting the energy of the fall), what is the minimum speed of the runner that will cause a fracture of the arm bone? Assume that the length of arm is 1 m and that the area of the bone is $4\ cm^2$.
20. Calculate the density of sea water at a depth of 1 000 m, where the water pressure is about $1.0 \times 10^7\ \mathrm{N/m^2}$. (The density of sea water is $1.03 \times 10^3\ \mathrm{kg/m^3}$ at the surface.)
21. If the shear stress exceeds about $4 \times 10^8\ \mathrm{N/m^2}$, steel ruptures. Determine the shearing force necessary (a) to shear a steel bolt $1.0\ cm$ in diameter and (b) to punch a $1.0\ cm$-diameter hole in a steel plate $0.5\ cm$ thick.
22. When water freezes, it expands by about 9%. What would be the pressure increase inside your automobile's engine block if the water in it froze? (The bulk modulus of ice is $2 \times 10^9\ \mathrm{N/m^2}$.)

Chapter 4

Thermal Properties of Matter





In this chapter we are concerned to study the thermal properties of matter which includes the concepts of heat, internal energy and temperature. The concept of temperature plays an important role in the physical and biological sciences. Any change of the temperature of the human skin is an indicator of the change of the internal energy of the biological cells, so the temperature concept plays the role of a label for such change.

4.1 Temperature and Heat

Temperature concept is related to the internal energy of the objects or reflects the average kinetic energy of the molecules or atoms composed the material. This quantity is a scalar quantity and tells how warm or cold an object is with respect to some standard. We express the temperature of a matter by a number that corresponds to the degree of hotness or coldness on some chosen scale. Therefore, the temperature can be defines as the property that determines whether or not the body in thermal equilibrium with its surroundings.

Heat it is a form of energy that is transferred from one body to another because of a temperature difference. It is related to temperature and describes the process of energy transfer from one object to another. Matter does not contain heat; however it contains molecular kinetic energy and possibly potential energy. If you touch a hot stove, energy enters your hand because the stove is warmer than your hand. The direction of spontaneous energy transfer is always from a warmer object to a neighboring colder one. Because of a temperature difference between the object, this transfer of energy is called heat. So, heat is energy in transient.

The energy resulting from heat flow is often called thermal energy, to make clear its link to heat and temperature. Scientists, however, often prefer to use internal energy term. The internal energy is the total of all energies inside the material, such as:

- Translational kinetic energy of atoms.
- Rotational and vibrational kinetic energy of molecules.
- Kinetic energy due to internal movements of atoms within molecules.
- Potential energy due to (attractive) forces between molecules.

So the substance does not contain heat, but it does contain internal energy. The internal energy of a system depends on its mass, or the number of molecules of the system.

4.2 Thermometers and Temperature Scales

A measure of temperature is obtained using a thermometer, a device constructed to make evident some property of a substance that changes with temperature. Many physical properties of materials change sufficiently with temperature to be used as the bases for thermometers:

- ⇨ The change in volume of a liquid
- ⇨ The change in length of a solid
- ⇨ The change in pressure of a gas held at constant volume
- ⇨ The change in electrical resistance of a conductor
- ⇨ The change in color of a very hot object.

By far the most obvious and commonly used property is thermal expansion, a change in the dimensions or volume of a substance that occurs when the temperature changes. A common thermometer is the liquid-in-glass type, which is based on the thermal expansion and contraction of a liquid, usually mercury or colored alcohol. These substances were chosen because of their relatively large thermal expansion and because they remain liquids over normal temperature ranges. Moreover, thermometers have to be sensitive enough to detect small changes in temperature, does show the temperature of the body in short time, and it doesn't take large quantity of heat for its own from the measured body.

Zeroth Law of Thermodynamics

To understand the basic principle on which the thermometer operates, we should define two concepts. First, when heat is transferred between two objects, whether or not they are touching, they are said to be in thermal contact. Second, when there is no longer a net heat transfer between objects in thermal contact, they are at the same temperature and are said to be in thermal equilibrium. This is called the zeroth law of thermodynamics, or the law of equilibrium, which state that:

If bodies A and B are separately in thermal equilibrium with a third body, C, then A and B will be in thermal equilibrium with each other if placed in thermal contact.

The Zeroth Law means we can measure whether two bodies have the same temperature without bringing them into thermal contact provided we have a thermometer. Thus, we can use a thermometer to take temperature readings of two different classrooms to establish whether they are at the same temperature; there is no need to physically bring the classrooms together. A thermometer registers its own temperature. When a thermometer is in thermal contact with something whose temperature we wish to know, energy will flow between the two until their temperatures are equal and thermal equilibrium is established. If we know the temperature of the thermometer, we then know the temperature of the something. The Zeroth Law implies that thermometer readings have a physical meaning attached to them, but we still need to calibrate a thermometer. This brings us to the topic of temperature scales.

The Celsius Scale

Thermometers are calibrated so that a numerical value may be assigned to a given temperature. For the definition of any standard scale or unit, two fixed reference points are needed. The ice point and the steam point of water are two convenient fixed points (temperatures at which water freezes and boils under a pressure of one atmosphere). A thermometer so calibrated is often called a centigrade thermometer (from centum: hundred; grades: degree). It is now called a Celsius thermometer, in honor of the Swedish astronomer Anders Celsius (1701-1744), who first suggested the scale (or at least, an inverted version of it). The space between the two reference points is divided into 100 equal parts called degrees. On the scale commonly used in laboratories, the number 0 is assigned to the temperature at which water freezes and the number 100 to the temperature at which water boils (at standard atmospheric pressure).

Temperature readings (T_C) of this scale are written as ℃ (degree Celsius); and temperature difference (ΔT_C) are commonly written as $C°$.

The Fahrenheit Scale

In the USA, the number 32 is assigned to the temperature at which water freezes, and the number 212 to the temperature at which water boils. On the Fahrenheit scale, there are 180 equal intervals, or degrees, between the two reference points. Such a scale makes up a Fahrenheit thermometer, named after its originator, the German physicist Gabriel Daniel Fahrenheit (1686-1736). The Fahrenheit scale has smaller degrees than the Celsius ($1°\mathrm{F} =$

$5/9$°C), which gives greater accuracy when reporting the weather in whole-number temperature readings.

Temperature readings (T_F) of this scale are written as °F (degree Fahrenheit); temperature differences (ΔT_F) are commonly written as F°.

The Kelvin Scale

The Kelvin scale, named after the British physicist Lord Kelvin (1824-1907), is a more fundamental temperature scale favored by scientists. The Kelvin scale is calibrated not in terms of the freezing and boiling points of water, but in terms of energy itself. The number 0 is assigned to the universal zero of temperature, i.e., the lowest possible temperature: absolute zero. The second reference (fixed) point is the triple point of water, which represents a unique set of conditions where water co-exists simultaneously in equilibrium as a solid, liquid and a gas. The conditions for the triple point are a pressure of $610\ Pa$, and a temperature taken to be $273.16\ K$ $(0.010$°C$)$. Note that this provides some connection with the other temperature scales, which are based on the properties of water. The Kelvin is defined as $1/273.16$ of the temperature at the triple point of water. Absolute zero corresponds to -2730°C on the Celsius scale $(-273.150$°C to be precise). Units on the Kelvin scale are the same size as degrees on the Celsius scale (so the temperature of melting ice is $+273\ Kelvin$). There are no negative numbers on the Kelvin scale.

Temperature (T_K) and temperature differences (ΔT_K) are stated in Kelvin (not degrees Kelvin), abbreviated as K (not °K).

Conversions

Arithmetic formulas are used to convert between the Fahrenheit, Celsius, and Kelvin temperature scales.

To convert between the Celsius and Fahrenheit temperature scales, consider we have a body its temperature level is at **M**, its temperature °C is, figure 1,

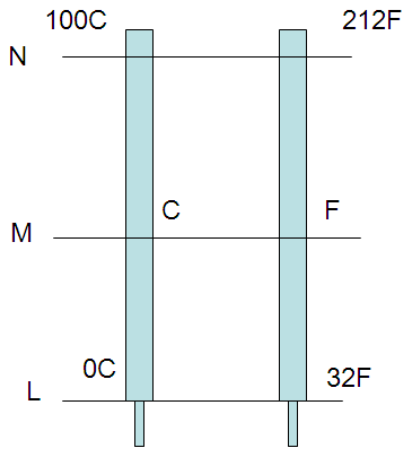


Figure 1: comparison between Fahrenheit Scale and Celsius Scale

$$\frac{ML}{NL}=\frac{C-0}{100-0}$$

and its equivalent temperature on the Fahrenheit scale is

$$\frac{ML}{NL}=\frac{F-32}{212-32}$$

Equate these equations, since they give the same temperature readings, figure 1, that is

$$\frac{F-32}{212-32}=\frac{C}{100}\Rightarrow\frac{F-32}{180}=\frac{C}{100}$$

So,

$$\begin{aligned} T_F &= \frac{9}{5}T_C+32 \quad \text{Celsius to Fahrenheit conversion} \\ T_C &= \frac{5}{9}\left(T_F-32\right) \quad \text{Fahrenheit to Celsius conversion} \end{aligned} \tag{4.1}$$

Note that $\Delta T_C = \frac{5}{9}\Delta T_F$

To convert between the Celsius and Kelvin temperature scales:

$$\begin{aligned} T_K &= T_C+273.15 \quad \text{Celsius to Kelvin conversion} \\ T_C &= T_K-273.15 \quad \text{Kelvin to Celsius conversion} \end{aligned} \tag{4.2}$$

- Note that $\Delta T_C = \Delta T_K$

Temperature Ranges

The Universe sustains an incredible temperature range. The highest temperatures likely to exist at this moment are found deep within stars: $\sim 4 \times 10^9 K$ seems to be a theoretical extreme. At a temperature only ten times higher matter fragments into subatomic particles. At the start of the Universe (10-20 billion years ago), the temperature is believed to have been $1039K$ (now it has cooled down to $3K$ - we are lucky to have a star called the Sun to keep us warm!). We live our delicate lives within a tiny band of hot and cold. The hottest thing you are likely to find around the house is a tungsten light bulb filament: $\sim 2800\ K$. Body temperature is about $310K\ (370C;\ 98.60F)$. Although temperature has a lower limit, there does not appear to be an upper limit. In the quest for absolute zero, experimenters have got bulk matter close to temperatures $\sim 0.00000002K$.

Example 4.1

Find the temperature at which the Fahrenheit and the Celsius scales coincide?

Solution

This will happen when $T_C = T_F$, so we can put in the conversion equation that:

$$T_C = \frac{5}{9}(T_C - 32) \Rightarrow 9T_C = 5T_C - 160 \Rightarrow 4T_C = -160 \Rightarrow so\ T_C = T_F = -40,$$

so the degree – 40 is the same in both scales.

4.3 Gas Laws: Macroscopic Description of an Ideal Gas

The experiments show that all gases have low densities, which mean larger intermolecular spacing in comparison with liquids or solids. The ideal gas can be described that gas which have very low density, a very weak intermolecular forces and almost no collisions between its molecules. The variables that describe the behavior of a given quantity (mass) of gas are:

$$Pressure\ P,\ \ Volume\ V,\qquad Temperature\ T,\ \ number\ of\ moles\ of\ a\ gas\ n$$

The equation that inter-relates all these quantities is called the equation of state.

Number of moles (n)

As you may have learned in a chemistry course, one mole of a gas is the quantity that contains $N_A = 6.02 \times 10^{23}$ molecules/mole (Avogadro's number) and equals the molecular weight of the substance expressed in grams. The number of moles, n, of a substance is related to its mass m, by

$$n = \frac{m}{M_w} = \frac{N}{N_A} \tag{4.3}$$

where N is the number of molecules, M_w is the molecular weight of a substance, usually expressed in grams per mole (e.g., the mass of one mole of oxygen is $32.0\ g$).

Temperature

The concept of temperature has been discussed in the previous section, however note that the temperature referred to in the equation of state or all related gases laws always in the absolute temperature or Kelvin scale.

Pressure

The pressure is defined as the normal forces (F_N) affect on a certain area (A), $P = F_N/A$. The pressure is measured by Newton per square meter (N/m^2), which is called a Pascal (Pa). Also the pressure can be measured using different units like atmospheric pressure (atm), bar, $torr\ or\ mmHg$. The following table represents the conversions between the different pressure units.

Table4.1: conversion between the different pressure unit systems.

	Torr, mmHg	Atm	Bar	Pascal
Pascal	0.0075	9.87×10^{-6}	10^{-5}	1
Bar	750	0.987	1	10^5
atm	760	1	1.013	1.013×10^5
Torr , mmHg	1	1.32×10^{-3}	1.33×10^{-3}	133.3

Volume

The volume of any solid, liquid or gas is how much three-dimensional space it occupies, often quantified numerically. The unit of volume is m^3 in SI system or cm^3 in cgs one. The liquids and gases are often measured by liter, where $1\ Liter\ =\ 10^{-3}\ m^3$.

Equation of state

This equation inter-relates the pressure, volume, temperature of a gas. Within a certain range of temperatures and pressures, many gases have been found to follow three simple laws. A gas that follows these laws completely is an idealization called an ideal gas.

Boyle's Law

When temperature of a gas is held constant, the pressure and volume of a quantity of gas are related as, figure 4.2, follows:

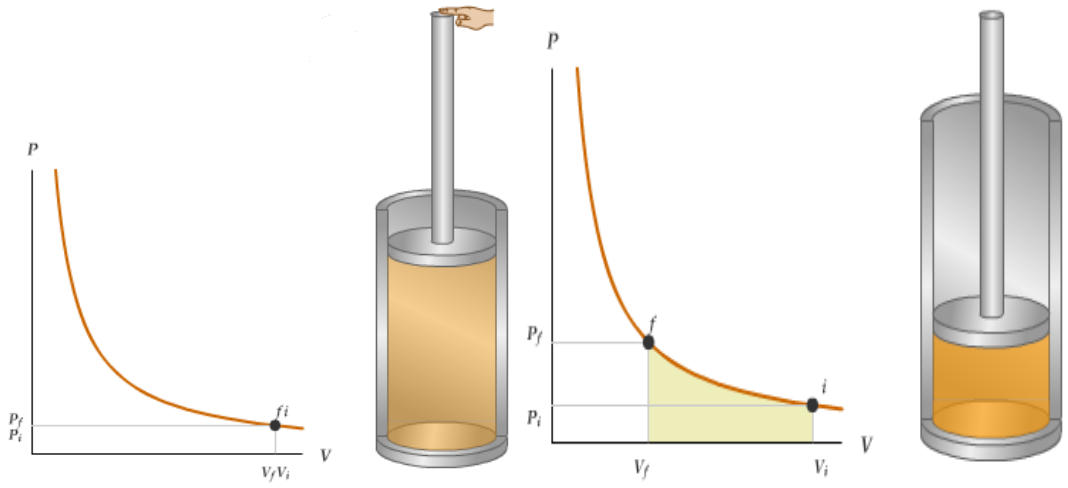


Figure 4.2: relation between the pressure and volume at constant temperature

$$PV = Constant, \qquad or \qquad P_1V_1 = P_2V_2 \tag{4.4}$$

This relation, the product of pressure and volume is constant, is known as Boyle's law. This product has the unit of energy ($N.m\ =\ Joule$) or the work exerted by the gas, which held constant at fixed temperature.

Charles' Law

When the pressure is held constant, the volume of a quantity of gas is related to the (absolute) temperature, figure 4.3, by:

$$\frac{V}{T} = constant \quad or \quad \frac{V_1}{T_1} = \frac{V_2}{T_2} \quad (at\ constant\ Pressure) \tag{4.5}$$

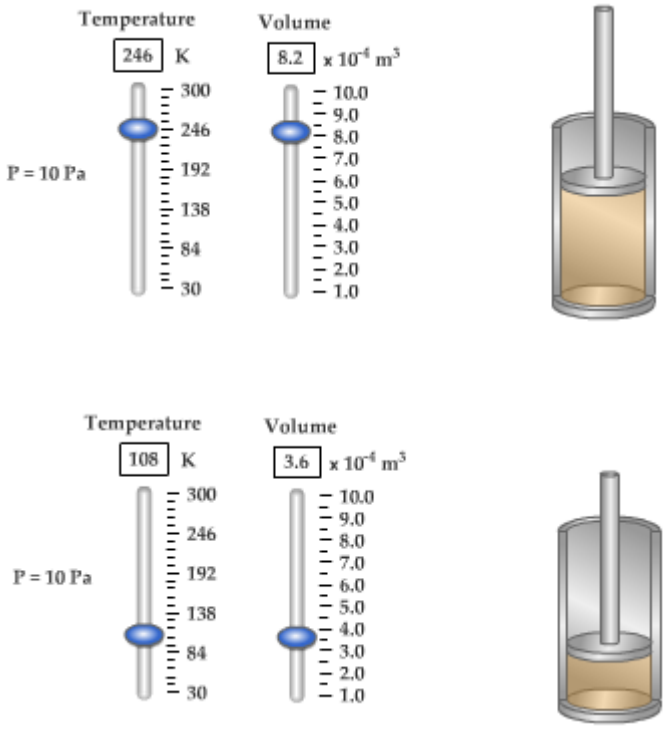


Figure 4.3: Volume temperature relation at a constant pressure, Charles' Law

Gay-Lussac's Law

The pressure exerted by a gas held at constant volume is directly proportional to the (absolute) temperature:

$$\frac{P}{T} = constant \quad or \quad \frac{P_1}{T_1} = \frac{P_2}{T_2} \quad (at\ constant\ Volume) \tag{4.6}$$

The Ideal Gas Law

Low-density gases obey the above mentioned laws, which can be combined into a single relationship, known as the ideal gas law. One version of the ideal gas law, i.e., the equation of state for an ideal gas, is that for a given quantity of gas:

$$\frac{PV}{T} = constant \quad or \quad \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad (fixed\ mass\ of\ gas) \tag{4.7}$$

This constant equal $n\,R$, where n is the number of moles, and R is a constant of proportionality called the universal gas constant, which has the value, $R = 8.314\,J/mole.K$. So the equation of state of an ideal gas can be written as:

$$PV = nRT \tag{4.8}$$

The total number of molecules in a gas N is equal to the product of the number of moles of gas and the number of molecules per mole: $N = nN_A$ and so the ideal gas law (sometimes called the perfect gas law) can also be written as

$$PV = Nk_BT\ (ideal\ gas\ law) \tag{4.9}$$

where $k_B = \frac{R}{N_A} = 1.38 \times 10^{-23} J/K$ is known as Boltzmann's constant.

Example 4.2

A 0.1 mole of O_2 gas at standard conditions. What will the pressure be if the volume is changed to 1 Liter at constant temperature, and find the density of the gas after change?

Solution

The standard condition means that the temperature of the gas $T = 273K$, and its pressure $P = 1\ atm = 1.013 \times 10^5\ Pa$. The gas is expanded at constant temperature, so that

$$P_1V_1 = nRT = P_2V_2,$$

So,

$$P_2 = \frac{nRT}{V_2} = \frac{0.1 \times 8.314 \times 273}{10^{-3}} = 2.27 \times 10^5 Pa = 2.24\ atm$$

To find the density of the O_2 gas, we can write the ideal gas law as

$$P_2 V_2 = nRT \rightarrow P_2 \frac{m}{\rho} = \frac{m}{M_w} RT \rightarrow \rho = \frac{P_2 M_w}{RT}$$

So the density is

$$\rho = \frac{P_2 M_w}{RT} = \frac{2.27 \times 10^5 \times 32 \times 10^{-3}}{8.314 \times 273} = 3.2 kg/m^3$$

Example 4.3

Pure helium gas (behave like an ideal gas) is admitted into a cylinder containing a movable piston. The initial volume, pressure, and temperature of the gas are 15 liters, $2\ atm$, and $300K$. If the volume of the gas is decreased to 12 liters by increasing its pressure to $3.5\ atm$, find the final temperature of the gas.

Solution

If the gas is sealed in the container, so the number of moles remain constants, so we can write the ideal gas equation as,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Thus,

$$T_2 = \left(\frac{P_2 V_2}{P_1 V_1}\right) T_1 = \frac{3.5\ atm \times 12\ liter}{2\ atm \times 15\ liter} \times 300K = 420\ K$$

Example 4.4

One mole of oxygen gas is at a pressure of 6 atm and a temperature of $7\ °C$. (a) If the gas is heated at constant volume until the pressure triples, what is the final temperature? (b) If the gas is heated until both the pressure and the volume are doubled, what is the final temperature?

Solution

(a) $T_1 = 273 + 7 = 280K$

The pressure became triples, so

$$P_2 = 3P_1 \rightarrow \frac{P_1}{T_1} = \frac{3P_1}{T_2} \rightarrow \frac{1}{T_1} = \frac{3}{T_2}$$

$$T_2 = 3T_1 = 3 \times 280 = 840\ K$$

(b) In this case, $P_2 = 2P_1, V_2 = 2V_1$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \rightarrow \frac{P_1V_1}{T_1} = \frac{4P_1V_1}{T_2}$$

$$T_2 = 4T_1 = 4 \times 280 = 1120\ K$$

4.4 Molecular Interpretation of Temperature

To make some insight into the meaning of temperature, by knowing that the product of the pressure and the volume (PV) is a form of energy, and according to the three dimensional kinetic theory of gases can be written as

$$PV = \frac{2}{3}NK_{av} \tag{4.10}$$

where K_{av} is the average kinetic energy of the molecule, which is equal

$$K_{av} = \frac{1}{2}m_0\bar{v}^2 \tag{4.11}$$

where m_0 is the mass of the molecule and $\bar{v}^2$ is the mean square speed of the molecule.

Now let us compare with the ideal gas equation

$$PV = \frac{2}{3}NK_{av} = \frac{2}{3}N\left(\frac{1}{2}m_0\bar{v}^2\right) \tag{4.12}$$

The kinetic energy is

$$\frac{1}{2}m_0\bar{v}^2 = \frac{3}{2}k_BT \tag{4.13}$$

k_BT is called the thermal energy of the gas. The important results from the previous equation say that the average kinetic energy of gas molecules is directly proportional to the absolute temperature of the gas. As the temperature increases, the molecules move with higher average speeds. The square root of $\bar{v}^2$ is called the root mean square (rms) speed of the molecules, which can be deduced as

$$v_{rms} = \sqrt{\bar{v}^2} = \sqrt{\frac{3k_BT}{m_0}} \tag{4.14}$$

We can use the fact that $k_B = R/N_A$ and that $m_0 = \frac{M_w}{N_A}$ to rewrite the previous equation as

$$v_{rms} = \sqrt{\frac{3k_BT}{m_0}} = \sqrt{\frac{3 \times \frac{R}{N_A}T}{m_0}} = \sqrt{\frac{3 \times \frac{R}{N_A}T}{\frac{M_w}{N_A}}}$$

$$v_{rms} = \sqrt{\frac{3RT}{M_w}} \tag{4.15}$$

This expression means that at a given temperature lighter molecules move faster, on the average, than heavier molecules. For example, hydrogen, with a molecular mass of $2\ gmole^{-1}$, moves four times as fast as oxygen, whose molecular mass is $32\ gmole^{-1}$ if they are at the same temperature. Table 2 lists the rms speeds for various molecules at $293K$.

Table 4.2: rms speeds for some gases

Gas	Molecular mass ($g \cdot mole^{-1}$)	v_{rms} at 293 K ($m\ s^{-1}$)
H_2	2.02	1902
He	4	1352
H_2O	18	637
Ne	20.1	603
N_2	28	511
NO	30	594
CO_2	44	408

Example 4.5

A tank of volume $0.3\ m^3$ contains $2\ moles$ of helium gas at $20°C$. Since the helium behaves like an ideal gas, (a) find the average kinetic energy per atom; (b) Determine the rms speed of the atoms.

Solution

(a) The average Kinetic energy per atom is,

$$K_{av} = \frac{1}{2} m_0 \bar{v}^2 = \frac{3}{2} k_B T$$

$$K_{av} = \frac{3}{2} k_B T = \frac{3}{2}(1.38 \times 10^{-23})(293) = 6.07 \times 10^{-21} J$$

(b) the molecular mass of He is $4 \times 10^{-3}\ kgmole^{-1}$, thus

$$v_{rms} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3 \times 8.314 \times 293}{4 \times 10^{-3}}} = 1350\ m/s$$

4.5 Thermal Expansion

When the temperature of a substance is increased, the molecules or atoms vibrate faster and tend to move further apart, on average. The result is an expansion of the substance. With few exceptions, all forms of matter - solids, liquids, gases and plasmas - generally expand when they are heated and contract when they are cooled (note: contraction can be considered as 'negative' expansion).

In many cases, the change in size is not very noticeable, but careful observation will detect them:

- Telephone wires are longer and sag more on a hot summer day than a cold winter day.
- Metal lids on glass fruit jars can often be loosened by heating them under hot water.
- If one part of a piece of glass is heated or cooled more rapidly than adjacent parts, the expansion or contraction that results may break the glass. This is especially true with thick glass; Pyrex glass is especially formulated to expand very little (three times less than ordinary glass) with increasing temperature (ovenware).
- A dentist uses filling material that has the same rate of expansion as teeth.
- The aluminum pistons of some car engines are made a little bit smaller in diameter than the steel cylinders to allow for the greater expansion rate of aluminum.
- A civil engineer uses reinforcing steel of the same expansion rate as concrete.
- Long steel bridges have one end of the bridge fixed, while the other end rides on rockers.
- The roadway is segmented with tongue-and-groove type gaps called expansion joints.

Linear Expansion of Solids

The change in one dimension of a solid (length, width or thickness) is called linear expansion. For small temperature changes, linear expansion is approximately proportional to the change

in temperature; $\Delta T = T - T_0$. The fractional change in length is called the thermal strain, which is equal

$$\frac{L - L_0}{L_0} = \frac{\Delta L}{L_0}, \tag{4.16}$$

Where L_0 is the original length at the original temperature T_0. This is related to the change in temperature by

$$\frac{\Delta L}{L_0} = \alpha \Delta T \quad \text{or} \quad \Delta L = \alpha L_0 \Delta T, \tag{4.17}$$

Where α is the thermal coefficient of linear expansion, which has units of K^{-1}. We can also write down an expression for the final length L after a temperature change:

$$\begin{aligned} \Delta L &= \alpha L_0 \Delta T \\ L - L_0 &= \alpha L_0 \Delta T \\ L &= L_0 + \alpha L_0 \Delta T \\ L &= L_0 (1 + \alpha \Delta T). \end{aligned} \tag{4.18}$$

The coefficient of expansion may vary slightly for different temperature ranges. Since this variation is negligible for most applications, we (usually) consider α to be a constant and independent of temperature. Different substances expand at different rates, which because they have different thermal expansion coefficient.

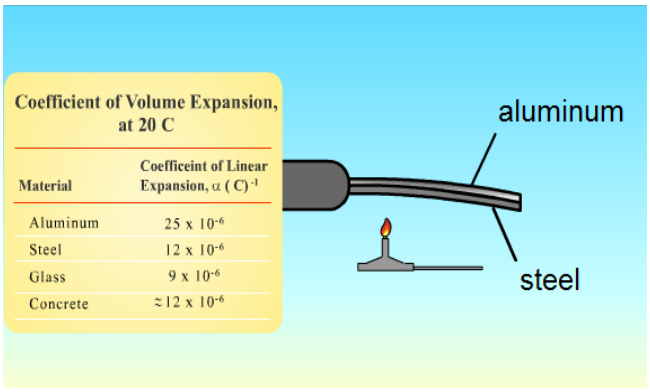


Figure 4.1: Bimetallic strip bends as the temperature changes because of the different expansion coefficient

This has the consequence that when two strips of different metals as shown in figure 4.1, such as brass and steel, is welded or riveted together, the greater the expansion of one metal results in a bending. Such a compound thin bar is called a bimetallic strip. When the strip is heated, one side of the double strip becomes longer than the other, causing the strip to bend into a curve. When the strip is cooled, it tends to bend in the opposite direction, because the metal that expands more (brass) also shrinks more. The movement of the strip may be used to turn a pointer, regulate a valve, or close a switch:

- A practical application is the thermostat. The back-and-forth bending of the bimetallic coil opens and closes an electric circuit. Refrigerators are equipped with thermostats to prevent them from becoming either too hot or too cold.
- Coils formed from such strips are used in dial thermometers.
- Bimetallic strips are used in oven thermometers, electric toasters, automatic chokes on carburetors, and various other devices.

Table 4.3: The average coefficient of linear expansion for some materials at room temperature

Material	$\alpha = (°C^{-1}) \times 10^{-6}$	Material	$\alpha = (°C^{-1}) \times 10^{-6}$
Aluminum	24	Ethyl alcohol	1120
Brass and bronze	19	Benzene	1240
Ordinarz Galss	9	Acetone	1500
Pyrex Glass	3.2	Glycerin	4850
Steel	11	Mercury	1820
Concrete	12	air	36700

Example 4.6

A copper telephone wire has essentially no sag between poles $35\,m$ apart on a winter day when the temperature is $-20.0°C$. How much longer is the wire on a summer day when $T_C = 35\,°C$?

Solution

The initial information for this question is that,

$$l_0 = 35\ m, at\ T_0 = -20\ ℃$$

We need to find the final length of the wire l when the temperature is $T_f = 35\ ℃$

But we know that from our class that,

$$\Delta l = \alpha l_0 \Delta T$$

and

$$\Delta T = (T_f - T_0) = 35 - (-20) = 55\ ℃$$

So the change in length is

$$\Delta l = \alpha l_0 \Delta T = 35\ m \times 1.7\ \times 10^{-5}\ ℃^{-1} \times 55\ ℃ = 0.0327\ m$$

Or,

$$\Delta l = 32.725\ mm$$

Example 4.7

A steel railroad track has a length of $30\ m$ when the temperature is $0°C$. (a) What is its length on a hot day when the temperature is $40°C$. (linear expansion coefficient for steel is $11 \times 10^{-6}\ °C^{-1}$) (b)Suppose the ends of the rail are rigidly clamped at $0°C$ as to prevent expansion. Calculate the thermal stress set up in the rail at $40\ °C$. Assuming the rail has a cross sectional area of $30\ cm^2$. (Young´s Modulus for steel = $200\ GPa$).

Solution

(a) The change in temperature is $\Delta T = 40 - 0 = 40℃$, so the increase in length

$\Delta L = \alpha L_0 \Delta T = (11\times10^{-6})(30)\,(40) = 0.013\ m$, so the new length is $30.013\ m$.

(b) The tensile stress is defined as $\sigma = Y\ \frac{\Delta L}{L_0} = (200\times10^{9})\,(\frac{0.013}{30}) = 8.7\times10^{7}\ Pa$, so the force of compression to prevent the elongation is given by

$$F = \sigma A = 8.7\times10^{7}\ (0.003) = 2.6\times10^{5}\ N$$

Area Expansion of Solids

The fractional change in the area of a solid is related to the temperature change by

$$\frac{\Delta A}{A_0} = \beta \Delta T \quad \text{or} \quad \Delta A = \beta A_0 \Delta T, \tag{4.19}$$

Where β is the thermal coefficient of area expansion, which has units of K^{-1}.

A solid may have different coefficients of linear expansion for different directions, but for simplicity we can assume that the same coefficient applies to all directions, i.e., the solids show isotropic expansion.

Since area, A, is length squared, L^2:

$$A = L^2 = L_0^{\,2}\left(1+\alpha \Delta T\right)^2 = A_0\left(1+2\alpha \Delta T + \alpha^2 \Delta T^2\right).$$

Since α have very small value (in fact, $\alpha \approx 10^{-6}$ for solids), the second-order term (containing α^2) can be dropped with negligible error. As a first-order approximation we then obtain

$$A \approx A_0\left(1+2\alpha \Delta T\right) \quad \text{or} \quad \frac{\Delta \mathrm{A}}{\mathrm{A}_0} \approx 2\alpha \Delta T. \tag{4.20}$$

Thus, the thermal coefficient of area expansion, β, is twice as large as the coefficient of linear expansion, i.e. $\beta \approx 2\alpha$.

Volumetric Expansion of Solids

Like the length and the area, the fractional change in the volume of a solid is related to the temperature change by

$$\frac{\Delta V}{V_0} = \gamma \Delta T \quad \text{or} \quad \Delta V = \gamma V_0 \Delta T, \tag{4.21}$$

where γ is the thermal coefficient of volume expansion, which has units of K^{-1}. Assuming isotropic expansion, a first-order expression for the volume expansion is

$$\frac{\Delta V}{V_0} \approx 3\alpha\,\Delta T \qquad \text{or} \qquad V \approx V_0\left(1+3\alpha\,\Delta T\right). \tag{4.22}$$

Thus, the thermal coefficient of linear expansion,α, is related to the thermal coefficient of volume expansion, γ by $\gamma \approx 3\alpha$.

Example 4.8

A pair of eyeglass frames are made of epoxy plastic (coefficient of linear expansion= $130 \times 10^{-6}\,°C^{-1}$). At $20\,°C$ the frames have circular lens holes $2.2\,cm$ in radius. To what temperature must the frame be heated in order to insert lenses $2.21\,cm$ in radius?

Solution

For area expansion we have

$$\frac{\Delta \mathrm{A}}{\mathrm{A}_0} \approx 2\alpha\,\Delta T \text{ , so } \Delta T = T_f - T_i = \frac{\Delta \mathrm{A}}{2\alpha \mathrm{A}_0}$$

At $20°C$, the initial area is

$$\mathrm{A}_0 = \pi r^2 = 3.14\times(2.2)^2 = 15.21\,cm^2,$$

And the area at the final temperature is

$$A = 3.14\times(2.21)^2 = 15.35\,cm^2,$$

Thus, the change in area is

$$\Delta \mathrm{A} = 15.35 - 15.21 = 0.14\ \mathrm{cm}^2,$$

Then,

$$\Delta T = T_f - 20 = \frac{\Delta \mathrm{A}}{2\alpha \mathrm{A}_0} = \frac{0.14}{2\,(130\times10^{-6})\times15.21} = 35.4\,, so\ T_f = 55.4\,℃$$

Example 4.9

The active element of a certain laser is made if a glass rod $30\ cm$ long by $1.5\ cm$ in diameter. If the temperature of the rode is increased by $65\ C^{\circ}$, find the increase in a. in its length, b. In its diameter, and c. in its volume, if alpha is $\alpha = 9 \times 10^{-6}\ ℃^{-1}$

Solution

a. the increase in length is

$$\Delta l = \alpha\ l_0 \Delta T = 9\times10^{-6}\times30\times(65) = 0.01755\ cm$$

b. the increase in the surface area is

$$\Delta A = 2\alpha\ A_0 \Delta T = 2\times9\times10^{-6}\times\pi\left(\frac{1.5}{2}\right)^2\times(65) = 2.068\times10^{-3}\ cm^2$$

The new area is

$$A_{new} = A_0 + \Delta A$$
$$A_{new} = \pi(0.75)^2 + 2.068\times10^{-3} = 1.769\,cm^2$$

Thus the new radius is

$$r_{new} = \sqrt{\frac{A_{new}}{\pi}} = \sqrt{\frac{1.769}{\pi}} = 0.7504\ cm$$

The increase in diameter is

$$\Delta d = 1.508 - 1.5 = 0.08\ cm$$

c. the increase volume is

$$\Delta V = 3\alpha\ (A_0\ l_0)\ \Delta T = 3 \times 9 \times 10^{-6}\ ℃^{-1} \times (\pi(0.75)^2 \times 30) \times 65\ C^{\circ}$$

$$\Delta V = 0.0904\ cm^3$$

For discussion: Expansion of Water (odd behavior)

- Increase the temperature of any common liquid and it will expand. But not water at temperatures near the freezing point: ice-cold water does just the opposite!
- Water at the temperature of melting ice contracts when the temperature is increased, and continues to do so until it reaches a temperature of $4.0℃$ ($3.980℃$, to be precise).
- With further increase in temperature, the water then begins to expand, and this continues until the boiling point.

4.6 Heat as a form of energy

We discussed the concept of heat as an energy transfer from a place to another that takes place as a consequence of temperature difference. The heat flow is always directed from hotter place to colder one. The common used unit of heat is Joule; however other units of heat are used nowadays. One of the most widely used of these units is the calorie (cal), which is defined as the amount of energy required to raise the temperature of 1 g of water from $14.5°C$ to $15.5°C$. A related unit is the kilocalorie ($kcal$ or written as $Cal = 103\ cal$) defined as the amount of energy required to raise the temperature of $1\ kg$ of water from $14.5°C$ to $15.5°C$. The mechanical equivalent of calorie is $4.18\ J$, so $1\ Cal = 4186\ J$.

In the British system there is a heat unit called British thermal unit (BTU), defined as the amount of energy required to raise the temperature of 1lb of water from $63°F$ to $64°F$. The equivalent of $1\ BTU = 252\ cal$.

Heat Capacity

To elucidate heat capacity concept, let us do the following simple experiment: take two samples with equal masses, $100\ grams$, one of them is water and the other is copper, both are at room temperature. Now we want to increase the temperature of both samples by, say $50\ degrees$. Both samples are heated by the same source of energy, gas stove. It is found that the water takes much longer time on the stove than the copper. Means, we have to put more heat (heat energy) into water sample than copper to reach to the same change in

temperature. This is referred to the meaning of the heat capacity of the materials. Heat Capacity of a system is the amount of heat required to change temperature of the whole system by one degree. It is noted by the capital letter C, and it is measured by JK^{-1}, $J\,°C^{-1}$ or cal.° C^{-1}.

The amount of heat (Q) needed to heat a subject from an initial temperature level to another final one can be expressed as:

$$Q = C\Delta T = C(T_f - T_i) \tag{4.23}$$

When ΔT is positive, this means that the system gains energy and it loses energy when ΔT is negative. Note that Q is dependent on the mass of the substance, so another quantity which is characteristic of the material should be defined. This quantity is called the specific heat capacity, which is defined as the amount of heat required to change temperature of one kilogram of a substance by one degree. It is notation is smaller letter c, which is the heat capacity per mass ($c = C/m$) and its unit is $J\,kg^{-1}K^{-1}$ or $Cal\,kg^{-1}K^{-1}$. So the last equation is written as,

$$Q = m\,c\,\Delta T = m\,c\,(T_f - T_i) \tag{4.24}$$

Table 4.4: the specific heat of some materials at atmospheric pressure.

Substance	Specific Heat ($Jkg^{-1}K^{-1}$)
Aluminum	900
water	4186
Copper	387
Glass	129
Ice	2090
Iron	448
Mercury	138
Silicon	703

For example the heat energy required to raise the temperature of $0.5\ kg$ of water by $10°C$ is equal to $(0.5kg)\ (4186\ J/kg\ °C)(10°C)\ =\ 20.93\ kJ\ =\ 5\ Cal.$

When two substances at different temperatures are mixed together in a closed system, thus the hot substance will lose energy and the cold one will gain heat energy. However the sum of heat gain and heat lost is zero (conservation of energy). Which means that $Q_{gain} + Q_{lost} = 0$. For example, assume that m_x is the mass of unknown substance, whose specific heat c_x is to be determined, and then it is heated to a certain temperature T_x. Likewise take an amount of water m_w which its specific heat is known c_w and its temperature T_w is known. If both substances are mixed and left until their temperature remain constant (final temperature T_f), so we can apply the conservation of energy as,

$$Q_{gain}(w) + Q_{lost}(x) = 0 \Longrightarrow m_w c_w\big(T_f - T_w\big) + m_x c_x\big(T_f - T_x\big) = 0$$

Solving this equation for c_x, we find

$$c_x = \frac{m_w c_w\,(T_f - T_w)}{m_x\,(T_x - T_f)} \tag{4.25}$$

Example 4.10

A $50\ g$ of ingot of metal is heated to $200°C$ and then dropped into a beaker containing a $0.4\ kg$ of water initially at $20°C$. If the final equilibrium temperature of the mixture is $22.4°C$, find the specific heat of the metal?

Solution

From the fact that the heat lost by the ingot equals the heat gained by water, we can write

$$c_x = \frac{m_w c_w\,(T_f - T_w)}{m_x\,(T_x - T_f)} = \frac{(0.4kg)(4186\,Jkg^{-1}°C^{-1})(22.4-20)}{(0.05\,kg)(200-22.4)} = 453\,J/kg°C$$

This value is near the value of iron, by comparing it with the data in the table.

Example 4.11

A quantity of hot water at $91°C$ and another cold one at $12°C$. How much kilogram of each one is needed to make an $800\ liter$ of water bath at temperature of $35°C$.

Solution

Assume the mass of hot water and cold one is m_H and m_C, respectively.

$800\ liter$ of water is equivalent to $800\ kg$, so $m_{hot} + m_{cold} = 800\ kg$,

From the conservation of energy

$$Q_{in} = Q_{out}$$

$$m_{cold}\, c_w \left(T_f - T_i\right)_{Cold} = m_{hot}\, c_w \left(T_f - T_i\right)_{hot}$$

Now, $\left(T_f - T_i\right)_{Cold} = (35 - 12) = 23\ ℃$, $\left(T_f - T_i\right)_{hot} = (35 - 91) = 56\ ℃$, and drop c_w from both sides, gives

$$m_{cold}\,(23℃)_{Cold} = m_{hot}\,(56\ ℃)_{hot}$$

$$m_{cold} = \frac{56}{23} m_{hot} = 2\, m_{hot}$$

Now, return to $m_{hot} + m_{cold} = 800\ kg \rightarrow\ m_{hot} + 2\, m_{hot} = 800\ kg$

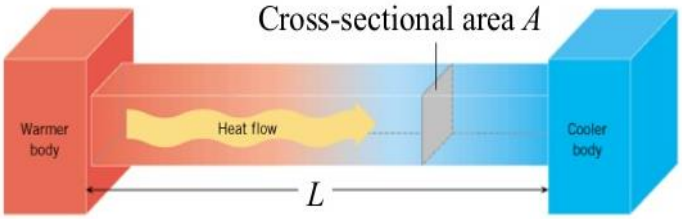
$$3\, m_{hot} = 800,\ \ m_{hot} = \frac{800}{3} = 266.67\ kg, and\ m_C = 800 - 266.67 = 533.33\ kg.$$

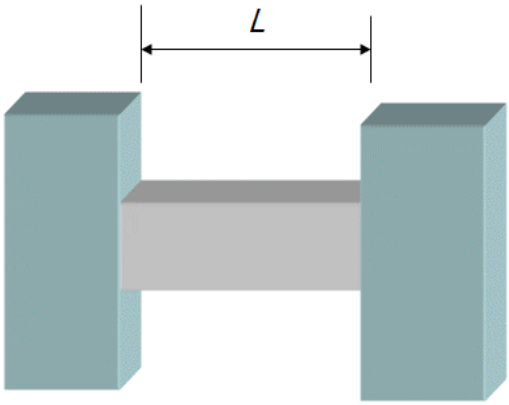
4.6 Heat Transfer

The heat is a transfer of the energy from a high temperature object to a lower temperature one. Heat transfer changes the internal energy of both systems involved according to the first law of thermodynamics. Heat can be transferred by three ways: conduction, convection and radiation.

Heat conduction

Conduction is heat transfer by means of molecular agitation within a material without any motion of the material as a whole. If one end of a metal rod is at a higher temperature, then energy will be transferred down the rod toward the colder end because the higher speed particles will collide with the slower ones with a net transfer of energy to the slower ones.





$$H \propto \Delta T,$$

$$H \propto \frac{1}{l},$$

$$H \propto A,$$

$$H = kA\frac{\Delta T}{l},$$

k is called thermal conductivity

For heat transfer between two plane surfaces, such as heat loss through the wall of a house, the rate of conduction heat transfer (H) is the energy transferred per unit of time:

$$H = \frac{Q}{t} = \kappa A \frac{T_{hot} - T_{cold}}{L} \qquad (4.26)$$

Where L is the separation distance (thickness) between the hot and the cold sides, A is the contact area, $(T_{hot} - T_{cold})$ is the temperature gradient, and κ is a constant called the thermal conductivity coefficient of the material measured in W/mK. This constant is a characteristic of the material. Table 4.5 shows some values of κ for different materials. Those with high κ like metals are good heat conductor and with small κ like gases and nonmetals are good insulator.

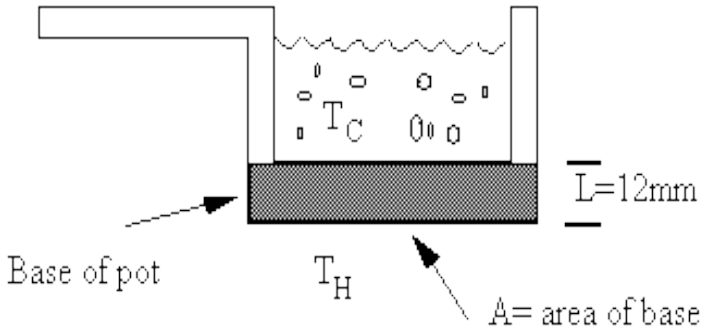
Table 4.5: The thermal conductivity coefficient for different materials

Substance	$(W\,m^{-1}K^{-1})$	Substance	$(W\,m^{-1}K^{-1})$
Silver	427	Ice	2
Copper	397	Water	0.6
Aluminum	238	Wood	0.08
Gold	314	Air	0.023
Concrete	0.8	Hydrogen	0.1
Glass	0.8	Helium	0.138

This is sometimes not obvious: Like when you shake hands with a person with cold hands. The conclusion that many people make is that cold has travelled from that person to you. It is only heat that travels. The coldness that you feel is simply the heat leaving your hand. Simple Experiment: Put a block of wood and a bowl of water in the fridge. Allow the water to freeze. Then take both of them out and feel them. Which feels "colder"? Most will say the ice. So which has the lowest temperature? If you say the ice, then you are wrong! They both have the same temperature. It feels colder because the ice conducts heat faster than wood. What you feel as "colder" simply means there is more heat leaving your hand every second than when touching the wood. So our concept of hot or cold does not just depend on temperature but also on how fast heat travels in different materials.

Example 4.12

An aluminum pot contains water that is kept steadily boiling ($100\ ℃$). The bottom surface of the pot, which is $12\ mm$ thick and $1.5 \times 10^4\ mm^2$ in area, is maintained at a temperature of $102\ °C$ by an electric heating unit. Find the rate at which heat is transferred through the bottom surface. Compare this with a copper based pot. The thermal conductivities for aluminum and copper are shown in the table.



Solution

The following is a schematic diagram of the pot. The rate of heat conduction across the base is given by,

$$H = \kappa A \frac{(T_{hot} - T_{cold})}{L}, \tag{4.27}$$

For the aluminum base:

$T_H = 102\,℃,\ T_C = 100\,℃, L = 12\,mm = 0.012\,m,\ \kappa_{Al} = 238\,Wm^{-1}K^{-1},$ and Base area $A = 1.5 \times 10^4\,mm^2 = 0.015\,m^2$.

Substituting these into the above equation:

$$H_{Al} = 238 \times 0.015 \times \frac{102 - 100}{0.012} = 588\,W$$

For the copper base $\kappa_{Cu} = 397\,Wm^{-1}K^{-1}$. So the rate of heat conduction across the base is

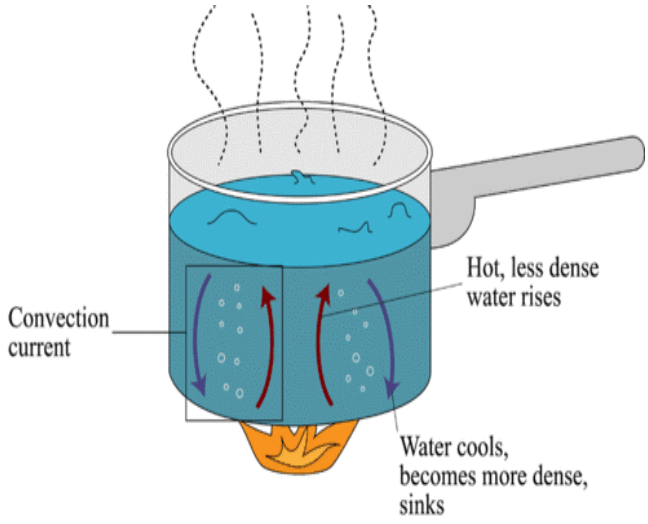
$$H_{Cu} = 397 \times 0.015 \times \frac{102 - 100}{0.012} = 1003\,W$$

So the copper based pot transfers $1.7\ times$ more energy every second compared with the aluminum pot. Generally copper bottom pots are more expensive. Are their prices $1.7\ times$ those of similar aluminum pots? Or this a simplistic way of looking at it?

Heat Convection

Convection is heat transfer by mass motion of a fluid such as air or water when the heated fluid is caused to move away from the source of heat, carrying energy with it. Convection above a hot surface occurs because hot air expands, becomes less dense, and rises (see Ideal Gas Law). Hot water is likewise less dense than cold water and rises, causing convection currents which transport energy.

Convection can also lead to circulation in a liquid, as in the heating of a pot of water over a flame. Heated water expands and becomes more buoyant. Cooler, denser water near the surface descends and patterns of circulation can be formed, though they will not be as regular as suggested in the drawing.



Heat Radiation

Energy is transferred by electromagnetic radiation. All of the earth's energy is transferred from the Sun by radiation. Our bodies radiate electromagnetic waves in a part of the spectrum that we can't see called the infra-red. However, there are some cameras that can actually see this radiation.

The color and texture of different surfaces determines how well they absorb the radiation.

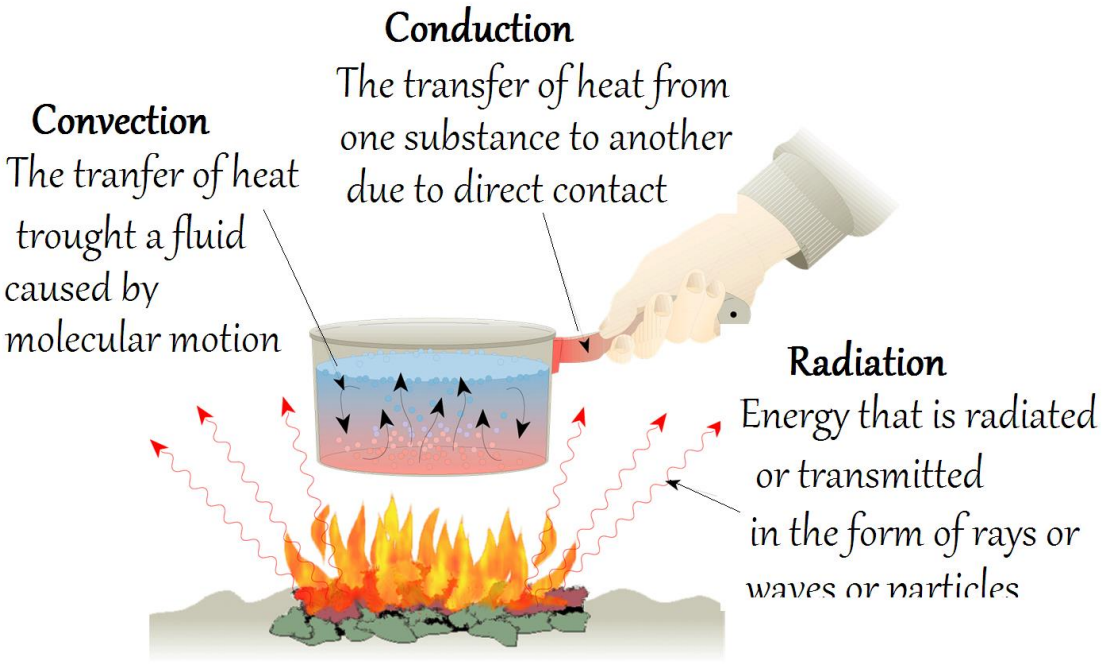
1. Black objects absorb more radiation than white objects.
2. Matt and rough surfaces absorb more than shiny and smooth surfaces.

If you are ever in the snow, take a black and a white piece of cardboard, both the same size. Lay them down on the snow side by side. Over time you will notice that the black cardboard sinks deeper into the snow because it absorbs more heat from the sun and therefore melts more snow underneath it

The relationship governing radiation from hot objects is called the Stefan-Boltzmann Law:

$$P = e\,\sigma\,A\,(T^4 - T_S^4) \tag{4.28}$$

Where P is the net radiated power measured in Watt, e is the emissivity ($= 1$ for ideal radiator), A is the radiation area in m^2, T is the temperature of the radiator in Kelvin, T_S is the temperature of the surroundings in Kelvin, and $\sigma = 5.67 \times 10^{-8} Watt/(m^2K^4)$ is a constant called Stefan-Boltzmann constant.



Example 4.13

A student tries to decide what to wear is staying in a room that is at $20°C$. If the skin temperature is $37°C$, how much heat is lost from the body in 10 minutes? Assume that the emissivity of the body is 0.9 and the surface area of the student is $1.5\ m^2$.

Solution

Using the Stefan-Boltzmann's law: $P_{net} = e\,\sigma\,A\,(T^4 - T_S^4)$

$$P_{net} = 0.9 \times 5.67 \times 10^{-8}\ \frac{Watt}{(m^2K^4)} \times 1.5\,m^2 \times \left(310^4 - 293^4\right)K^4 = 142.77\,Watt$$

Notice how the units cancel up with each other. The total energy lost during $10\ min$ is

$$Q = P_{net}\Delta t = 142.77\,W \times 600s = 85.7\,kJ$$

Generally, every object at non-absolute-zero temperature radiates radiation at all wavelengths (theoretically speaking). But the amount of energy radiated at each wavelength depends on the body temperature. A body at temperature $800℃$ will look red, but at temperature $3000℃$ would look White. The wavelength at which the radiation is most intense is given by the Wien Displacement law

$$\lambda = \frac{B}{T} \tag{4.29}$$

Where B is a constant and equals to $2.898 \times 10^{-3}\ mK$

Example 4.14

Sun surface temp is $6000K$. What is wavelength of maximum radiation?

Solution

$$\lambda = \frac{B}{T} = \frac{2.898 \times 10^{-3}\ mK}{6000K} = 4.83 \times 10^{-7}\ m$$

Example 4.15

Sun surface temperature is 6000K. What is wavelength of maximum radiation?

$$\lambda = \frac{2.898\times10^{-3}\,mK}{6000K} = 4.83\times10^{-7}\,m$$

Cooling of the Human Body

Hypothermia: Abnormally low body temperature. The condition needs treatment at body temperatures of $35℃\ (95\ °F)$ or below. And hypothermia becomes life threatening below body temperatures of $32.2\ ℃\ (90\ °F)$.

Stage 1

Body temperature drops by $1–2\ °C$ below normal temperature (down to $35–37\ °C$). Mild to strong shivering occurs. The victim is unable to perform complex tasks with the hands; the hands become numb. Breathing becomes quick and shallow. Victim may feel sick to their stomach, and very tired. Often, a person will experience a warm sensation, as if they have recovered, but they are in fact heading into Stage 2. Another test to see if the person is entering stage 2 is if they are unable to touch their thumb with their little finger; this is the first stage of muscles not working. They might start to have trouble seeing.

Stage 2

Body temperature drops by $2–4\ °C$ below normal temperature ($33–35\ °C$). Shivering becomes more violent. Muscle miscoordination becomes apparent. Movements are slow and labored, accompanied by a stumbling pace and mild confusion, although the victim may appear alert. Surface blood vessels contract further as the body focuses its remaining resources on keeping the vital organs warm. The victim becomes pale. Lips, ears, fingers and toes may become blue.

Stage 3

Body temperature drops below approximately $32\ °C$. Shivering usually stops. Difficulty speaking, sluggish thinking, and amnesia (memory loss) start to appear; inability to use hands and stumbling is also usually present. Cellular metabolic processes shut down. Below $30\ °C$, the exposed skin becomes blue and puffy, muscle coordination becomes very poor, walking becomes almost impossible, and the victim exhibits incoherent/irrational behavior including terminal burrowing or even a stupor. Pulse and respiration rates decrease significantly, but fast heart rates can occur. Major organs fail. Clinical death occurs. Because of decreased cellular activity in stage 3 hypothermia, the body will actually take longer to undergo brain death.

Wind Chill Factor

A surface loses heat through conduction, convection, and radiation. The rate of convection depends on both the difference in temperature between the surface and the fluid surrounding it and the velocity of that fluid with respect to the surface. As convection from a warm surface heats the air around it, an insulating boundary layer of warm air forms against the surface. Moving air disrupts this boundary layer, allowing for cooler air to replace the warm air against the surface. The faster the wind speed, the more readily the surface cools.

The effect of wind chill is to increase the rate of heat loss and reduce any warmer objects to the ambient temperature more quickly. Dry air cannot, however, reduce the temperature of these objects below the ambient temperature, no matter how great the wind velocity. For most biological organisms, the physiological response is to generate more heat in order to maintain a surface temperature in an acceptable range. The attempt to maintain a given surface temperature in an environment of faster heat loss results in both the perception of lower temperatures and an actual greater heat loss. In other words, the air 'feels' colder than it is because of the chilling effect of the wind on the skin. In extreme conditions this will increase the risk of adverse effects such as frostbite.

Cooling power of wind
expressed as an equivalent chill temperature (under calm conditions)

ESTIMATED WIND SPEED (IN MPH)	ACTUAL THERMOMETER READING (F)											
	50	40	30	20	10	0	-10	-20	-30	-40	-50	-60
	EQUIVALENT TEMPERATURES (F)											
Calm	50	40	30	20	10	0	-10	-20	-30	-40	-50	-60
5	48	37	27	16	6	-5	-15	-26	-36	-47	-57	-68
10	40	28	16	4	-9	-24	-33	-46	-58	-70	-83	-95
15	36	22	9	-5	-18	-32	-45	-58	-72	-85	-99	-112
20	32	18	4	-10	-25	-39	-53	-67	-82	-96	-110	-124
25	30	16	0	-15	-29	-44	-59	-74	-88	-104	-118	-133
30	28	13	-2	-18	-33	-48	-63	-79	-94	-109	-125	-140
35	27	11	-4	-21	-35	-51	-67	-82	-98	-113	-129	-145
40	26	10	-6	-21	-37	-53	-69	-85	-100	-116	-132	-148
Winds greater than 40 MPH have little additional effect.	LITTLE DANGER (for properly clothed person) Maximum danger of false sense of security.				INCREASING DANGER Danger from freezing of exposed flesh.			GREAT DANGER				
	Trench foot and immersion foot may occur at any point on this chart.											

Summary

Two bodies are in thermal equilibrium with each other if they have the same temperature. The zeroth law of thermodynamics states that if objects A and B are separately in thermal equilibrium with a third object C, then objects A and B are in thermal equilibrium with each other.

To measure temperature, there are many scales can be used:

The Celsius Scale

The space between the two reference points is divided into 100 equal parts called degrees.

The Fahrenheit Scale

Fahrenheit scale takes the number 32 to the temperature at which water freezes, and the number 212 to the temperature at which water boils

The converges between these two scales

$$\begin{aligned} T_F &= \frac{9}{5}T_C + 32 \quad \text{Celsius to Fahrenheit conversion} \\ T_C &= \frac{5}{9}(T_F - 32) \quad \text{Fahrenheit to Celsius conversion} \end{aligned} \tag{4.1}$$

The Kelvin Scale

The SI unit of absolute temperature is the Kelvin, which is defined to be the fraction $1/273.16$ of the temperature of the triple point of water.

To convert between the Celsius and Kelvin temperature scales:

$$\begin{aligned} T_K &= T_C + 273.15 \quad \text{Celsius to Kelvin conversion} \\ T_C &= T_K - 273.15 \quad \text{Kelvin to Celsius conversion} \end{aligned} \tag{4.2}$$

- Note that $\Delta T_C = \Delta T_K$

Boyle's Law

When temperature of a gas is held constant, the pressure and volume of a quantity of gas are related

$$PV = \text{const.} \quad \text{or} \quad P_1V_1 = P_2V_2 \quad \text{(at constant temperature)}$$

Charles' Law

When the pressure is held constant, the volume of a quantity of gas is related to the (absolute) temperature

$$\frac{V}{T} = \text{const.} \qquad \text{or} \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{(at constant pressure)} \tag{4.5}$$

Gay-Lussac's Law

The pressure exerted by a gas held at constant volume is directly proportional to the (absolute) temperature:

$$\frac{p}{T} = \text{const.} \qquad \text{or} \qquad \frac{p_1}{T_1} = \frac{p_2}{T_2} \quad \text{(at constant volume)} \tag{4.6}$$

The Ideal Gas Law

the equation of state for an ideal gas, is that for a given quantity of gas:

$$\frac{pV}{T} = \text{constant} \qquad \text{or} \qquad \frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2} \quad \text{(fixed mass of gas)} \tag{4.7}$$

$$pV = nRT \tag{4.8}$$

Where n is the number of moles, and R is a constant of proportionality called the universal gas constant, which has the value, $R = 8.314\, J/mole.K$.

By knowing that the product of the pressure and the volume (pV) is a form of energy, and according to the three dimensional kinetic theory of gases can be written as

$$PV = \frac{2}{3}NK_{av} \tag{4.10}$$

where K_{av} is the average kinetic energy of the molecule, which is equal

$$K_{av} = \frac{1}{2}m_0\bar{v}^2 \tag{4.11}$$

When the temperature of an object is changed by an amount ΔT, its length changes by an amount ΔL that is proportional to ΔT and to its initial length

$$\frac{\Delta L}{L_0} = \alpha \Delta T \quad \text{or} \quad \Delta L = \alpha L_0 \Delta T, \tag{4.17}$$

Where α is the thermal coefficient of linear expansion, which has units of K^{-1}.

The fractional change in the area of a solid is related to the temperature change by

$$\frac{\Delta A}{A_0} = \beta \Delta T \quad \text{or} \quad \Delta A = \beta A_0 \Delta T, \tag{4.19}$$

The fractional change in the volume of a solid is related to the temperature change by

$$\frac{\Delta V}{V_0} = \gamma \Delta T \quad \text{or} \quad \Delta V = \gamma V_0 \Delta T, \tag{4.21}$$

where γ is the thermal coefficient of volume expansion, which has units of K^{-1}. Assuming isotropic expansion, a first-order expression for the volume expansion is

$$\frac{\Delta V}{V_0} \approx 3\alpha \Delta T \quad \text{or} \quad V \approx V_0 (1 + 3\alpha \Delta T). \tag{4.22}$$

Thus, the thermal coefficient of linear expansion, α, is related to the thermal coefficient of volume expansion, γ by $\gamma \approx 3\alpha$.

The amount of heat (Q) needed to heat a subject from an initial temperature level to another final one can be expressed as:

$$Q = C \Delta T = C (T_f - T_i) \tag{4.23}$$

Heat Transfer:

Conduction: For heat transfer between two plane surfaces, such as heat loss through the wall of a house, the rate of conduction heat transfer (H) is the energy transferred per unit of time:

$$H = \frac{Q}{t} = \kappa A \frac{T_{hot} - T_{cold}}{L} \tag{4.26}$$

Convection: Convection is heat transfer by mass motion of a fluid such as air or water when the heated fluid is caused to move away from the source of heat, carrying energy with it

Heat Radiation: Energy is transferred by electromagnetic radiation

$$P = e\,\sigma\,A\,(T^4 - T_S^4) \tag{4.28}$$

Wien Displacement law

$$\lambda = \frac{B}{T} \tag{4.29}$$

PROBLEMS

1. Liquid nitrogen has a boiling point of $-195.81°C$ at atmospheric pressure. Express this temperature in (a) degrees Fahrenheit and (b) Kelvin's.
2. The temperature difference between the inside and the outside of an automobile engine is $450°C$. Express this temperature difference on the (a) Fahrenheit scale and (b) Kelvin scale.
3. $2.50\ g$ of $XeF4$ gas is placed into an evacuated $3.00\ liter$ container at $80°C$. What is the pressure in the container?
4. A hydrogen gas thermometer is found to have a volume of $100.0\ cm^3$ when placed in an ice-water bath at $0°C$. When the same thermometer is immersed in boiling liquid chlorine, the volume of hydrogen at the same pressure is found to be $87.2\ cm^3$. What is the temperature of the boiling point of chlorine?
5. (a) Show that the density of an ideal gas occupying a volume V is given by $\rho = p\,M\,/\,R\,T$ where M is the molar mass. (b) Determine the density of oxygen gas at atmospheric pressure and $20.0°C$.
6. An ideal gas with volume of $3\ L$ is compressed at fixed temperature by increasing its pressure to $250\ kPa$. Find the bulk modulus of this gas.
7. (a) How many atoms of helium gas fill a balloon of diameter $30.0\ cm$ at $20.0°C$ and $1.00\ atm$? (b) What is the average kinetic energy of the helium atoms? (c) What is the root-mean-square speed of each helium atom?

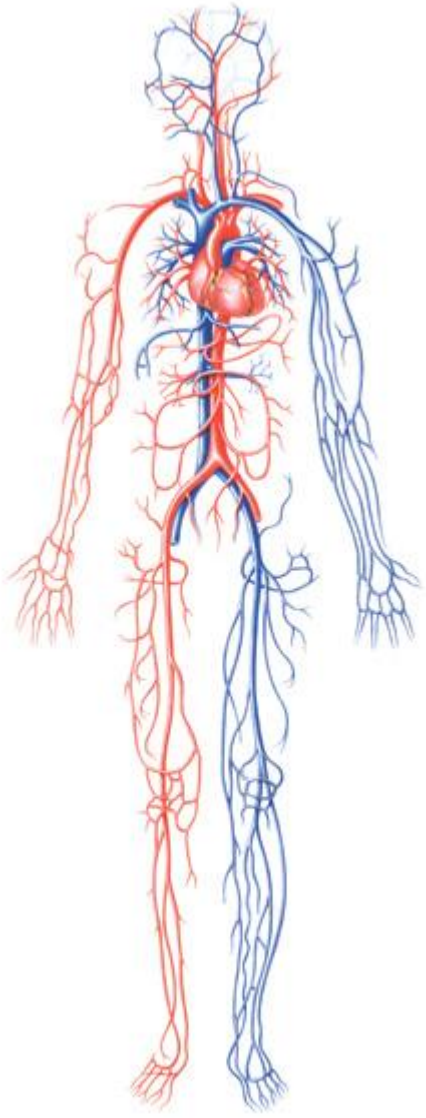
8. A cube $10.0\ cm$ on each edge contains air (with equivalent molar mass $28.9\ g/mol$) at atmospheric pressure and temperature $300\ K$. Find (a) the mass of the gas, (b) its weight, and (c) the force it exerts on each face of the cube. (d) Comment on the underlying physical reason why such a small sample can exert such a great force.
9. A steel rod $4.00\ cm$ in diameter is heated so that its temperature increases by $70.0°C$. It is then fastened between two rigid supports. The rod is allowed to cool to its original temperature. Assuming that Young's modulus for the steel is $20.6 \times 10^{10}\ N/m^2$ and that its average coefficient of linear expansion is $11 \times 10^{-6}\ °C^{-1}$, calculate the tension in the rod.
10. A steel ball bearing is $4\ cm$ in diameter at $20.0°C$. A bronze plate has a hole in it that is $3.9\ cm$ in diameter at $20.0°C$. What common temperature must they have so that the ball just squeezes through the hole?
11. A concrete slab has a length of $12\ m$ at $-5\ ºC$ on a winter's day. What is the change in length from winter to summer, when the temperature is $35\ ºC$? The linear expansion coefficient of concrete is $1 \times 10^{-5}°C^{-1}$.
12. A steel rod is initially at $20\ °C$ and has a length of $2\ m$ and a cross sectional area of $10\ cm^2$.

    a) If it is heated to 120 °C, by how much does its length increase. b) How large a force must be applied to its end to restore the original length? (Young's Modulus for steel = $2 \times 10^{11}\ N/m^2$, the coefficient of thermal expansion for steel = $1.72 \times 10^{-5}\ K^{-1}$.
13. A clock with a brass pendulum has a period of $1.000\ s$ at $20.0°C$. If the temperature increases to $30.0°C$, (a) by how much does the period change, and (b) how much time does the clock gain or lose in one week?
14. A thermometer has a mercury-filled glass bulb with a volume of $2 \times 10^{-7}\ m^3$ attached to a thin glass capillary tube with an inner radius of $5 \times 10^{-5}\ m$. If the temperature increases by $100°C$, how far will the mercury rise in the tube? (volume thermal expansion coefficient of mercury= $1.82 \times 10^{-4}\ K^{-1}$).
15. If $46.6\ kJ$ is required to heat $0.15\ kg$ of helium gas from $20°C$ to $80°C$ at constant pressure, find the specific heat of helium?

16. Sphere of iron is heated to $900K$ and then cooled by placing it in a container filled of $5\ kg$ water of $300K$, what is the mass of the sphere if the final temperature of the mixture is $340K$, the specific heat of iron is $490\ J/kgK$ and that of water is $4200\ J/kgK$. Find the heat loss by the iron in Calorie.
17. A box with a total surface area of $1.20\ m^2$ and a wall thickness of $4.00\ cm$ is made of an insulating material. $A\ 10.0 - W$ electric heater inside the box maintains the inside temperature at $15.0°C$ above the outside temperature. Find the thermal conductivity k of the insulating material.
18. The surface of the Sun has a temperature of about $5800\ K$. The radius of the Sun is $6.96 \times 10^8\ m$. Calculate the total energy radiated by the Sun each second. (Assume that $e = 0.96$)
19. A person walking at a modest speed generates heat at rate of $280\ W$. If the surface area of the body is $1.5\ m^2$ and if the heat is assumed to be generated $0.03\ m$ below the skin, what temperature difference between the skin and interior of body would exist if the heat were conducted to the surface? Assume that the thermal conductivity coefficient is $0.2\ Wm^{-1}K^{-1}$.
20. A $2\ m$ length of copper pipe of $10\ cm$ diameter containing hot water at $80°C$. If the surroundings are at $20°C$, at what rate does the pipe lose thermal energy due to radiation? (take $e = 1$)
21. A glass window pane has an area of $3.00\ m^2$ and a thickness of $0.600\ cm$. If the temperature difference between its surfaces is $75.0°F$, what is the rate of energy transfer by conduction through the window? $\kappa = 0.8\ W\ m^{-1}K^{-1}$

Chapter 5

Fluid Mechanics



A fluid is a subset of the states of matter, consisting of liquids, gases and plasmas. This is because they have common properties that are distinct from solids. They are playing a very important role in many fields of sciences and medical sciences. Fluid movement for Solute transport in soft connective tissue is a fundamental process, involving many physiological phenomena, such as nutrient supply, removal of metabolic waste product and movement of newly-synthesized molecules. In this chapter we will recognize the properties of the fluids and their behavior. The flow of the fluids will be studied for nonviscous and viscous fluids.

5.1 Fluid Characteristics

There are essentially three states of ordinary matter: solid, liquid, and gas. Solid objects are characterized by their rigidity and incompressibility. That is, they cannot easily be deformed into different shapes without breaking. (Rubber is a somewhat special solid: it can be deformed, but as soon as the applied force is removed, it returns to its original state).

On the other hand, Liquids have no shape of their own: they take on the shape of whatever container they happen to be in. However, liquids, like solids, are difficult to compress. Finally, gases have no definite shape, and can be compressed. They tend to fill whatever container they occupy, as long as the container is sealed. Otherwise the gases leak out and spread out as far as they can. All types of matter are made of molecules. The difference between solids, liquids and gases has to do with how tightly the molecules are held together by electromagnetic forces. In solids, the intermolecular bonds so strong, which give them hard structure, the molecules are very tightly bound into rigid structures, whereas in liquids the intermolecular bonds are somehow moderate, which allow the molecular layers to side over each other (flow). In liquids the binding forces are looser, allowing the molecules to slide over each other (flow). Gas molecules on the other hand have virtually no binding forces between them, so that they tend to get as far away from each other as their container allows.

To study the fluids, we should introduce the following definitions:

Pressure: It is defined before in chapter 3 as the force per unit area. It is usually more convenient to use pressure rather than force to describe the influences upon fluid behavior. For a rigid body, an external force cause it to change its position from a place to another, however it is common in fluid

mechanics to say that a pressure difference cause the flowing of the fluid from point to another. So if a fluid flow through a tube of cross sectional area A from a point 1 to 2, then the force exerted on the fluid is given by:

$$F = (P_1 - P_2)A = \Delta P\, A \tag{5.1}$$

The standard unit for pressure is the Pascal, which is a Newton per square meter (N/m^2). Pressure in a fluid can be seen to be a measure of energy per unit volume by means of the definition of work, that is, this energy is related to other forms of fluid energy by the Bernoulli equation.

$$P = \frac{Force}{Area} = \frac{F}{A} = \frac{F.\Delta x}{A.\Delta x} = \frac{W}{\Delta V} = \frac{Energy}{volume}$$

Where ΔV is the unit volume, which related to the unit mass Δm, of the fluid by the density ρ, as $\Delta m = \rho \Delta V$.

Kinetic energy density: The kinetic energy of a moving fluid has a great importance when dealing with fluids at motion: more useful in applications like the Bernoulli equation when it is expressed as kinetic energy per unit volume. For a unit mass Δm of a fluid moving with average speed $\bar{v}$, its kinetic energy is given by

$$K.E = \frac{1}{2}\Delta m\overline{v^2} = \frac{1}{2}\rho \Delta V \overline{v^2}$$

So that the kinetic energy density is written as

$$K.E.density = \frac{kinetic\ energy}{volume} = \frac{\frac{1}{2}\rho \Delta V \overline{v^2}}{\Delta V} = \frac{1}{2}\rho \overline{v^2} \tag{5.2}$$

Potential energy density: Similarly, like the kinetic energy density, it is the potential energy per unit volume, for a certain mass of fluid at distance h above the ground; then the potential energy density is

$$P.E.density = \frac{potential\ energy}{volume} = \frac{\Delta m\, g\, h}{\Delta V} = \frac{\rho \Delta V g h}{\Delta V} = \rho g h \tag{5.3}$$

Compressibility and Incompressibility: If the density of the fluid is constant ($\rho = constant$) everywhere through the flow, it is called incompressible and it is called compressible if the density is not fixed.

Viscous and Nonviscous: The viscosity is the frictional forces originated inside the fluids and which is considered as the resistance of flow. Some description of fluids neglects these forces and the fluid is called nonviscous fluid.

Types of flow: If we consider that the fluid consists of laminas or layers, so we can represent each layer by an imaginary line called a streamline. This line describes the flow of the fluid and its direction. There are two main types of flow:

Laminar flow: In this type of flow, the speed of flow is low and the streamlines of flow are parallel, figure 5.1a.

Turbulent flow: The flow has a high speed, where the streamlines intersect each other, figure 5.1b. Later we will discuss how to distinguish between both types of flow.

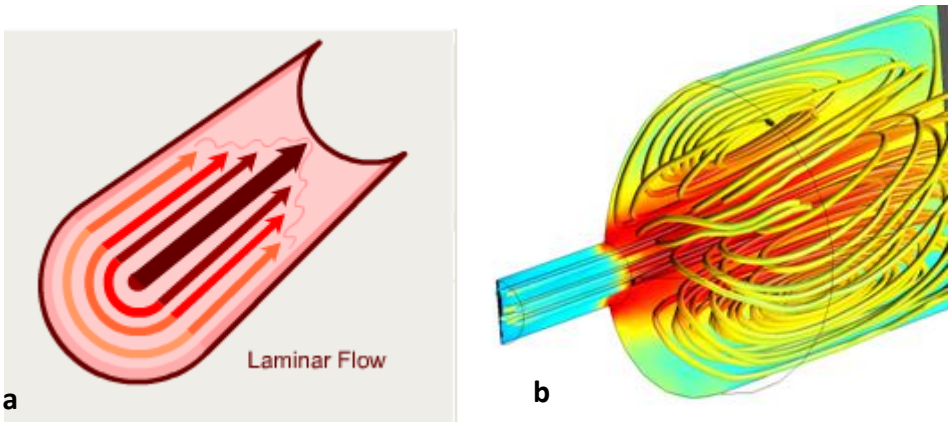


Figure 5.1: a. laminar and b. Turbulent flow of fluids

5.2 Fluid Flow and the Continuity Equation

Fluids, by definition can flow, but are essentially incompressible. This provides some very useful information about how fluids behave when they flow through a pipe, or a hose. Consider a hose with a decreasing diameter decreases along its length, as shown in the figure 5.2 below. The ``continuity equation is a direct consequence of the rather trivial fact that what goes into the hose must come out.

$$Q_{in} = Q_{out} \tag{5.4}$$

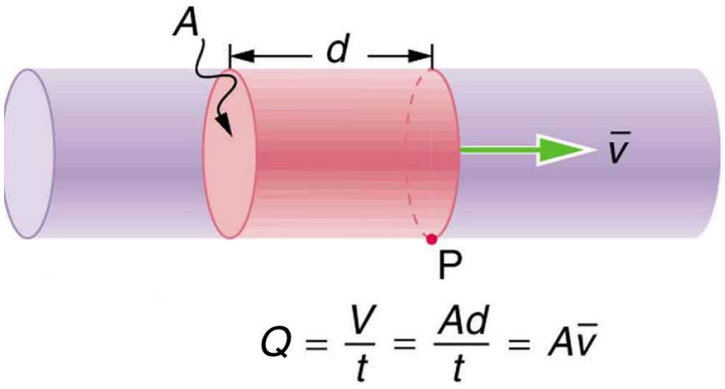
The volume of water flowing through the hose per unit time (i.e. the flow rate (Q) at the left must be equal to the flow rate at the right, or in fact anywhere along the hose.

$$Q = \frac{\Delta V}{\Delta t} = constant \tag{5.5}$$

The flow rate is measured in the units of volume per unit time, m^3/s.

Moreover, it can be shown that the flow rate at any point in the hose or tube is equal to the area of the hose at that point times the speed with which the fluid is moving. Consider a fluid is flowing in a tube as shown in figure 5.2, where the radius of the tube is decreasing, thus at a certain point at the tube the fluid flow rate is

$$Q = \frac{\Delta V}{\Delta t} = \frac{A\,\Delta x}{\Delta t} = A\bar{v} = constant \tag{5.6}$$



Where $\bar{v} = \frac{\Delta x}{\Delta t}$, is the average speed of flow at any point. Take $A = \pi r^2$, so the continuity equation can be rewritten as

$$A_1\,\overline{v_1} = A_2\overline{v_2}$$

Or,

$$r_1^2\overline{v_1} = r_2^2\overline{v_2} \tag{5.7}$$

This means that the speed of flow increases obviously by a small decrease of the diameter of the tube.

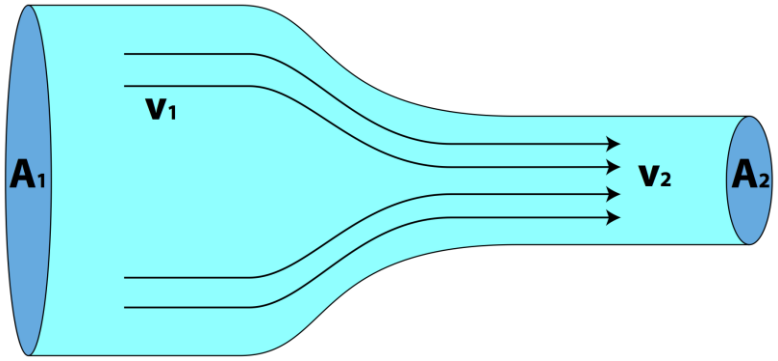


Figure5 .2: The flow of a fluid in a tube of different cross sections

Example 5.1

A water pipe leading up to a hose has a radius of $1\ cm$. Water leaves the hose at a rate of $3\ liter$ per minute. (a) Find the velocity of water in the pipe. (b) The hose has a radius of $0.5\ cm$. What is the velocity of water in the hose?

Solution

The velocity can be found by using the flow rate and the area of the pipe or the hose. The flow rate is

$$Q = \frac{\Delta V}{\Delta t} = \frac{0.003}{60} = 5 \times 10^{-5}\, m^3/s$$

So the velocity is given by:

$$v = \frac{Q}{A} = \frac{Q}{\pi r^2} = \frac{5\times10^{-5}\, m^3 s^{-1}}{3.14\times(0.01m)^2} = 0.159\, ms^{-1}$$

The flow rate is constant, so $r_1^2\overline{v_1} = r_2^2\overline{v_2}\ , \Rightarrow \frac{v_2}{v_1} = \frac{r_1^2}{r_2^2}$

And so the velocity in the hose v_2 can be calculated as:

$$v_2 = v_1 \frac{r_1^2}{r_2^2} = 0.159\ \frac{1}{0.25} = 0.636\, ms^{-1}$$

The water flow faster in the narrower channel

5.3 Bernoulli's Equation

Bernoulli derived in 1730s an important equation to describe the flow of fluids. This equation is stated that *the work done on a fluid as it flows from one place to another is equal to the change in its mechanical energy*. This equation is applicable for incompressible fluids, nonviscous fluids (where no energy loss), laminar flow, and for steady state flow (when the flow speed at any point is constant with time). To derive Bernoulli's equation, we consider the flow of a fluid in a tube of cross sectional area A from section 1 to section 2 as shown in figure 5.3. The pressure, speed of flow, and height of the fluid at cross section 1 is denoted as $P_1,\ v_1,\ y_1$respectively. For the same at section 2, we have $P_2,\ v_2,\ y_2$. The net force, on the fluid in the tube causing the fluid to transfer from section 1 to section 2 is $F = (P_1 - P_2)A = \Delta P\ A$. If the fluid in the section moves a short distance Δx, so that the work done on the fluid is given by

$$W\ =\ F.\ \Delta x\ = (P_1 - P_2)A\ \Delta x \tag{5.8}$$

Since the product $\mathrm{A}\ \Delta\mathrm{x}$ is the volume $\Delta\mathrm{V}$ of the fluid leaving the section, thus equation 5.8 becomes:

$$W\ = (P_1 - P_2)\ \ \Delta V \tag{5.9}$$

On the other hand, the change in the kinetic energy of a volume $\Delta\mathrm{V}$ of the fluid flowing from section 1 to 2 is $\frac{1}{2}\ \rho\ \Delta V\ \overline{v_2^2}\ -\ \frac{1}{2}\ \rho\ \Delta V\ \overline{v_1^2}$, and the same for the change in the potential energy, which is $\rho\ \Delta V g\ y_2\ - \rho\ \Delta V g\ y_1$. Since the energy has to be conserved, the work in equation 5.9 must equal the change in the kinetic energy and the change in the potential energy, so that we can write:

$$(P_1 - P_2)\Delta V\ = \frac{1}{2}\ \rho\ \Delta V\ \overline{v_2^2}\ -\ \frac{1}{2}\ \rho\ \Delta V\ \overline{v_1^2}\ +\ \rho\ \Delta V g\ y_2\ - \rho\ \Delta V g\ y_1$$

By eliminating $\Delta\mathrm{V}$ from both sides and rearrange the similar terms in one side, we get:

$$P_1 + \frac{1}{2}\ \rho\ \ \overline{v}_1^2\ + \rho\ g\ y_1 = P_2\ +\ \frac{1}{2}\ \rho\ \overline{v}_2^2\ \ +\ \rho\ g\ y_2 \tag{5.10}$$

This can be generalized for any two points through the flow of the fluid, so Bernoulli's equation can be written as

$$P+\frac{1}{2}\,\rho\,\overline{v^2}+\rho\,g\,y=\ constant \tag{5.11}$$

Thus Bernoulli's equation can be stated as the pressure of the fluid plus its mechanical energy density ($kinetic\ energy\ density+\ potential\ energy\ density$) is the same everywhere in the flow.

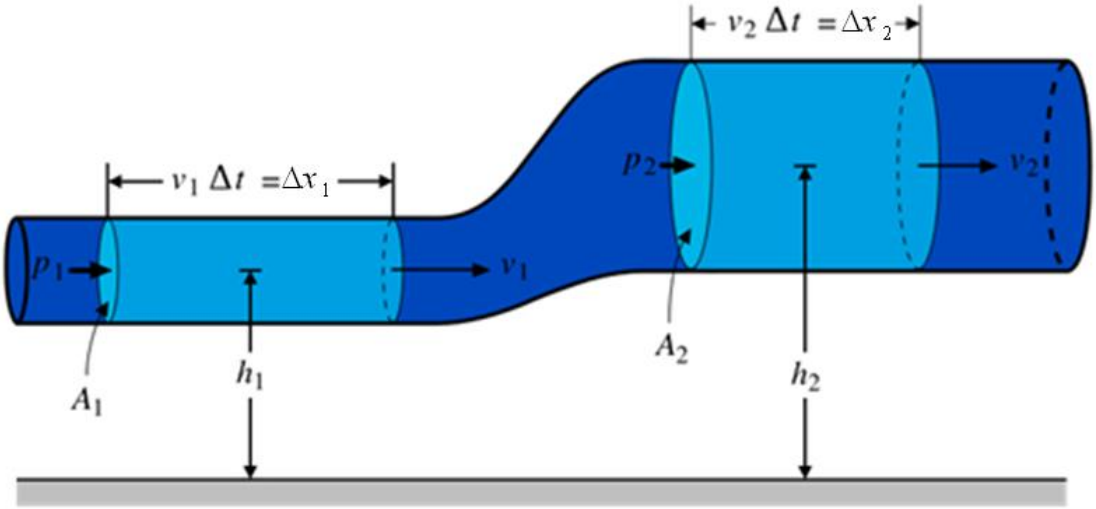


Figure5.3: The flow of a liquid from point 1 to point 2 via a pressure difference $\mathbf{p_1-p_2}$

Note: For solving Bernoulli's equation problems, choose always two points in the flow line and at each point determine the variables p, v, y. In some cases if you have two unknowns, you should then use the equation of continuity, which help you to solve two equations with two unknowns.

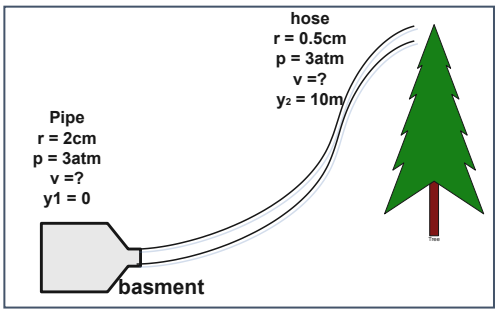
Example 5.2

Water enters the basement through a pipe $2\ cm$ in radius at an absolute pressure of $3\ atm$. A hose with a $0.5\ cm$ radius is used to water plants $10\ m$ above the basement. Find the speed of water as it leaves the hose?

Solution

We have two points of interest. Point 1 in the pipe at the basement, where $\mathrm{p_1} = 3\mathrm{atm}, \mathrm{y_1} = 0\mathrm{m}, \mathrm{v_1} = ?$ Point 2 is in the hose just at the moment the water leaving the hose for planting the tree, where $\mathrm{p_2} = 1\mathrm{atm}, \mathrm{y_2} = 10\mathrm{m}, \mathrm{v_2} = ?$ By applying Bernoulli's equation, we have

$$p_1 - p_2 = \frac{1}{2}\rho\,(\overline{v_2^2} - \overline{v_1^2}) + \rho\, g\,(y_2 - y_1)$$



We have two unknowns v_1 *and* v_2 , so we can reduce them to only one unknown by using the continuity equation, where $v_1 = \frac{r_2^2}{r_1^2}\, v_2$

So $v_1 = \frac{1}{16}\, v_2$,

$$\overline{v_2^2} - \overline{v_1^2} = \overline{v_2^2} - \left(\frac{1}{16}\, v_2\right)^2 = \overline{v_2^2} - \frac{1}{256}\,\overline{v_2^2} = \frac{256\,\overline{v_2^2} - \overline{v_2^2}}{256} = \frac{(256-1)\,\overline{v_2^2}}{256} = \frac{255}{256}\,\overline{v_2^2}$$

and $\overline{v_2^2} - \overline{v_1^2} = \frac{255}{256}\, v_2^2$, then we substitute in the last equation to get:

$$(3-1) \times 1.013 \times 10^5 = \frac{1}{2}\,(1000)\,\frac{255}{256}\,\mathrm{v_2^2} + 1000(10)\,(10-0)$$

Solve for v_2 , we get $v_2 = 14.35\, ms^{-1}$.

Example 5.3

Water is flowing from a hole of $1\ cm$ radius at the bottom of a closed cylindrical container of $2m$ diameter. If the height of the water in the container is $2\ m$ and the pressure over the surface of water is $3\ atm.$, calculate how much time it took until the container became empty?

Solution

The container is cylindrical in shape with radius $\mathrm{R} = 1\ \mathrm{m}$, and the height of water in it is $H = 2m$, so that we can calculate the amount of water in the container

$$\Delta V \;=\; \pi R^2 H \;= 3.14 \times\; 1^2 \times 2 \;= 6.28\, m^3$$

If we could calculate the flow rate $Q = \pi R^2 \bar{v}$ from the hole and which is equal $\Delta V/\Delta t$, we can compute the time needed to empty the container, so we should calculate $\bar{v}$. To apply Bernoulli's equation we assume two points, one at the surface of water with $P_s \;=\; 3atm,\; h_s \;= 2\, m,\; v_s \;=?$ and the other point at the exit of the hole at the bottom of the container with $P_h \;=\; 1atm,\; h_h \;= 0\, m,\; v_h \;=?$. Writing Bernoulli's equation

$$P_s - P_h = \frac{1}{2}\,\rho\;(\,\overline{v_h^2} - \overline{v_s^2}\,) + \rho\; g\;(y_h \;- y_s)$$

By comparing the radius of the container to the radius of the hole

$$\frac{R}{r} \;= \frac{1}{1\times 10^{-2}} \;= 100,$$

So that the speed of water at the surface related with that at the hole

$$\frac{v_s}{v_h} \;=\; \frac{r^2}{R^2} \;= \frac{1}{10000}.$$

This indicates that $v_s \;\ll v_h$, which leads that v_s^2 is much smaller than v_h^2. We can consider $\overline{v_h^2} - \overline{v_s^2} \;\approx \overline{v_h^2}$, and $y_h \;- y_s \;= y_s \;=\; 2\, m.$ Substitute in Bernoulli's equation, we get:

$$p_s - p_h = \frac{1}{2}\,\rho\,\overline{v_h^2} \;- \rho\, g y_s$$

By solving for v_h, we get:

$$\overline{v_h^2} = \frac{2\big(\,(p_s \;-\; p_h) + \rho\, g y_s\big)}{\rho} \;= \frac{2 \;\times 2 \times 1.013 \times 10^5 Pa + 10^3\,\frac{kg}{m^3} \times 10\,\frac{m}{s^2} \times 2m}{10^3\,\frac{kg}{m^3}}$$

$$\overline{v_h^2} = 444.40\; m^2/s^2, \qquad so, \qquad v_h = \sqrt{444.40} = 21.08\; ms^{-1}$$

So the time required to empty the container

$$\Delta t \;= \frac{\Delta V}{Q} \;= \frac{\Delta V}{\pi r^2 v_h} \;=\; \frac{6.283}{3.14 \;\times 10^{-4} \times 21.08} \;= 948.73\; s \sim 15.81\; min.$$

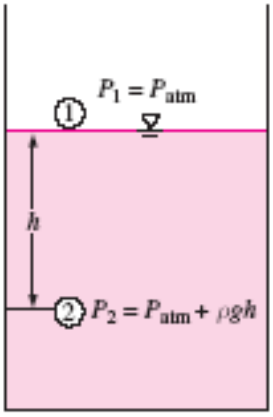
5.4 Applications of Bernoulli's equation

The Bernoulli equation can be applied to a great many situations not just the pipe flow we have been considering up to now. There are two main consequences of this equation

Static consequence

Static Consequence: In this consequence we have zero speed of flow, where the fluid is settle in its container. In this situation the kinetic energy density term leads to zero, and Bernoulli's equation becomes

$$p+ \rho\ g\ y = \ constant \tag{5.12}$$



or

$$p_1 + \rho\ g\ y_1 = p_2 \ + \ \rho\ g\ y_2$$

This means that the sum of the pressure and the potential energy density is constant everywhere inside a fluid in static. For example, if we have a liquid in an open air container as show in the figure and it is required to measure the pressure at a point at depth h inside the container. Take two points: point 1 at the surface of the liquid, where the pressure is the atmospheric pressure and point 2 at depth $h = \left(y_1 - y_2\right)$ inside the container, also $v_1 = \ v_2 = 0$, and then by applying Bernoulli's equation, we have:

$$p_a + \rho\, g\, y_1 = p_2 + \rho\, g\, y_2$$

so

$$p_2 = p_a + \rho\, g(y_1 - y_2) = p_a + \rho\, gh \tag{5.13}$$

So the pressure inside any container is equal the pressure at the surface plus the potential energy density at that point.

Note: The difference between the absolute pressure at any point and the atmospheric pressure $(p - p_a)$ is called the gauge pressure.

Example 5.4

What is the pressure on a swimmer $5\ m$ below the surface of a lake?

Solution

Using the depth of the swimmer is $h = 5\ m$, and the density for water is $\rho = 1000\ kgm^{-3}$, and the atmospheric pressure is $1.013 \times 10^5\ Pa$. So using equation 5.11 to calculate the pressure on the swimmer to be:

$$p = p_a + \rho\, gh = 101300 + (1000)(10)5 = 1.5 \times 10^5\ Pa.$$

Manometer

One of the most common applications of the static consequence of Bernoulli's equation is the manometer. A common simple manometer consists of a U shaped tube of glass filled with some liquid. Typically the liquid is mercury because of its high density. With both ends of the tube open, the liquid is at the same height in each leg, figure 5.4. When positive pressure is applied to one leg, the liquid is forced down in that leg and up in the other. The difference in height, "h," which is the sum of the readings above and below zero, indicates the gauge pressure ($p = \rho gh$). When a vacuum (low pressure) is applied to one leg, the liquid rises in that leg and falls in the other. The difference in height, "h," which is the sum of the readings above and below zero, indicates the amount of vacuum.

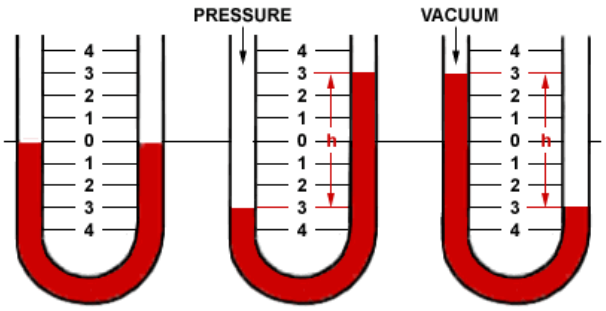


Figure5.4: A U shaped tube called Manometer is used to measure the pressure of unknown gases

The manometer is a part of a device called a sphygmomanometer, figure 5.5, used to measure the blood pressure. The measurements are carried out at the upper arm of the human (the brachial artery at the elbow), since as it is about at the same level as the heart. During a complete heart pumping cycle, the pressure in the heart goes through maximum value (systolic blood pressure: pounding sound is heard), where the blood is pumped from the heart, and a minimum value (diastolic blood pressure: the sound can no longer be heard), where the heart relaxes and the blood returned from the veins filled the heart. Blood pressures are usually presented as $\text{systolic/diastolic}$ ratios. Typically readings for a resting healthy adult are about $120/80$ in torr ($1\ atm\ =\ 760\ mmHg = 760\ torr$ for The Italian physicist Evangelista Torricelli) and $16/11$ in kPa. The borderline for high blood pressure (hypertension) is usually defined to be $140/90$ in $torr$.



Figure5.5: Typical sphygmomanometer used to measure the blood pressure

Horizontal flow consequence

The second consequence of Bernoulli's equation is that when the flow of a fluid is horizontally. In this case, the potential energy density term will vanish since all points on the flow line have the same height, so that the dynamic energy term will be considered and the equation will be given as:

$$p+\frac{1}{2}\,\rho\,\bar{v}^2 \;=\; constant \tag{5.14}$$

Or we can write for two points on the same plane of flow, as:

$$P_1+\frac{1}{2}\,\rho\,\bar{v}_1^2 = P_2 \;+\; \frac{1}{2}\,\rho\;\bar{v}_2^2$$



This equation has many interesting applications. The simple demonstration is by blowing air between two half sheets of paper. We can apply Bernoulli's equation by taking two points at the same plane. The first point is outside the sheet, where the pressure and the speed of flow are noted as P_{out} , v_{out} respectively and the other point is between the sheets of paper, where the pressure and the speed of flow are noted as P_{in} , v_{in} respectively, so that we have

$$P_{out}+\frac{1}{2}\,\rho\;\bar{v}_{out}^2 \;= P_{in} \;+\; \frac{1}{2}\,\rho\,\bar{v}_{in}^2$$

Rearrange the equation leads to the following

$$P_{out} \;- P_{in} \;= \frac{1}{2}\,\rho\,(\;\bar{v}_{in}^2 - \bar{v}_{out}^2\,)$$

When a person blows between the two sheets, so that the speed of air flowing inside will be larger than that outside ($v_{in} > v_{out}$)*:* this means that the left hand side is positive and therefore $P_{out} > P_{in}$. This pressure difference results in the sheets moving closer toward

one another. Thus, the pressure drops when the velocity of the flow increases for a fluid moving at a constant height.

This pressure drop associated with increasing fluid velocities has many everyday implications. For example, concerning flight, Bernoulli's Principle has to do with the shape of an airplane's wing as shown in Figure 5.6. The bottom is flat, while the top is curved. Air travels across the top and bottom at the same time, so air travels slower on the bottom (creating more pressure) and faster on top (creating less pressure).

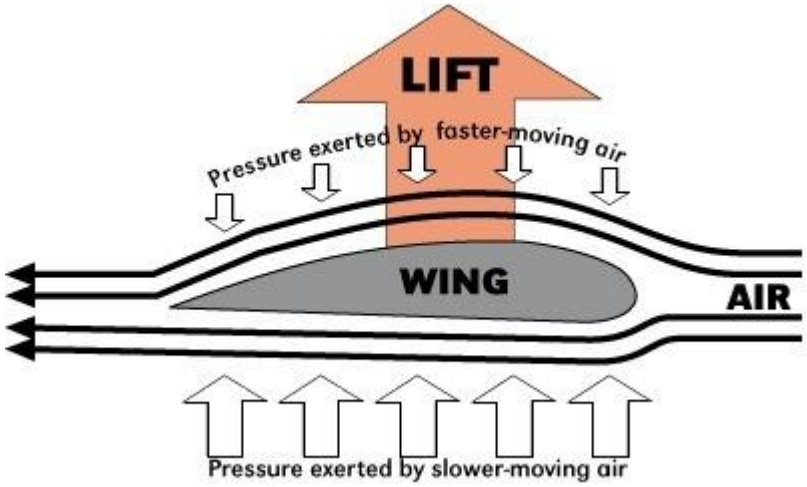


Figure5.6: Form of airplanes wing and the direction of the wind around it

Bernoulli's principle can explain the clogging of arteries when the blood flows through an artery section of smaller cross sectional area. According to Bernoulli the pressure of blood within this section will drop inside the arterial wall, and on the other hand the pressure on the outside arterial wall will be larger than inside causing the clogging of the blood vessel.

Example 5.5

The diameter of a horizontal blood vessel is reduced from $12mm$ to $4\ mm$. What is the flow rate of blood in the vessel, if the pressure at the wide part is 8 kPa and 4 kPa at the narrow one. (Take the density of blood to be $1060\ kgm^{-3}$.)

Solution

By applying Bernoulli's equation for horizontal flow and by taking one point in the wider section and the other at the narrower one, so we get:

$$P_{wid} - P_{narr} = \frac{1}{2}\,\rho\,(\,\bar{v}_{narr}^2 - \bar{v}_{wid}^2\,)$$

Using the continuity equation,

$$v_{wid} = \frac{r_{narr}^2}{r_{wid}^2}\,v_{narr} = \left(\frac{4}{12}\right)^2 v_{narr} = \frac{1}{9}\,v_{narr}$$

Then substitute and solve for v_{narr} to get

$$4 \times 10^3 = \frac{1}{2}\,(1060)\;\bar{v}_{narr}^2\,(\,1 - \frac{1}{81}\,)\,,$$

so then $v_{narr}^2 = \frac{81\times2\times4\times10^3}{80\times1060} = 7.64\,,$

which gives $v_{narr} = = \sqrt{7.64} = 2.76\,ms^{-1}\,.$

The flow rate is constant everywhere and can be calculated from the relation

$$Q = \pi r_{narr}^2 v_{narr} = 3.14 \times (2 \times 10^{-3})^2\;\; 2.76$$

$$Q = 34.68 \times 10^{-6}\,m^3 s^{-1}$$

Venturi Tube

A practical instrument which makes use of the Bernoulli principle and a manometer pressure gauge is the Venturi flow meter. The figure 5.7 shows that you can express the fluid velocity v_1 at the inlet of the device in terms of the difference in pressure measured by the manometer. This device can be used to measure the speed of flow of fluids and hence the flow rates. The fluid flows through different cross sectional areas in different portion of the tube. At the narrower section, the speed of flow increases and according to Bernoulli's principle the pressure drops. The pressure drops can be measured using a gauge manometer and by

knowing the sizes of the cross sectional area of the tube, then the speed of flow can be easily calculated. For point 1 and 2 in the figure 5.7, we have

$$P_1 - P_2 = \frac{1}{2}\rho\left(\bar{v}_2^2 - \bar{v}_1^2\right)$$

And from the continuity equation we have $v_2 = \frac{A_1}{A_2} v_1$, which can be substituted to get

$$v_1 = \sqrt{\frac{2\left(p_1 - p_2\right)}{\rho\left[\left({}^{A_1}/_{A_2}\right)^2 - 1\right]}}$$

Thus, a measurement of the pressure drop $(P_1 - P_2)$ and knowledge of the areas of the tube determines the speed the blood flow v_1.

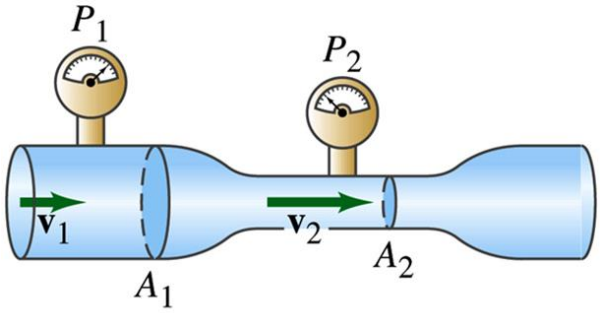


Figure5.7: Venturi tube with different cross sectional parts

Example 5.6

The flow of blood through a large artery in a dog is diverted through a Venturi flow meter. The wider part of the flow meter has an area of $0.08\ cm^2$, which equals the cross sectional area of the artery. The narrower part of the flow meter has an area of $0.04\ cm^2$. If the speed of the blood in the artery was $12.5\ \text{cm s}^{-1}$, find the drop in pressure through the flow meter?

Solution

The speed of blood in the artery is the speed of the blood in the wider section,

so $v_1 = \ 0.125\, ms^{-1}$, then from the equation, we have:

$$P_1 - P_2 = \frac{1}{2}\,\rho\,(\,\bar{v}_2^2 - \bar{v}_1^2\,)$$

From the continuity equation $v_2 = \frac{A_1}{A_2}\,v_1 = \frac{0.08}{0.04} \times 0.125 = 0.25\, ms^{-1}$, so that the pressure drop is

$$P_1 - P_2 = \frac{1}{2}\,(1060)\left(\,0.25^2 - \,0.125^2\,\right) = \,25\,Pa.$$

5.5 The Role of Gravity on blood circulation

We learned from Bernoulli's principle that the pressure of the fluid change according to its kinetic energy density and as well as it potential energy density. Because of that, the blood pressure in human organs is affected by its location from earth. During the blood circulation, the venous system is used to return the blood from the lower extremities to the heart. It is expected to have a problem of lifting blood long distances to the heart against the force of gravity.

If we have a person in the reclining (laying down) position, the measurement of blood pressure in the large arteries are almost the same everywhere. The small drop in pressure between the heart and the feet or the brain is due to the viscous forces. According to Bernoulli's equation $P + \frac{1}{2}\,\rho\,\bar{v}^2 + \rho\,g\,y = \,constant,$ we can analyze the situation in the reclining position. The velocities in the three main arteries (Brain, heart, and feet) are small and roughly equal, so that the term $\frac{1}{2}\,\rho\,\bar{v}^2$ can be ignored. Furthermore, in this position also the height of the brain, heart and feet are almost equal, so that the term $\rho\,g\,y$ can be ignored from the formula. This

results in equal blood pressure in the three parts $P_B = P_H = P_F$. Note that $\boldsymbol{B}, \boldsymbol{H}$ and $\boldsymbol{F}$ refer to the brain, heart and feet, respectively.

In the standing position, the situation is different, where only the term $\frac{1}{2}\rho\,\overline{v}^2$ can be ignored and the term $\rho\, g\, y$ has a significant effect. Hence the gauge pressures at the brain p_B, at the heart p_H and at the foot p_F are related by:

$$P_F = P_H + \rho g h_H = P_B + \rho g h_B \tag{5.15}$$

Note that $h_F = 0$ in the standing position.

Typical values for adults standing upward $h_H = 1.3\ m$ *and* $h_B = 1.7\ m$. Typical value of the blood pressure at the heart is $P_H = 13.3\ kPa$, and take the blood density to be $1060\ kg\ m^{-3}$, we find:

$$P_F = P_H + \rho g h_H = 13.3 \times 10^3 + (1060)(10)(1.3) \sim 27.1\ kPa$$

In a similar way, we find that:

$$P_B = P_H + \rho g(h_H - h_B) = 13.3 \times 10^3 + (1060)(10)(-0.4) = 9.06\ kPa$$

This explains why the pressures in the lower and upper parts of the body are very different when the person is standing, although they are about equal in the reclining. The high blood pressure at the foot explain the possibility of lifting blood uphill to the heart, and in addition the muscles surrounding the veins contract and cause constriction.

Example 5.7

When a $1.7\ m$ tall man stands, his brain is $0.5\ m$ above his heart. If he bends so that his brain is $0.4\ m$ below his heart, by how much does the blood pressure in his brain changes?

Solution

We know that the blood pressure of the organ change by changing its position from the earth. The blood pressure at the brain in the standing case is given by:

$$P_{B_{stand.}} = p_H + \rho g(h_H - h_B)_{stand.}$$

Where $(h_H - h_B)_{stand} = -0.5\, m$,

so

$$P_{B_{stand.}} = 13.3 \times 10^3 + 1060 \times 10 \times (-0.5) = 8\, kPa$$

The blood pressure at the brain in the bending position is given by:

$$P_{B_{bending.}} = p_H + \rho g(h_H - h_B)_{bending.}$$

Where $(h_H - h_B)_{stand} = 0.4\, m$, this results in

$$p_{B_{bending}} = 13.3 \times 10^3 + 1060 \times 10 \times (0.4) = 17.54\, kPa$$

So the blood pressure at the brain will increase by bending, so the change in blood pressure is $\Delta P = P_{B_{bending}} - P_{B_{stand.}} = 17.54 - 8 = 9.54\, kPa.$

5.6 Effect of acceleration on Blood pressure

It is a common symptom for some people having hypotension to feel dizziness when they exist in an elevator of upward acceleration. So that the question arise, is the blood pressure at the organs affected when man under upward or downward acceleration. Indeed, when a person in an erect position experiences an upward or downward acceleration, his weight will be different.

To understand the effect of acceleration on the scale tightened in the ceiling of moving (upward or downward) elevator cabinet with acceleration a, see figure 5.8.

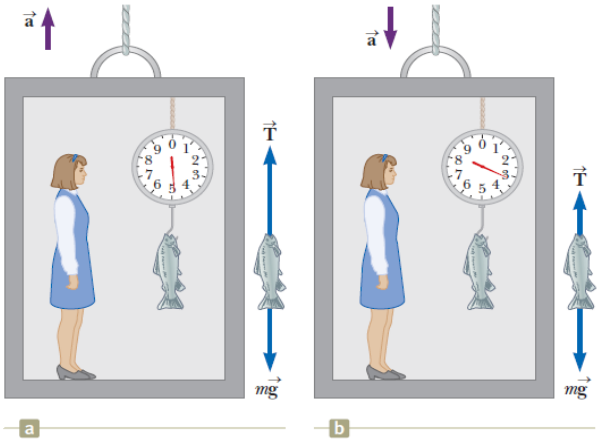


Figure 5.8: the effect of acceleration on the Tension in the wire

We all agree that the tension in the wire when the elevator is not moving is

$$\sum F_y = 0 = T - mg \rightarrow T = mg$$

In other words the tension the wire equals the fish weight (mg).

Now, if the elevator accelerating upward, then

$$\sum F_y = ma = T - mg \rightarrow T = mg + ma = m(g + a)$$

Thus the effective acceleration is $(a + g)$, and the apparent weight is greater than the actual weight.

On other hand, if the elevator in moving downward with acceleration a, then

$$\sum F_y = -ma = T - mg \rightarrow T = mg - ma = m(g - a)$$

Thus the effective acceleration is $(g - a)$, and the apparent weight is less than the actual weight.

Upward acceleration: If a man in an erect position experience **upward** acceleration **a**, then his effective weight becomes $m\ (g + a)$. Applying Bernoulli's equation to the foot, brain and heart with $\boldsymbol{g}$ replaced by $\boldsymbol{g + a}$, so we have:

$$P_B = P_H + \rho\ (g + a)(\ h_H - h_B)$$

or

$$P_B = P_H - \rho\,(g + a)(\,h_B - h_H) \qquad (5.16)$$

It is noted here that for standing person the term $(\,h_B - h_H)$ is positive and also the same for $(g + a)$, thus the blood pressure at the brain will be reduced even farther by increasing the upward acceleration $(\boldsymbol{a})$. At certain value of $\mathbf{a}$, the human will lose consciousness because the collapse of the arteries in the brain when the blood pressure at the brain equal zero. Put $P_B = 0$ in equation 5.16, so we get:

$$0 = P_H - \rho\,(g + a)(\,h_B - h_H)$$

This can results in

$$(g + a) = \frac{P_H}{\rho\,(\,h_B - h_H)}$$

Take

$$(\,h_B - h_H) = 0.4\,m,\ P_H = 13.3\,kPa, \text{ and } \rho = 1060\,kgm^{-3},$$

We get:

$$(g + a) = \frac{13.3 \times 10^3}{1060\ (0.4)} = 31.4\,ms^{-2} = 3.2\,g$$

So the value of the upward acceleration causing consciousness is $\mathbf{3.2\,g}$. This factor should limit the speed with which a pilot can pull out of dive. A related experience is the feeling of light headache that sometimes occurs when one suddenly stands up.

We can also show the change of the blood pressure at the foot by the upward acceleration situation, by putting $g + a$ instead of

$$P_F = P_H + \rho(g + a)h_H$$

This relation shows that the blood pressure at the foot will increase by increasing the upward acceleration.

Downward acceleration

If a man in an erect position experience downward acceleration, then his effective weight becomes $m\ (g-a)$. Applying Bernoulli's equation to the foot, brain and heart with *g* replaced by $g-a$, so we have:

$$P_B = P_H + \rho\ (g-a)(\ h_H - h_B)$$

Or

$$P_B = P_H - \rho\ (g-a)(\ h_B - h_H) \qquad \textbf{(}5.17)$$

Thus the blood pressure at the brain will increase even farther by increasing the downward acceleration (**a**), which opposite to what occurs by the upward acceleration. This increase should be controlled and observed, where at certain value of **a** the blood pressure at the brain may cause an explosion of the arteries in the brain, which is so dangerous. The same calculation for the blood pressure at the foot results in a decrease of the blood pressure by increasing the downward acceleration.

Example 5.8

A $1.8\,m$ tall man stand in an elevator accelerating upward at $12\,ms^{-2}$, what is the blood pressure in the brain and foot. Take the height difference between the heart and the brain to be $0.35\,m$?

Solution

The elevator accelerating upward, thus the effective acceleration on man is

$(g+a) = 10 + 12 = 22\,ms^{-2}$,

Substitute in the equation 5.16, we get

$$P_B = P_H + \rho\ (g+a)(\ h_H - h_B) = 13300 + 1060 \times 22 \times -0.35 = 5.14\,kPa$$

The pressure at the brain decrease

$$P_F = P_H + \rho(g+a)h_H = 13300 + 1060 \times 22 \times 1.45 = 47.11\,kPa$$

An increase of the blood pressure at the foot is observed.

5.7 Viscous Fluid Flow

In the discussions of the fluid flow according to Bernoulli's principle, we ignored the friction forces among fluid layers; the effect of viscosity on the flow of fluid. In this section, the coefficient of viscosity factor will be taken into consideration. The viscosity in fluids is originated from the frictional forces between the fluids laminas and their container. The viscosity hence can be considered as the resistance of flow of fluids, like current resistance. Fluids resist the relative motion of immersed objects through them, as well as, to the motion of layers with differing velocities within them. Viscosity in gases is originated from the successive collision between the gas molecules, and it is expected that the viscosity is a temperature dependent. The viscosity of gases becoming more by increasing their temperature, because of the increase of their kinetic energy and the probability of collisions will be more considerable. In liquids, the viscosity decrease by increasing the temperature.

To be more convenient with the viscosity, put a small amount of liquid between two plates of glass separated by a distance Δy as shown in figure 5.9. The upper plate is free to move and the lower plate is fixed. If the upper plate was forced to move with a velocity, Δv , away horizontally, there will be a resistance for this motion. It is found that there will be a lamina or layer of the liquid which moves with the upper plate, and other lamina which is stationary. There is a gradient of velocity as you move from the stationary plate to the moving one and the liquid tends to move in parallel layers, which is called laminar flow.

It is found that the force **F** required to move the upper plate at constant average speed is proportional directly with the speed gradient and the surface area of the plate and inversely proportional with the separation distance between the plates, which means that:

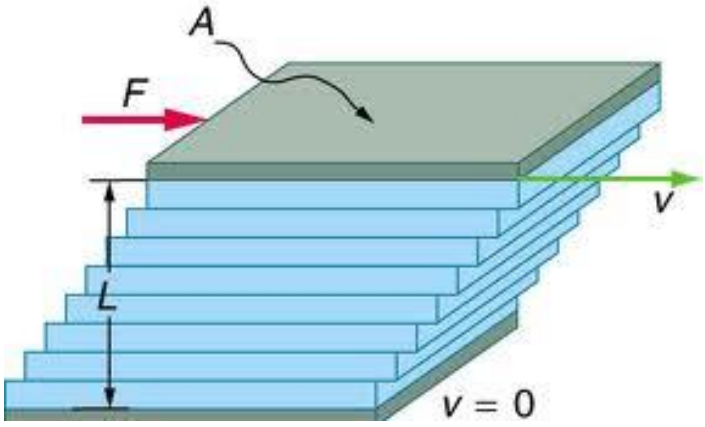


Figure5.9: Two parallel glass plates separated by a thin layer of liquid

$$F \propto \Delta v\,,\; F \propto A \; and \; F \propto \frac{1}{\Delta y}\,, so \; F \propto \frac{A\,\Delta v}{\Delta y}$$

The constant of proportion is called the viscosity coefficient, which is represented by the Greek symbol $\boldsymbol{\eta}$ "eta". So we can write

$$F = \eta \frac{A\,\Delta v}{\Delta y} \tag{5.18}$$

Rearrange equation 5.18 for the coefficient of viscosity, that is

$$\eta = \frac{F/A}{\Delta v/\Delta y} \tag{5.19}$$

The viscosity can be defines as the ratio between the shearing stress ($\mathbf{F/A}$) to the rate of shearing strain or the gradient of velocity. The dimension of the viscosity coefficient can be deduced as:

$$[\eta] = \left[\frac{F/A}{\Delta v/\Delta y}\right] = \left[\frac{MLT^{-2}/L^2}{LT^{-1}/L}\right] = ML^{-1}T^{-1}$$

The S.I. unit of the viscosity coefficient is Pascal second, where $\boldsymbol{kgm^{-1}s^{-1} = 1\,Pa.s}$. The Pascal second is rarely used in technical and scientific publications. The common used unit is called poise ($\boldsymbol{P}$), where $1\;poise = \frac{dyne.s}{cm^2} = 10^{-1}\;Pa.\mathbf{s}$, so the abbreviation centipoises is equal $10^{-3} Pa.s$.

Table 5.1 lists some typical coefficient of viscosity measures in $Pa.s$ for some fluids, which are usually related by the temperature as the viscosity coefficient is temperature dependent.

Table 1: Typical values of viscosity coefficient for some fluids in units of $\boldsymbol{Pa.s}$.

Temperature ℃	Castor Oil	Water	Air	Normal blood	Blood Plasma
0	5.3	1.792×10^{-3}	1.71×10^{-5}		
20	0.986	1.005×10^{-3}	1.81×10^{-5}	3.015×10^{-3}	1.81×10^{-3}
37	-	0.695×10^{-3}	1.87×10^{-5}	2.08×10^{-3}	1.257×10^{-3}
40	0.231	0.656×10^{-3}	1.9×10^{-5}		
60	0.08	0.469×10^{-3}	2.00×10^{-5}		
80	0.03	0.357×10^{-3}	2.09×10^{-5}		
100	0.017	0.284×10^{-3}	2.18×10^{-5}		

5.8 Laminar Flow in a Tube

One of the most interesting applications in fluid dynamics is the laminar flow in a cylindrical tube such as the pipes or human blood vessels. Consider a fluid moving through a tube of length **L** and cross sectional area, $A = \pi R^2$. The pressure difference across the segment of the tube is $\Delta P = P_2 - P_1$ as shown in figure 5.10. Because of the existence of the viscosity inside the fluid, the layer of the fluid adjacent to the cylindrical wall moves very slowly and the inward successive layers move at increasing velocities. The maximum velocity will be for the fluid at the central axis of the tube, $\boldsymbol{v_{max}}$ and the minimum velocity will be for the layer adjacent to the wall, $\boldsymbol{v_{min}} = \mathbf{0}$. So the average velocity, $\overline{\boldsymbol{v}}$, is half the maximum velocity at the center of the tube, $\overline{\boldsymbol{v}} = \frac{\mathbf{1}}{\mathbf{2}}\,\boldsymbol{v_{max}}$. From the continuity equation, the flow rate then $Q = A\bar{v} = \frac{1}{2}\,Av_{max}$.

The pressure drop $\Delta p = p_2 - p_1$ along the tube of length L is directly proportional to the average velocity of flow and to the length L of the tube. Thus, it can be understood that the average velocity of flow and the flow rate of the fluid are proportional to the pressure gradient $\Delta P/L$. Another factor affect the average velocity of flow is the radius of the tube and the coefficient of viscosity of the moving fluid. In general we can write such proportionalities as:

$$\bar{v} \;\propto\; \frac{\Delta P}{L}\, R^m\, \eta^n$$

These factors are the only quantities that can enter the formula of $\overline{\boldsymbol{v}}$, and we assume that R and η have unknown powers. To find an exact expression for $\overline{\boldsymbol{v}}$, we need mathematics which beyond the level of this textbook, so we prefer to use the dimensional analysis method to find the values of the unknown powers **m** and **n**. The formula is correct if both sides has the same dimension, so it is required that:

$$[\bar{v}] \;= \left[\frac{\Delta P}{L}\, R^m\, \eta^n\right]$$

So,

$$L\,T^{-1} \;=\; \frac{MLT^{-2} \times L^{-2}}{L}\, L^m \left(M\,L^{-1}T^{-1}\right)^n$$

$$L\,T^{-1} = (M \times M^n) \times \left(\frac{L \times L^{-2} \times L^m \times (L^{-1})^n}{L}\right) \times (T^{-2} \times (T^{-1})^n)$$

Collect terms,

$$M^0 \times L \times T^{-1} = M^{n+1} \times L^{-2+m-n} \times T^{-2-n}$$

Equate the power of each dimension,

$$n + 1 = 0 \rightarrow n = -1, and$$

$$-2 + m - n = 1 \rightarrow -2 + m - (-1) = 1 \rightarrow m = 2$$

Thus the formula for the average velocity becomes:

$$\bar{v} \propto \frac{\Delta P}{L} R^2 \eta^{-1} = constant. \frac{\Delta P\, R^2}{\eta\, L}$$

From the mathematical calculation, we have the same formula, where the value of the constant is $\mathbf{\frac{1}{8}}$. So the average velocity and the flow rate of laminar flow of a fluid through a tube is given by:

$$\bar{v} = \frac{\Delta p\, R^2}{8\, \eta\, L} \tag{5.20}$$

$$Q = A\bar{v} = \pi R^2 \bar{v} = \frac{\pi\, \Delta p\, R^4}{8\, \eta\, L} \tag{5.21}$$

The formula for Q is called *Poiseuille's law* after the physician, Jean Luis Poiseuille (1799-1869). It indicates that high viscosity leads to low flow rate and speed of flow, which is. It also shows that the flow rate is proportional to the 4th power of **R**, which is extremely dependent. This indicates that for blood vessel, any small change in the radius of the vessel results in a considerable change of the flow rate. For example, if the radius of an artery is halved, so the flow rate will be reduced to $\left(\mathbf{\frac{1}{2}}\right)^{\mathbf{4}} = \mathbf{\frac{1}{16}}$.

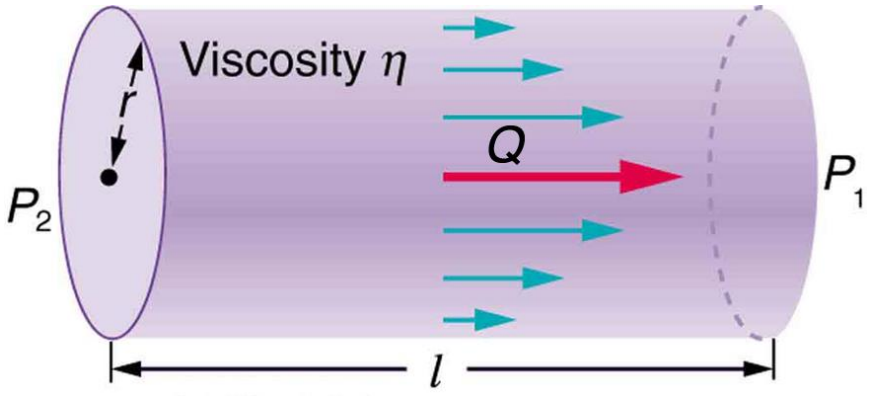


Figure5.10: A laminar flow in a tube for viscous liquid

Example 5.9

What is the pressure drop in the blood as it passes through a capillary $5\ mm$ long and $3\mu m$ in radius if the speed of the blood at the center of the capillary is $0.6\ ms^{-1}$. Take the viscosity of blood to be $2.08\ cp$. If the radius of the capillary is reduced by $20\ \%$, find the change in the flow rate?

Solution `

The speed of blood at the center of the capillary is the maximum speed of flow, $v_{max} = 2\bar{v} = 0.6\, m/s$.

So that

$$\bar{v} = 3 \times 10^{-1} = \frac{\Delta p\, R^2}{8\, \eta\, L,}$$

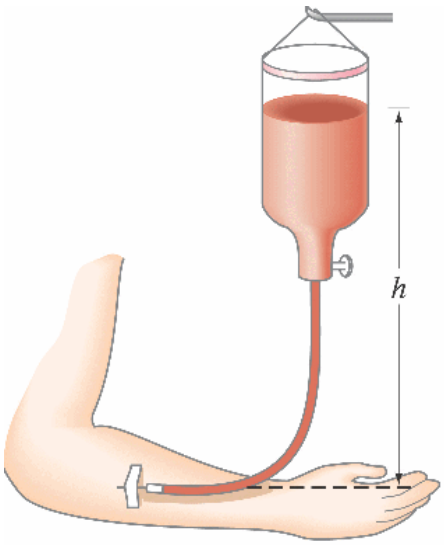
We have $R = 3 \times 10^{-6}\, m, L = 5 \times 10^{-3}\, m$ and $\eta = 2.08 \times 10^{-3} Pa.s,$ substitute to get

$$\Delta p = \frac{8\bar{v}\eta L}{R^2} = \frac{8 \times 3 \times 10^{-1} \times 2.08 \times 10^{-3} \times 5 \times 10^{-3}}{(3 \times 10^{-6})^2} = 2.77 \times 10^6\ Pa$$

The flow rate is proportional ***to*** $R^4,$ when $R_2 = 0.8\, R_1\,,$ ***so*** $Q_2 = (0.8)^4 Q_1 = 0.407\, Q_1$

Example 5.10

A patient is to be given a blood transfusion. If the inside diameter of the $4.0\ cm$ long needle is $0.40\ mm$, the blood pressure in the arm is $2400\ Pa$, and the required flow is $4.0\ cm^3/min$, how high h should the bottle be placed? ($\rho = 1060\ kg/m^3, \eta = 0.0028\ Pa \cdot s$)



Solution

Use Poiseuille's equation to calculate the excess pressure beyond $2400\ Pa$ that is required to overcome the viscosity of the blood, and then use equation $P = \rho gh$ to find h needed to provide such a pressure.

$$Q = \frac{\pi}{8}\frac{\Delta P}{L}\frac{R^4}{\eta} \Rightarrow \Delta P = \frac{8QL\eta}{\pi R^4} \Rightarrow P_2 - P_1 = \frac{8QL\eta}{\pi R^4} \Rightarrow$$

$$P_2 = P_1 + \frac{8QL\eta}{\pi R^4} \quad \text{and} \quad P_2 = \rho_{blood} gh$$

Equate both equations gives

$$\rho_{blood} gh = P_1 + \frac{8QL\eta}{\pi R^4}$$

$$h = \frac{1}{\rho_{blood} g}\left(P_1 + \frac{8QL\eta}{\pi R^4}\right)$$

Now, substitute your factors

$$h = \frac{1}{1060kg/m^3 \times 9.8m/s^2}\left(2400N/m^2 + \frac{8\left(\frac{4 \times 10^{-6}m^3}{60\ s}\right) \times 0.04m \times 0.0028Pa.s}{\pi(2 \times 10^{-4}m)^4}\right)$$

$h \cong 1.40\, m$

Or,

$$P_2 = P_1 + \frac{8QL\eta}{\pi R^4}$$

$$P_2 = \frac{2400N}{m^2} + \frac{8\left(\frac{4 \times 10^{-6} m^3}{60\ s}\right) \times 0.04m \times 0.0028Pa.s}{\pi(2 \times 10^{-4} m)^4} = 14580.66Pa$$

Then

$$P_2 = \rho g h = 14580.66Pa$$

Solving for h,

$$h = \frac{14580.66Pa}{1060kg/m^3 \times 9.8m/s^2} \cong 1.40m$$

Power dissipation

The power dissipated during the flow of a fluid is the rate of energy required to maintain the flow. In general, the power is defined as the net force $\boldsymbol{F}$ times the average speed, $\mathbb{p} = F\,\bar{v}$, $\left(\mathbb{p} = \frac{Work}{\Delta t} = \frac{F \cdot \Delta x}{\Delta t} = F\frac{\Delta x}{\Delta t} = F\,v\right)$ But the force on a segment is the pressure drop times the cross sectional area, $F = \Delta pA$

Thus the power is given as:

$$\mathbb{p} = \Delta pA\,\bar{v} = \Delta pQ \qquad (5.22)$$

Where Q is the flow rate measured in $\boldsymbol{m^3 s^{-1}}$ and $\Delta p = (p_2 - p_1)$ is the pressure difference measured in N/m^2, so the unit of the power is $\frac{N}{m^2}\ \frac{m^3}{s} = \frac{N.m}{s} = \frac{J}{s} = Watt.$

Example 5.11

Determine the power dissipated to maintain the flow of blood in the capillary as described in the last example?

Solution

From the data given in the last example, the power is given as:

$$\mathbb{P} = \Delta p A\, \bar{v} = \pi R^2 \bar{v}\, \Delta p = 3.14\,(3\times10^{-6})^2(0.3)(1.15\times10^{6}) = 9.75\times10^{-6} W\,.$$

Flow Resistance

The viscosity of fluids is defined as the flow resistance, which is originated from the frictional forces inside the fluid. The flow resistance can be defined in general as *the ratio of the pressure drop through a segment and the flow rate,* (notice the analogy to electric resistance, the pressure difference like the potential difference, and the flow rate like the electric current)

$$\mathcal{R}_f = \frac{\Delta p}{Q} \tag{5.23}$$

When the flow is Laminar, and from Poiseuille's equation, equation 5.21, we get:

$$\mathcal{R}_f = \frac{\Delta p}{\pi \Delta p R^4/8\eta L} = \frac{8\eta L}{\pi R^4} \tag{5.24}$$

The unit of $\mathcal{R}_f$ is $\frac{Pa}{m^3/s} = Pa.s/m^3$, which is the unit of the viscosity per volume or, in general, we can define the flow resistance as the viscosity density. It is observed that the flow resistance is directly proportional with the coefficient of viscosity, and inversely proportional with the 4th power of the radius of the tube. This means that most of flow resistances and pressure drops occur in smaller arteries and vascular beds of the body.

Example 5.12

Compare the flow resistance in a capillary of 5 **μ**m in radius and that in an artery of 5 cm in radius?

Solution

Since the flow resistance is inversely proportional to the 4th power of the radius, so that:

$$\frac{\mathcal{R}_{fcapillary}}{\mathcal{R}_{fartery}} = \frac{R^4_{artery}}{R^4_{capillary}} = \frac{(5\times10^{-2})^4}{(5\times10^{-6})^4} = 10^{16}$$

The flow resistance in small capillaries is much larger than that of the arteries.

From the definition of the flow resistance, the power dissipation given in equation 5.22 can be expressed in terms of $\mathcal{R}_f$ as:

$$\mathbb{P} = \Delta pQ = \mathcal{R}_f Q^2 = \frac{\Delta p^2}{\mathcal{R}_f} \quad (5.25)$$

Example 5.13

A large artery has an inner radius of $4\,mm$. Blood flows through the artery at the rate of $1\,cm^3 s^{-1}$. Find

a) The average and maximum speed of the blood in the artery
b) The pressure drop in a 10 cm long segment of the artery
c) The flow resistance of blood over the 10 cm segment
d) The power dissipated through the flow

Solution

The artery has a radius $R = 4 \times 10^{-3}\,m$. The flow rate of blood is $Q = 1 \times 10^{-6}\,m^3/s$

a) the average velocity can be found from continuity equation as:

$$\bar{v} = \frac{Q}{\pi R^2} = \frac{1 \times 10^{-6}}{3.14\,(4 \times 10^{-3})^2} = 0.02\,ms^{-1}$$

So the maximum speed, $v_{max} = 2\,\bar{v} = 0.04\,ms^{-1}$

b) For a segment of the artery $L = 0.1\,m$, and take viscosity coefficient of blood to be $\eta = 2.08 \times 10^{-3} Pa.s$. ***From*** equation 5.20, we can write

$$\Delta p = \frac{8\eta L\bar{v}}{R^2} = \frac{8\,(2.08 \times 10^{-3})(0.1)(0.02)}{(4 \times 10^{-3})^2} = 2.08\,Pa$$

c) The flow resistance is given by:

$$\mathcal{R}_f = \frac{\Delta p}{Q} = \frac{2.08}{1 \times 10^{-6}} = 2.08 \times 10^6\ Pa.s/m^3$$

d) The power dissipation through the flow is

$$\mathbb{P} = \Delta pQ = 2.08 \times 1 \times 10^{-6} = 2.08\,\mu W$$

5.9 Turbulent Flow

We learned that when flow speed becomes high the streamlines of the flow start to intersect each other; and this flow is called turbulent flow. In such flow, the mechanical energy

dissipated is much larger than that in the laminar flow, so that it is often desirable to ensure that the flow does not become turbulent. Poiseuille's law is applicable for laminar flow, so it is necessary to determine whether the flow is laminar or turbulent. There is a dimensionless quantity called Reynolds Number ($\mathbf{N_R}$) used to distinguish the type of the flow. Consider a fluid of density $\boldsymbol{\rho}$ and viscosity coefficient $\boldsymbol{\eta}$ flows with an average velocity $\bar{\mathbf{v}}$ through a tube of radius **R**, hence the Reynolds number is defined by

$$N_R = \frac{2\rho\bar{v}R}{\eta} \qquad (5.26)$$

It is found experimentally that if

$N_R < 2000$ $\quad$ $flow\ is\ laminar$

$N_R > 3000$ $\quad$ $flow\ is\ turbulent$

$2000 < N_R < 3000$ $\quad$ $flow\ is\ unstable$

Example 5.14

The flow rate of blood in a blood vessel of $2\ cm$ in diameter is 1 Liter per minute, Determine whether the flow is laminar or turbulent, if the density of blood is $1060\ kgm^{-3}$ and the coefficient of viscosity of the blood is $2.1 \times 10^{-3}\ Pa.s$

Solution

The flow rate of the blood

$$Q = \pi R^2 \bar{v} = \frac{1\times10^{-3}}{60} = 1.66 \times 10^{-5}\ m^3/s,$$

so that

$$\bar{v} = \frac{Q}{\pi R^2} = \frac{1.66 \times 10^{-5}}{3.14\ (1 \times 10^{-2})^2} = 0.053\ ms^{-1}$$

The Reynolds number

$$N_R = \frac{2\rho\bar{v}R}{\eta} = \frac{2\ (1060)\ 0.053(1 \times 10^{-2})}{2.1 \times 10^{-3}} = 535$$

$N_R < 2000$***, so the flow is laminar.***

Example 5.15

A small human capillary of 100 **μ**m radius has a length of 2 cm, calculate

a) The blood flow resistance across this capillary
b) If the pressure drop across the capillary is 2.3 kPa, what is the flow rate
c) What is the maximum speed of blood through the capillary
d) The power dissipated across the capillary
e) Reynolds's number and then determine the type of flow.

Solution

a) Take the viscosity of blood to be $\eta = 2.1 \times 10^{-3} Pa.s$, the length of the capillary $L = 2 \times 10^{-2}$ and its radius $R = 10^{-4} m$.

$$R_f = \frac{8\,\eta L}{\pi R^4} = \frac{8 \times 2.1 \times 10^{-3} \times 0.02}{3.14 \times 10^{-16}} = 1.07 \times 10^{12}\ Pa.sm^{-3}$$

b) $If \Delta p = 2.3\ kPa,\ so\ Q = \frac{\Delta p}{R_f} = \frac{2.3 \times 10^3}{1.07 \times 10^{12}} = 2.15 \times 10^{-9}\ m^3/s$

c) The flow rate is defined as $Q = \pi R^2 \bar{v} = \frac{1}{2}\pi R^2 v_{max}$, so the maximum speed

$$v_{max} = \frac{2\,Q}{\pi R^2} = \frac{2\,(2.15 \times 10^{-9})}{3.14 \times 10^{-8}} = 0.136\ m/s$$

d) The power dissipation is given as:

$$\mathbb{P} = Q\Delta p = 2.15 \times 10^{-9} \times 2.3 \times 10^3 = 4.95\ \mu W$$

e) Reynolds's number

$$N_R = \frac{2\rho \bar{v} R}{\eta} = \frac{2\,(1060)\,0.136/2 \times (100 \times 10^{-6})}{2.1 \times 10^{-3}} \cong 6.9 < 2000$$

So, the flow is laminar

Summary

The volume of water flowing through the hose per unit time (i.e. the flow rate (Q) at the left must be equal to the flow rate at the right, or in fact anywhere along the hose.

$$Q = \frac{\Delta V}{\Delta t} = constant \quad (5.5)$$

the continuity equation can be rewritten as

$$A_1 \overline{v_1} = A_2 \overline{v_2}$$

Or,

$$r_1^2 \overline{v_1} = r_2^2 \overline{v_2} \quad (5.7)$$

The sum of the pressure, kinetic energy per unit volume, and gravitational potential energy per unit volume has the same value at all points along a streamline for an ideal fluid. This result is summarized in Bernoulli's equation:

$$p + \frac{1}{2} \rho \overline{v^2} + \rho g y = constant \quad (5.11)$$

The pressure in a fluid at rest varies with depth h in the fluid according to the expression

$$P = P_0 + \rho g h \quad (14.4)$$

where P_0 is the pressure at $h = 0$ and r is the density of the fluid, assumed uniform. Pascal's law states that when pressure is applied to an enclosed fluid, the pressure is transmitted undiminished to every point in the fluid and to every point on the walls of the container.

In the standing position, the situation is different, where only the term $\frac{1}{2} \rho \overline{v^2}$ can be ignored and the term $\rho g y$ has a significant effect. Hence the gauge pressures at the brain p_B, at the heart p_H and at the foot p_F are related by:

$$p_F = p_H + \rho g h_H = p_B + \rho g h_B \quad (5.15)$$

Viscosity coefficient, which is represented by the Greek symbol **η** "eta". So we can write

$$F = \eta \frac{A\,\Delta v}{\Delta y} \quad (5.18)$$

Rearrange equation 5.18 for the coefficient of viscosity, that is

$$\eta = \frac{F/A}{\Delta v/\Delta y} \quad (5.19)$$

The average velocity and the flow rate of laminar flow of a fluid through a tube is given by:

$$\bar{v} = \frac{\Delta p\, R^2}{8\,\eta\, L} \quad (5.20)$$

Poiseuille's law: It indicates that high viscosity leads to low flow rate and speed of flow, which is. It also shows that the flow rate is proportional to the 4th power of **R**, which is extremely dependent.

$$Q = A\bar{v} = \pi R^2 \bar{v} = \frac{\pi\,\Delta p\, R^4}{8\,\eta\, L} \quad (5.21)$$

The power dissipated during the flow of a fluid is the rate of energy required to maintain the flow.

$$\mathbb{P} = \Delta p A\, \bar{v} = \Delta p Q \quad (5.22)$$

The viscosity of fluids is defined as the flow resistance

$$\mathcal{R}_f = \frac{\Delta p}{Q} \quad (5.23)$$

Using Poiseuille's equation,

$$\mathcal{R}_f = \frac{\Delta p}{\pi \Delta p R^4/8\eta L} = \frac{8\eta L}{\pi R^4} \quad (5.24)$$

Problems

1. As a woman walks, her entire weight is momentarily placed on one heel of her high-heeled shoes. Calculate the pressure exerted on the floor by the heel if it has an area of $1.50\ cm^2$ and the woman's mass is $55.0\ kg$. Express the pressure in Pa. (In the early days of commercial flight, women were not allowed to wear high-heeled shoes because aircraft floors were too thin to withstand such large pressures.)
2. The pressure exerted by a phonograph needle on a record is surprisingly large. If the equivalent of $1.00\ g$ is supported by a needle, the tip of which is a circle $0.200\ mm$ in radius, what pressure is exerted on the record in N/m^2?
3. What depth of mercury creates a pressure of 1.00 atm?
4. The greatest ocean depths on the Earth are found in the Marianas Trench near the Philippines. Calculate the pressure due to the ocean at the bottom of this trench, given its depth is 11.0 km and assuming the density of seawater is constant all the way down.
5. The aqueous humor in a person's eye is exerting a force of 0.300 N on the 1.10-cm^2 area of the cornea. (a) What pressure is this in mm Hg? (b) Is this value within the normal range for pressures in the eye?
6. How much force is exerted on one side of an 8.50 cm by 11.0 cm sheet of paper by the atmosphere? How can the paper withstand such a force?
7. The left side of the heart creates a pressure of 120 mm Hg by exerting a force directly on the blood over an effective area of $15.0\ cm^2$. What force does it exert to accomplish this?
8. (a) Convert normal blood pressure readings of 120 over 80 mm Hg to newtons per meter squared using the relationship for pressure due to the weight of a fluid $(P = h\rho g)$ rather than a conversion factor. (b) Discuss why blood pressures for an infant could be smaller than those for an adult. Specifically, consider the smaller height to which blood must be pumped.

9. How tall must a water-filled manometer be to measure blood pressures as high as 300 mm Hg?
10. During forced exhalation, such as when blowing up a balloon, the diaphragm and chest muscles create a pressure of $60.0\ mm\ Hg$ between the lungs and chest wall. What force in newtons does this pressure create on the $600\ cm^2$surface area of the diaphragm?
11. You can chew through very tough objects with your incisors because they exert a large force on the small area of a pointed tooth. What pressure in pascals can you create by exerting a force of $500N$ with your tooth on an area of $1.00\ mm^2$?
12. If the pressure in the esophagus is $-2.00mmHg$ while that in the stomach is$+20.00mmHg$, to what height could stomach fluid rise in the esophagus, assuming a density of $1.10\ g/mL$? (This movement will not occur if the muscle closing the lower end of the esophagus is working properly.)
13. A large storage tank open to the atmosphere at the top and filled with water develops a small hole in the side at a point $16\ m$ below the water level. If the flow rate from the hole is $2.5 \times 10^3\ m^3/min$, determine a) the speed at which water leaves the hole and b) the diameter of the hole
14. How high can water rise in the pipe of a building if the gauge pressure at the ground floor is 2×10^5 Pa?
15. A large artery is cannulated and a saline solution of density $1.3\ gcm^{-3}$ is used as the manometer fluid. What is the gauge pressure of the blood if the high difference in the manometer tube is $0.67\ m$?
16. A horizontal tube of radius $1\ cm$ is joined to a second horizontal tube of radius $0.5\ cm$. There is a pressure difference of $6660\ Pa$ between the two tubes. a) Which tube has the higher water pressure b) What volume of water flows through the tubes per second?
17. A garden hose has an inner radius of 1 cm. The hose has a nozzle with an inside radius of 1.5 mm. Water flows through the hose with an average speed of 25 cm/s. a) what is the speed of water as it emerges from the nozzle? b) How long does it take to spray $4\ liter$ of water from the hose?

18. Water flows through a pipe of radius 30 cm at the rate $0.2\ m^3\ /\ s$. The pressure in the pipe is atmospheric. The pipe slants downhill and feeds into a second pipe of radius $15\ cm$ which is located at 60 cm lower. What is the gauge pressure at the lower part?
19. Calculate the flow rate in an aorta with a cross sectional area of $2\ cm^2$ if the flow speed is $40\ cm\ /s$.
20. A needle of radius 0.3 mm and length 3 cm is used to give a patient a blood transfusion. Assume the pressure differential across the needle is achieved by elevating the blood 1 m above the patient´s arm. a) What is the rate of flow of blood through the needle? b) At this rate of flow, how long it take to inject 500 cm^3 of blood into patient?
21. A jet pilot pulls his plane out of dive in such a way that his upward acceleration is 3 g's. What you predict as the blood pressure in the brain?
22. If an elevator accelerates upward at $10\ ms^{-2}$, what is the average blood pressure in the brain? What is the average blood pressure in the feet? If the elevator accelerates downward with the same acceleration, what is the average blood pressure in the brain and feet? take $g\ = 10\ ms^{-2}$.
23. The pulmonary artery, which connects the heart to the lungs, has an inner radius of 2.6 mm and is 8.4 cm long. If the pressure drop between the heart and lungs is 400 Pa, what is the average speed of blood in the pulmonary artery?
24. The aorta in humans has a diameter of about 2 cm and at certain times, the blood speed through it is about 55 cm /s. What is the type of flow? Find also the flow resistance over 30 cm long segment and the power for dissipating the blood through this segment.
25. A standing $1.8\ m$ tall person in an elevator accelerates downward with $1.5\ g$. The height difference between the brain and heart is $40\ cm$. Find the pressure at the brain and foot if the pressure at the heart is $13.3\ kPa$. If this person is suddenly bending so his brain is $30\ cm$ below his heart, calculate the change of the pressure at the brain.
26. An artery has an inner radius of 2 mm. If the flow is laminar and the average flow velocity is $0.03\ ms^{-1}$, What is the (a) maximum velocity (b) the flow rate, (c) and the pressure drop in 0.05 m, if the artery is horizontal?

27. The pressure drop along a length of a horizontal artery is 100 Pa. The radius of the artery is 10 cm, and the flow is laminar. (a) what is the net force on blood in this portion of the artery? (b) If the average speed of blood is $0.015\ ms^{-1}$, find the power expended in maintaining the flow?

28. The radius of an artery is increased by a factor of 1.5 (a) The pressure drop remains the same, what happens to the flow rate? (b) If the flow rate stays the same, what happens to the pressure drop? (Assume laminar flow).

29. The average velocity of water at 20C in a tube of radius 0.1 m is 0.2 ms^{-1}. (a) Is the flow laminar or turbulent? (b) what is the flow rate?

30. A small artery has a length of 0.11 cm and a radius of 25 μm. (a) calculate its resistance.(b) If the pressure drop across the artery is 1.3 kPa, what is the flow rate?

31. The heart of a resting adult pumps blood at a rate of $5.00\ L/min.$ (a) Convert this to . cm^3/s(b) What is this rate in m^3/s ?

32. Blood is pumped from the heart at a rate of 5.0 L/min into the aorta (of radius 1.0 cm). Determine the speed of blood through the aorta.

33. Blood is flowing through an artery of radius 2 mm at a rate of 40 cm/s. Determine the flow rate and the volume that passes through the artery in a period of 30 s.

34. A major artery with a cross-sectional area of $1.00\,\text{cm}^2$ branches into 18 smaller arteries, each with an average cross-sectional area of $0.400\,\text{cm}^2$. By what factor is the average velocity of the blood reduced when it passes into these branches?

35. (a) As blood passes through the capillary bed in an organ, the capillaries join to form venules (small veins). If the blood speed increases by a factor of 4.00 and the total cross-sectional area of the venules is $10.0\,\text{cm}^2$, what is the total cross-sectional area of the capillaries feeding these venules? (b) How many capillaries are involved if their average diameter is $10.0\,\mu\text{m}$?

36. The human circulation system has approximately 1×10^9 capillary vessels. Each vessel has a diameter of about $8\,\mu\text{m}$. Assuming cardiac output is 5 L/min, determine the average velocity of blood flow through each capillary vessel.

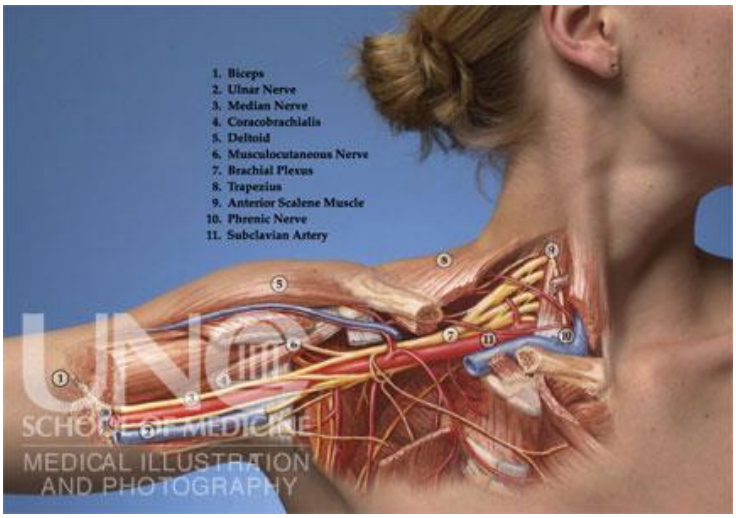
37. (a) What is the pressure drop due to the Bernoulli effect as water goes into a 3.00-cm-diameter nozzle from a 9.00-cm-diameter fire hose while carrying a flow of 40.0 L/s? (b) To what maximum height above the nozzle can this water rise? (The actual height will be significantly smaller due to air resistance.)

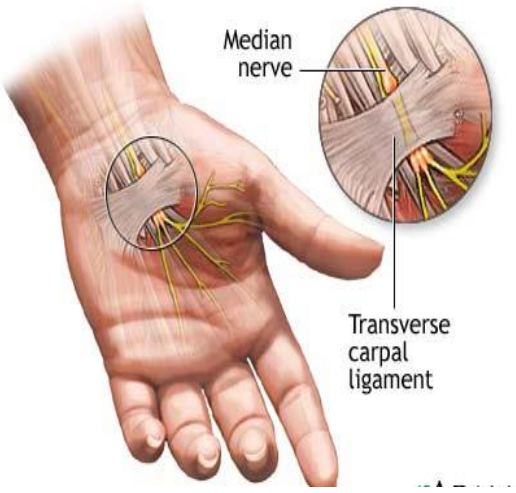
38. A glucose solution being administered with an IV has a flow rate of $4.00\ \text{cm}^3/\text{min}$. What will the new flow rate be if the glucose is replaced by whole blood having the same density but a viscosity 2.50 times that of the glucose? All other factors remain constant.

39. A small artery has a length of $1.1 \times 10^{-3} m$ and a radius of $2.5 \times 10^{-5} m$. If the pressure drop across the artery is $1.3\ kPa$, what is the flow rate through the artery? (Assume that the temperature is $37°\text{C}$.)

40. Calculate the Reynolds numbers for the flow of water through (a) a nozzle with a radius of 0.250 cm and (b) a garden hose with a radius of 0.900 cm, when the nozzle is attached to the hose. The flow rate through hose and nozzle is 0.500 L/s. Can the flow in either possibly be laminar?

Chapter 6
Bioelectricity
"Signals in the Body"





Bioelectricity refers to electrical potentials and currents occurring within or produced by living organisms. It results from the conversion of chemical energy into electrical energy. Bioelectric potentials are generated by a number of different biological processes, and are used by cells to govern metabolism, to conduct impulses along nerve fibers, and to regulate muscular contraction. In most organisms bioelectric potentials vary in strength from one to several hundred millivolts. The most important difference between bioelectric currents in living organisms and the type of electric current used to produce light, heat, or power is that a bioelectrical current is a flow of ions (atoms or molecules carrying an electric charge), while standard electricity is a movement of electrons. Moreover, Bioelectricity deals with the ability of tissue to generate electricity, as for instance done by the heart (electrocardiography (ECG)). This electricity is endogenic, that is generated by the tissue itself. Bioimpedance and bioelectricity methods use electrodes with galvanic coupling to tissue. In general, bioimpedance and bioelectricity is about biomaterials in a broad sense: materials that are living, have lived or are potential building blocks for living tissue. The tissue of interest may be plant, fruit, egg, fish, animal or the human body. But it may also be dead biological material such as hair or nail; or excised material such as beef. The basic building block is the living cell, and a prerequisite for its life is that it is surrounded by an electrolyte solution.

6.1 Electrostatics

In this modern era, most people know the atomic structure: the atom consists of a nucleus where the positively charged particles called protons and neutral particles called neutron; revolve around the nucleus negatively charged particles called electrons. In equilibrium, these atoms are neutral (have no excess charge). So, the negative charges or positive charges come in discrete quantities, each quantity has the charge of electron/proton which equals to $1.6 \times 10^{-19} C$, where C is the SI unit of charge; $1\ C$ of charge is approximately equal to the charge of 6.24×10^{18} electrons or protons.

Charles Coulomb (1736–1806) measured the magnitudes of the electric forces between charged objects. He confirmed that the electric force between two small charged fixed spheres is

- is inversely proportional to the square of the separation r between the particles and directed along the line joining them;
- is proportional to the product of the charges q_1 and $q_2$on the two particles;
- is attractive if the charges are of opposite sign and repulsive if the charges have the same sign.

Coulomb expressed his law as

$$F_e = k_e \frac{|q_1||q_2|}{r^2} \tag{6.1}$$

Where F_e is the magnitude of Coulomb's electric force, and k_e is a constant called the Coulomb constant, which has a value in the SI units of $k_e \cong 9 \times 10^9 N \cdot m^2/C^2$

This constant is also written in the form

$$k_e = \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \frac{N \cdot m^2}{C^2}$$

where the constant ε_0 (lowercase Greek epsilon) is known as the permittivity of free space and has the value $\varepsilon_0 = 8.8542 \times 10^{-11} C^2/N \cdot m^2$.

Electric field

From equation 6.1, we can see that the force between the charged particles is instantaneous, which means if we move one particle closer of farther the electric force between them has to change immediately. Michael Faraday introduced a new concept to explain what happens. He introduced the electric field into science. The function of a field is to transmit a force. Faraday developed an illustrative idea called lines of force, as shown in Figure 6.1.

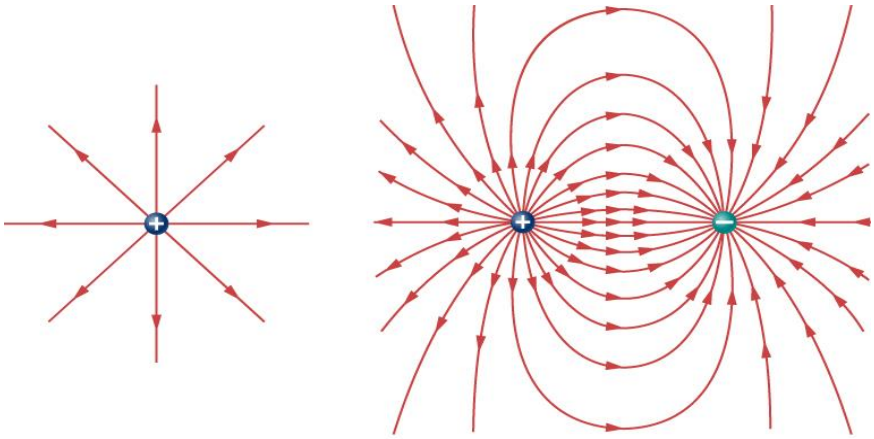


Figure 6.1: The lines of the electric field around two charge configurations. In (a), the field lines emanate from the single positive charge. In (b), the field lines emanate from the positive charge and terminate on the negative charge.

The electric filed lines represented by arrow which emanates uniformly from positive charge, figure 6.1a, and terminate at the negative charge, figure 6.1b. The number of lines describes how strong the filed is. In that sense, the farther the region from the charge the less number of lines you are going to encounter (weak force effect), but the close you are to the charge you are going to encounter larger number of electric field lines (strong force effect). Thus, when an electric charge, say the test charge q_0, is placed in the field of another charge Q , then the q_0 would "feels" and electric force that is proportional to the electric field it enters, which is given by

$$E = F_e/q_0 = k_e \frac{|q|}{r^2} \tag{6.2}$$

Electrical Potential and Potential Differences

Until now, we are dealing with fixed charges. If we want to move a positive charge from negative charge vicinity, we have to do some work against the attractive force between them, and vice versa. This work stored as energy added to the charge system stored in the relative position. The potential energy per unit charge U/q_0 is independent of the value of q_0 and has a unique value at every point in an electric field. This quantity U/q_0 is called the electric potential (or simply the potential) V. Thus, the electric potential at any point in an electric field is

$$V = U/q_0 \tag{6.3}$$

Electrical potential difference is measured in terms of the energy that must be added to a unit of charge to move it a given distance. In the SI system of units, the unit of electrical potential difference is the joule per coulomb "volt". Thus, the electrical potential difference is called "voltage." For example, a nine volt battery has a potential difference of nine volts between its terminals.

Now suppose that a test charge q_0 moves from A to B, figure 6.2. We the potential energy is given by

$$U = -q_0 \Delta V = -q_0 E d \tag{6.4}$$

or

$$E = |V_B - V_A|/d \tag{6.5}$$

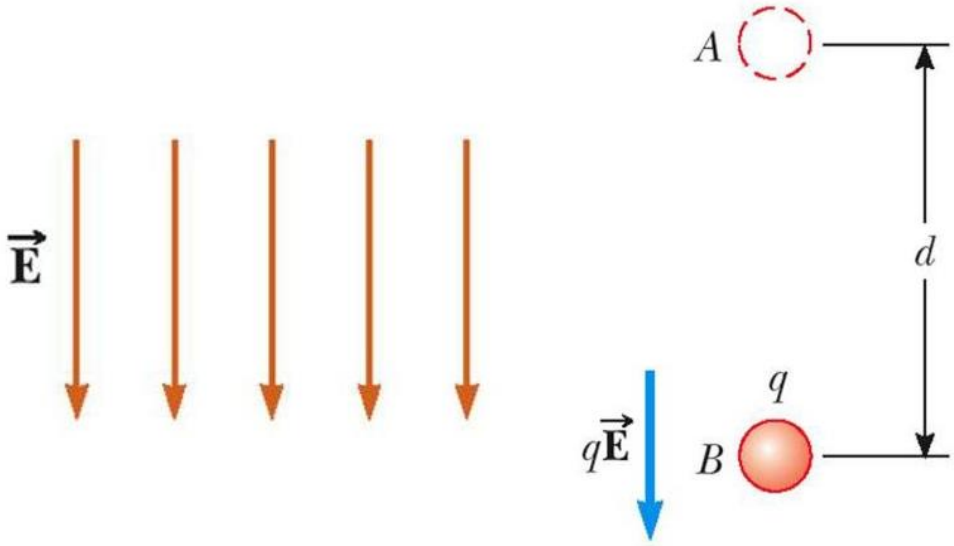


Figure 6.2: When the electric field E is directed downward, point B is at a lower electric potential than point A. A positive test charge that moves from point A to point B loses electric potential energy.

We see that if q_0 is positive, and then ΔU is negative. We conclude that a positive charge loses electric potential energy when it moves in the direction of the electric field. This means that an electric field does work on a positive charge when the charge moves in the direction of the electric field. If q_0 is negative, then ΔU is positive and the situation is reversed: A negative charge gains electric potential energy when it moves in the direction of the electric field.

6.2 Electric Current

Now consider a system of electric charges in motion. Figure 6.3, shows electric charges random motions when there is no an applied external electric potential (the upper figure), and the charge motion when there is an applied electric potential.

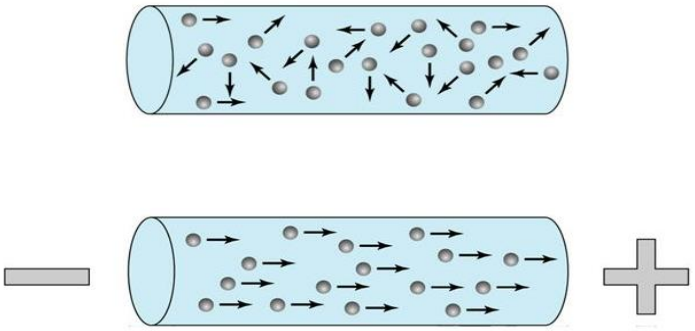


Figure 6.3: shows the effect of the electric potential on the motion of the charges. The upper figure with no electric potential, and the lower with electric potential.

Whenever there is a net flow of charge through some region, a current is said to exist. To define current more precisely, suppose that the charges are moving perpendicular to a surface of area A, as shown in Figure 6.4. (This area could be the cross-sectional area of a wire, for example.) The current is the rate at which charge flows through this surface. If ΔQ is the amount of charge that passes through this area in a time interval Δt, the average current I_{av} is equal to the charge that passes through A per unit time:

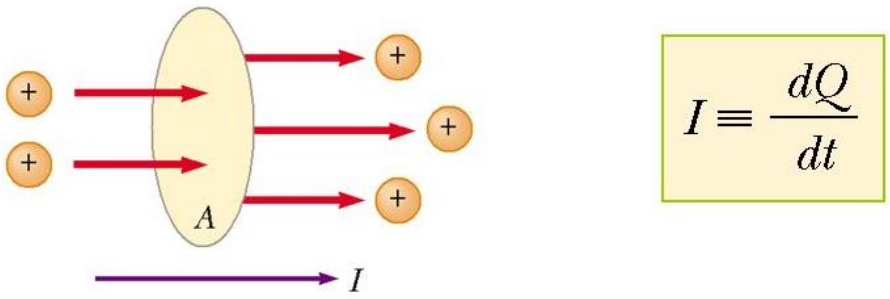


Figure 6.4: Charges in motion through an area A. The time rate at which charge flows through the area is defined as the current *I*. The direction of the current is the direction in which positive charges flow when free to do so.

$$I_{av} = \frac{\Delta Q}{\Delta t} \tag{6.6}$$

If the rate at which charge flows varies in time, then the current varies in time; we define the instantaneous current I as the differential limit of average current:

$$I = \frac{dQ}{dt} \tag{6.7}$$

The SI unit of current is the ampere (A):

$$1A = \frac{1\,C}{1\,sec} \tag{6.8}$$

That is, $1\,A$ of current is equivalent to $1\,C$ of charge passing through the surface area in $1\,s$. The charges passing through the surface in Figure 6.3 can be positive or negative, or both. It is conventional to assign to the current the same direction as the flow of positive charge. In electrical conductors, such as copper or aluminum, the current is due to the motion of negatively charged electrons. Therefore, when we speak of current in an ordinary conductor, the direction of the current is opposite the direction of flow of electrons. The current density is defined as the current per unit area.

$$J = I/A$$

Example 6.1

A silver wire is $1\ mm$ in diameter transfers charge of $90\ C$ in $1\ hr$ and $15\ min$. Silver contains $5.8 \times 10^{28} e/m^3$. What is the current in the wire? What is the drift velocity of the electrons in the wire?

Solution

$$I = \frac{Q}{t} = \frac{90C}{(60+15)\times 60\,\text{sec}} = \frac{90}{4500} = 0.02A = 20\ mA$$

$$a = \pi r^2 = \pi\left(\frac{10^{-3}}{2}\right)^2 = 7.854\times10^{-7} m^2, n = 5.8\times10^{28} e/m^3, e = 1.602\times10^{-19} C, v = ?$$

$$I = neva = 0.02\ A$$

$$v = \frac{I}{nea} = \frac{0.02}{5.8\times10^{28}\times1.602\times10^{-19}\times7.854\times10^{-7}} = 2.741\times10^{-6}\ m/s$$

6.3 Resistance and Ohm's Law

In order to produce current in a wire, an electric field must exits along the wire. This can be achieved by maintaining a potential difference between the two ends of the wire. Usually, the magnitude of the current flow in a conductor is proportional to the voltage difference between the two ends of the wire. Under this condition we characterize the electrical behavior of the body, wire, by its resistance R, which is defined by the following formulae,

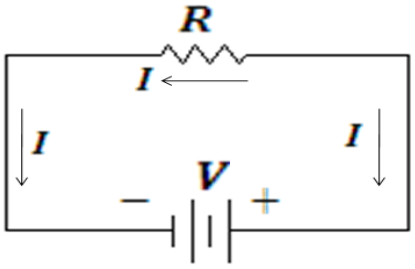


Figure 6.5: Simple resistive circuit

$$V = IR \quad Volts \tag{6.9}$$

where V is the voltage difference in volts, between the two ends of the wire, I is current in amperes and R is a parameter called resistance in ohms. Equation 6.9 is the well known as Ohm's Law. The resistance of a conducting body can have any value between wide limits depending on the material type and geometry of the body. The resistance of a conducting body usually varies with temperature. Devices used in circuits to provide fixed resistance of known values are called resistors. Electrical energy is converted into heat when current flow through a resistor. Thus resistor is used as heaters elements. They are also used to control current and voltage in circuit. If the relation between the volt and current linear relation, for certain resistance, we call this resistance an Ohmic resistance. Resistors are usually represented in the circuit diagram as ⁓⁓⁓ . The resistance of materials varies with temperature: the resistance of metals increases with temperature increase but the semiconductors resistance decrease with temperature increase (why!). Over a moderate range the resistance of the metal increase approximately linear! We expect the increase in resistance would be proportional to the resistance value and the temperature increase, and the constant of proportionality is called the temperature coefficient of resistance α, that is

$$R_T = R_0\left(1+\alpha\left(T - T_0\right)\right) \tag{6.10}$$

where α is the temperature coefficient of resistance's material. We know the relation between the resistance dimension and its resistivity with the resistance is

$$R = \rho\frac{l}{A} \tag{6.11}$$

where l is the length of the wire, A is the cross sectional area, and ρ is the resistivity. So equation 6.10, can be rewritten as,

$$\rho_T = \rho_0\left(1+\alpha\left(T - T_0\right)\right) \tag{6.12}$$

Example 6.2

Calculate the resistance of an aluminum cylinder that is $10.0\ cm$ long and has a cross-sectional area of $2.0 \times 10^{-4} m^2$. Repeat the calculation for a cylinder of the same dimensions and made of glass having a resistivity of $3.0 \times 10^{10} \Omega \cdot m$. Aluminum Resistivity is $\rho_{Al} = 2.82 \times 10^{-8} \Omega \cdot m$

Solution

$$R_{Al} = \rho \frac{l}{A} = 2.82 \times 10^{-4} \Omega \cdot m \ \frac{0.1m}{2.0 \times 10^{-4} m^2} = 1.41 \times 10^{-4} \Omega$$

$$R_{glass} = \rho \frac{l}{A} = 3.0 \times 10^{10} \Omega \cdot m \ \frac{0.1m}{2.0 \times 10^{-4} m^2} = 1.5 \times 10^{13} \Omega$$

Notice the huge difference between Aluminum and glass in resistance. Aluminum is used as a good conductor while glass is used as a good insulator.

Example 6.3

Calculate the resistance per unit length of a 22-gauge Nichrome wire, which has a radius of $0.321\ mm$. If a potential difference of $10\ V$ is maintained across a $1.0\ m$ length of the Nichrome wire, what is the current in the wire?

Solution

The cross-sectional area of this wire is

$$A = \pi r^2 = \pi (0.321 \times 10^{-3} m)^2 = 3.24 \times 10^{-7} m^2$$

The resistivity of Nichrome is $\rho = 1.5 \times 10^{-6} \Omega m$

$$\frac{R}{l} = \frac{\rho}{A} = \frac{1.5 \times 10^{-6} \Omega m}{3.24 \times 10^{-7} m^2} = 4.6 \Omega / m$$

Because a 1.0-m length of this wire has a resistance of $4.6\ \Omega / m$

$$I = \frac{\Delta V}{R} = \frac{10V}{4.6\Omega} = 2.2A$$

Electric Power

Imagine electric circuit like the one in figure 6.4, to carry charge ΔQ around the circuit, work has to be done by the emf, battery. The emf keeps the charge ΔQ moving around the circuit all the time, as long as the battery is on. The power is defined as the work done per unit time, thus the power done by the emf is

$$\mathcal{P} = \frac{\Delta U}{\Delta t} = \frac{\Delta Q}{\Delta t}\Delta V = I\Delta V \tag{6.13}$$

Using Ohm's law Equation 6.9 can be written in different form, which are

$$\mathcal{P} = I\Delta V = I^2R = \frac{V^2}{R} \tag{6.14}$$

6.4 Electromotive Force (emf)

The emf source is a device that cause charges to move around a circuit because of the potential difference at its terminals. Thus, the emf $\boldsymbol{\mathcal{E}}$ describes the work done per unit charge, and hence the SI unit of emf is the volt. Besides that, the potential difference across the emf is called the terminal voltage, in case there is no load connected across its terminals. Generally speaking, there is no ideal emf source; most of these sources have internal resistance which affects its performance. When you operate your radio for a long time and if you touched the batteries, you'll find them hot! This is simply because apart of the battery power is drained internally (because of the internal resistance) and appeared as a heat. In figure 6.6, the emf $\boldsymbol{\mathcal{E}}$ is distributed on the internal resistance r and the load resistance R, that is

$$\mathcal{E} = V_R + V_r \tag{6.15}$$

For ohm's law we know that

$$V_R = IR \quad and \quad V_r = Ir \tag{6.16}$$

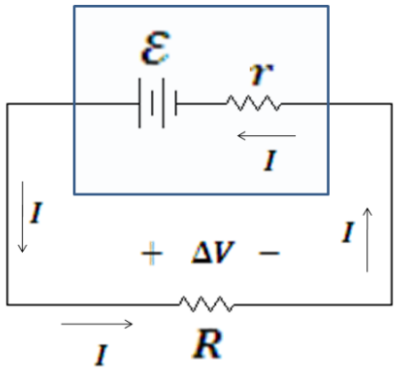


Figure 6.6: A circuit consisting of a resistor connected to the terminals of a battery.

So, the potential difference across the resistance R or at the external terminals of the emf is given by

$$\Delta V = \mathcal{E} - Ir \tag{6.17}$$

Substitute equation 6.16 into equation 6.15 we have,

$$\mathcal{E} = IR + Ir \tag{6.18}$$

Solving for the current gives

$$I = \frac{\mathcal{E}}{R+r} \tag{6.19}$$

This equation shows that the current in this simple circuit depends on both the load resistance R external to the battery and the internal resistance r. If R is much greater than r, as it is in many real-world circuits, we can neglect r.

Example 6.4

A battery has an emf of $12.0\,V$ and an internal resistance of $0.05\,\Omega$. Its terminals are connected to a load resistance of $3.00\,\Omega$. (a) Find the current in the circuit and the terminal voltage of the battery. (b) Calculate the power delivered to the load resistor, the power delivered to the internal resistance of the battery, and the power delivered by the battery.

Solution

The circuit current is

$$I = \frac{\mathcal{E}}{r+R} = \frac{12.0V}{(0.05+3)\Omega} = 3.93A$$

$$\Delta V = \mathcal{E} - Ir = 12 - 3.94 \times 0.05 = 11.8Volt$$

This voltage should appear across the 3Ω resistance, to make sure apply Ohm's law, that is

$$V_R = IR = 3.93 \times 3 = 11.8\, Volt$$

The power

The power delivered to the load resistance

$$\mathcal{P} = IV = 3.94 \times 11.8 = 46.5\, W$$

The power delivered to the internal resistance

$$\mathcal{P} = I^2 r = (3.94)^2 \times 0.05 = 0.78\, W$$

The power delivered by the battery

$$\mathcal{P} = IV = 3.94 \times 12 = 47.28\, W$$

Notice that the power delivered by the battery is equal to the power delivered to the load resistance and the internal resistance.

Example 6.5

A battery has an emf of 15.0 V. The terminal voltage of the battery is 11.6 V when it is delivering 20.0 W of power to an external load resistor R. (a) what is the value of R? (b) What is the internal resistance of the battery?

Solution

To find the load resistance, just apply the power law

$$P = \frac{V^2}{R} \rightarrow R = \frac{V^2}{P} = \frac{(11.6)^2}{20} \Rightarrow R = 6.73\Omega$$

Using ohm's law and equation 6.19 we can find the internal resistance

The circuit is

$$I = \frac{V}{R} = \frac{11.6}{6.73} = 1.72A \implies I = \frac{\mathcal{E}}{R + r} \Rightarrow 1.72 = \frac{15}{6.73 + r} \Rightarrow r = 1.97\Omega$$

6.5 Simple Resistive Circuits

Resistances in Series

If we have more than one resistance connected end to end in electric circuit, we call this wiring series-connected resistor. The equivalent resistance of the circuit can be found by noticing that for a series combination of resistors, the currents in the two resistors are the same because any

charge that passes through R_1 must also pass through R_2, as in figure 6.7. As you can see from the circuit,

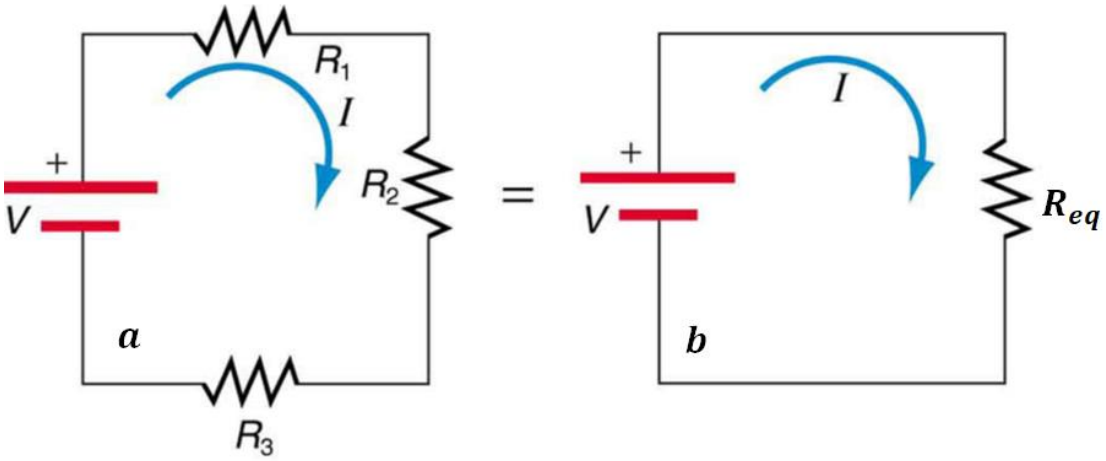


Figure 6.7: a. Resistors in series, **b.** Series equivalent circuit

the volt of the battery distributed on the two resistors $R_1$and R_2. Let us assume the voltage across R_1 is V_1 and the voltage across R_2 is V_2.

$$V = V_1 + V_2 \tag{6.20}$$

Since the voltage distributed on R_1 and R_2 thus one current, I, can flow in the circuit. Apply Ohm's law equation 6.20 becomes,

$$IR_{eq} = IR_1 + IR_2 \tag{6.21}$$

where R_{eq} is the equivalent resistance of the circuit. Divide both sides of the equation 6.21 by i we have,

$$R_{eq} = R_1 + R_2 \tag{6.22}$$

Equation 6.22 implies that the equivalent resistance of resistors connected in series is the sum of these resistors.

Voltage Divider Circuit

Sometimes, electronic circuits need more than one voltage level from a single voltage supply. One way of achieving this goal is the voltage divider circuit, which is depicted in figure 6.8.

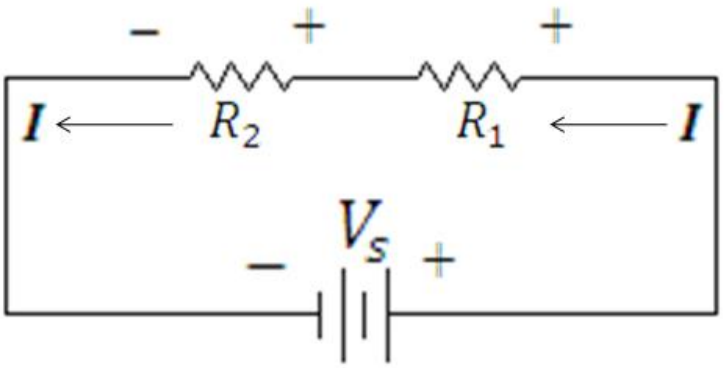


Figure 6.8: voltage divider circuit

We have noticed that the current in both resistances carry the same current. Thus,

$$V_s = IR_1 + IR_2$$

or

$$I = \frac{V_S}{R_1+R_2} \tag{6.23}$$

Now we can use Ohm's law to calculate V_1 across R_1 and V_2 across R_2.

$$V_1 = IR_1 = V_s \frac{R_1}{R_1+R_2} \tag{6.24}$$

$$V_2 = IR_2 = V_s \frac{R_2}{R_1+R_2} \tag{6.25}$$

Equations 6.24 and 6.25 show that the V_1 and V_2 are fractions of V_s.

Resistances in Parallel

When two elements or more are connected at a single pair node pair, they said to be in parallel. Parallel-connected circuit elements have the same voltage across their terminals. Figure 6.9 illustrates resistors connected in parallel. When the current I_s reaches point a in Figure 6.9a, called a junction, it splits into three parts, with I_1 going through R_1, I_2 going through R_2 and I_3 going through R_3. A junction is any point in a circuit where a current can split. This split results in less current in each individual resistor than the current leaving the battery. Because charge must be conserved, the current I_s that enters point a must equal the total current leaving at the point b:

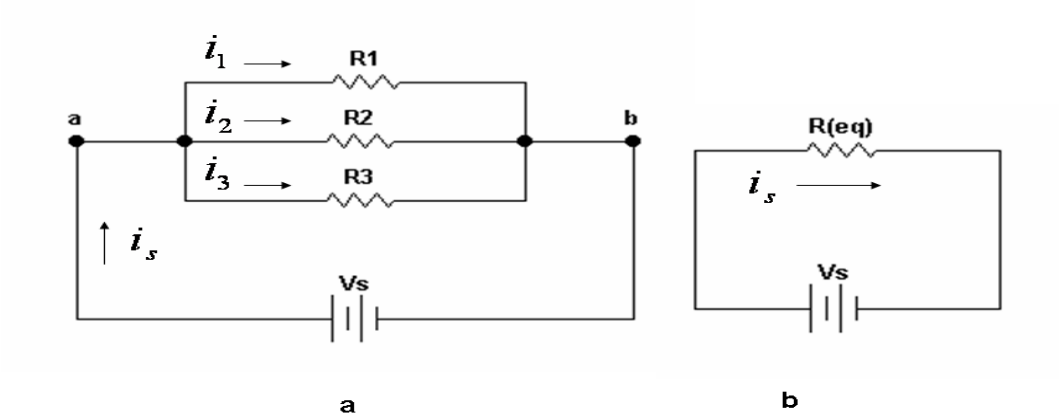


Figure 6.9: a. Three resistors connected in parallel; **b.** The equivalent resistance

$$I_s = I_1 + I_2 + I_3 \tag{6.26}$$

The parallel connection of the resistors means that the voltage across each resistor must be the same. Hence, form Ohm's law,

$$I_1R_1 = I_2R_2 = I_3R_3 = V_s \tag{6.27}$$

Therefore,

$$I_1 = \frac{V_s}{R_1}, I_2 = \frac{V_s}{R_2}, \text{ and } I_3 = \frac{V_s}{R_3} \tag{6.28}$$

Substituting in equation 6.28 into equation 6.26, we have

$$I_s = V_s\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) \tag{6.29}$$

from which we can get

$$\frac{I_s}{V_s} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$$

Which can be rearranged for the equivalent resistance, that is

$$\frac{1}{R_{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) \tag{6.30}$$

If we have just two resistors the equivalent resistance is given by

$$\frac{1}{R_{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) \rightarrow R_{eq} = \frac{R_1R_2}{R_1+R_2} \tag{6.31}$$

Current Divider law

For Simplicity the current divider circuit is shown in figure 6.10 consist of two resistors connected in parallel across a voltage source. The current divider is designed to divide the total circuit current I_s between the R_1 and R_2. The voltages across the resistance are,

$$I_1R_1 = I_2R_2 = V_s \text{ (The connection in parallel)} \tag{6.32}$$

and the total circuit current and the power supply voltage, V_s, is

$$I_s = \frac{V_s}{R_{eq}} \text{ or } V_s = I_sR_{eq} \tag{6.33}$$

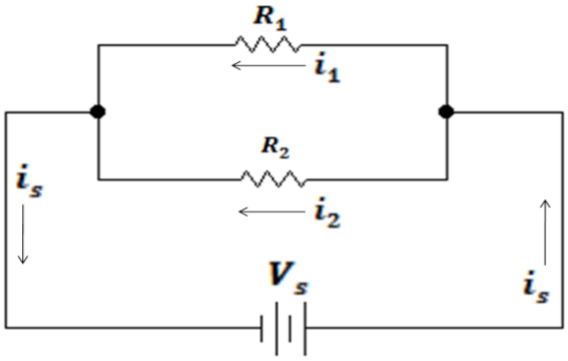


Figure 6.10: Two resistances current divider circuit

Use equation 6.31 for R_{eq} in equation 6.36, we have

$$V_s = I_s\frac{R_1R_2}{R_1+R_2} \tag{6.34}$$

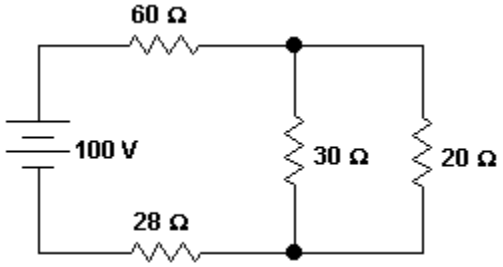
Use V_s in equation 6.35 in equation 6.33 we find

$$I_1 = I_s\frac{R_2}{R_1+R_2}, \text{ and } I_2 = I_s\frac{R_1}{R_1+R_2} \tag{6.35}$$

Equation 6.35 shows that the current divides between two resistors in parallel such the current in one resistor equals the current entering the parallel pair multiplied by the other resistance and divided by the sum of the resistors.

Example 6.6

For the following circuit find the equivalent resistance, total current, and the current through 20Ω resistor.



Solution

$20\ //\ 30$ means that the 20Ω resistor is in parallel with the 30Ω resistor.

$$20//30 = \left(\frac{1}{30}+\frac{1}{20}\right)^{-1} = \frac{20\times 30}{50} = 12\Omega,$$

$$R_{eq} = 60+12+28 = 100\Omega,$$

$$I_{total} = \frac{V}{R} = \frac{100V}{100\Omega} = 1A,$$

$$I_{20\Omega} = 1\times\frac{30}{20+30} = 0.6A$$

6.6 The Capacitor

Consider two-conductor carrying charges of equal magnitude but of opposite sign, as shown in Figure 6.11. Such a combination of two conductors is called a capacitor. The conductors are called plates. A potential difference ΔV exists between the conductors due to the presence of the charges. Because the unit of potential difference is the volt, a potential difference is often called a voltage. We shall use this term to describe the potential difference across a circuit element or between two points in space.

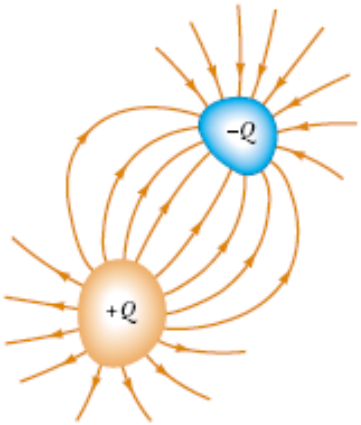


Figure 6.11: A capacitor consists of two conductors carrying charges of equal magnitude but opposite sign.

What determines how much charge is on the plates of a capacitor for a given voltage? In other words, what is the capacity of the device for storing charge at a particular value of ΔV? Experiments show that the quantity of charge Q on a capacitor is linearly proportional to the potential difference between the conductors; that is, $Q \propto \Delta V$. The proportionality constant depends on the shape and separation of the conductors. We can write this relationship as $Q = C\Delta V$ if we define capacitance as follows:

The capacitance C *of a capacitor is the ratio of the magnitude of the charge on either conductor to the magnitude of the potential difference between them*:

$$C = \frac{Q}{\Delta V} \tag{6.36}$$

Note that by definition capacitance is always a positive quantity. Furthermore, the potential difference ΔV is always expressed in Equation 6.36 as a positive quantity. Because the potential difference increases linearly with the stored charge, the ratio $Q/\Delta V$ is constant for a given capacitor. Therefore, capacitance is a measure of a capacitor's ability to store charge and electric potential energy.

From Equation 6.36, we see that capacitance has SI units of coulombs per volt. The SI unit of capacitance is the farad (F), which was named in honor of Michael Faraday:

$$F = 1C/V$$

The farad is a very large unit of capacitance. In practice, typical devices have capacitances ranging from microfarads ($10^{-6}F \equiv \mu F$) to picofarads $10^{-12}(F)$. For practical purposes,

capacitors often are labeled "mF" for microfarads and "mmF" for *micro-microfarads* or, equivalently, "pF" for pico-farads, see table 1.1 for details on prefixes.

Parallel plate capacitor

Two parallel metallic plates of equal area A are separated by a distance d, as shown in Figure 6.12. One plate carries a charge $+Q$, and the other carries a charge $-Q$.

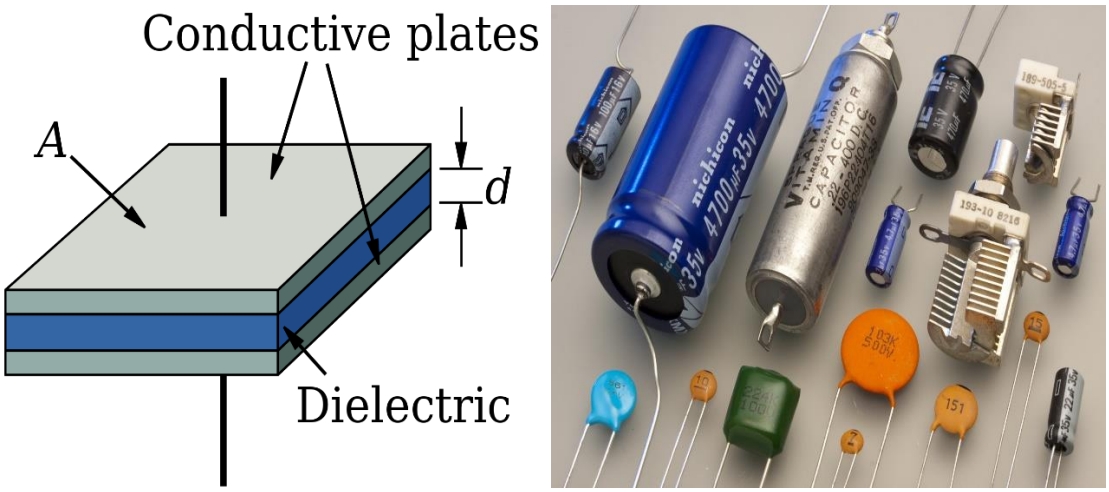


Figure 6.12: schematic of parallel plate capacitor and types of real life capacitors

Experimentally we found that the capacitance of a parallel-plate capacitor is proportional to the area of its plates and inversely proportional to the plate separation, just as we expect from our conceptual argument. So,

$$C = \frac{\kappa \varepsilon_0 A}{d} \tag{6.37}$$

Where κ is the dielectric constant for the filling material between the plates, its value equals to one for air.

Example 6.7

How much charge is on each plate of a $4.00\mu F$ capacitor when it is connected to a $12.0\,V$ battery? (b) If this same capacitor is connected to a $1.50 - V$ battery, what charge is stored?

Solution

$$Q = C\Delta V = 4.0 \times 10^{-6} \times 12.0 = 48 \times 10^{-6} = 48\mu C$$

$$Q = C\Delta V = 4.0 \times 10^{-6} \times 1.5 = 6.0 \times 10^{-6} = 6\mu C$$

Example 6.8

Two conductors having net charges of +10.0 μC and -10.0 μC have a potential difference of $10.0\ V$. Determine (a) the capacitance of the system and (b) the potential difference between the two conductors if the charges on each are increased to +100 μC and -100 μC.

Solution

$$C = \frac{Q}{V} = \frac{10\mu C}{10 Volt} = 1.0\ \mu F \Longrightarrow V = \frac{Q}{C} = \frac{100\mu C}{1.0\mu F} = 100.0\ Volt$$

6.7 *RC* circuit

So far we have been analyzing steady-state circuits, in which the current is constant. In circuits containing capacitors, the current may vary in time. A circuit containing a series combination of a resistor and a capacitor is called an RC circuit.

Charging a Capacitor

Let us assume that the capacitor in Figure 6.13 is initially uncharged. There is no current while switch S is open (Fig. 6.13b). If the switch is closed at however, charge begins to flow, setting up a current in the circuit, and the capacitor begins to charge. Note that during charging, charges do not jump across the capacitor plates because the gap between the plates represents an open circuit. Instead, charge is transferred between each plate and its connecting wire due to the electric field established in the wires by the battery, until the capacitor is fully charged. As the plates become charged, the potential difference across the capacitor is increased. The value of the maximum charge depends on the voltage of the battery. Once the maximum charge is reached, the current in the circuit is zero because the potential difference across the capacitor matches that supplied by the battery.

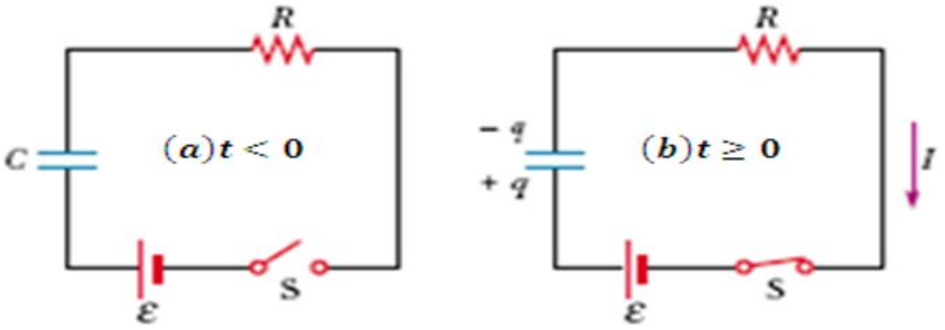


Figure 6.13: (a) A capacitor in series with a resistor, switch, and battery, circuit diagram representing this system at time $t < 0$ before the switch is closed. (b) Circuit diagram at time $t > 0$, after the switch has been closed.

During the charging process the battery voltage is divided between the Capacitor C and the resistor R. The voltages across the resistor and the Capacitor change with time, so we can write,

$$\mathcal{E} = V_R(t) + V_C(t) \tag{6.38}$$

As you may noticed, the voltages is time dependent, which can be rewritten as,

$$\mathcal{E} = \frac{q(t)}{C} + I(t)R \tag{6.39}$$

Solving this equation for $q(t)$, we find

$$q(t) = Q(1 - \exp(-t/RC)) \tag{6.40}$$

where $exp \equiv e$ is the base of the natural logarithm and $Q = C\mathcal{E}$ which is the final charge on the capacitor plates. We can find an expression for the charging current by differentiating Equation 6.40 with respect to time. Using $I = \frac{dq}{dt}$ we find that

$$I(t) = \frac{\mathcal{E}}{R} e^{-t/RC} \tag{6.41}$$

Plots of capacitor charge and circuit current versus time are shown in Figure 6.14. Note that the charge is zero at $t = 0$ and approaches the maximum value $C\mathcal{E}$ as $t \longrightarrow \infty$. The current has its maximum value $I_0 = \mathcal{E}/R$ at $t = 0$ and decays exponentially to zero as $t \longrightarrow \infty$. Equation 6.40 can be written in terms of the potential different across the capacitor, that is

$$V_C(t) = \mathcal{E}\left(1 - e^{-t/RC}\right) \tag{6.42}$$

Equation 6.42 tells us that when the $t \longrightarrow \infty$ the potential across the capacitor reaches $\mathcal{E}$. The quantity RC, which appears in the exponents of Equations 6.40 and 6.41, is called the time constant τ of the circuit. It represents the time it takes the current to decrease to 1/e of its

initial value; that is, in a time $\boldsymbol{\tau}, I = I_0 e^{-1} = \mathbf{0.37 I_0}$. In a time $2\tau, I = I_0 e^{-2} = 0.135 I_0$ and so forth. Likewise, in a time τ, the charge increases from zero to $C\mathcal{E}(1 - e^{-1}) = \mathbf{0.63 C\mathcal{E}}$. The following dimensional analysis shows that τ has the units of time:

$$R = \frac{Volt}{Amp} = \frac{Volt}{Coulomb/\sec} \quad \text{and} \quad C = \frac{Q}{V} = \frac{Coulomb}{Volt} \ ,$$

Thus, $RC = \frac{Volt}{Coulom/\sec} \times \frac{Coulomb}{Volt} = \sec$

Since the resistance R and the capacitance C have constant values, and their multiplication is constant also, and in the units of time. Thus, it is called time constant.

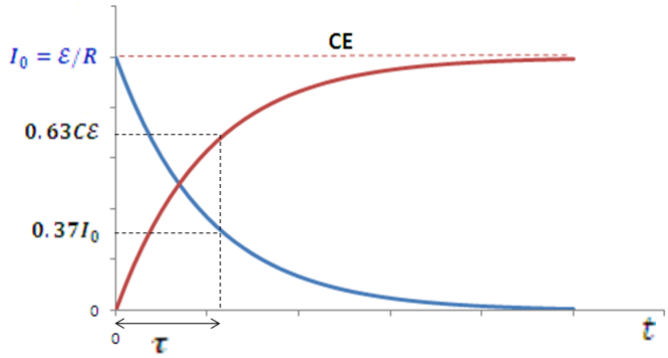


Figure 6.14: (a) Plot of capacitor charge versus time for the circuit shown in Figure 6.12. **(b)** Plot of current versus time for the circuit shown in Figure 6.12.

Example 6.9

An uncharged capacitor and a resistor are connected in series to a battery. If $\mathcal{E} = 12.0 Volt, C = 5.0 \mu F, and\ R = 0.8\ M\Omega$ find the time constant of the circuit, the maximum charge on the capacitor, the maximum current in the circuit, and the charge and current as functions of time

Solution

The time constant

$$\tau = RC = 0.8 \times 10^6 \times 5.0 \times 10^{-6} = 4 sec$$

The maximum charge

$$Q = C\mathcal{E} = 5.0 \times 10^{-6} \times 12.0 = 60 \mu C$$

The maximum charging current

$$I_0 = \frac{\mathcal{E}}{R} = \frac{12}{0.8 \times 10^6} = 15 \mu A$$

Charge and current as function of time

$$q(t) = Q\left(1 - e^{-t/RC}\right) \qquad and\ \ I(t) = I_0 e^{-t/RC}$$

$$q(t) = 60\mu C\left(1 - e^{-t/4}\right)\ \ and\ \ I(t) = 15\mu A e^{-t/4}$$

Example 6.10

An uncharged capacitor and a resistor are connected in series to a battery. If $\mathcal{E} = 12.0Volt, C = 3.0\mu F, and\ R = 1\ M\Omega$. Find how long it will take before the capacitor receives 99% of its final charge.

Solution

The time constant is

$$\tau = RC = 1.0 \times 10^6 \times 3.0 \times 10^{-6} = 3sec$$

$$q(t) = Q\left(1 - e^{-t/RC}\right)$$

$$\frac{q(t)}{Q} = 0.99 = \left(1 - e^{-t/3}\right)$$

Now take the natural logarithm for both side and solve for t, we will find

$$t = 13.8\ sec$$

Example 6.11

A resistance R and a $10\mu f$ capacitor are connected in series across a $200\ volt$ source. Across the capacitor is a neon lamp that strikes (glows) at $120\ volt$. Calculate the value of R to make the lamp glows after 5.0 seconds after the switch has been closed.

Solution This means that the voltage across the capacitor has to rise to $120\ Volt$ is 5.0 seconds,

$$V_C(t) = \mathcal{E}\left(1 - e^{-t/RC}\right)$$

$$120 = 200\left(1 - e^{-5/RC}\right)$$

Now solve for RC, we find

$$RC = 5.464\ sec \longrightarrow R = 5.464sec/10\mu F$$

$$R = 1.366M\Omega$$

Discharging a Capacitor

Now let us consider the circuit shown in Figure 6.15, which consists of a capacitor carrying an initial charge Q, a resistor, and a switch. The initial charge Q is not the same as the maximum charge Q in the previous discussion, unless the discharge occurs after the capacitor is fully charged (as described earlier). When the switch is open, a potential difference Q/C exists across the capacitor and there is zero potential difference across the resistor because $I = 0\ A$. If the switch is closed at $t = 0$ the capacitor begins to discharge through the resistor. At some time t during the discharge, the current in the circuit is I and the charge on the capacitor is q (Fig. 6.15b). The circuit in Figure 6.15 is the same as the circuit in Figure 6.13 except for the absence of the battery. Thus, we eliminate the emf $\mathcal{E}$ from Equation 6.39 to obtain the appropriate loop equation for the circuit in Figure 6.15:

$$0 = \frac{q(t)}{C} + I(t)R \tag{6.43}$$

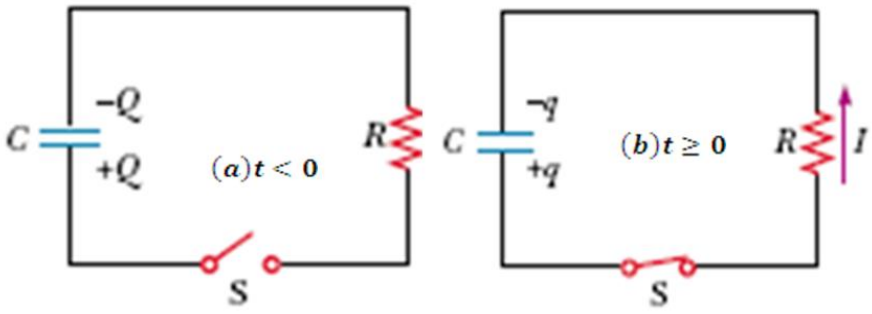


Figure 6.15: (a) A charged capacitor connected to a resistor and a switch, which is open at $t < 0$ **(b)** After the switch is closed, a current that decreases in magnitude with time is set up in the direction shown, and the charge on the capacitor decreases exponentially with time.

Solving equation 6.43 the current through the resistance R, we found

$$q(t) = Qe^{-t/RC} \tag{6.44}$$

Differentiating this expression with respect to time gives the instantaneous current as a function of time:

$$I = \frac{dq(t)}{dt} = -\frac{Q}{RC}e^{-\frac{t}{RC}} = -I_0 e^{-\frac{t}{RC}} \tag{6.45}$$

where $I_0 = Q/RC$ is the initial current. The negative sign indicates that discharging current direction is opposite to that of charging.

Solving equation 6.43 gives us the equation which describes the voltage across a capacitor with time during discharging:

$$V_C(t) = V_0 e^{-t/RC} \tag{6.46}$$

At $t = RC$ the potential difference across the capacitor equals to

$$V_C(t)|_{t=RC} = \frac{V_0}{e} = 0.37V_0 \tag{6.47}$$

This behavior of equation 6.46 is depicted in figure 6.16, which shows that the voltage across the capacitor decreases with time according to equation 6.47.

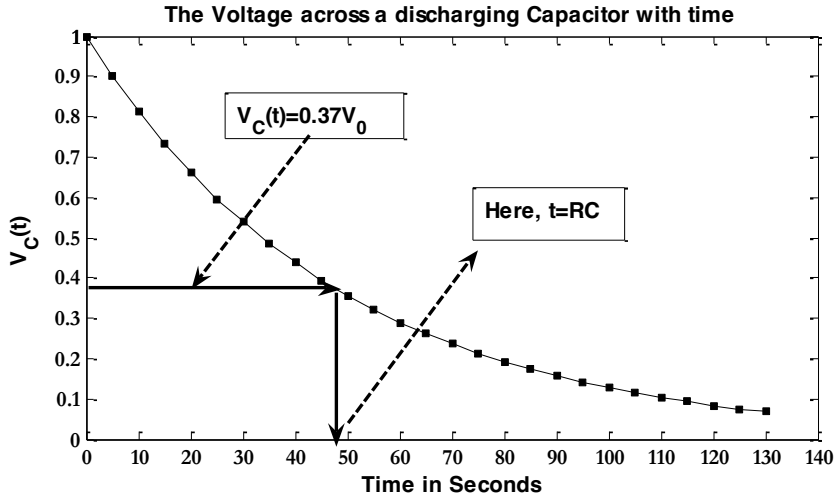


Figure 6.16: voltage across the capacitor while discharging. The figure shows how we can find the capacitor capacitance from its discharging behavior.

Example 6.12

Consider a capacitor of capacitance C that is being discharged through a resistor of resistance R. After how many time constants is the charge on the capacitor one-fourth its initial value?

Solution

We need to find the time required for the capacitor's charge to drop to its one fourth, that is $q(t) = Q/4$,

$$q(t) = Qe^{-t/\tau} \rightarrow Q/4 = Qe^{-t/\tau} \rightarrow 1/4 = e^{-t/\tau},$$

Now solve for t in terms of $\tau = RC$ (the time constant),

$$t = 1.39\tau$$

Example 6.13

A $20\mu F$ capacitor initially charged to potential difference of 500 Volt is discharged through an unknown resistance. After $60\ seconds$, the potential difference at the capacitor terminal is $185\ volt$. What is the magnitude of the resistance?

Solution

The voltage across the capacitor as a function of time is given by,

$$\boldsymbol{V_C(t) = V_0\, e^{-t/RC}}$$

Since the ratio $^{185}/_{500} = 0.37$. This means the Voltage drops after a time equals one time constant, that is $\tau = 60\ sec$

Thus,

$$R \times 20\mu F = 60 \rightarrow R = \frac{60sec}{20\mu F} = 3M\Omega$$

Example 6.14

A $0.1\mu F$ capacitor initially charged to potential difference of 100 Volt is discharged through $1M\Omega$ resistance. Find the following: Initial value of the discharged current; its value 0.1 second later; charge of the capacitor plates after 0.1 second.

Solution

The initial discharge current, at $t = 0$, is given by

$$I_0 = \frac{V}{R} = \frac{100}{1M\Omega} = 100\mu A$$

Check the RC circuit time constant

$$\tau = RC = 0.1\mu F \times 1M\Omega = 0.1 \text{ sec}$$

That means the current, charge, and voltage will drop to 0.37 of its initial value.

The current after 0.1 sec is equal to

$$I = 0.37I_0 = 0.37 \times 100\mu A = 37\mu A$$

The charge after 0.1 sec is also 0.37 of its initial value, that is

$$Q = C\mathcal{E} = 01.\mu F \times 100Volt = 10\mu C$$

$$q = 0.37Q = 0.37 \times 10\mu C = 3.7\mu C$$

6.8 Nerve Conduction

The human nervous system contains roughly $100\ billion$ neurons, connected in elaborate networks that transmit information from one location in the body to another. Consisting of the brain and spinal cord, the central nervous system interprets sensory input, initiates muscle contraction, and carries out all other cognitive tasks. The nerves that communicate messages between the central nervous system and the rest of the body compose the peripheral nervous system.

Despite the enormous complexity of the nervous system, there are aspects of neuron function that can be understood from simple physical principles. One of those aspects is the propagation of electrical impulses along neurons. Since neurons send information to one another via electrical signals, we can treat them like classical electrical circuits. In this topic we will review basic concepts in neurobiology and then describe the circuit model.

Neurobiology Review

Neurons can be divided into three main parts: the soma, or cell body, which contains the nucleus and other organelles, and two types of fiber-like extensions called dendrites and axons, figure 6.17. Dendrites receive inputs from other cells and conduct signals towards the cell body. Axons conduct signals away from the cell body towards their tips, where they are then passed on to other neurons or to muscle cells. A neuron may have many dendrites but usually has only one axon, which can be as long as $1\ m$. The junction between the axon of one neuron and the dendrite or cell body of another is called the synapse. Dendrites and axons increase the distance over which cells can communicate and allow for complex neural networks that enable intelligence

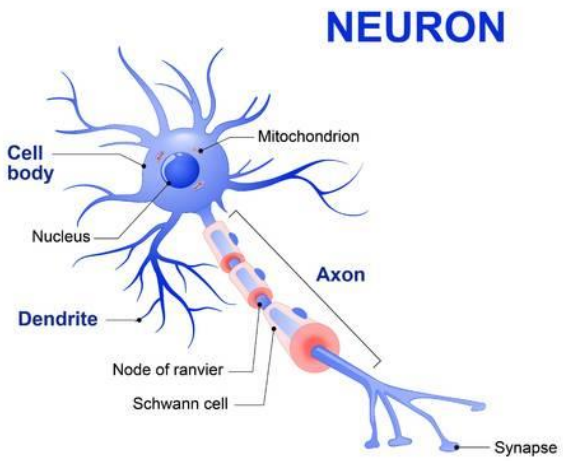


Figure 6.17: Structure of the neuron. Soma: Cell body, Dendrites: receive messages, Axon: sends messages**,**

Information travels through a neuron in the form of a combination of chemical and electrical conduction.

Resting and Acting Potential

Enclosed in the membrane of any cell is a jellylike substance that contains both inorganic and organic matter. In the cell body, this substance is called cytoplasm, but in the axon it is called axoplasm. For an inactive neuron, the axoplasm has an overall negative charge. This is because proteins, amino acids, phosphates, and other negatively-charged entities inside the cell cannot cross the selectively-permeable cell membrane. The resting potential is originated from the differences in concentration of ions inside and outside the membrane. It depends also on the difference in permeability of the cell wall to the different ions. Two types of positively-charged ions, potassium (K^+) and sodium (Na^+), can cross the cell membrane through selective ion channels. Normally there are more potassium ions inside the cell than outside, whereas there are more sodium ions outside the cell than inside. Due to these ionic effects, the resting potential of the axoplasm is about $-70\ mV$ relative to the extracellular fluid. Because the resting potential is negative, the electrical field is directed from outside to inside. Therefore it drives additional ions through the membrane into the cell. The migration of the ions cause a potential difference, V, leading to the formation of a potential hill against the movement of extra ions till equilibrium takes place. When the thermal energy of the ions equal the height of

the potential hill, then the potential difference across the membrane is given by Nernst equation:

$$V = \pm 2.3\ \frac{k_B T}{q}\ log\left(\frac{C_o}{C_i}\right) \tag{6.48}$$

Where $k_B T$ is the thermal energy, q is the electronic charge, C_0 and C_i is the concentration of the ions outside and inside the axon respectively. For example $T\ =\ 37°C\ =\ 310K$, so

$$\frac{k_B T}{q} = \frac{1.38 \times 10^{-23} J/K \times 310K}{1.6 \times 10^{-19} C} = 0.0267V = 26.7\ mV$$

Then equation 6.48 can be rewritten as:

$$V = \pm 61.4\ mV\ log\left(\frac{C_o}{C_i}\right) \tag{6.49}$$

For a nerve cell, the intercellular has K^+ concentration, $C_i\ =\ 0.141\ mol/liter$, whereas the extracellular fluid has a K^+ concentration, $C_0\ =\ 0.005\ mol/liter$, which give the Nernst potential to be:

$$V = \pm 61.4\ mV\ log\left(\frac{0.005}{0.141}\right) = -89.2\ mV$$

In practice, the actual resting potential measured by the electrophysiological units is about $-85\ mV$. The model membrane system, just discussed and the Nernst potential were based on the assumption that the permeability of the membrane to the flow of ions is constant. However, in practice, the permeability of the cell membrane to the diffusion of ions changes according to the cellular metabolic function and its temperature.

Excitable cells have special Na and K channels with gates that open and close in response to the membrane voltage (voltage-gated channels). Between Schwann cells are small regions of exposed axon called nodes of Ranvier. These nodes contain the voltage-gated ion channels that allow action potentials to propagate down the axon, so that the signal jumps from node to node as shown in figure 6.19. When a nerve cell receives a stimulating action such as electric signal, thermal mechanical or chemical, its membrane suddenly become permeable to the Na^+. This will permit the Na^+ to diffuse through the cell membrane to the interior and combine with the Cl^-. Thus membrane potential will depolarize (becomes more positive) producing spike as shown in figure **6.18**. The rate of flow through the cellular membrane increases rapidly to 100 times that of K^+ after short time $\sim\ 2ms$. During this period the interior potential

changes from $-\ 90\ mV$ to $65\ mV$. After this period, the sodium gate is closed, where the membrane become impermeable to sodium ions and in the same time K^+ gate is opened diffusing it out of the cell. This makes the membrane repolarized (becomes more negative) till it returns to its normal state. This change of the cell potential is called the action potential.

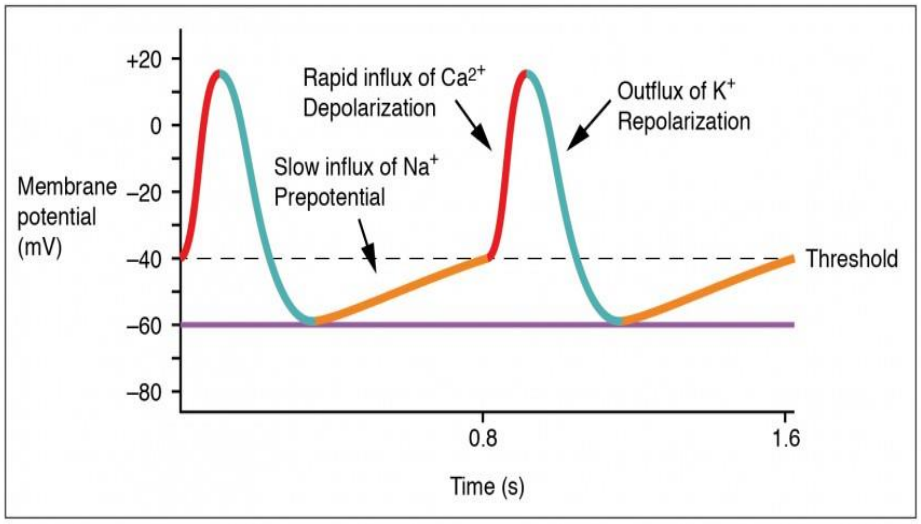


Figure 6.18: The depolarization and repolarization process resulted from external stimulation of the nerve.

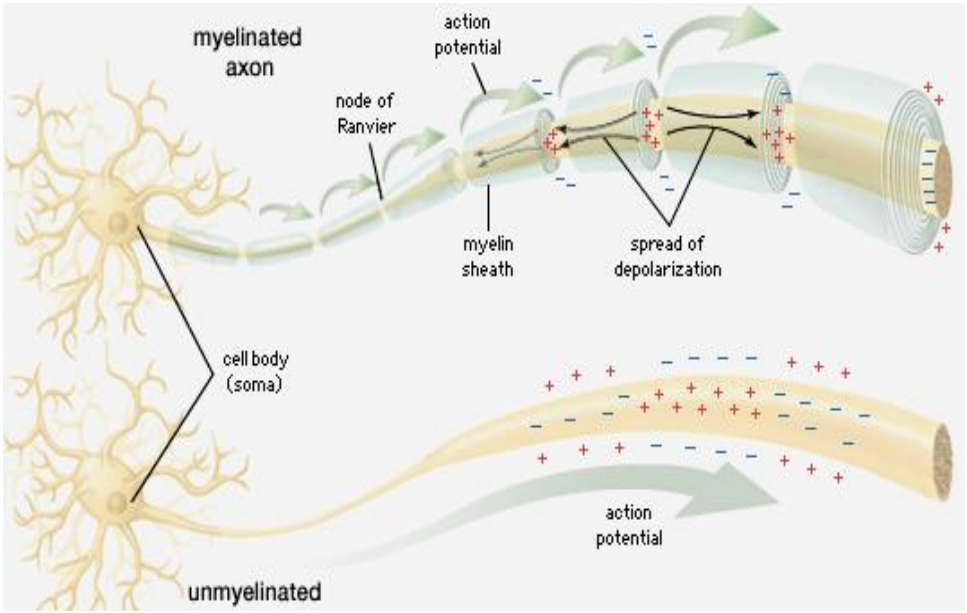


Figure 6.19: Structure of a neuron with myelinated axon and unmyelinated axon

Electrical Properties of Neurons

Axon acts as cables that transmit bioelectric impulses from one nerve cell to another cells or the central nervous system. A nerve fiber consists of a thin hollow tube containing positive and negative ions. The fiber is immersed in an extracellular fluid which contains the same ionic composition. The wall of the axon tube is semi-permeable membrane which, although a dielectric, allows ions to migrate into and out of the fiber. The axon can be considered as a cable that transmits nerve impulses as shown in figure 6.20.

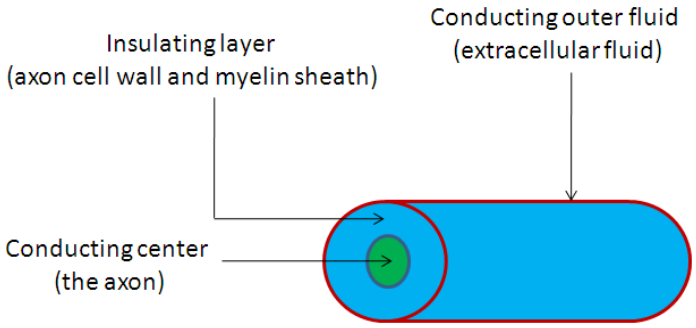


Figure 6.20: Axon as an insulated wire

Examination of the axons of various neurons with an electron microscopy indicates that there are two different types of nerve fibers. As shown in figure 6.20 the membrane of some axons are covered with a fatty insulating layer called myelin and they are called myelinated axons (MA). The membrane of other axons has no myelin sheath and they are called unmyelinated axons (UA).

Resistance

In the neuron, there are two substances that exhibit electrical resistance: the axoplasm itself and the cell membrane plus myelin sheath, if present.

The electrical resistance R along the length of the axon follows the same principles as a wire: equation 6.11

$$R = \rho_a \frac{l}{\pi r^2} \tag{6.11}$$

Both myelinated and unmyelinated neurons has a resistivity ρ_a of the axoplasm is $2.0\ \Omega \cdot m$. If the average neuron has an axon $1\ mm$ long and a $5\mu m$ radius, we can use Equation 6.11 to

find that the resistance of the axoplasm $R_{axoplasm} = 2.5 \cdot 10^7 \Omega$. This huge value indicates that axons are actually poor electrical conductors.

The cell membrane is also permeable to charge; its resistance is not infinite, even when myelinated. Rather than depending on cross-sectional area, the resistance through the membrane (R_m) depends on the surface area of the axonal membrane $2\pi rl$:

$$R_m = \frac{\rho_m}{2\pi\, r\, l} \tag{6.50}$$

Where ρ_m is the membrane resistivity measured in $\Omega.m^2$. For an unmyelinated axon (UA), $\rho_{mUA} = 0.2\ \Omega.m^2$. So, again for an average axon $1\ mm$ long with radius $5\ \mu m$, $R_{mUA} = 6.4 \cdot 106\ \Omega$. Myelinated axons (MA) have a much higher resistivity, $\rho_{mMA} = 40\ \Omega.m^2$, so $R_{mMA} = 1.3 \cdot 107\ \Omega$. The membrane resistance R_m represents the resistance of the leakage current. There is an operator called the space parameter λ, which represent how can the impulse signal travel along the axon before the leakage is complete. This occurs when the axon resistance R is equal to the membrane resistance R_m. So that at $R = R_m$, then put $l = \lambda$, which gives us:

$$\frac{\rho_a \lambda}{\pi r^2} = \frac{\rho_m}{2\pi r \lambda}$$

So, we can solve for λ, which gives us

$$\lambda = \sqrt{\frac{\rho_m r}{2\rho_a}} \tag{6.51}$$

The space parameter ranges from 0.05 to $0.7\ cm$ for unmyelinated to myelinated axons.

Example 6.15

A myelinated axon has a space parameter of $1\ cm$, find its radius? $\rho_{mMA} = 40\ \Omega.m^2$ and $\rho_a = 2\ \Omega.m$.

Solution

From equation 6.51, we can write r in terms of the space parameter as:

$$r = \frac{2\rho_a \lambda^2}{\rho_m} = \frac{2\ (2\Omega.m).10^{-4}m^2}{40\ \Omega.m^2} = 10^{-5}m = 10\mu m$$

Capacitance

From the previous section, we know that the capacitor is an electrical device that stores electric charge. It consists of two conductors side by side, separated by some insulating substance called the dielectric. The Capacitance of parallel plate capacitor is given by equation 6.37, that is

$$C = \frac{\kappa\varepsilon_0 A}{d} = \frac{\epsilon A}{d}$$

Where $\epsilon = \kappa\varepsilon_0$

Again, the equation makes sense intuitively because the larger the surface area between the plates the more charge can be stored. Furthermore, the smaller plate separation the greater the attraction between the charges, which also increases the capacity for charge storage.

For a lipid bilayer, $\epsilon = 5 \cdot 10^{-11} F/m$ and $d = 50\text{Å} = 5 \cdot 10^{-9}\, m$. Thus, the capacitance per unit area for an unmyelinated axon of $5\ nm$ thickness is:

$$C_{um} = \frac{C}{A} = \frac{\epsilon}{d} = \frac{5\times10^{-11}}{5\times10^{-9}} = 10^{-2} F/m^2 \text{ (Unmyelinated axon)}$$

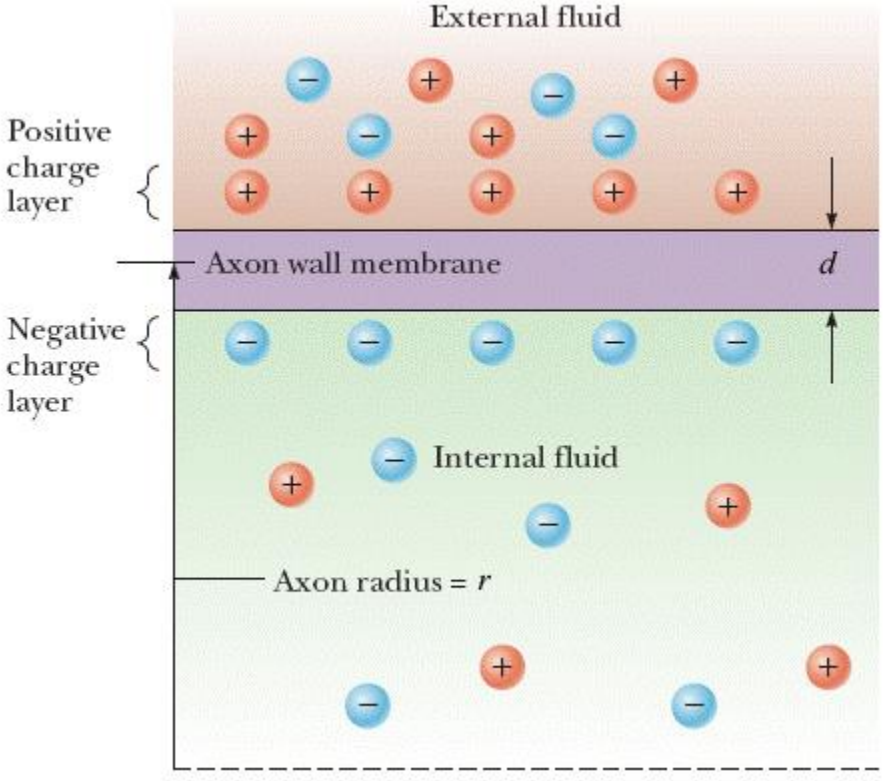
For myelinated axons, the myelin sheath contains a membrane that wraps around the axon a couple of hundred times. This multilayer arrangement effectively increases the thickness of the lipid bilayer by a factor of 200 ($1\ \mu m$ total thickness), so capacitance per unit area for a myelinated axon is:

$$C_m = \frac{C}{A} = \frac{\epsilon}{d} = \frac{5\times10^{-11}}{1\times10^{-6}} = 5 \times 10^{-5} F/m^2 \text{ (Myelinated axon)}$$

Example 6.16

The voltage across a membrane forming a cell wall is $80.0\ mV$ and the membrane is $9.00\ nm$ thick. What is the electric field strength? You may assume a uniform electric field.

Solution



$$E = \frac{V_{AB}}{d} = \frac{80.0 \times 10^{-3}\ \text{V}}{9.00 \times 10^{-9}\ \text{m}} = \underline{8.89 \times 10^{6}\ \text{V/m}}$$

Example 6.17

Membrane walls of living cells have surprisingly large electric fields across them due to separation of ions. What is the voltage across an $8.00\ nm$–thick membrane if the electric field strength across it is $5.50\ MV/m$? You may assume a uniform electric field.

Solution

$$V_{AB} = Ed = (5.50 \times 10^{6}\ \text{V/m})(8.00 \times 10^{-9}\ \text{m}) = 4.40 \times 10^{-2}\ \text{V} = \underline{44.0\ \text{mV}}$$

Example 6.18

An axon membrane is $7.5\ nm$ thick. In the resting state the axon potential is $-90\ mV$, what is the direction and magnitude of the electric field in the membrane. If the membrane has a capacitance of $0.01\ F/m^2$, what is its dielectric constant?

Solution

From equation 6.5, the electric field E is given as

$$E = \frac{V}{d} = \frac{-90\times10^{-3}V}{7.5\times10^{-9}m} = 1.2\times10^{7}\ V/m.$$

The capacitance per unit area

$$C_m = \frac{C}{A} = \frac{\epsilon}{d} = 0.01\ F/m^2, \text{ so } \varepsilon = C_m d = 0.01\times7.5\times10^{-9} = 7.5\times10^{-11}\ F/m.$$

Neuron's Equivalent Circuit

The electrical properties of neurons are summarized by the diagram, figure 6.21, and table 6.1.

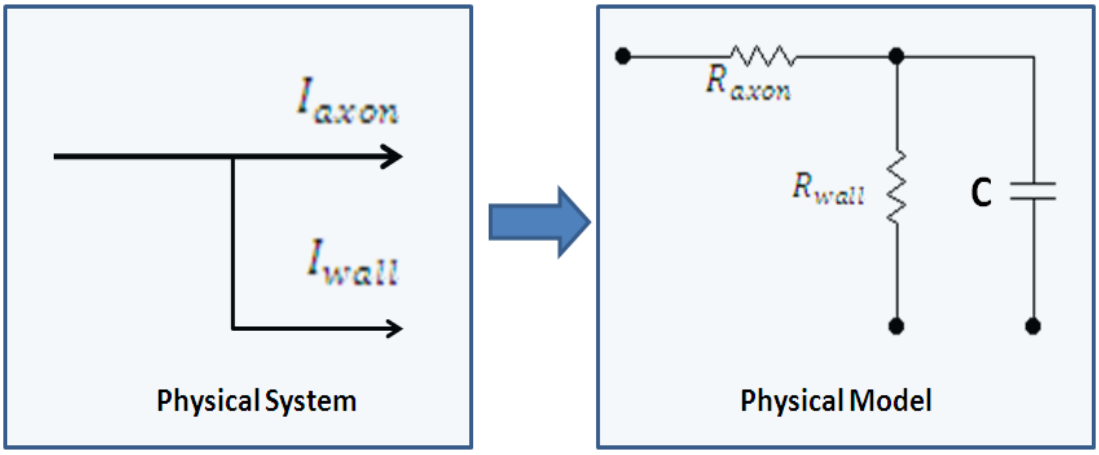


Figure 6.21: The physical model shows wires, two resistors, and a capacitor that approximate the physical flow of charge through real axons.

Table 6.1: Useful constants

	Unmyelinated Axon (UA)	Myelinated Axon (MA)
Axoplasm resistivity	$\rho_{axoplasm} = 2.0\Omega\cdot m$	$\rho_{axoplasm} = 2.0\Omega\cdot m$
Wall resistivity	$\rho_{UA} = 0.20\Omega\cdot m^2$	$\rho_{MA} = 40.0\Omega\cdot m^2$
Wall capacitance/area	$\frac{C}{A} = 10^{-2}F/m^2$	$\frac{C}{A} = 5\times10^{-5}F/m^2$

Impulse Propagation

As explained in the Introduction, neurons are connected so that action potentials travel between them in only one direction. The electrical properties of nerve cells discussed in the last section suggest that impulse propagation can be modeled as an array of resistors and capacitors, as shown below in Figure 6.22.

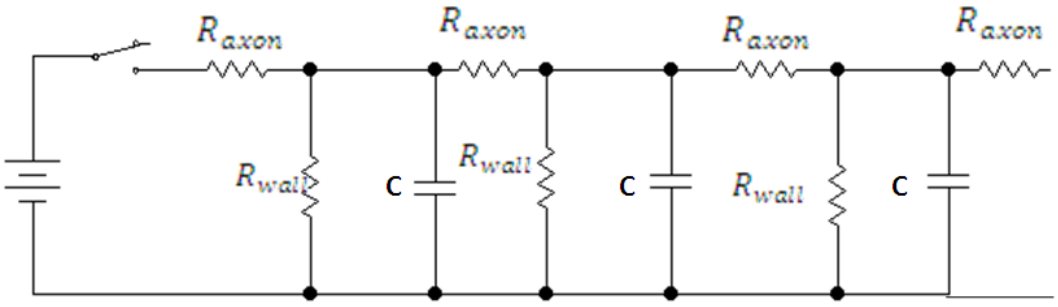


Figure 6.22: Physical model of neural connections.

The analysis of this electric circuit is complex, requiring the use of differential equations (considering the cylindrical cross-section of the neuron), but we can gain insight by considering this simplified version which shows how the electrical signal propagate in the nervous system; which is modeled as a sequence of charging and discharging circuit.

Propagation Speed

For simplicity, we will at first ignore the wall resistance. For one circuit unit (i.e. one neuron), the voltage changes over time according to the equation

$$V(t) = V_0\left(1 - e^{-t/R_{\text{axoplasm}}\,C}\right) \tag{6.52}$$

where V_0 is the resting potential. Equation 6.49 is graphed below.

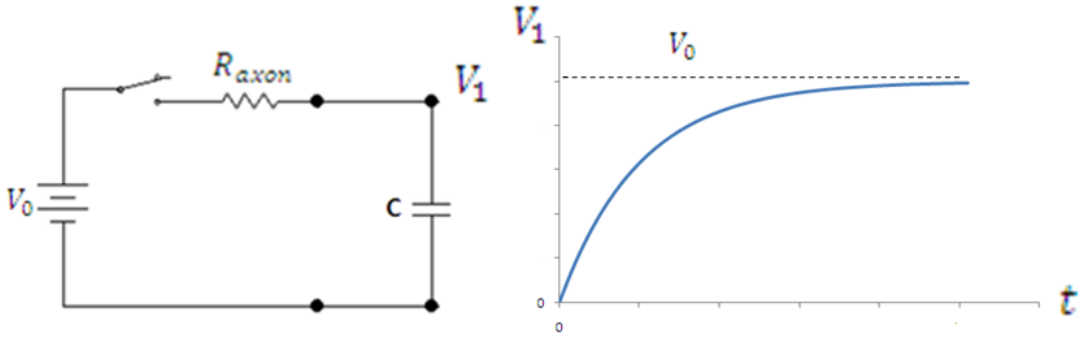


Figure 6.23: Voltage V_1 of a single neuron as a function of time.

The charging time is $\tau = R_{axoplasm}C$. When we add another unit, the problem gets a bit more complicated. According to the model, the first unit charges up before the second unit begins to charge. With every additional unit, there is an additional delay of $\tau = R_{axoplasm}C$

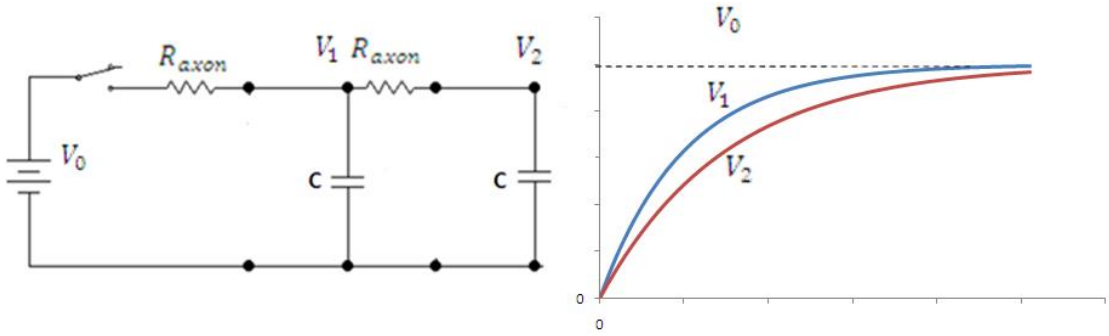


Figure 6.24: The circuit with two neurons charges twice as slow as the circuit with one.

Since a unit must charge completely before it can discharge to the next unit (like the all-or-none action potential), there is a time delay equal to τ in the propagation of the electrical signal between two units. If the length of each unit is x, then the speed of propagation is given by.

$$v = \frac{x}{t} = \frac{x}{R_{axon}C} \tag{6.53}$$

Substituting the expressions for the resistance of the axoplasm as $R_a = \frac{\rho_a x}{\pi r^2}$ and the capacitance C equals the capacitance per unit area of the membrane C_m multiplied by the surface area of a cylinder $(A = 2\pi rx)$ into the expression for the velocity we obtain:

$$v = \frac{x}{\left(\frac{\rho_a x}{\pi r^2}\right)(C_m 2\pi r x)} = \frac{r}{2C_m \rho_a\, x} \tag{6.54}$$

The r dependence in the numerator of equation 6.52 accounts for a fundamental rule of neurobiology: the wider the axon, the faster the axonal speed of propagation. For myelinated neurons, the myelin sheath covers the axon in $1\,mm - long$ sections. Thus, within each myelinated section, one would predict that. For unmyelinated neurons, for which $v = 5 \times 10^{-6} m/(2 \times 2.0 \Omega m \times 5 \times 10^{-5} F/m^2 \times 1 \times 10^{-3} m \approx 20 m/sec)$ the speed of propagation is 200 times slower or $0.1\,m/s$. While most neurons share the same basic structure, they vary greatly in length and speed of signal propagation. In the brain where axons are as short as $0.01\,mm$, signals travel $0.5 - 2.0\,m/s$. In the limbs, however, axons can be

up to $1.0\ m$ in length and carry signals at $100\ m/s$. By examining the electrical properties of neurons, we can see what factors determine the speed of propagation

Clearly the latter would not be suited for nerve communication over long distances. Recall that our calculations do not take into account the resistance of the cell membrane and myelin sheath. Including it leads to leakage of the electrical signal through the wall, called signal attenuation. For a single unit, the effect is profound, as shown below.

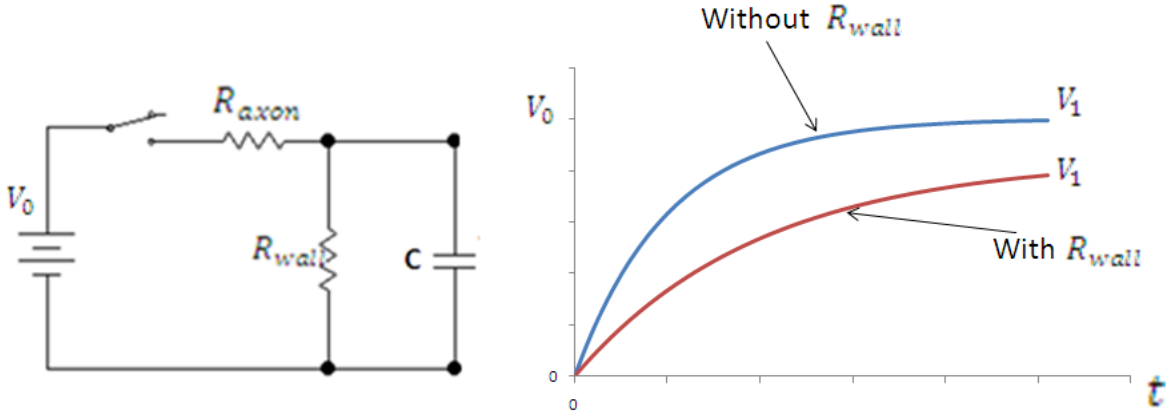


Figure 6.25: The voltage is much lower if we consider the wall resistance

Note that the two resistors in Figure 6.25 are in series. Such a sequence of resistors acts as a voltage divider, that is, the voltage across R_{wall} is a fraction of V_0 given by $R_{wall}/(R_{axon} + R_{wall})$. You can infer from this that the signal decreases by this factor every time it propagates through one of these segments. Therefore, as the signal propagates through several such segments, the voltage decreases geometrically with the number of units traveled i.e.

$$v = v_0 e^{-x/\lambda} \tag{6.55}$$

where x is the distance traveled down the circuit. The value of λ is $0.05\ cm$ for unmyelinated neurons and $0.7\ cm$ for myelinated neurons. When we graph Equation 6.52 we see that myelinated nerve fibers carry nerve impulses farther, figure 6.23

According to our estimates, nerve impulses cannot travel much more than about $1\ cm$. But we know that some axons in the body can be up to $1\ m$ long! In order to allow their signals to travel greater distances, neurons amplify their signals chemically.

Example 6.19

A myelinated nerve of $10\ \mu m$ thick with a space parameter of $0.5\ cm$. Calculate the speed of the pulse propagation along $1\ mm$. Find the dielectric constant of the membrane material?

Solution

Using equation 6.54 the speed of propagation is $v = \frac{r}{2C_m\rho_a\, x}$ where for myelinated axon $\rho_a = 2\Omega.m,\ \rho_m = 40\ \Omega.m^2\ and\ C_m = 5\times10^{-5}\ F/m^2$. As shown in example 6.18, we can find r in terms of the space parameter λ, as $r = \frac{2\rho_a\lambda^2}{\rho_m}$ then substitute in the formula of v to give

$$v = \frac{\lambda^2}{C_m\rho_m x} = \frac{(5\times10^{-3})^2 m^2}{\left(5\times10^{-5}\ \frac{F}{m}\right)(40\ \Omega.m^2)(1\times10^{-3}m)} = 12.5\ m/s$$

Role in human health

Bioelectricity is one of the fundamental forms of energy in the human body. In the form of moving action potentials, it is the basis for such central bodily functions as conduction of motor, autonomic, or sensory messages along the nerves; muscle contraction; and brain function. Specifically, motor nerve signals result in muscle contractions. Autonomic nervous signals control such basic functions of the body as breathing and heartbeat. Sensory nerve signals collect input from the outside world, including warnings of damage to the body in the form of pain.

Bioelectrical signals in humans

There are three types of electrical signals in human beings, two of which are routinely monitored or analyzed for diagnostic purposes. The first is the electroencephalogram, which is a relatively weak, fluctuating signal that originates in the brain. The second is the electrocardiogram, which is about 100 times stronger than the electroencephalogram, and is produced by the contractions of the heart muscle. The third type of electrical signal in humans, the surface electrical potential, is about as strong as the electrocardiogram but changes more slowly over time. The origin and significance of the surface electrical potential in humans are not yet known.

Common diseases and disorders

A large number of diseases and disorders are related to disturbances of the bioelectrical system. These conditions can be classified according to the component of the nerve cell/muscle cell group, or motor unit, which is affected. The motor unit can be divided into the motor neuron, the nerve root (paired bundles of nerves coming from the spinal cord), the nerve plexus (bundles of nerves further removed from the spinal cord), the peripheral nerve, the neuro-muscular junction, and the muscle fiber. Defects in any of these components may disrupt bioelectrical signals.

6.9 Application

$ECG\ or\ EKG$

Many heart problems change the electrical "signature" of the heart in distinct ways. The device that records this electrical signature can help reveal these heart problems is called electrocardiogram. An electrocardiogram, also called an $EKG\ or\ ECG$, is a simple test that detects and records the electrical activity of the heart. It is used to detect and locate the source of heart problems. Electrical signals in the heart trigger heartbeats. These signals start at the top of the heart in an area called the right atrium. The electrical signals travel from the top of the heart to the bottom. They cause the heart muscle to contract as they travel through the heart. As the heart contracts, it pumps blood out to the rest of the body. An EKG shows how fast the heart is beating. It shows the heart's rhythm (steady or irregular) and where in the body the heartbeat is being recorded. It also records the strength and timing of the electrical signals as they pass through each part of the heart. Figure 6.26, shows an EKG is sometimes called a $12-lead\ EKG$ (or $12-lead\ ECG$) because the electrical activity of the heart is most often recorded from 12 different places on the body at the same time.

Example 6.20

An ECG monitor must have an RC time constant less than $1.00\times10^{2}\mu s$ to be able to measure variations in voltage over small time intervals. (a) If the resistance of the circuit (due mostly to that of the patient's chest) is $1.00\ k\Omega$, what is the maximum capacitance of the circuit? (b) Would it be difficult in practice to limit the capacitance to less than the value found in (a)?

Solution

(a) $C = \frac{\tau}{R} = \frac{1.00 \times 10^{-4}\ \text{s}}{1.00 \times 10^{3}\ \Omega} = \underline{1.00 \times 10^{-7}\ \text{F}}$

(b) No, in practice it would not be difficult to limit the capacitance to less than $100\ nF$, since typical capacitors range from fractions of a picofarad (pF) to milifarad (mF).

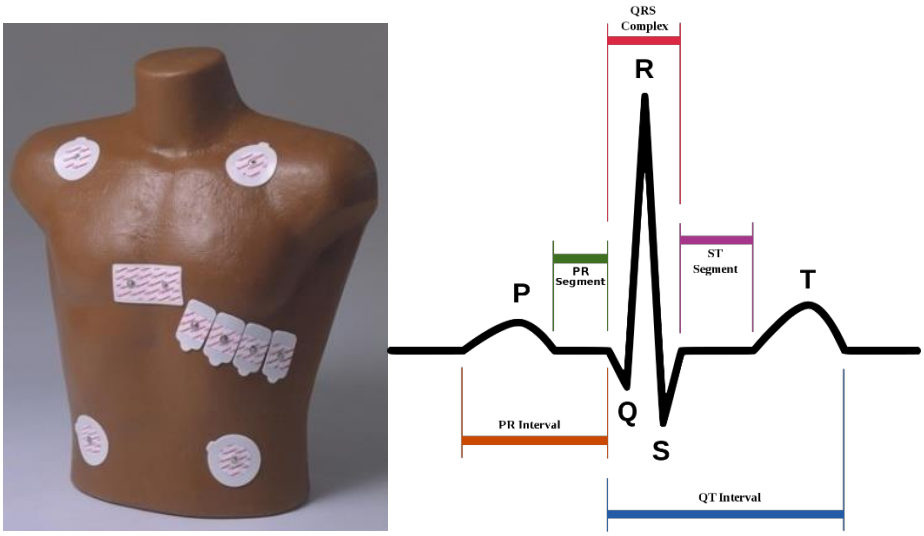


Figure 6.26: a. ECG contact points, b. the ECG reading

What EKG Reveals

Many heart problems change the electrical signature of the heart in distinct ways. EKG recordings of this electrical activity can help reveal a number of heart problems, including:

- Heart attack
- Lack of blood flow to the heart muscle
- A heart that is beating irregularly, or too fast or too slow
- A heart that does not pump forcefully enough

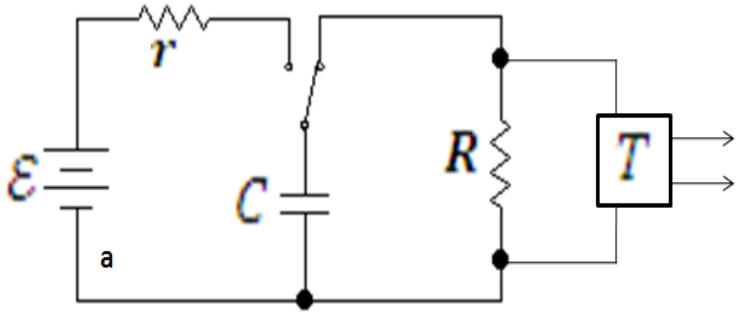
EKG recordings can help doctors diagnose a heart attack that is happening now or has happened in the past. This is especially true if doctors can compare a current EKG recording to an older one. EKG recordings can also reveal: Heart muscle that is too thick or parts of the heart that is too big Birth defects in the heart. Disease in the heart valves between the different heart chambers: An EKG also reveals whether the heartbeat starts at the top right part of the

heart like it should. It shows how long it takes for the electrical signals to travel through the heart.

Pacemaker

Your heart has its own internal electrical system that controls the speed and rhythm of your heartbeat. With each heartbeat, an electrical signal spreads from the top of the heart to the bottom. As it travels, the electrical signal causes the heart to contract in an organized manner and pump blood. Electrical signals normally begin in a group of cells called the sinus node. As these signals spread from the top to the bottom of the heart, they coordinate the timing of heart cell activity. First, the two upper chambers of the heart, called atria (AY-tree-uh), contract. This contraction squeezes blood into the lower chambers of the heart, which are called ventricles (VEN-trih-kuls). The ventricles then contract and send blood to the rest of the body. The combined contraction of the atria and ventricles is a heartbeat. These heart beats has to follow a certain rhythm to sustain life. Sometimes due to some heart complications heart loses this rhythm. So, some heart patients are being helped by surgically implanted artificial pacemakers, which are battery-powered circuits that pulse if the person's pacemaker fails to do so. Most pacemaker batteries last between seven and eleven years before they need replacing. After the pacemaker signals the battery is weakening, patient will have time to see his doctor and schedule replacement surgery. The battery will not immediately quit working.

One model has pulses triggered 75 times per minute by a $0.4\ \mu F$ capacitor, which rapidly charges through a very small resistance r and then slowly discharged through a large resistance R, figure 6.27. When the charge drop to 37% of the initial charge, transistor delivers a short pulse to the heart and then recharge the capacitor almost immediately through r



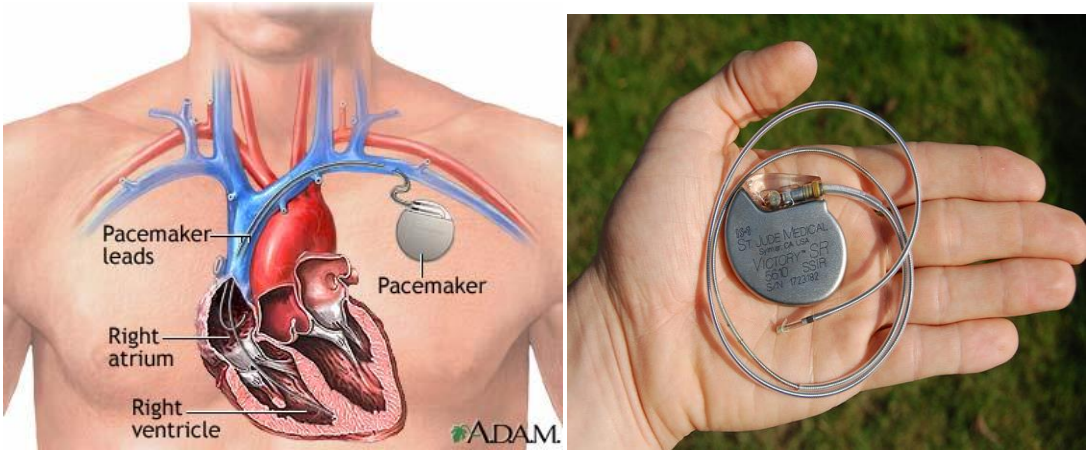


Figure 6.27: a. The basic pacemaker timing circuit**. b.** pacemaker implemented

Example 6.21

Find the time constant of the discharged RC circuit, neglecting the small time needed to recharge the capacitor through r, Find the resistance R.

Solution

There are 75 pulses per minute, which means we need a pulse every

$$\mathbf{60/75 = 0.8\ second}$$

The time constant of the discharged RC circuit is $\boldsymbol{\tau = RC = 0.8\ second}$

The resistance R is $\boldsymbol{R = \frac{\tau}{C} = \frac{0.8sec}{0.4\times10^{-6}F} = 2\times10^{6}\Omega = 2M\Omega}$

6.10 Electrotherapy

Wound Healing

Many controlled studies have shown that a small microampere current as a long term treatment leads to accelerated healing. There are two classes of cases: accelerated healing of bone fractures, and of skin surface wounds. Ischemic dermal ulcers are treated with DC, and the healing rate is approximately doubled. It has been found that a monopolar cathodic application during the first days followed by an anodic application, gives the best results. In a skin wound it is believed that positive charge carriers (ions and proteins) are transported to the liquid wound zone by endogenic migration. An increased rate of bone formation has also been found when small currents are applied to each side of a bone fracture (bipolar electrode system). This is of particular interest in cases of bone fractures that will not grow the natural way: so-called

non-union). Nordenstrom (1983) described the use of DC treatment by applied needles into tumors for the treatment of cancer.

Vasodilation

This effect is usually linked to pain in cold extremities. Increased blood flow may increase the temperature from the range $22-24\,℃$ to $31-34\,℃$, also in the extremities not stimulated. The effect must therefore be elicited in higher centers of the CNS. The afferent pain nerves have a higher threshold and rheobase than sensory and motor nerves. Thus it is possible to stimulate sensory and motor nerves without eliciting pain. Very short pulses of duration $10-400\,s$ are used, with constant amplitude current up to about 50 mA and treatment duration of 15 minutes or more. The skin electrodes may be bipolar or monopolar. The position is in the pain region; an electrode pair may, for example, be positioned on the skin on the back of the patient, or implanted with thin leads out through the skin. The electrode pair may also be positioned outside the pain area (e.g. at regions of high afferent nerve fiber densities in the hand).

Electroacupuncture

The secretion of endorphins is obtained with low frequency ($2-4\,Hz$) stimulation, corresponding to the rhythmic movement of an acupuncture needle. Instead of, or in addition to the mechanical movement, the needle is used as monopolar electrode, pulsed by a low frequency in the same frequency range ($1-4\,Hz$). This is not a TENS method strictly speaking, as the electrode is invasive and the current not transcutaneous.

Summary

The electric current I in a conductor is defined as

$$I = \frac{dQ}{dt}$$

where dQ is the charge that passes through a cross-section of the conductor in a time dt. The SI unit of current is the ampere (A), where $1\,A\ =\ 1\,C/s$.

The resistance R of a conductor is defined either in terms of the length of the conductor or in terms of the potential difference across it:

$$R = \rho\frac{l}{A} = \frac{\Delta V}{I}$$

Where l is the length of the conductor, ρ is the resistivity of the material of which it is made, A is its cross-sectional area, ΔV is the potential difference across it, and I is the current it carries. The SI unit of resistance is volts per ampere, which is defined to be $1\ ohm = \Omega$

If a potential difference ΔV is maintained across a resistor, the ***power***, or rate at which energy is supplied to the resistor, is

$$\mathcal{P} = I\Delta V$$

The emf of a battery is equal to the voltage across its terminals when the current is zero. That is, the emf is equivalent to the open-circuit voltage of the battery.

The equivalent resistance of a set of resistors connected in series is

$$R_{eq} = R_1 + R_2 + R_3 + \cdots$$

The equivalent resistance of a set of resistors connected in parallel is

$$\frac{1}{R_{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

The capacitance C of a capacitor is the ratio of the magnitude of the charge on either conductor to the magnitude of the potential difference between them:

$$C = Q/\Delta V$$

where Q is the excess charge on the capacitors plates and ΔV is the potential difference across the capacitor.

If a capacitor is charged with a battery through a resistor of resistance R, the charge on the capacitor and the current in the circuit vary in time according to the expressions

$$q(t) = Q\left(1 - e^{-t/RC}\right) \qquad and\ \ I(t) = I_0 e^{-t/RC}$$

where $\mathrm{Q} = \mathrm{CE}$ is the maximum charge on the capacitor, and $I_0 = \frac{\mathcal{E}}{R}$. The product RC is called the time constant τ of the circuit. If a charged capacitor is discharged through a resistor of resistance R, the charge and current decrease exponentially in time according to the expressions

$$I = -I_0 e^{-\frac{t}{RC}} \ \text{ and } q(t) = Qe^{-t/RC}$$

where Q is the initial charge on the capacitor and $I_0 = \frac{Q}{RC}$ is the initial current in the circuit

Resting and Acting Potential

$$V = \pm 61.4\, mV \log\left(\frac{C_o}{C_i}\right) \tag{6.49}$$

Where $k_B T$ is the thermal energy, e is the electronic charge, C_0 and C_i is the concentration of the ions outside and inside the axon respectively.

Electrical Properties of Neurons

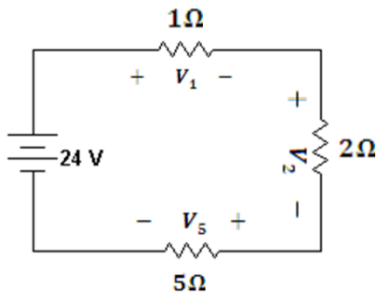
The electrical resistance R along the length of the axon follows the same principles as a wire

$$R = \rho_a \frac{l}{\pi r^2}$$

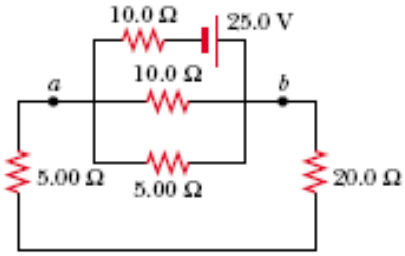
Problems

1. What is the current in milliamperes produced by the solar cells of a pocket calculator through which 4.00 C of charge passes in 4.00 h?
2. During open-heart surgery, a defibrillator can be used to bring a patient out of cardiac arrest. The resistance of the path is $500\,\Omega$ and a 10.0-mA current is needed. What voltage should be applied?
3. (a) A defibrillator passes 12.0 A of current through the torso of a person for 0.0100 s. How much charge moves? (b) How many electrons pass through the wires connected to the patient?
4. A clock battery wears out after moving 10,000 C of charge through the clock at a rate of 0.500 mA. (a) How long did the clock run? (b) How many electrons per second flowed?
5. What is the resistance of a 20.0-m-long piece of 12-gauge copper wire having a 2.053-mm diameter?
6. (a) To what temperature must you raise a copper wire, originally at $20.0°C$, to double its resistance, neglecting any changes in dimensions? (b) Does this happen in household wiring under ordinary circumstances?
7. What power is supplied to the starter motor of a large truck that draws 250 A of current from a 24.0-V battery hookup?

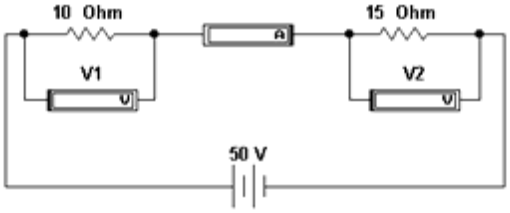
8. (a) What is the resistance of ten $275\text{-}\Omega$ resistors connected in series? (b) In parallel?
9. a) Given a 48.0-V battery and $24.0\text{-}\Omega$ and $96.0\text{-}\Omega$ resistors, find the current and power for each when connected in series. (b) Repeat when the resistances are in parallel.
10. What is the output voltage of a 3.0000-V lithium cell in a digital wristwatch that draws 0.300 mA, if the cell's internal resistance is $2.00\,\Omega$?
11. What is the internal resistance of an automobile battery that has an emf of 12.0 V and a terminal voltage of 15.0 V while a current of 8.00 A is charging it?
12. A platinum coil has a resistance of $3.146\,\Omega$ at $40°C$ and $3.767\,\Omega$ at $100°C$. Find the resistance at $0°C$ and then find R_0
13. A resistance thermometer, which measures temperature by measuring the change in resistance of a conductor, is made from platinum and has a resistance of $50.0\,\Omega$ at $20.0°C$. When immersed in a vessel containing melting indium, its resistance increases to $76.8\,\Omega$. Calculate the melting point of the indium. Hint $\boldsymbol{\alpha = 3.92 \times 10^{-3}}$
14. A certain resistor has a resistance of 150.4Ωat $20℃$and resistance of 162.4Ω at $40℃$. What is its temperature coefficient of resistivity?
15. For the circuit shown, calculate $i_5;\ v_1; v_2; v_5;$ and the power delivered by the source.



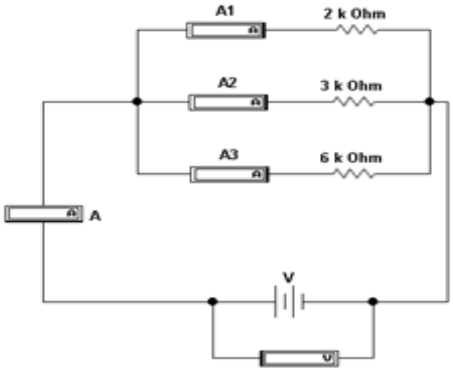
16. Consider the circuit shown in Figure below. Find (a) the current in the 20.0Ω resistor and (b) the potential difference between points a and b.



17. In the circuit given below, find the following: the reading of the ammeter (A); voltmeter (V_1); voltmeter (V_2).



18. In the circuit given below, if the reading of $A_1 = 5\,mA$, then find the following: the reading of A_2; the reading of A_3; the reading of A; and the reading of V.



19. The timing device in an automobile's intermittent wiper system is based on an RC time constant and utilizes a $0.500\text{-}\mu\text{F}$ capacitor and a variable resistor. Over what range must R be made to vary to achieve time constants from 2.00 to 15.0 s?
20. A heart pacemaker fires 72 times a minute, each time a 25.0-nF capacitor is charged (by a battery in series with a resistor) to 0.632 of its full voltage. What is the value of the resistance?
21. The duration of a photographic flash is related to an RC time constant, which is $0.100\,\mu\text{s}$ for a certain camera. (a) If the resistance of the flash lamp is $0.0400\,\Omega$ during discharge, what is the size of the capacitor supplying its energy? (b) What is the time constant for charging the capacitor, if the charging resistance is $800\,\text{k}\Omega$?
22. A parallel-plate capacitor has an area $A = 2.0 \times 10^{-4} m^2$ and a plate separation $d = 1.0\,mm$. Find its capacitance.
23. After two time constants, what percentage of the final voltage, emf, is on an initially uncharged capacitor C, charged through a resistance R?

24. When two unknown resistors are connected in series with a battery, $225\ W$ is delivered to the combination with a total current of $5.00\ A$. For the same total current, 50.0 W is delivered when the resistors are connected in parallel. Determine the values of the two resistors.
25. A $500\text{-}\Omega$ resistor, an uncharged $1.50\text{-}\mu F$ capacitor, and a 6.16-V emf are connected in series. (a) What is the initial current? (b) What is the RC time constant? (c) What is the current after one time constant? (d) What is the voltage on the capacitor after one time constant?
26. A $2.00 - nF$ capacitor with an initial charge of $5.10\ \mu C$ is discharged through a $1.30k\Omega$ resistor. (a) Calculate the current through the resistor $9.00\mu s$ after the resistor is connected across the terminals of the capacitor. (b) What charge remains on the capacitor after $8.00\ \mu s$? (c) What is the maximum current in the resistor?
27. Consider a series RC circuit for which $R = 1.0M\Omega\ , C = 5.0\mu F$ and $\mathcal{E} = 30.0Volt$ Find (a) the time constant of the circuit and (b) the maximum charge on the capacitor after the switch is closed. (c) If the switch is closed at $t = 0$ find the current in the resistor $10.0\ s$ later.
28. In RC circuit the voltage and current expressions are

$$v = 100\exp(-1000t)\ V, \qquad t \geq 0;$$
$$i = 5\exp(-1000t)\quad mA, \qquad t \geq 0^{+}$$

Find $R, C,$ and the time constant τ

29. A heart defibrillator being used on a patient has an RC time constant of $10.0\ ms$ due to the resistance of the patient and the capacitance of the defibrillator. (a) If the defibrillator has an$8.00\mu F$ capacitance, what is the resistance of the path through the patient? (You may neglect the capacitance of the patient and the resistance of the defibrillator.) (b) If the initial voltage is $12.0\ kV$, how long does it take to decline to $6.00 \times 10^2 V$?
30. Using the exact exponential treatment, find how much time is required to discharge a $250\text{-}\mu F$ capacitor through a $500\text{-}\Omega$ resistor down to 1.00% of its original voltage
31. Using the exact exponential treatment, find how much time is required to charge an initially uncharged $100 - pF$ capacitor through a $75.0\text{-}M\Omega$ resistor to 90.0% of its final voltage.

Chapter 7

Waves sound and optics



The physics of waves is important in diverse areas of science from simple radio system, to laser systems, to quantum computing, to seismology and oceanography, and to living cell processes and human communication. Understanding waves is of crucial importance to all science fields**.** Most of the data communications and transfer depends on the understanding of wave nature. The data can be transmitted via two kinds of waves namely: sound waves and electromagnetic waves such as light pulses. Although sound and light are very different phenomena, but both sharing the same nature, which is, the wave nature. A wave can be defined as a disturbance that carries energy from one place to another without mass transfer. The energy carried by the waves stimulates our sensory mechanisms.

In this chapter, we will first explain briefly some general properties of wave motion applicable to both sound and light and then review the nature of sound. Using this background we will examine the process of hearing and some other biological aspects of sound. In the rest of the chapter we are going to discuss the electromagnetic spectrum, light propagation in various media, lens law, eye sight, and eyesight corrections.

7.1 Introduction

Greeks began their study of waves by thinking about music, but now some of physics branches use waves in such away, but may one wander what are the characteristics of the wave? What are their properties? How can waves be described mathematically in a manner that allows us to understand their basic nature and qualities?

The waves are examples of "periodic motion," in that they involve a cyclical pattern of motion. Waves can be classified into two categories namely: mechanical (sound, water waves, etc) waves; and electromagnetic (light, radio, television, laser, x-ray waves, etc) waves. In mechanical waves some physical medium is being disturbed according to wave's source. Electromagnetic waves do not require a medium to propagate.

A wave can be described as a disturbance that travels through a medium from one point to another. For example, when a spring is hanged from one of its ends, and the other end a mass m is attached to the other end. This position is called the equilibrium position. When the spring is stretched (due to outside force, pulled by somebody's hand) from bob and held at rest, the spring is stretched and its turns displaced equally apart, but the spring stores energy within. When the external force is removed the potential energy stored in

the spring tries to restore the equilibrium position, which converts the potential energy to kinetic energy as shown in figure 7.1. Thus, there is a continuing exchange between the potential and kinetic energies; this exchange would create a disturbance in the spring. Since disturbance repeats itself in periodic manner we call this motion a simple harmonic motion. You have to notice that when the bob is moving away from the equilibrium position the spring's coils are moving apart from each other, and when the bob moves toward the equilibrium position the spring's coils are moving close to each other. This fashion is of great importance to explain how sound waves propagate through medium.

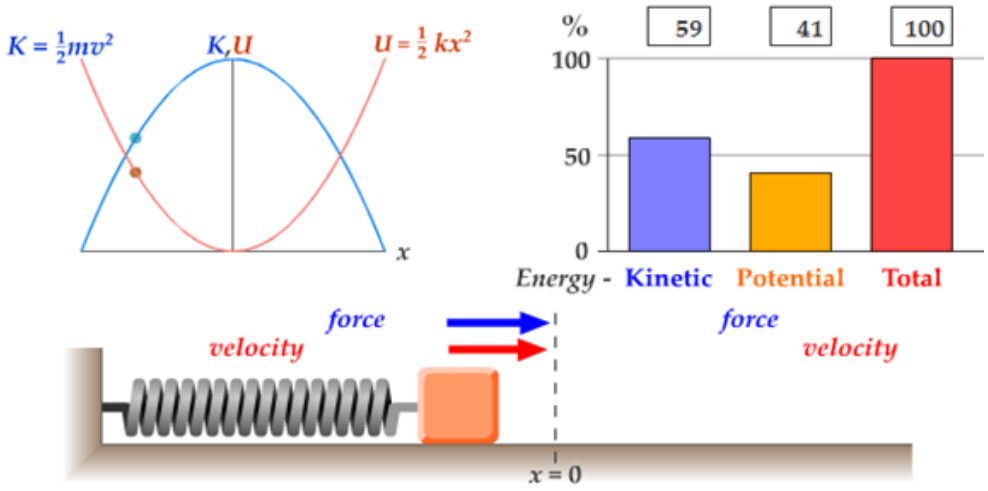


Figure 7.1: Simple harmonic motion by spring

Sound waves are mechanical waves and to demonstrate how these waves propagates in medium, consider a spring is stretched from the equilibrium position, as shown in figure 7.2. Then the holding hand to the spring starts move inward and outward with respect to the spring. Disturbance is observed in the spring, which leads the spring's coils to move toward/away from each others, figure 7.2. Eventually, a disturbance travels through the medium. However, if the spring is long enough we will notice that first coil of the spring is continuously and periodically vibrated in a back-and-forth manner, we would observe a repeating disturbance moving within the spring which endures over some prolonged period of time. This periodic repetition disturbance, which moves through a medium from one location to another, is referred to as a **wave**.

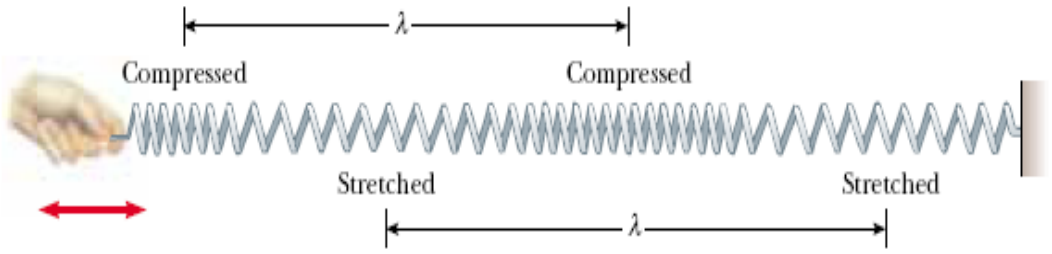


Figure 7.2: A wave along a stretched spring. Each compressed region is followed by a stretched region.

Any medium can be depicted (imagined) as a series of particles connected by tiny springs. As one particle moves, the spring connected to the next particle begins to stretch and applies a force to its adjacent neighbor. Similarly, its neighbor begins to move and stretch its neighbor particle and this process continuo throughout the medium. This interaction one particle of the medium with the next adjacent particle allows the disturbance to travel through the medium. In the case of the spring wave, the particles or interacting parts of the medium are the individual coils of the spring. In the case of a sound wave in air, the particles or interacting parts of the medium are the individual molecules of air.

7.2 Waves Characteristics

All harmonic waves, whether they are sound, light, or water waves, have some things in common. Wave is a disturbance which travels in space and time. This means each particle in the medium oscillates about its equilibrium position without moving around (time) and the wave action (disturbance) also travels throughout the medium (place), see figure 7.3. Thus, the mathematical form, that would represent the wave, should describe these characteristics in both space and time.

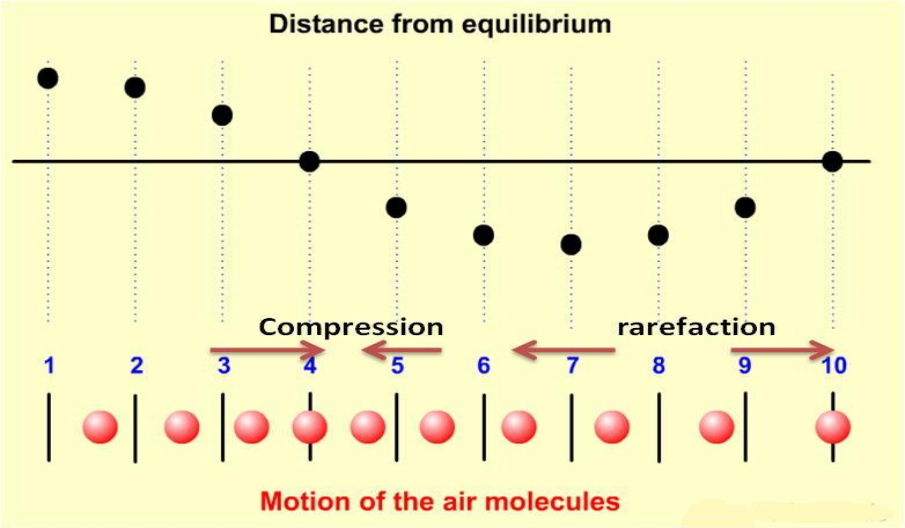


Figure 7.3: Disturbance in the medium, space and time

Fig. 7.4a shows a general form of a wave represented in terms of space, and 7.4b shows it in time.

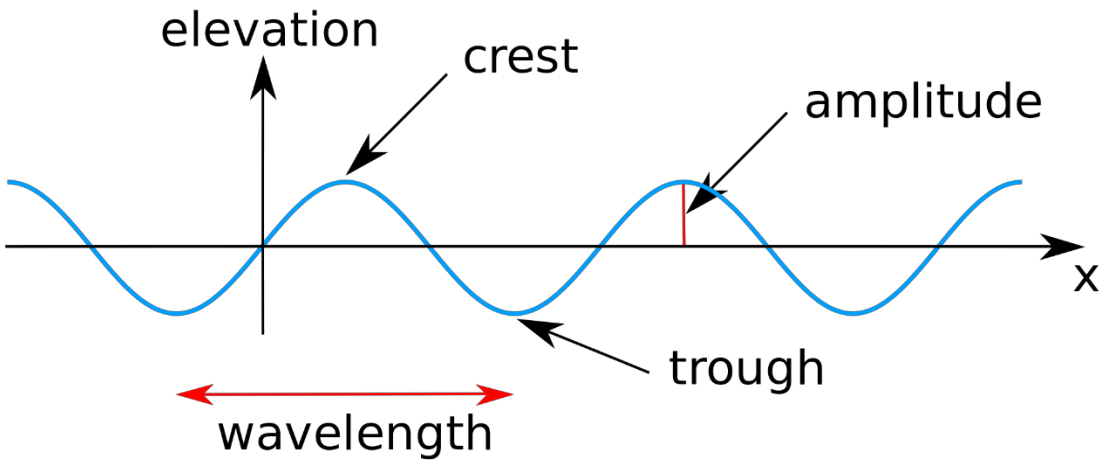


Figure 7.4: General diagram of a wave in terms of space X.

Amplitude and Equilibrium (A): The amplitude (A) of the wave is the maximum displacement of a particle from its equilibrium position (rest position) or the height of the wave. Equilibrium for a pendulum would be standing still: there would be no oscillation occurring at that point in time. However, once a pendulum starts swinging, it crosses this equilibrium point as it swings back and forth. A pendulum is in equilibrium when it is at its lowest point. If it swings $5\ cm$ to either side of this position, its amplitude is said to be $5\ cm$.

Wavelength and Wave Properties (λ): The length of the wave is the wavelength (λ), and is simply the length of one cycle of the wave. The wavelength is shown as the distance between two successive wave crests. The wavelength can also be measured between successive troughs as shown in Fig. 7.4a, or between any two equivalent (have the same phase and displacement) points on the wave.

The period of an oscillation (T): is time interval needed to make one complete a full cycle (back and forth or up and down), figure 7.4b. If two children play on a seesaw bouncing back and forth; the period is the time taken by one of them to start from ground, flies, and then returns back to ground again.

The frequency of a wave (f): is the number of wave cycles completed by one point along the wave in a given time period, usually one second. Wave's frequency is related to the period of a wave by

$$f = \frac{1}{T} \tag{7.1}$$

where f is the frequency and T is the period. The frequency is measured in cycles per second, or hertz (Hz). If the period of a wave is 10 seconds (i.e., it takes 10 seconds for the wave to complete one cycle), then the frequency is $0.1\ Hz$. In other words, the wave completes 0.1 cycles every second.

Wave speed

Remember that a wave is a traveling disturbance. Wave speed is a description of how fast a wave travels. The speed of wave depends on the properties of the medium being disturbed (remember the springs). For instance, sound waves travel through room temperature air with a speed of about $343\ \mathrm{m/s}$, whereas they travel through most solids with a speed greater than 343 m/s. The speed of a wave (ν) is related to the frequency, period, and wavelength by the following simple equations:

$$\nu = \frac{\lambda}{T} \tag{7.2}$$

$$\nu = \lambda f \tag{7.3}$$

Where ν is the wave speed, λ is the wavelength, T is the period, and f is the frequency. Wave speed is commonly measured in units of meters per second ($\mathrm{m/s}$).

Example 7.1

The musical note "A" is a sound wave with a frequency of $440\ Hz$. The wavelength of the wave is $78.4\ cm$. What is the speed of the sound wave?

Solution

To determine the speed of the wave, we can use equation 3 and substitute the given values for wavelength and frequency, making sure we are using the standard units.

$f = 440\ Hz, \qquad \lambda = 78.4cm = 0.784\ m$

$v = \lambda f = (0.784)(440) = 345\ m/s$

This value ($345\ \mathrm{m/s}$) is the approximate value of the speed of sound in air.

7.3 Mathematical Representation of Waves

Any travelling wave can be represented mathematically as a periodic function, like sine or cosine function. After a time t, the wave travel with a speed v a distance $x = vt$, so that we write the wave function in the form

$$y(x,t) = A\,sin[k\,(x - vt)] \tag{7.4}$$

Where k is called the wave number, which is the reciprocal of the wavelength, $\mathrm{k} = \frac{2\pi}{\lambda}$ and it is measured in m^{-1}. A is the amplitude of the wave measured in m. It can be written as

$$y(x,t) = A\,sin[k\,x - \omega t] \tag{7.5}$$

Where ω is the angular frequency measured in Hz or s^{-1};

$$\omega = 2\pi f = 2\pi\frac{v}{\lambda} = kv\ , \tag{7.6}$$

where v is the wave speed measure in m/s. For example, if we have the function

$$y(x,t) = 20\,sin[12\,x - 3t] \quad m, \tag{7.7}$$

We can deduce the parameter of the wave from the previous representation. The amplitude of the wave is $20\ \mathrm{m}$, its wave number $k = 12\ m^{-1}$ which means that its wavelength $\lambda =$

$2\pi/k = 2\pi/12 = \pi/6\ m$. The angular frequency is $3\ \mathrm{s}^{-1}$, so that its frequency f is $\omega/2\pi\ = 3/2\pi\ \mathrm{s}^{-1}$. The speed of propagation of the wave is $v = \lambda f = \frac{\pi}{6} \times \frac{3}{2\pi}\ = 0.25\ \mathrm{ms}^{-1}$.

Example 7.2

A wave can be represented in the form $y = 2\ sin[0.1(x - 5t)]\ m$, find the followings: a) its amplitude, b) its speed of propagation, c) its wavelength, d) its frequency.

Solution

Comparing the representation with the form $y(x,t)\ =\ A\ sin[k\ (x - vt)]$, we can find that:

The amplitude is: $2\ \mathrm{m}$,

The speed of propagation: $v =\ 5\ m/s$,

The wavelength: $\lambda = \frac{2\pi}{k}\ = \frac{2\pi}{0.1} = 20\pi\ m$,

The frequency of the wave: $f = \frac{v}{\lambda} = \frac{5}{20\pi} = \frac{1}{4\pi}\ s^{-1}$

7.4 Types of waves

The waves are classified into two main types, mechanical and electromagnetic waves. Commonly known mechanical waves are sound waves and that for electromagnetic waves are the light waves.

Sound waves are mechanical waves consisting of the vibration of the material or medium through which the wave propagates. For example, when you speak, your vocal cords vibrate air molecules in your throat in a regular pattern; the resulting sound is transmitted through the air as regular vibrations to the listener as a wave travelling at the speed of, $v_s\ =\ 340\ m/s$ (speed of sound in air)

Light waves are electromagnetic waves (EM waves) that need no medium (i.e. material) in which to propagate: they can travel in a vacuum! We know this: light crosses space from the Sun to Earth. All EM waves are oscillating electric and magnetic fields that travel at a speed $\mathrm{c}\ =\ 3 \times 10^8\ \mathrm{m/s}$ (300 million meters per second).

Sound waves

Humans blessed with sensory systems to detect Sound and light. Human eyes are designed to detected light (images), and human ears are designed to detect vibrations like voices, music, etc. We seldom take the time to wander about the characteristics and behaviors of sound and the mechanisms by which sounds are produced, propagated, and detected. The basis for an understanding of sound, music and hearing is the physics of waves. Sound is a wave which is created by vibrating objects and propagated through a medium from one location to another. In this subsection, we will investigate the nature, properties and behaviors of sound waves and apply basic wave principles towards an understanding of hearing process.

Sound waves are divided into three categories that cover different frequency ranges:

1. Audible waves: waves that lie within the range of sensitivity of the human ear. They can be generated in a variety of ways, such as by musical instruments, human vocal cords, and loudspeakers.
2. Infrasonic waves: waves having frequencies below the audible range. Elephants can use infrasonic waves to communicate with each other, even when separated by many kilometers.
3. Ultrasonic waves: waves having frequencies above the audible range. Ultrasonic waves are used widely in many medical applications, such as in medical imaging.

The sound wave can be represented mathematically as a sinusoidal function. *If* $s(x,t)$ is the displacement of a small volume element from its equilibrium position, we can express this harmonic displacement function as

$$s(x,t) = s_{max}\ cos(kx - \omega\, t) \tag{7.8}$$

where s_{max} is the maximum displacement of the medium from equilibrium (in other words, the displacement amplitude of the wave), k is the angular wavenumber or just simply the wavenumber, and ω is the angular frequency. Note that the displacement of the medium is along x, in the direction of motion of the sound wave, which means we are describing a longitudinal wave.

The speed of sound wave c which travels in the medium is determined by the strength of the forces among the molecules. At the macroscopic level these forces are characterized by the bulk modulus B. This quantity is a measure of how hard it is to compress a substance as it was

explained in chapter 3. When the pressure on an object is increased its volume decreases, its density, which is its mass to volume ratio, increases. The bulk modulus relates the fractional change in density to the pressure change as follows

$$\Delta P = B\frac{\Delta\rho}{\rho} \tag{7.9}$$

Where ρ is the medium's density and $\Delta\rho$ is the change in density. The velocity of sound v is found to depend only on the bulk modulus and the density of the medium. Using dimensional analysis one finds

$$v = \sqrt{\frac{B}{\rho}} \tag{7.10}$$

A temperature change accompany the compression and rarefaction in the sound wave and little heat flow occurs so it is necessary to use the adiabatic, not the isothermal, bulk modulus. Table 7.1 lists densities and sound velocity of some materials.

Table 7.1: Densities and sound velocity of different materials

Material	Density (kgm^{-3})	Sound velocity (ms^{-1})
Air	1.20	344
Carbon dioxide($0^{\circ}C$)	1.98	259
Hydrogen ($0^{\circ}C$)	0.0899	1284
Alcohol	790	1207
Benzine	870	1295
Water	998	1498
Aluminum	2700	5000
Copper	8930	3750
Glass (pyrex)	2320	5170
Iron	7900	5120
Fat	900	1450
Muscle	1060	1590
Bone	1500	4080
Blood ($37^{\circ}C$)	1030	1570

Example 7.3

Find the speed of sound in water, which has a bulk modulus of $2.1 \times 10^9\ \mathrm{N/m^2}$ and a density of $10^3\ \mathrm{kgm^{-3}}$. (b) Dolphins use sound waves to locate food. For a bit of food at a distance of $110\ m$, how much time passes between the moment the dolphin emits a sound pulse and the moment the dolphin hears its reflection and thereby detects the distant target?

Solution

$$v_{\text{water}} = \sqrt{\frac{\text{B}}{\rho}} = \sqrt{\frac{2.1 \times 10^9}{1.00 \times 10^3}} = 1.4\text{km/s}$$

–

The total distance covered by the sound wave as it travels from dolphin to target and back is $2 \times 110 = 220\text{m}$ but we know

$$\Delta\text{t} = \frac{\Delta\text{x}}{v} = \frac{220}{1.4 \times 10^3} = 0.16\ \text{s}$$

Example 7.4

Two children are at opposite ends of an iron pipe. One strikes an end of the pipe with a stone. What is the ratio of times it takes the sound waves in air and in the iron to reach the second child?

Solution

The time necessary for a sound wave of velocity v to travel the length of the pipe d is found from $d = vt$ or $t = \frac{d}{v}$ using table 7.1

$$\frac{\text{t}_1}{\text{t}_2} = \frac{\text{d}/v_{\text{air}}}{\text{d}/v_{\text{iron}}} = \frac{\text{c}_{\text{air}}}{\text{c}_{\text{iron}}} = \frac{5120}{344} = 14.9$$

7.5 Sound Power and Intensity

Sound waves, like other waves, transport energy. For a sound wave, the amount of energy transported is proportional to the square of the amplitude of the wave. The relevant physical quantity related to loudness is the sound power produced by the source, or the rate at which sound energy is produced versus time. We know that power is the time rate of change of energy and has units of Joules per second or alternatively $1\ \text{Watt} = 1\ \text{Joule/second.}$

The loudness, we hear, depends on the ratio of the area of our sound collector to the total area of the sphere surrounding the sound source. This motivates us to introduce of another physical quantity associated with sound waves, which is called the sound intensity. The

intensity of a sound wave (I) is the amount of power in the wave per unit area and has units of $\frac{W}{m^2}$.

$$\mathrm{I} = \frac{\text{power}}{\text{area}} = \frac{\mathrm{P}}{4\pi \mathrm{r}^2} \tag{7.11}$$

A sound wave emanated from a source spread in space as spherical wave, which covers a spherical area given by $4\pi R^2$. It follows the inverse square law is given by:

$$\frac{I}{I_0} = \frac{R_0^2}{R^2} \tag{7.12}$$

Here $\mathrm{I_0}$ and I are the sound intensities at radii $\mathrm{R_0}$ and R, respectively. According to the inverse square law, increasing the distance from the source by a factor 2, leads to a decrease of the intensity by a factor of 4.

Example 7.5

A sound source of intensity $10\ \mathrm{W/m^2}$ at a distance of $10\ \mathrm{m}$, what is the power emitted by the source, and what is the density will be at a distance of $2\ \mathrm{m}$?

Solution

The intensity of sound is given by

$\mathrm{I} = \frac{\mathrm{P}}{4\pi \mathrm{r}^2}$, so that $\mathrm{P} = \mathrm{I} \times 4\pi \mathrm{r}^2 = 10 \times 4 \times 3.14 \times (10)^2 = 1.256 \times 10^4 \mathrm{W}$

The intensity at 2 m, is given as

We will introduce a logarithmic scale for sound power, because our perception of sound loudness is logarithmic. We define the sound power level from a source as

$$\frac{\mathrm{I}}{\mathrm{I_0}} = \frac{\mathrm{R_0^2}}{\mathrm{R^2}} \ggg \mathrm{I} = \mathrm{I_0}\frac{\mathrm{R_0^2}}{\mathrm{R^2}} = 1.256 \times 10^4 \times \frac{100}{4} = 3.14 \times 10^5 \mathrm{W}$$

7.6 Sound Level

The human ear can detect sounds with intensity as low as $10^{-12}\mathrm{W/m^2}$ and as high as $1\ \mathrm{W/m^2}$ and above that value it is painful. This range of intensity is called the audible sound which is a pressure wave with frequency between $20\ \mathrm{Hz}$ and $20{,}000\mathrm{Hz}$ For an air temperature of $20°\mathrm{C}$ where the sound speed is $344\ \mathrm{m/s}$, the audible sound waves have wavelengths from

$0.0172\ \text{m}$ to $17.2\ \text{m}$. Usually the relation between sound intensities is specified by the sound intensity levels using a logarithmic scale. The unit of this scale is called the decibel (dB). The intensity level β, of any sound is defined in term of intensity, I, as follows:

$$\beta = 10\ log\frac{I}{I_0} \tag{7.13}$$

Where I_0 is the intensity of some reference level and the logarithm is to the base 10, $\log_{10}$. I_0 is taken as the minimum intensity audible to an average person, which is the threshold of hearing, is equal $1 \times 10^{-12}\ \text{W/m}^2$. For example, the intensity level of a sound of intensity $1 \times 10^{-10}\ \text{W/m}^2$. It will be

$$\beta = 10\ log\frac{1 \times 10^{-10}}{1 \times 10^{-12}} = 20\ dB$$

The intensity level over the audible region ranges between $0dB$ to $120\ dB$. Some sources quote $120\ \text{dB}$ as the pain threshold and define the audible sound frequency range as ending at about $20{,}000\ \text{Hz}$ where the threshold of hearing and the threshold of pain meet. An increase of the intensity by a factor of 100 corresponds to a level increase by $10\ \text{dB}$.

Example 7.6

A loudspeaker is advertised to produce range of frequencies with uniform intensity $\pm 3\text{dB}$. By what factor does the intensity change for the maximum intensity level of 3dB?

Solution

Let us call the average intensity I_1 and the average intensity level is β_1. Then the maximum intensity, I_2, corresponds to level of $\beta_2 = \beta_1 + 3\ \text{dB}$. Thus

$$\beta_2 - \beta_1 = 10\log\frac{\text{I}_2}{\text{I}_0} - 10\log\frac{\text{I}_1}{\text{I}_0}$$

$$3\text{dB} = 10\log\frac{\text{I}_2}{\text{I}_1} \gg\gg\gg\gg\gg\gg\gg \log\frac{\text{I}_2}{\text{I}_1} = 0.3 \gg\gg\gg\gg\gg\gg \frac{\text{I}_2}{\text{I}_1} = 10^{0.3} = 2$$

Or I_2 is twice as intense as I_1.

Example 7.7

The intensity level of the sound from a jet plane at a distance of $30\ \text{m}$ is $140\ \text{dB}$. What is the intensity level at $300\ \text{m}$?

Solution

$$140\ \text{dB} = 10\log\left(\frac{\text{I}}{10^{-12}}\right)$$

Solving the equation we have $10^{14} = \frac{\text{I}}{10^{-12}}$ »»»»»»»»» so $\text{I} = 10^2 \text{W/m}^2$

At 300 m, 10 times as far away, so the intensity will be $\frac{1}{100}$ of I. so at $300\ \text{m}$, the intensity is $1\ \text{W/m}^2$. So the intensity level for $1\ \text{W/m}^2$ is

$$\beta = 10\log\frac{1}{1\times 10^{-12}} = 120\ \text{dB}$$

Even at 300 m, the sound is at the threshold of pain.

7.7 Hearing the Sound

For humans hearing is limited to frequencies between about 20 Hz and 20 kHz, with the upper limit generally decreasing with age. Sound detection requires the conversion of the mechanical vibration of sound waves into a form that permits the analysis of their frequency and intensity. The human ear, figure 7.5, is remarkable sensor as it responds accurately to any variation in sound frequency and intensity. Fig.7.2 shows the anatomy of the human ear. The ear canal, also known as the auditory canal or outer ear, collects sound waves from air and transmits them to the eardrum. The ear canal, because of its tubular shape, acts as a resonator that allows the ear to hear sound frequencies within the resonance range better than others. The middle ear, is designed to increase the transmission of sound waves as they pass from the air to the aqueous fluid of the inner ear, finally the inner ear, in turn, contains sensors that convert water waves into nerve impulses. The following sections describe the physical principles behind these functions.

The Auditory Canal

The outer ear collects sound waves and transmits them to the eardrum. The outer ear acts as a filter, restricting the passage of certain frequencies to the inner ear. This performance can be understood in terms of acoustical resonances discussed in the previous section.

The Middle Ear

Sound waves funneled through the auditory canal cause the tympanic membrane (the eardrum) to vibrate. The vibrations are conducted through the three tiny bones (collectively called the ossicles) of the middle ear to their point of contact on the inner ear known as the oval window.

These ossciles, commonly known as the hammer, anvil, and stirrup, function as a hydraulic press and amplify the force on the oval window to nearly 15 times that of the force on the eardrum. In addition muscle connected to the ossciles control the amplitude of their motion so that loud sounds do not damage the sensitive inner ear.

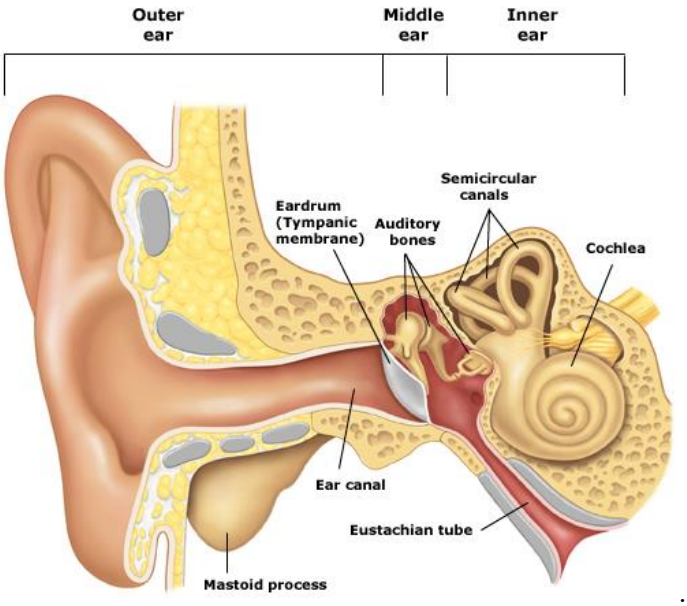


Figure 7.5: Human ear anatomy

The inner ear has two channels filled with fluid. The cochlear duct, containing nerve ending in the organ of Corti, divides the two champers except at the end farthest from the windows. The oval window is driven by one of the ossciles. A flexible round window in the other chamber flexes as the perilymph moves, so the volume of the inner ear stays constant. Because the cochlear duct is thicker near the narrow end of the cochlea, different frequencies of vibration of the perilymph cause the partition to flex at different points along its length. The flexing is sensed by the nerve hairs in the region of excitation, and nerve impulses travel towards the brain.

7.8 The Doppler Effect

You have probably heard of the Doppler Effect in association with recent improvements in radar technology: Metrologies uses "Doppler Radar" to predict weather changes, since Doppler Rader is capable of measuring the velocities of winds, and is instrumental in the identification of tornados. I think, you might have notices the pitch (frequency) of the horn

or siren of an approaching vehicle is higher than when it passes you and when it moves away. This "Doppler Shift" in the frequency also has an important medical usage in the measurement of the speeds of moving fluids inside the body.

When an object which emits a wave (sound or light), the frequency is determined by the object itself. The wavelength, however, is a function of the speed of propagation in the medium as well as the motion of the object within the medium. When the source of the wave approaches you at a speed u_s, figure 7.6, the wavelength is shortened, observer A, by an amount u_sT, where T is the period of the wave. This is simply due to the motion of the source: when the wave cycle started, the source was at point "a", but when the cycle ends the source has moved u_sT closer. Since the "received" wavelength is related to the "source" wavelength by

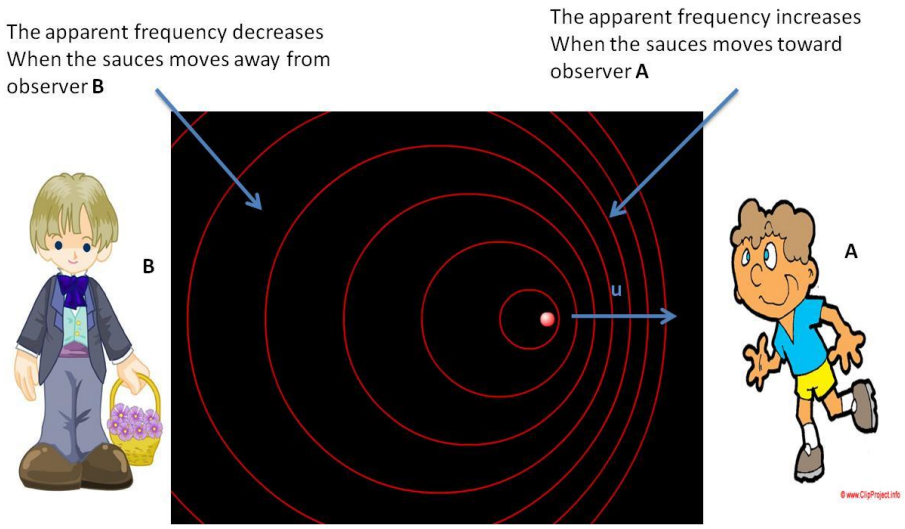


Figure 7.6: Doppler effect

$$\lambda_r = \lambda_s - u_sT = \lambda_s - u_s\frac{\lambda_s}{v} = \lambda_s\left(\frac{v-u_s}{v}\right) \qquad (7.14)$$

Where v is the speed of the wave. The received frequency $\nu_r = \frac{v}{\lambda_r}$ is related to the source frequency by

$$\nu_r = \nu_s\left(\frac{v}{v-u_s}\right) \qquad (7.15)$$

This received frequency is heard, when the source is moving towards the receiver. However when it is moving away from the receiver, observer B in figure 7.6, then the received frequency is given by

$$\nu_r = \nu_s\left(\frac{v}{v+u_s}\right) \qquad (7.16)$$

Hence the frequency you hear is higher than the frequency emitted by the approaching source. As the source passes you and recedes, the "speed of approach" us becomes negative, and the frequency you hear becomes lower than the frequency emitted by the moving away source. This shift in frequency is the Doppler Shift.

The same principle applies when the source is stationary but you are approaching it at a speed $\mathrm{u_r}$. Now the received wavelength is related to the source wavelength by

$$\lambda_r = \lambda_s - u_r T = \lambda_s - u_r \frac{\lambda_r}{v} \tag{7.17}$$

$$\lambda_r = \lambda_s \left(\frac{v}{v+u_r}\right) \tag{7.18}$$

(since the moving receiver now determines the period of the wave) and the received frequency is related to the source frequency by

$$\nu_r = \nu_s \left(\frac{v+u_r}{v}\right) \tag{7.19}$$

This received frequency is heard, when the receiver is moving towards the source. However when it is moving away from the source, then the received frequency is given by

$$\nu_r = \nu_s \left(\frac{v-u_r}{v}\right) \tag{7.20}$$

Example 7.8

A stationary civil siren has a frequency of $1000\mathrm{Hz}$. What frequency will be heard by drivers of cars moving at $15\mathrm{m/s}$ (a) away from the siren (b) toward the siren

Solution

With u_r = 15ms-1, drivers moving away from the siren hear a frequency

$$\nu_r = \nu_s \left(\frac{v-u_r}{v}\right)$$

$$= 1000 \frac{(344-15)}{344} = 956\mathrm{Hz}$$

using u_r = -15ms-1 for a driver moving toward the source

$$\nu_r = \nu_s \left(\frac{v+u_r}{v}\right) = 1000 \frac{(344+15)}{344} = 1044Hz$$

If both the source and receiver are moving and $\mathrm{u_s}$ and $\mathrm{u_r}$ are the speeds with which they are approaching each other (respectively), the Doppler Shift is

$$\nu_r = \nu_s \left(\frac{v+u_r}{v-u_s}\right) \tag{7.21}$$

How can this be used to measure (for instance) the speed of flowing blood in vivo ((Latin word for "within the living")? If u is the speed of the blood and ν_s is the frequency of an ultrasonic source, two Doppler Shifts occur as the source is reflected from the moving blood. First the frequency is shifted as the blood "receives" it, since the blood is moving toward the source. Then, when the echo is "sent" from the blood to the monitor the frequency is shifted again, this time due to the motion of the source. The resultant shift is

$$\nu_r = \nu_s \left(\frac{v+u}{v-u}\right) \tag{7.22}$$

This allows us to determine the speed of the blood as

$$\mathrm{u} = \mathrm{v}\left(\frac{\nu_r - \nu_s}{\nu_r + \nu_s}\right) \tag{7.23}$$

So if the speed of sound in blood is 1500 m/s and a transmission of 1 MHz results in a 1.05 MHz echo, the speed of the blood is 36.6 m/s. Since the measurement is instantaneous, the speed of blood during laminar flow can be measured as a function of time.

7.9 Clinical Uses of Sound

The most familiar clinical use of sound is in the analysis of body sounds with a stethoscope. This instrument consists of a small bell-shaped cavity attached to a hollow flexible tube. The bell is placed on the skin over the source of the body sound (such as the heart or lungs). The sound is then conducted by the pipe to the ears of the examiner who evaluates the functioning of the organ.

A modified version of the stethoscope consists of two bells that are placed on different parts of the body. The sound picked up by one bell is conducted to one ear, and the sound from the other bell is conducted to the other ear. The two sounds are then compared. With this device, it is possible, for example, to listen simultaneously to the heartbeats of the fetus and of the pregnant mother.

Ultrasound is cyclic sound pressure with a frequency greater than the upper limit of human hearing. Although this limit varies from person to person, it is approximately 20 kilohertz ($20{,}000\ \mathrm{Hz}$) in healthy, young adults and thus, $20\ \mathrm{kHz}$ serves as a useful lower limit in

describing ultrasound, figure 7.7. The production of ultrasound is used in many different fields, typically to penetrate a medium and measure the reflection signature or supply focused energy. The reflection signature can reveal details about the inner structure of the medium. Over the past two decades ultrasound has undergone numerous advances in technology such as gray-scale imaging, real-time sonography, high resolution $7.5 - 10\ \text{MHz}$ transducers, and color-flow Doppler and more. The most well its applications of this technique is its use in sonography to produce pictures of fetuses in the human womb. There are a vast number of other applications as well.

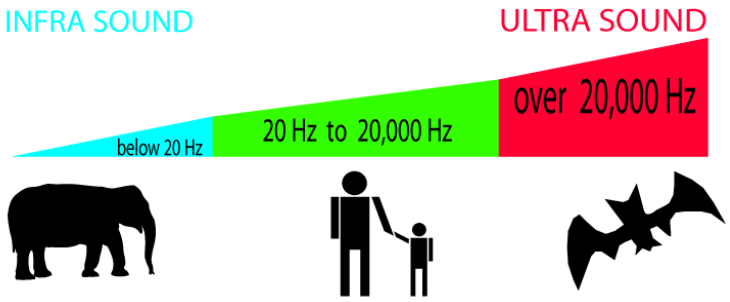


Figure 7.7: Sound waves frequency Spectrum

Ultrasound used for medical purposes is from one MHz (one million cycles per second) to 20 MHz. Ultrasound imaging does not usually use higher than 10 MHz. Higher frequency ultrasound waves can form sharper images, but the images are fainter because tissues absorb higher frequency energy more readily. Just like any other type of sound, the higher the frequency of ultrasound, and the shorter the wavelength. Ultrasound has a wavelength of about $1.5\ \text{mm}$.

As we have already learnt the speed of ultrasound, like any other sound wave, does not depend on its frequency. The speed of ultrasound depends on what material or tissue it is traveling in. The mass and spacing of the molecules of the material and the attracting force between the particles of the material all have an effect on the speed of the ultrasound as it passes through. Ultrasound travels faster in dense materials and slower in compressible materials. In soft tissue sound travels at $1500\ \text{m/s}$, in bone about $3400\ \text{m/s}$, and in air $330\ \text{m/s}$.

Ultrasound is reflected at the boundaries between different materials. Ultrasound reflects very well wherever soft tissue meets air, or soft tissue meets bone, or where bone meets air. Frequency is unchanged as sound travels through various tissues. That means that in tissues where sound travels more slowly, wavelength decreases. Just as the spacing between cars on a

highway narrows when they slow down for construction, the compression areas of a wave get jammed together when sound slows down.



Fig. 7.8: An ultrasound image of a human fetus in the womb after 20 weeks of development, showing the head, body, arms, and legs in profile.

Ultrasound waves are produced by a transducer. A transducer is a device that takes power from one source, converts the energy into another form, and delivers the power to another target. In this case the transducer acts like a loudspeaker and a microphone. The transducer converts electrical signals to ultrasound waves, and picks up the reflected waves converting them back into electrical signals. Ultrasound imaging is based on the same principles involved in the sonar used by bats, ships and fishermen. When a sound wave strikes an object, it bounces backward, or echoes. By measuring these echo waves it is possible to determine how far away the object is and its size, shape, consistency (whether the object is solid, filled with fluid, or both) and uniformity.

In medicine, ultrasound is used to detect changes in appearance and function of organs, tissues, or abnormal masses, such as tumors.

In an ultrasound examination, a transducer both sends the sound waves and records the echoing waves. When the transducer is pressed against the skin, it directs a stream of inaudible, high-frequency sound waves into the body. As the sound waves bounce off of internal organs, fluids and tissues, the sensitive microphone in the transducer records tiny changes in the sound's pitch and direction. These signature waves are instantly measured and used to form pictures on a television screen. Therefore, structures within living organisms can be examined with ultrasound, as with X-rays. Ultrasonic examinations are safer than X-rays and often can

provide as much information. In some cases, such as in the examination of a fetus and the heart, ultrasonic methods can show motion, which is very useful in such displays.

7.10 Electromagnetic Spectrum

Light is an electromagnetic wave An Electromagnetic disturbance that propagates through space as a wave may be monochromatic characterized by single wavelength (distinct color), or polychromatic (many colors) in which it is represented by many wavelengths. The distribution of energy ($E = h\nu$, h is Plank's constant and ν is the frequency) among various constituent waves is called spectrum of the radiation, as we can say that the visible light spectrum extends between 380 nm to 770nm. Thus all wavelengths in this range are visible to us. Various regions of the electromagnetic spectrum are referred to by particular names, figure 7.9, such as radio waves, cosmic rays, visible light, and ultraviolet radiations. These categories based on the way how they are produced or detected.

Radio waves are the result of charges accelerating through conducting wires. Ranging from more than $10^4 m$ to about $0.1\ m$ in wavelength they are generated by such electronic devices as LC oscillators and are used in radio and television communication systems.

Microwaves have wavelengths ranging from approximately $0.3\ m$ to $10^{-4} m$ and are also generated by electronic devices. Because of their short wavelengths, they are well suited for radar systems and for studying the atomic and molecular properties of matter.

Infrared waves have wavelengths ranging from $10^{-3} m$ to the longest wavelength of visible light, $7 \times 10^{-7} m$. These waves, produced by molecules and room-temperature objects, are readily absorbed by most materials. Infrared radiation has practical and scientific applications in many areas, including physical therapy, IR photography, and vibrational spectroscopy.

Visible light, the most familiar form of electromagnetic waves, is the part of the electromagnetic spectrum that the human eye can detect. Light is produced by the rearrangement of electrons in atoms and molecules. The various wavelengths of visible light, which correspond to different colors, range from red (770 nm) to violet (380 nm). The sensitivity of the human eye is a function of wavelength, being a maximum at a wavelength of about 550 nm.

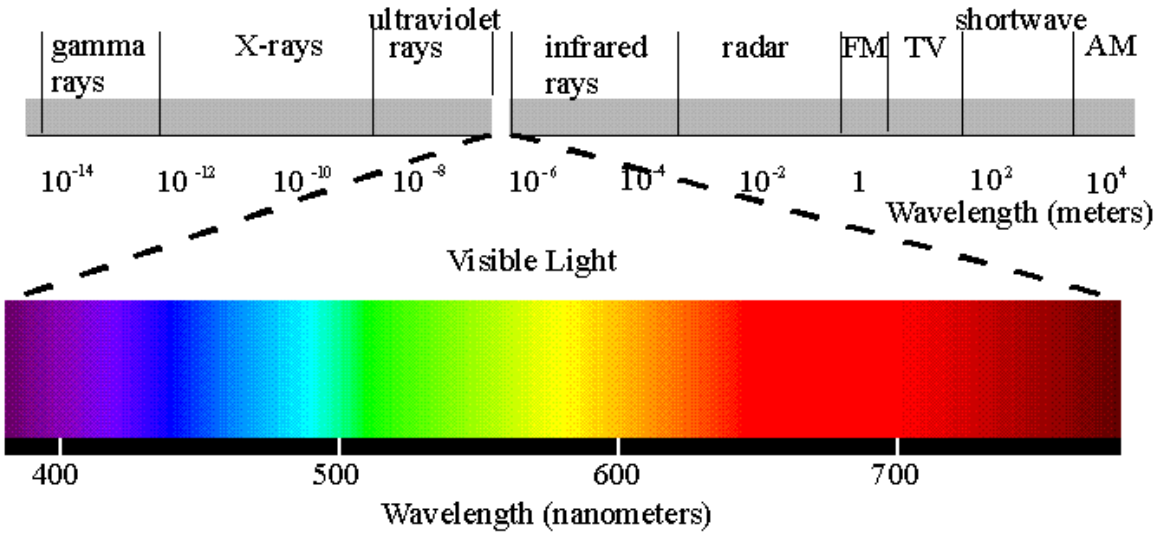


Figure 7.9: Electromagnetic spectrum

Ultraviolet waves cover wavelengths ranging from approximately $4 \times 10^{-7}\text{m}$ to $6 \times 10^{-10}\text{m}$. The Sun is an important source of ultraviolet (UV) light, which is the main cause of sunburn. Ultraviolet rays have also been implicated in the formation of cataracts, a clouding of the lens inside the eye. Most of the UV light from the Sun is absorbed by ozone (O3) molecules in the Earth's upper atmosphere, in a layer called the stratosphere. This ozone shield converts lethal high-energy UV radiation to infrared radiation, which in turn warms the stratosphere.

X-rays have wavelengths in the range from approximately $10^{-8}\ \text{m}$ to $10^{-12}\ \text{m}$. The most common source of x-rays is the deceleration of high-energy electrons bombarding a metal target. X-rays are used as a diagnostic tool in medicine and as a treatment for certain forms of cancer. Because x-rays damage or destroy living tissues and organisms, care must be taken to avoid unnecessary exposure or overexposure. X-rays are also used in the study of crystal structure because x-ray wavelengths are comparable to the atomic separation distances in solids (about 0.1 nm).

7.11 Geometrical Optics: A review

Properties of Light

In a vacuum, all electromagnetic waves, including visible light, travel at a speed about, $3.0 \times 10^{8}\ \text{m/s}$. The speed of light in air is approximately the same, but in water and other dense

transparent media, it travels significantly slower. The index of refraction is denoted by n, which is the ratio of the speed of light c in vacuum to the speed v in the material, that is

$$n = c/v \tag{7.24}$$

Note that the larger the value of n, the slower the speed of light in the medium. The speed of light waves in a medium v is related to frequency f and wavelength λ according to the well-known equation

$$v = f\lambda \tag{7.25}$$

Note that the value of n for a given material is smaller for long wavelengths than it is for short ones. It is worthwhile to mention that the frequency of the wave doesn't change in vacuum or in medium thus we can write the refractive index as,

$$n = \frac{c}{v} = \frac{\lambda_0 f}{\lambda_m f} = \frac{\lambda_0}{\lambda_m} \tag{7.26}$$

From equation 7.25 and 7.26, we notice that the dependence of v and n on wavelength. This dependence is called dispersion. We will approximate the index of refraction in air by unity and in water by 1.33; these values are measured using light with a wavelength of $589\ \text{nm}$ in a vacuum.

Example 7.9

If the North Star, Polaris, were to burn out today, in what year would it disappear from our vision? Take the distance from the Earth to Polaris as 6.44×10^{18} m.

Solution

Since the light from this star travels at 3.00 × 108 m/s, the last bit of light will hit the Earth in

$$\frac{(6.44 \times 10^{18}\ \text{m})}{\left(3.00 \times 10^{8}\ \frac{\text{m}}{\text{s}}\right)} = 2.15 \times 10^{10}\ \text{s} = 680\ \text{years}$$

Therefore, it will disappear from the sky in the year $2009 + 680 = 2689\ \text{A.D.}$

Example 7.10

A half-wave antenna works on the principle that the optimum length of the antenna is one-half the wavelength of the radiation being received. What is the optimum length of a car antenna when it receives a signal of frequency $94.0\ \text{MHz}$?

Solution

The wavelength of the signal is

$$\lambda = \frac{c}{f} = \frac{(3.0 \times 10^8 \text{m/sec})}{(94 \times 10^6 \text{sec}^{-1})} \cong 3.2\ \text{m}$$

Thus, to operate most efficiently, the antenna should have a length of $(3.2\ \text{m})/2 = 1.60\ \text{m}$. For practical reasons, car antennas are usually one-quarter wavelength in size.

Reflection

When a narrow ray of light is incident on certain shiny surface, this ray is defined as the incident ray. The point of incidence is where the incident ray strikes a surface. The normal is a construction line drawn perpendicular to the surface at the point of incidence. The reflected ray is the portion of the incident ray that leaves the surface at the point of incidence. The angle of incidence is the angle between the incident ray and the normal. The angle of reflection is the angle between the normal and the reflected ray, see figure 8.2

Al-Hasan Ibn Al-Haytham, is regarded as the "father of modern optics" (called Alhazen, or Al-Basri), stated the law of reflections as follow:

The incident ray, the reflected ray, and the normal to the reflecting interface at the point of incidence all lie in the same plane.

The angle of incidence equals the angle of reflection

Snell's law: Refraction

If a light ray traveling through medium a strikes medium b with an angle θ_a relative to the normal, figure 7.10, the angle θ_b relative to the normal at which the ray passes through medium b is predicted by the law of refraction

$$n_a \sin\theta_a = n_b \sin\theta_b \qquad (7.27)$$

The change in direction that takes place at the interface of the two media results from the change in wavelength and wave speed of the electromagnetic wave.

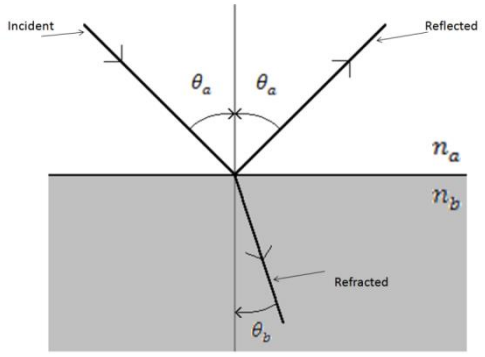


Figure 7.10: shows an incident ray from less-dense medium n_a on more-dense medium n_b. The refracted ray gets closer to the normal

When $n_a < n_b$, light is bent towards the normal. When $n_b < n_a$, light is bent away from the normal. Then at a certain angle of incidence the refraction angle, θ_b , is 90°. Thus the angle of incidence is called the critical angle, θ_c. Any angle of incidence greater than the critical angle the incident ray is reflected by an angle equals the angle of incidence, figure 7.11. That is

$$n_a \sin\theta_a = n_b \sin 90^\circ = n_b$$

$$\sin\theta_c = n_b/n_a \qquad (7.28)$$

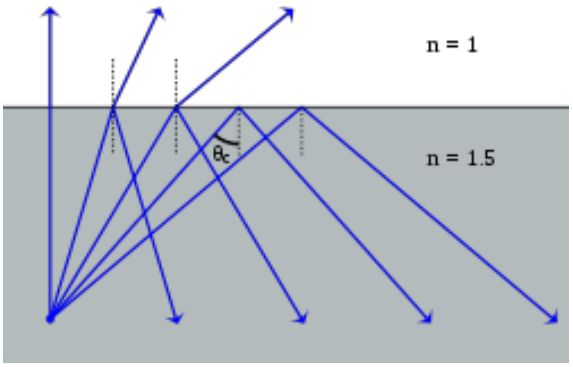


Figure 7.11: critical angle, notice any ray incident with an angle greater than the critical it exhibits total internal reflected, TIR.

As we shall see, different parts of the eye have different indices of refraction. These components are able to bend light in just the right way to produce a clear image on the retina of a normal eye.

Example 7.11

An underwater scuba diver sees the Sun at an apparent angle of 45.0° from the vertical. What is the actual direction of the Sun?

Solution

$n_1 \sin\theta_1 = n_2 \sin\theta_2 \Longrightarrow 1 \times \sin\theta_1 = 1.33 \times \sin 45$

$\sin\theta_1 = 1.33 \times \sin 45 = 0.94 \Longrightarrow \theta_1 = \sin^{-1} 0.94$

$\theta_1 = 70.13^{\circ}$

Thus, the angle to the horizontal is $90^{\circ} - 70.13^{\circ} = 19.87^{\circ}$

Example 7.12

A glass fiber $n = 1.5$ is submerged in water $n = 1.33$. What is the critical angle for light to stay inside the optical fiber?

Solution

$$\sin\theta_c = \frac{n_b}{n_a} \Longrightarrow \frac{1.33}{1.5} = 0.8866$$

$\theta_c = \sin^{-1} 0.8866 = 62.46^{\circ}$

Thin Lenses

Generally, thin lenses are the most used lenses in the practice. Most lenses are spherical lenses: their two surfaces are parts, with the same axis as each other, of the surfaces of spheres. Each surface can be convex (bulging outwards from the lens) figure 7.12a, concave (depressed into the lens) figure 7.12b. The line joining the centers of the spheres making up the lens surfaces is called the axis of the lens. Typically the lens axis passes through the physical centre of the lens, because of the way they are manufactured. Lenses may be cut or ground after manufacturing to give them a different shape or size. The lens axis may then not pass through the physical centre of the lens.

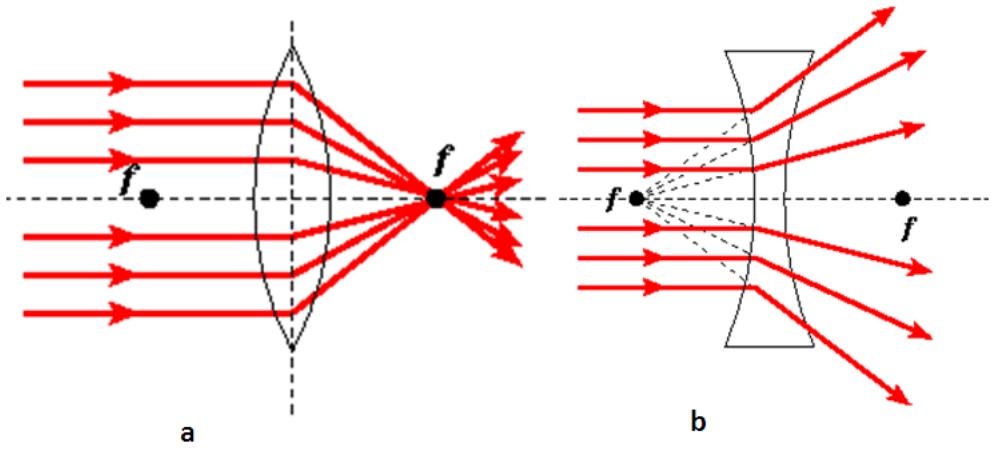


Figure 7.12: a. converging and **b.** diverging lens

When parallel incoming rays are incident one side of a thin lens, these rays will collimate at a certain point on the other side. This point is called the focal point f, and the lens focal length is the shortest distance from f to the center of the lens, figure 8.3. There are two main types of thin lenses: converging (positive) lenses and diverging (negative) lenses. Converging lenses, also called focusing lenses, are thicker at their center than at their edges and have a positive focal length (forms real images), figure 7.12a; diverging lenses are thicker at their edges than at their centers and have a negative focal length (forms imaginary images), figure 7.12b. Determining the focal length of a diverge lens: parallel light rays entering a diverging lens, if we were to extend the diverging rays back to the left, they would intersect (converge) at the focal point f on the incoming side of the lens figure 7.12b. This is why we define the focal length to be negative. Similarly, rays initially aimed at the focal point on the outgoing side of the lens would pass through the lens and exit parallel to one other.

As we mentioned before, a thin lens is formed from two spherical surfaces. The radius of curvature of these surfaces controls the focal length of the lens and the refractive index of both the lens material and the surrounding medium, that is

$$\frac{1}{\mathrm{f}} = (\mathrm{n_{lens}} - \mathrm{n_{medium}})\left(\frac{1}{\mathrm{R_1}} - \frac{1}{\mathrm{R_2}}\right) \qquad (7.29)$$

Here, R_1 is the radius of curvature of the first surface light encounters as it passes through the lens, while R_2 is the radius of curvature of the exiting surface.

Sign convention of lens radii R_1 and R_2

R_1 is positive the first surface is convex, and if R_1 is negative the surface is concave. The signs are reversed for the back surface of the lens: if R_2 is positive the surface is concave, and if R_2 is negative the surface is convex. If either radius is infinite, the corresponding surface is flat.

Example 7.13

A contact lens is made of plastic with an index of refraction of 1.50. The lens has an outer radius of curvature of +2.00 cm and an inner radius of curvature of +2.50 cm. What is the focal length of the lens?

Solution

R_1 is the outer radius and R_2 is the inner radius, substitute in the lens maker's formula gives

$$\frac{1}{f} = (1.5 - 1)\left(\frac{1}{2cm} - \frac{1}{2.5cm}\right) = 0.5 \times 0.1 = 0.05 \rightarrow f = 20cm$$

Example 7.14

A converging glass $(n = 1.52)$ lens has a focal length of 40.0 cm in air. Find its focal length when it is immersed in water, which has an index of refraction of 1.33.

Solution

We can use the lens makers' equation in both cases, noting that R1 and R2 remain the same in air and water

$$\frac{1}{f_{air}} = (n_{lens} - n_{air})\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{i}$$

$$\frac{1}{f_{water}} = (n_{lens} - n_{water})\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{ii}$$

Divide equation i by equation ii, we have

$$\frac{f_{water}}{f_{air}} = \frac{(n_{lens} - n_{air})}{(n_{lens} - n_{water})} = \frac{1.52 - 1}{1.52 - 1.33} = 2.73$$

Thus,

$$f_{water} = 2.73 f_{air} = 2.73 \times 40cm \cong 109.5cm$$

Imaging with a thin lens

Now, if a is the distance from the object to the lens, and b is the distance from the lens to the image. Then a and b are related to f by the thin lens formula, figure 7.13

$$1/f = 1/a + 1/b \qquad (7.30)$$

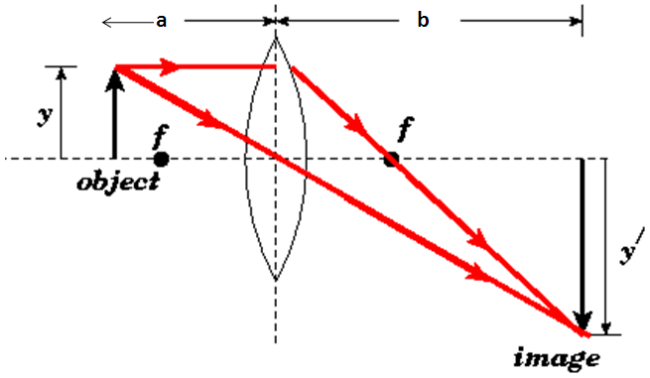


Figure 7.13: Ray diagram of a thin lens

What this means is that, if an object is placed at a distance a along the axis in front of a positive lens of focal length f, a screen placed at a distance b behind the lens will have a sharp image of the object projected onto it, as long as $a > f$.

Sign convention

Figure 8.5 is useful for obtaining the signs of a and b. Applying these rules to a convex lens, we see that when $a > f$, the quantities a, b, and R_1 are positive, and R_2 is negative. Therefore, a, b, and f are all positive when a converging lens forms a real image of an object. For a concave lens, a and R_2 are positive and b and R_1 are negative, with the result that f is negative.

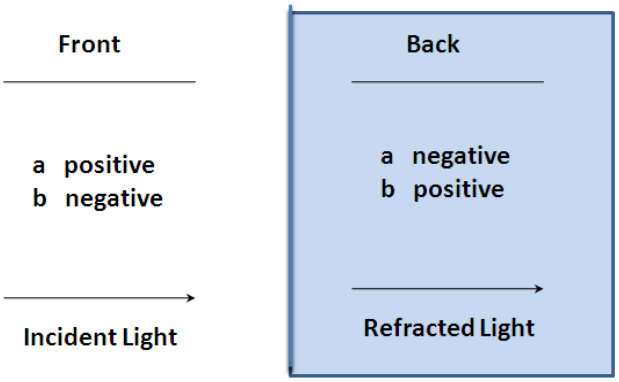


Figure 7.14: The lenses sign convention for thin lens formula

Magnification of Images

Consider a thin lens through which light rays from an object pass, as in figure 7.13. The lateral magnification of the lens is defined as the ratio of the image height $\acute{y}$ to the object height y:

$$M = -\frac{\acute{y}}{y} \tag{7.31}$$

From this expression, it follows that when M is positive, the image is upright and on the same side of the lens as the object. When M is negative, the image is inverted and on the side of the lens opposite the object.

Optical power

Optical power (refractive power, focusing power, or convergence power) is the degree to which a lens, mirror, or other optical system converges or diverges light. It is equal to the reciprocal of the focal length of the device.

$$\mathbb{P} = \frac{1}{f} \tag{7.32}$$

The dioptre is the most common unit of measurement of optical power. The SI unit for optical power is the inverse meter (m^{-1}).

The optical power of a single lens is roughly equal to the sum of the powers of each surface. These approximations are commonly used in optometry. When a lens is immersed in a refractive medium, its optical power and focal length change.

An eye that has too much or too little refractive power to focus light onto the retina has a refractive error. A myopic eye has too much power so light is focused in front of the retina. Conversely, a hyperopic eye has too little power so when the eye is relaxed, light is focused behind the retina. An eye with a refractive power in one meridian that is different from the refractive power of the other meridians has astigmatism. Anisometropia is the condition in which one eye has a different refractive power than the other eye.

Example 7.15

A converging lens of focal length 10.0 cm forms an image of each of three objects placed (a) 30.0 cm, (b) 10.0 cm, and (c) 5.00 cm in front of the lens. In each case, find the image distance and describe the image.

Solution

Using thin lens formula

$$\frac{1}{30}+\frac{1}{b}=\frac{1}{10}\Longrightarrow b=15.0\text{cm}$$

The positive sign indicates that the image is in back of the lens and real. The magnification is

$$M=-\frac{b}{a}=-\frac{15.0}{30.0}=-0.5$$

The image is reduced in size by one half, and the negative sign for M means that the image is inverted

When the object is placed at the focal point, the image is formed at infinity.

We now move inside the focal point, to an object distance of 5.00 cm:

$$\frac{1}{5.0\text{cm}}+\frac{1}{b}=\frac{1}{10}\Longrightarrow b=-10\text{cm}$$

The negative image distance indicates that the image is in front of the lens and virtual. The magnification is

$$M=-\frac{b}{a}=-\frac{-10}{5.0}=2.0$$

The image is enlarged, and the positive sign for M tells us that the image is upright

Example 7.16

A diverging lens has a focal length of -20.0 cm. An object 2.00 cm tall is placed 30.0 cm in front of the lens. Locate the image.

Solution

Using the thin-lens equation with $p=30.0\text{ cm}$ and $f=-20\text{cm}$ we obtain

$\frac{1}{30}+\frac{1}{b}=\frac{1}{-20}$ Solving for b,

$b=-120\text{cm}$

The negative sign tells us that the image is in front of the lens and virtual,

Magnifying glass

Magnifying glass consists of one converging lens. This device increases the apparent size of an object. Suppose an object is viewed at some distance p from the eye, figure 7.14. The size of the image formed at the retina depends on the angle θ subtended by the object at the eye.

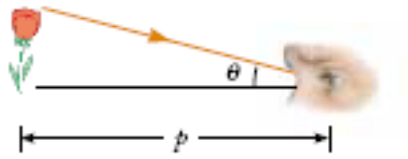


Figure 7.14: The size of the image formed on the retina depends on the angle θ subtended at the eye.

As the object moves closer to the eye, θ increases and a larger image is observed. However, an average normal eye cannot focus on an object closer than about 25 cm, the near point (Fig. 7.15a). Therefore, θ is maximum at the near point. To further increase the apparent angular size of an object, a converging lens can be placed in front of the eye as in Figure 7.15b, with the object located at point O, just inside the focal point of the lens. At this location, the lens forms a virtual, upright, enlarged image. We define angular magnification m as the ratio of the angle subtended by an object with a lens in use (angle θ in Fig. 7.15b) to the angle subtended by the object placed at the near point with no lens in use (angle θ_0 in Fig. 7.15a):

$$m = \theta/\theta_0 \qquad (7.33)$$

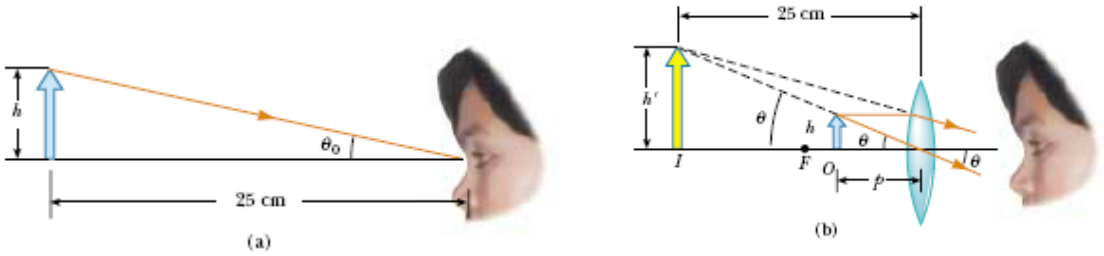


Figure 7.15: (a) An object placed at the near point of the eye ($p = 25\ cm$) subtends an angle $\theta_0 = h/25$at the eye. (b) An object placed near the focal point of a converging lens produces a magnified image that subtends an angle $\theta = h'/25$ at at the eye.

The angular magnification is a maximum when

$$\frac{1}{p}+\frac{1}{-25cm}=\frac{1}{f}\Longrightarrow p=\frac{25f}{25+f} \quad (7.34)$$

Then the maximum angular magnification is

$$m_{max}=1+\frac{25cm}{f} \quad (7.35)$$

Since the eye can focus on images formed anywhere: between the near point and infinity. It is most relaxed when the image is at infinity. When the object is at the focal length of the magnifying glass the image formed at infinity, in that sense the eye is relaxed, the angular magnification is

$$m_{min}=\frac{25cm}{f} \quad (7.36)$$

Example 7.17

What is the maximum magnification that is possible with a lens having a focal length of 10 cm, and what is the magnification of this lens when the eye is relaxed?

Solution

The maximum magnification occurs when the image is located at the near point of the eye. Under these circumstances, Equation 7.35 gives

$$m_{max}=1+\frac{25cm}{f}=1+\frac{25cm}{10cm}=3.5$$

When the eye is relaxed, the image is at infinity. In this case, we use Equation 7.36:

$$m_{min}=\frac{25cm}{f}=\frac{25cm}{10cm}=2.5$$

The Compound Microscope

In practice, the magnification of the magnifying glass is not enough, i.e. human cell structure cannot be seen by the aid of the magnifying glass. Thus a bigger magnification factor is needed. To achieve this higher magnification factor, microscope is designed to do so. Generally, Microscope constructed by combining two lenses, a schematic diagram of which is shown in Figure 7.16a. The first lens is called the objective lens, because it is close to the objects to be seen, which has a very short focal length $\mathrm{f_0} < 1\ cm$; the second lens is called the eyepiece,

because it is close to the observer eye, which has a focal length f_e of a few centimeters. The distance of separation between the two lenses is L, which is much greater than either f_0 or f_e . In order to operate microscope properly, the object has to be placed at a distance greater than the focal length of the objective lens f_0(why: to form a real image between the objective lens and the eyepiece). This image has lateral magnification $M_o \cong -L/f_0$. When you design a microscope, you make sure that the objective lens real image is located at a distance less than the focal length of the eyepiece (why: review the magnify glass).

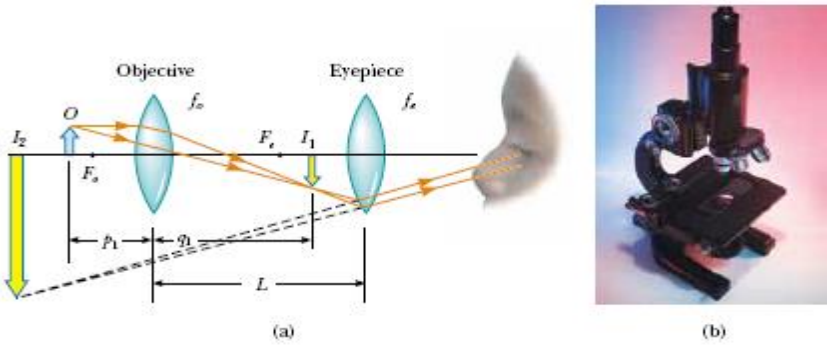


Figure 7.16: (a) Diagram of a compound microscope, which consists of an objective lens and an eyepiece lens. (b) A compound microscope

The object of the eyepiece is the real image formed from the objective lens. So, the observer sees from the eyepiece inverted virtual image with angular magnification $M_e = 25cm/f_e$. Thus the overall magnification is

$$M = M_o\, M_e = \frac{25cm}{f_e} \times -\frac{L}{f_0} \qquad (7.37)$$

The negative sign indicates that the image is inverted.

Example 7.18

The distance between the eyepiece and the objective lens in a certain compound microscope is 23.0 cm. The focal length of the eyepiece is 2.50 cm, and that of the objective is 0.400 cm. What is the overall magnification of the microscope?

Solution

$$M = M_o\, M_e = \frac{25cm}{f_e} \times -\frac{L}{f_0} = \frac{25cm}{2.5cm} \times \frac{23cm}{0.4cm} = -575$$

7.11 Human Eye

The components of the eye are as follows, figure 7.17:

The cornea is an optically clear membrane that holds in place the fluids inside the eye. It is roughly spherical in shape.

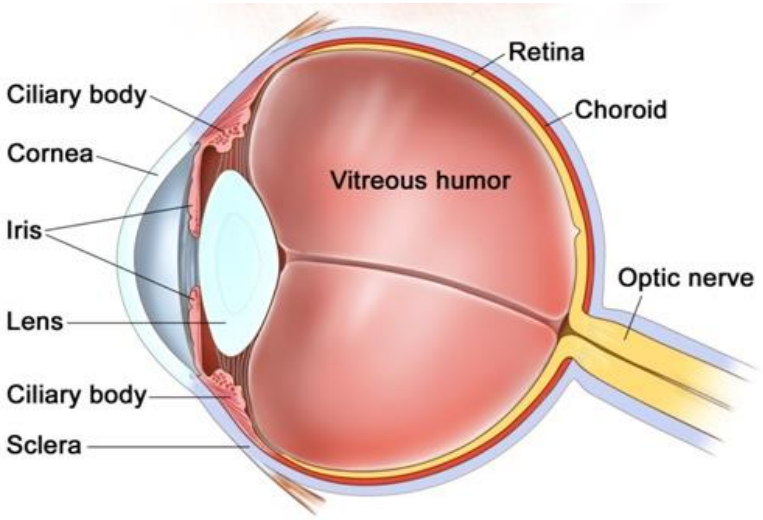


Figure 7.17: The human eye anatomy

The pupil is the aperture through which light passes. Its diameter can vary from 1.5 mm to 8.0 mm, controlling the amount of light reaching the back of the eye by a factor of about 30. As we will see, the diameter of the pupil has a significant impact on the perceived field depth and clarity of an object.

Iris. The iris is the colored muscle surrounding the pupil. When it contracts, the pupil dilates (becomes bigger), allowing more light to enter the eye. When it relaxes, the pupil constricts. The iris is often compared to the diaphragm of a camera.

The aqueous and vitreous humors are the fluids that keep the eye inflated. The aqueous humor, between the cornea and the lens, consists mainly of water. It is continuously generated and absorbed by the body at a rate or 2-3 microliters per minute. The vitreous humor is a gelatin-like substance that fills the rest of the eye. Diffusion of other substances through the vitreous humor occurs very slowly.

Crystalline lens: The lens is composed of a strong elastic membrane that encapsulates a viscous, protein-rich substance. The outer edge of the lens is attached to the ciliary muscle, which exerts a transverse tension on the lens. When the muscle contracts, the lens thickens,

increasing the refractive power of the lens. This allows us to see objects up close. When the ciliary muscle relaxes, the lens flattens, making distance vision possible. This function is extremely important to vision; we will discuss it further in a later section.

The retina is the layer of tissue lining the back of the eye. It is covered with two types of light receptors called rods and cones. These receptors convert light into nerve impulses that travel to the visual cortex where they are interpreted.

Accommodation

To focus clearly on nearby objects, the eye must increase its total refractive power. As mentioned earlier, it accomplishes this by changing the shape of the lens, a process called accommodation. Figure 7.18 illustrates how accommodation occurs.

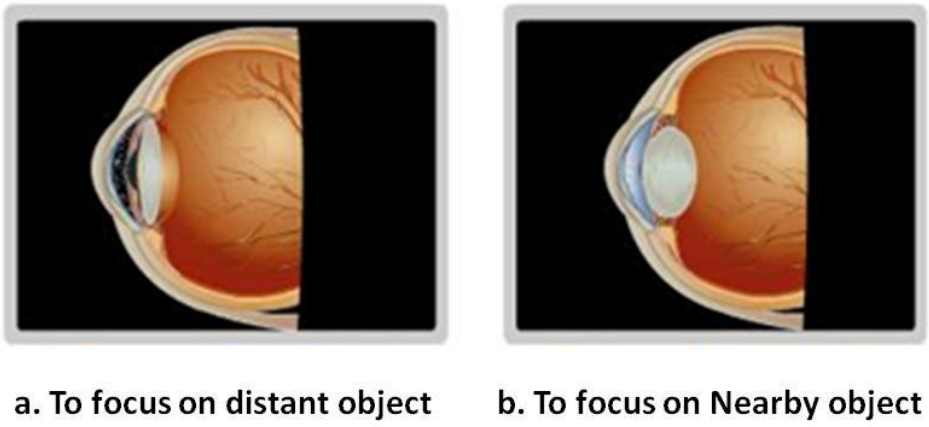


Figure 7.18: Eye accommodation, notice the difference lens shape

The ciliary muscle is attached to the suspensory ligaments, which hold the lens in place. When the ciliary muscle relaxes, the ligaments exert tension on the lens, causing it to assume the flattened shape needed for distance vision. When the ciliary muscle contracts, the ligaments are forced inwards, causing the lens to assume a spherical shape in which the radii of curvature decrease. This allows the eye to focus on nearby objects. Through accommodation, the refractive power of the lens can increase from 17 diopters up to 31 diopters, a 14 diopter increase from normal refractive power. Thus, it is said that the lens has an accommodation power of 14 diopters. As we age, the lens slowly grows and loses its elasticity. When this occurs, the ciliary muscle cannot cause it to assume a spherical shape as easily, and accommodation power diminishes. The average accommodation power for several age groups:

Children 14 diopters; age 445-50 about 2 diopters; and age of 70 about zero diopters. This phenomenon is inevitable: it is the reason people with excellent vision must begin to wear reading glasses when they reach middle age.

Eye Sight problems and corrections

The last section on the effects of pupillary diameter touched upon how the shape of the eye affects vision. Emmetropia is the state of the eye when the image is focused exactly on the retina, eye with no problems. Hyperopia, or farsightedness, occurs when the eye is too short: the image is focused behind the retina and appears fuzzy, figure 7.19b. Myopia, or nearsightedness, happens when the eye is elongated so that the image is focused just before the retina, figure 7.19c.

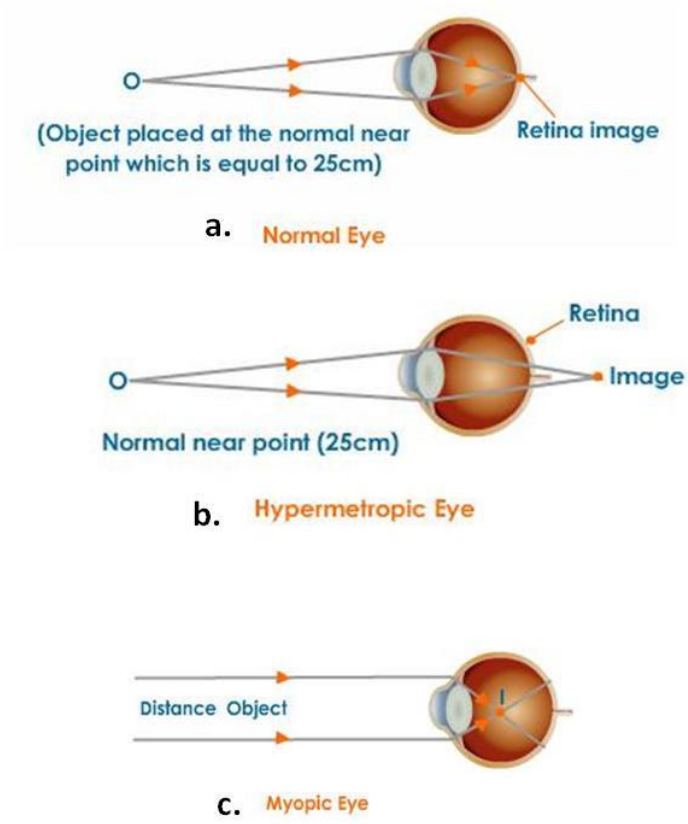


Figure 8.11: eye sight for a. normal eye, b. Hypermetropic (far) eye, and c. Myopic (near) eye

The long sighted eye can be corrected by placing a converging lens in front of the eye, figure 8.12a. The lens refracts the incoming rays more toward the principal axis before entering the eye, allowing them to converge and focus on the retina.

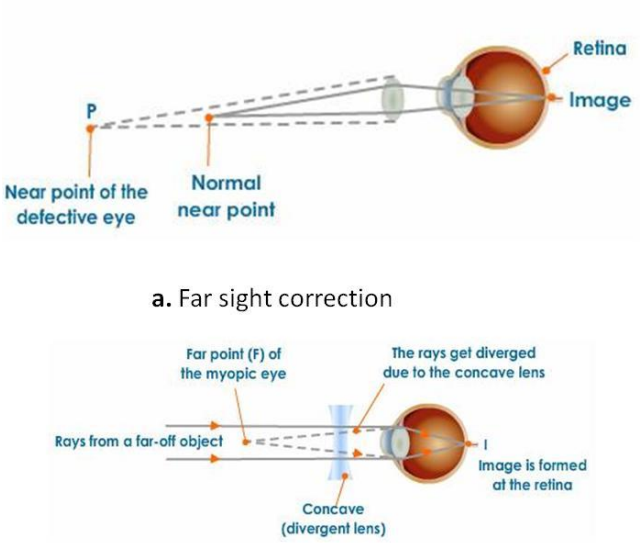


Figure 7.20: far and near sight correction

Nearsightedness can be corrected with a diverging lens, figure 8.12b. The lens refracts the rays away from the principal axis before they enter the eye, allowing them to focus on the eye. Another common vision problem, astigmatism, arises most often when the cornea is not perfectly spherical. As a result, the radii of curvature of the cornea's surface are different when viewed from the top and side, figure 7.21, which can be corrected by a cylindrical lens.

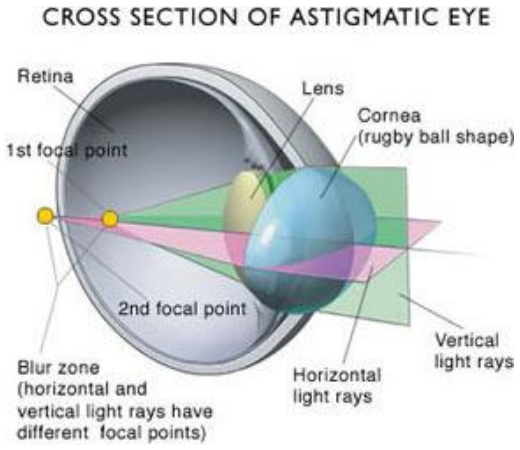


Figure 7.21: The cornea of an astigmatic eye has different radii of curvature when viewed from the top and side.

Summary

Any travelling wave can be represented mathematically as a periodic function,

$$y(x,t) = A\,sin[k\,x - \omega t] \tag{7.5}$$

The velocity of sound v is found to depend only on the bulk modulus and the density of the medium.

$$v = \sqrt{\frac{\mathrm{B}}{\rho}} \tag{7.10}$$

Usually the relation between sound intensities is specified by the sound intensity levels using a logarithmic scale. The unit of this scale is called the decibel (dB). The intensity level β, of any sound is defined in term of intensity, I, as follows:

$$\beta = 10 \log \frac{\mathrm{I}}{\mathrm{I_0}} \tag{7.13}$$

Where $\mathrm{I_0}$ is the intensity of some reference level

Doppler Effect: The wavelength is a function of the speed of propagation in the medium as well as the motion of the object within the medium.

1. when the receiver is moving towards the source

   $$\nu_r = \nu_s \left(\frac{v+u_r}{v}\right) \tag{7.19}$$

2. when it is moving away from the source, then the received frequency is given by

   $$\nu_r = \nu_s \left(\frac{v-u_r}{v}\right) \tag{7.20}$$

3. if both the source and receiver are moving and $\mathrm{u_s}$ and $\mathrm{u_r}$ are the speeds with which they are approaching each other (respectively), the Doppler Shift is

   $$\nu_r = \nu_s \left(\frac{v+u_r}{v-u_s}\right) \tag{7.21}$$

The radius of curvature of these surfaces controls the focal length of the lens and the refractive index of both the lens material and the surrounding medium, that is

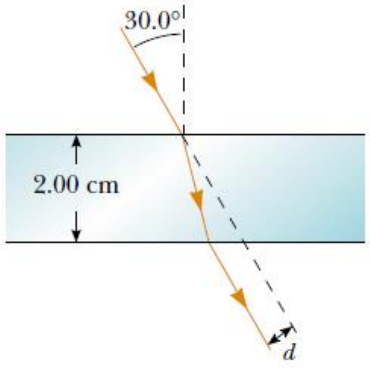
$$\frac{1}{\mathrm{f}} = (\mathrm{n_{lens}} - \mathrm{n_{medium}}) \left(\frac{1}{\mathrm{R_1}} - \frac{1}{\mathrm{R_2}}\right) \tag{7.29}$$

Problems

1. Find the wavelength of a water wave of frequency 40 Hz traveling at 120 cm/s.
2. How long should an antenna be for a receiver with frequency of 30 MHz?
3. Two waves of the same wavelength propagating in different mediums, where the first wave has a speed half of the second. What is the ratio between their frequencies?
4. If the wave length of the visible light ranges between 400 and 800 nm, find the maximum and minimum frequencies of the visible light?
5. An average adult has a hearing range from 20 to 20 kHz. What are the wavelengths of sound waves corresponding to the two frequencies?
6. The depth of an object under water is detected by transmitting light to the object and detecting the reflected sound pulse, which take an interval of 2 s, find the depth of the object?
7. A wave of amplitude 2 m, propagates to the right along the x/axis with a speed of 3 m/s and has a wavelength of 0.4 m. Write down the mathematical representation of such wave.
8. If a wave represented by the following formula $Y = 0.2\,cos\,3(x - t)\,m$, find a) its amplitude b) its wavelength c) its speed and d) its frequency.
9. What is the total output power of a loudspeaker, where its intensity over a hemisphere 20 m away is 0.001 Wm^{-2}?
10. The intensity of a guitar is double that of a flute. What is the ratio of the pressure amplitude produced buzz the guitar and the flute?
11. If the pressure difference across the eardrum is 30 Pa, How large the force acting on the eardrum wit a surface area of 0.9 cm2?
12. If the intensity of a sound wave is half of another sound wave, what is the difference in the intensity level? If the intensity level of a sound is 5 dB louder than that of a sound of intensity $5 \times 10^{-8}\,Wm^{-2}$, what is the intensity of the first sound?
13. A jet plane flying at an altitude of 4000 m produces a loudness of 40dB at the ground. What would be the intensity level at one fourth of this altitude?
14. If the intensity level of a radio station is 50 dB at a distance of 10 m, what is the intensity at this distance?
15. The security alarm on a parked car goes off and produces a frequency of 735 Hz. The speed of sound is 343 m/s. As you drive toward this parked car, pass it, and drive away, you observe the frequency to change by 78.4 Hz. At what speed are you driving?
16. A police car is moving at 29 m/sand is behind a speeding car moving faster in the same direction as the police car. The radar gun in the police car emits an 8.0x109 Hz

electromagnetic wave. The frequency of the returning wave is measured to be 318Hz less than the emitted frequency. Find the speeder's speed.

17. The passenger in a car traveling 8 m/s judges the car's horn to have a frequency of 600Hz. If the sound of the horn gets reflected from a building ahead, (a) what is the frequency of the echo the passenger hears? (b) What beat frequency will the passenger hear?
18. A stationary observer hears a frequency of 560 Hz from an approaching car. After the car passes, the observed frequency is 460 Hz. What is the speed of the car? (speed of sound in air = 343 m/s)
19. A car approaching a stationary observer emits 450 Hz from its horn. If the observer detects a frequency of 470 Hz, how fast is the car moving?
20. A narrow beam of sodium yellow light, with wavelength 589 nm in vacuum, is incident from air onto a smooth water surface at an angle $\theta_1 = 35.0^\circ$. Determine the angle of refraction θ_2 and the wavelength of the light in water.
21. When the light illustrated in Figure below passes through the glass block, it is shifted laterally by the distance d. If $n = 1.5$ what is the value of d?



22. A glass fiber, $n = 1.5$, is submerged in water, $n = 1.33$. What is the critical angle for light to stay inside the optical fiber?
23. A contact lens is made of plastic with an index of refraction of 1.50. The lens has an outer radius of curvature of +2.00 cm and an inner radius of curvature of +2.50 cm. What is the focal length of the lens?
24. A thin lens has a focal length of 25.0 cm. Locate and describe the image when the object is placed (a) 26.0 cm and (b) 24.0 cm in front of the lens.
25. An object positioned 32.0 cm in front of a lens forms an image on a screen 8.00 cm behind the lens. (a) Find the focal length of the lens. (b) Determine the magnification.(c) Is the lens converging or diverging?

26. The accommodation limits for Nearsighted Nick's eyes are 18.0 cm and 80.0 cm. When he wears his glasses, he can see faraway objects clearly. At what minimum distance can he see objects clearly?
27. A nearsighted person cannot see objects clearly beyond 25.0 cm (her far point). If she has no astigmatism and contact lenses are prescribed for her, what power and type of lens are required to correct her vision?
28. A philatelist examines the printing detail on a stamp, using a biconvex lens with a focal length of 10.0 cm as a simple magnifier. The lens is held close to the eye, and the lens-to-object distance is adjusted so that the virtual image is formed at the normal near point (25.0 cm). Calculate the magnification.
29. A lens that has a focal length of 5.00 cm is used as a magnifying glass. (a) Where should the object be placed to obtain maximum magnification? (b) What is the magnification?
30. The distance between the eyepiece and the objective lens in a certain compound microscope is 23.0 cm. The focal length of the eyepiece is 2.50 cm, and that of the objective is 0.400 cm. What is the overall magnification of the microscope?

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